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ARITHMÉTIQUE DES DROITES DE KUMMER

ARITHMETIC OF KUMMER LINES

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ARITHMÉTIQUE DES DROITES DE KUMMER

Les courbes elliptiques sont utilisées dans de nombreux protocoles cryptographiques depuis quarante ans et leur arithmétique a été largement étudiée. Dans de nombreuses applications, effectuer une multiplication scalaire efficacement sur une courbe elliptique est essentiel. Ce calcul s'effectue avec les coordonnées XZ de Montgomery et le Montgomery ladder. Ce système de coordonnées est un cas particulier de droite de Kummer.

D'un autre côté, les variétés abéliennes de dimension supérieure sont de plus en plus populaires en cryptographie, mais leur arithmétique a été moins étudiée. Le système de coordonnées universel en dimension supérieure est basé sur les fonctions thêta, et donne un autre exemple de droite de Kummer lorsque appliqué à une courbe elliptique.

Dans cette thèse, nous introduisons un cadre pour les droites de Kummer qui unifie les différents systèmes de coordonnées et les différentes formules déjà connues dans la littérature. Ce point de vue plus général permet de donner une méthode algorithmique pour déterminer des formules pour l'arithmétique des droites de Kummer, notamment en utilisant des outils en dimension 2. Nous introduisons des variantes du ladder pour effectuer la multiplication scalaire. Cela nous permet également de classifier les courbes elliptiques à partir de critères simples sur leur droite de Kummer. Nous donnons des applications aux volcans d'isogénies et à la factorisation d'entiers via ECM.

Les idées développées dans cette thèse, bien qu'orientées sur l'arithmétique des courbes elliptiques et des droites de Kummer, essaient d'utiliser au plus le cadre général des variétés abéliennes et du système de coordonnées universel des fonctions thêta, ce qui laisse la place à la généralisation de ces idées en dimension supérieure.

Mots-clés : Courbes elliptiques, droites de Kummer, cryptographie, coordonnées thêta, variétés abéliennes

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ARITHMETIC OF KUMMER LINES

Elliptic curves have been used in numerous cryptographic protocols for the past forty years and their arithmetic has been widely studied. In several applications, it is essential to perform the scalar multiplication on an elliptic curve efficiently. This computation is achieved with Montgomery XZ-coordinates and the Montgomery ladder. This system of coordinates is a particular case of Kummer lines.

On the other hand, abelian varieties of higher dimension are more and more popular in cryptography, but their arithmetic has been less studied. The universal system of coordinates in higher dimension is based on theta functions, and gives another example of Kummer lines when used on an elliptic curve.

In this thesis, we introduce a framework for Kummer lines which unifies the system of coordinates and the various formulas found in the literature. This global framework enables us to give an algorithmic method to derive formulas for the arithmetic of Kummer lines, using in particular some tools in dimension 2. We also introduce some variants of the ladder to perform the scalar multiplication. This also enables us to classify elliptic curves via simple criteria on their Kummer line. We give applications to isogeny volcanoes and integer factorization via ECM.

The ideas developed in this thesis, although focused on the arithmetic of elliptic curves and Kummer lines, try to use at most the general context of abelian varieties and the universal system of coordinates from theta functions, leaving the opportunity to generalize the ideas in higher dimension.

Keywords: Elliptic curves, Kummer lines, cryptography, theta coordinates, abelian varieties

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1 Caractérisée entre autre par une sur-utilisation des notes de pied de page.
2 Certaines personnes me rappelleraient que je n'y ai pas passé tant de temps que ça et que je suis tout juste capable de faire visiter la ville :'). Aussi, je m'attendais à moins de pluie, mais c'est un piège classique apparemment…
3 Oui, je me suis relu. Non, il n'y a pas de faute.
4 Une pensée pour l'abonnement TGVMax également.

RÉSUMÉ ÉTENDU EN FRANÇAIS

Cette thèse se concentre sur l'étude des droites de Kummer, en particulier leur arithmétique. Étant donné une courbe elliptique E, sa droite de Kummer est le quotient $E/\pm 1$, isomorphe à la droite projective $\mathbb{P}^1$ en tant que courbe. Puisqu'un point P et son opposé sur une courbe elliptique sont symétriques par rapport à l'axe des abscisses, cela revient à travailler uniquement avec la coordonnée $x(P)$ du point, l'élément neutre étant envoyé sur le point à l'infini de la droite projective. Bien que la droite de Kummer n'ait plus une structure de groupe, elle peut être utilisée pour calculer efficacement la multiplication scalaire $n \cdot P$. En effet, Montgomery a introduit le « Montgomery ladder » [Mon87] qui calcule $x(n \cdot P)$ et $x((n+1) \cdot P)$ pour n'importe quel point $P \in E$ et entier $n \in \mathbb{Z}$. Appliqué à une courbe de Montgomery donnée par une équation

$$E : By^2 = x(x^2 + \mathcal{A}x + 1)$$

où $\mathcal{A}$, B sont des éléments du corps de base k, ces formules sont bien plus efficaces que faire la multiplication complète avec la loi de groupe de la courbe étant donné que l'on ne travaille qu'avec une seule coordonnée. De fait, le « Montgomery ladder » est le standard utilisé en cryptographie à base de courbes elliptiques depuis bientôt 40 ans. Par ailleurs, la donnée de $x(n \cdot P)$ et $x((n+1) \cdot P)$ permet de récupérer complètement $n \cdot P$ sur la courbe. Ainsi, la méthode privilégiée pour effectuer le produit scalaire sur une courbe elliptique utilise les droites de Kummer.

Si l'on s'intéresse tant à l'arithmétique des droites de Kummer, c'est parce que les courbes elliptiques sont très populaires en cryptographie à clé publique comme mentionné précédemment, et la multiplication scalaire y est centrale. C'est l'objet du Chapitre 1, qui présente le problème du logarithme discret, au cœur de nombreux protocoles comme l'échange de clés de Diffie-Hellman ou le schéma de signature d'ElGamal. Les courbes elliptiques ont été introduites en cryptographie indépendamment par Koblitz et Miller [Mil86 ; Kob91] car le logarithme discret y est résolu en temps exponentiel sur un ordinateur classique, ce qui correspond au meilleur cas que l'on puisse espérer. L'échange de clés (ECDH) et la signature (ECDSA) à base de courbes elliptiques sont des algorithmes encore utilisés aujourd'hui et dans de nombreux contextes, comme la biométrie des passeports, les puces RFID, l'authentification SSH, ….

Bien sûr, ces méthodes reposant sur le logarithme discret sont vulnérables à l'ordinateur quantique et l'algorithme de Shor. On introduit alors brièvement dans ce chapitre quelques notions de cryptographie à base d'isogénies, qui sont des morphismes entre courbes elliptiques à noyau fini, et permettent de construire des protocoles résistants à l'ordinateur quantique. Une dernière application des courbes elliptiques à la cryptographie est la factorisation d'entiers via ECM, efficace pour trouver des facteurs premiers de « petite » ou « moyenne » taille.

Au vu de leur usage, il est essentiel d'avoir une multiplication scalaire rapide sur les courbes elliptiques. Le « Montgomery ladder » couplé à des courbes de Montgomery est la méthode privilégiée depuis plusieurs décennies du fait de l'efficacité des formules, mais s'agit-il des « meilleures » ? L'algorithme de Montgomery repose sur deux types de formules :

- les formules de doublement, qui calculent $x(2 \cdot P)$ à partir de la donnée de $x(P)$;
- les formules d'addition différentielle, qui calculent $x(P + Q)$ à partir de la donnée de $x(P)$, $x(Q)$ et $x(P - Q)$.

Le Chapitre 2 s'attarde sur les aspects arithmétiques et comment mesurer l'efficacité de formules en discutant du coût relatif des différentes opérations élémentaires. Évaluer l'efficacité d'une formule revient donc à compter le nombre de multiplications et carrés qui interviennent et à comparer. Le « Montgomery ladder » y est également introduit. Faisons par exemple la comparaison avec le modèle thêta carré / twisté, qui est un autre cas de droite de Kummer avec des formules alternatives. Le doublement en thêta carré peut être plus rapide qu'en Montgomery, là où l'addition différentielle est toujours plus lente, voir le Tableau 1.

Opération	Montgomery	Thêta carré / twisté
Doublement	$2\mathbf{M} + 2\mathbf{S} + 1\mathbf{m}_0$	$4\mathbf{S} + 2\mathbf{m}_0$
Addition différentielle	$2\mathbf{M} + 2\mathbf{S} + 1\mathbf{m}$	$2\mathbf{M} + 2\mathbf{S} + 1\mathbf{m} + 1\mathbf{m}_0$

Table 1 : Comparaison des formules entre deux modèles de droites de Kummer

Une autre question que l'on peut se poser sur les droites de Kummer est comment les décrire efficacement et de façon unifiée. La droite de Kummer d'une courbe de Montgomery E est obtenue via le quotient $E/\pm 1$, ce qui revient à ne considérer que la coordonnée x. Le modèle thêta carré / twisté est dérivé de la théorie des fonctions thêta. Une définition combinatoire est introduite dans la Section 4.3 du chapitre sur les courbes elliptiques, une droite de Kummer est la donnée d'un revêtement $\pi : E \to \mathbb{P}^1$ de degré 2 avec exactement 4 points de ramification dont un « marqué » (qui correspond à l'élément neutre de la courbe elliptique), où E est une courbe elliptique, ou de manière équivalente π est une application à travers laquelle la multiplication par -1 se factorise. Cette description unifiée est le point de départ des différentes contributions de cette thèse :

- peut-on classifier les différents modèles de droites de Kummer, et peut-on avoir des critères simples sur la courbe elliptique pour qu'elle ait un modèle de droite de Kummer prescrit ? C'est l'objet du Chapitre 8 ;
- les formules de doublement sur une courbe de Montgomery sont rapides notamment car une courbe de Montgomery a un point de 2-torsion rationnel, et donc une 2-isogénie rationnelle. Pour un modèle de droite de Kummer donné avec un point de 2-torsion rationnel, peut-on obtenir algorithmiquement des formules de 2-isogénies efficaces, et par conséquent des formules de doublement ? C'est l'objet du Chapitre 6 ;

- peut-on exploiter l'existence d'un point de 2-torsion rationnel pour accélérer l'addition différentielle, avec une approche similaire que pour le cas du doublement, et peut-on de même obtenir ces formules algorithmiquement ? C'est l'objet du Chapitre 7.

Le Chapitre 3 introduit les éléments importants de la théorie des variétés abéliennes, particulièrement les isogénies et la polarisation. Ces éléments sont réutilisés dans le Chapitre 4 qui traite des courbes elliptiques, qui correspondent aux variétés abéliennes de dimension 1. Enfin, le Chapitre 5 est le dernier chapitre de contenu préliminaire et présente la théorie du groupe thêta et des fonctions thêta, qui sont des éléments centraux dans la seconde partie de cette thèse. Bien que cette thèse porte sur l'arithmétique des droites de Kummer et donc se concentre sur les courbes elliptiques, il est utile d'avoir la théorie des variétés abéliennes en plus, d'une part car l'on va être amené à utiliser des outils de dimension 2 pour améliorer l'arithmétique en dimension 1, et d'autre part car certains éléments de notre approche générale devraient pouvoir se généraliser en dimension supérieure.

Cette thèse a donné lieu à deux publications [RS24 ; RS25] et une pré-publication [BRS25], qui correspondent au contenu des Chapitres 6, 7 et 8 respectivement. Nous allons désormais détailler le contenu de ces chapitres.

CHAPITRE 6 : 2-ISOGÉNIES ENTRE DROITES DE KUMMER

Dans ce chapitre, on utilise la théorie du groupe thêta pour trouver de manière systématique des formules de 2-isogénies à partir d'une droite de Kummer avec un point de 2-torsion rationnel. Si $f : E \to E'$ est une 2-isogénie entre courbes elliptiques dont le noyau est donné par un point de 2-torsion rationnel T, alors f est notamment invariante par translation par T. C'est cette propriété que l'on souhaite exploiter sur l'application induite entre les droites de Kummer.

Ce que l'on souhaite donc, c'est obtenir des sections X' et Z' au-dessus du diviseur $2(0_{E'})$ invariantes par translation par T et dérivées des sections X et Z au-dessus de $2(0_E)$. En effet, les coordonnées projectives de la courbe elliptique nous fournissent un système de coordonnées sur la droite de Kummer via les générateurs des fibrés en droites suivants :

$$\Gamma((0_E)) = \langle Z_0 \rangle,\ \Gamma(2(0_E)) = \langle X, Z := Z_0^2 \rangle,\ \Gamma(3(0_E)) = \langle XZ_0, Y, ZZ_0 \rangle.$$

La théorie du groupe thêta de Mumford va nous permettre de répondre à ces questions. Si D' est un diviseur de E', et D un diviseur de E tels que f^*D' soit linéairement équivalent à D, alors on dit que D' est une descente de D. D'une part, on a une bijection entre les descentes de D et les possibles relevés du noyau K au groupe thêta $G(D)$. D'autre part, ce groupe agit sur les sections au-dessus de D, et si $\widetilde{K}$ est un relevé de K au groupe thêta, on a un isomorphisme canonique $\Gamma(D)^{\widetilde{K}} \simeq \Gamma(D')$, ces deux résultats font partie du théorème 5.1.7.

En spécialisant aux descentes de $2(0_E)$ et $4(0_E)$, on peut ainsi retrouver des sections X' et Z' comme l'on souhaite. En effet, si $4(0_E)$ descend sur

$2(0_{E'})$, alors on a des sections X' et Z' invariantes par un relevé $\widetilde{K}$ du noyau K, lui-même généré par le point de 2-torsion rationnel T.

On fait alors le lien entre l'action du groupe thêta et la translation par T. D'une part, comme T est de 2-torsion, la translation par T descend en une homographie $t_T : \mathbb{P}^1 \to \mathbb{P}^1$ sur les droites de Kummer. Si l'on considère un relevé affine de cette homographie, on obtient un scalaire λ_T tel que $t_T^2 = \lambda_T \, \mathrm{id}$. Ce scalaire, appelé le type de T, est bien défini à un carré près, et sera important dans la suite notamment pour des questions de classification. C'est aussi pour cette raison que l'on étudie les descentes de $2(0_E)$, le choix d'un relevé du noyau au groupe thêta permet de retrouver ce type. De fait, projectivement le relevé du noyau va agir sur les sections comme la translation par T, ce qui mène à la méthodologie suivante pour trouver des formules :

1. calculer l'action de la translation par T sur X^2, XZ et Z^2 (qui sont des sections au-dessus de $4(0_E)$) ;

2. déterminer via l'isomorphisme $\Gamma(4(0_E))^{\widetilde{K}} \simeq \Gamma(2(0_{E'}))$ une base de sections invariantes X', Z' et définir l'application $f : (X : Z) \mapsto (X' : Z')$;

3. récupérer le modèle de la droite de Kummer décrit par X' et Z' en déterminant les 4 points de ramification à partir de ceux de la droite de Kummer associée à X et Z.

Ce cadre fournit donc une méthode qui permet de façon systématique et algorithmique de retrouver des formules de 2-isogénies entre droites de Kummer, on récupère ainsi de manière unifiée différentes formules de la littérature, on pourra voir les Théorèmes 6.3.3, 6.3.7 et 6.3.10. Cela fournit également des formules de doublement en utilisant la duale des 2-isogénies calculées.

Une application que l'on propose est le « hybrid ladder ». Grâce au cadre unifié, on sait changer facilement via des homographies de modèle de droite de Kummer. En exploitant ces changements, on peut ainsi essayer de bénéficier de la meilleure addition différentielle du modèle de Montgomery en la couplant au meilleur doublement du modèle thêta twisté. Notre variante coûte $3\mathbf{M} + 6\mathbf{S} + 2\mathbf{m}_0$ par bit, tandis que le « Montgomery ladder » coûte lui $5\mathbf{M} + 4\mathbf{S} + 1\mathbf{m}_0$ (pour un point de base normalisé). Ainsi, lorsque $\mathbf{m}_0$ est suffisamment petit, on peut espérer battre le « Montgomery ladder » *lorsque la courbe de Montgomery associée a toute sa 2-torsion rationnelle*. Par ailleurs, notre variante est plus rapide que le « theta ladder » (« Montgomery ladder » appliqué au modèle thêta twisté), qui coûte $3\mathbf{M} + 6\mathbf{S} + 3\mathbf{m}_0$ par bit.

CHAPITRE 7 : DEMIES ADDITIONS DIFFÉRENTIELLES SUR DES DROITES DE KUMMER

Ce chapitre est la suite directe du précédent, où l'on s'est intéressé à avoir des formules de doublement efficaces via des 2-isogénies. On veut exploiter à nouveau ces dernières pour obtenir de meilleures formules d'addition

différentielle. Notons $\varphi : E \to E'$ une 2-isogénie donnée par un noyau rationnel $\langle T \rangle$. On veut étudier l'isogénie entre *surfaces abéliennes*

$$F : E \times E \to E \times E, (P, Q) \mapsto (P + Q, P - Q).$$

Obtenir des formules pour F donne directement des formules d'addition différentielle. Idéalement, on souhaiterait relier F à l'isogénie diagonale $\Phi = (\varphi, \varphi)$, ce afin de ré-utiliser le calcul de φ nécessaire pour le doublement. Cela nous amène à introduire une nouvelle famille de formules, appelées les « demies additions différentielles », qui permettent de calculer $P + Q$ sur la droite de Kummer à partir de $P - Q$, $\varphi(P)$ et $\varphi(Q)$.

Une obstruction à la méthode du chapitre précédent est que l'on ne peut pas factoriser directement F par Φ puisque l'on n'a pas d'inclusion des noyaux. En revanche, la théorie du groupe thêta nous permet quand même de relier les sections au-dessus de $2(0_E) * 2(0_E)$ aux sections au-dessus de $2(0_{E'}) * 2(0_{E'})$, où le produit de diviseurs est donné en tirant en arrière les diviseurs selon la première et deuxième composante de $E \times E$. On exploite alors ici le groupe thêta entre surfaces abéliennes pour obtenir des formules sur des courbes elliptiques. Cette méthode nous permet de dériver des formules de demies additions différentielles, on pourra voir les Sections 7.4.2 et 7.5 pour des exemples de telles formules.

L'application principale est le « half ladder », une autre variante du « Montgomery ladder ». Dans le ladder classique, on effectue à chaque étape une addition différentielle complète et un doublement complet, ce qui fait que l'on reste toujours sur la droite de Kummer initiale. Nous proposons plutôt de d'abord considérer une chaîne de 2-isogénies, et à l'aide des duales et des demies additions différentielles, on peut remonter « en arrière » jusqu'à la droite de Kummer initiale tout en effectuant la multiplication scalaire.

Sans pré-calcul et sur une courbe de Montgomery possédant toute sa 2-torsion rationnelle (c'est-à-dire une courbe elliptique avec un modèle thêta twisté), nous achevons une complexité de $4\mathbf{M} + 4\mathbf{S} + 2\mathbf{m}_0$ par bit, ce qui correspond toujours au meilleur cas du « Montgomery ladder » où le point de base est normalisé avec une faible coordonnée x. De plus, la chaîne initiale de 2-isogénies peut rentrer dans un pré-calcul, on peut alors se comparer à la variante avec pré-calcul du « Montgomery ladder » proposée dans [Oli+17, Alg. 4]. Si l'on sauve du temps sur le pré-calcul (on a juste à faire une 2-isogénie par bit, là où les auteurs ont besoin d'un doublement complet), on perd $1\mathbf{m}_0$ à chaque étape.

Par ailleurs, on explique brièvement comment l'on pourrait généraliser ce « half ladder » en dimension supérieure, où l'on espère voir un gain significatif par rapport à la multiplication scalaire actuelle.

CHAPITRE 8 : MODÈLES DE DROITES DE KUMMER ET REPRÉSENTATIONS GALOISIENNES

Les chapitres précédents s'intéressaient à trouver de manière exhaustive des formules efficaces, et permettaient d'avoir une bonne compréhension

unifiée des modèles de droites de Kummer. Cette étude permet alors de s'intéresser à des résultats de classification dans ce dernier chapitre. Bien que l'article correspondant avait été commencé avant ceux sur les 2-isogénies et les demies additions différentielles, il a été le dernier à être soumis, plus de recul ayant été pris sur les droites de Kummer entre temps et permettait d'apporter également plus d'éléments.

Ce chapitre commence par rentrer plus en détails sur les spécificités des différents modèles de droites de Kummer, particulièrement celui associé à une courbe de Montgomery, mais aussi ceux obtenus à partir des fonctions thêta, de leur carré et de leur twist par les thêta constantes. On s'intéresse aussi aux différentes formules de conversion entre ces modèles, et quand est-ce que celles-ci sont rationnelles. C'est un premier pas vers notre classification.

On s'intéresse ensuite à différentes approches pour trier nos modèles. Tout d'abord, on utilise des critères de représentations galoisiennes et de courbes modulaires. On montre ainsi par exemple qu'une courbe elliptique a un modèle de Montgomery si et seulement si elle a un noyau rationnel généré par un point de 4-torsion (ce dernier n'étant pas nécessairement rationnel), tandis qu'elle admet un modèle thêta si et seulement elle possède deux noyaux cycliques d'ordre 4 rationnels et indépendants. Ces conditions en termes de noyaux rationnels se reformulent en l'existence de 4-isogénies cycliques et rationnelles sur les modèles concernés.

Cette interprétation en termes de 4-isogénies donne une application naturelle aux volcans d'isogénies : on peut déterminer le modèle d'une courbe elliptique ordinaire en lisant directement son volcan de 2-isogénies associé, il suffit de vérifier l'existence ou non de 4-isogénies cycliques.

Une dernière classification et interprétation que l'on propose se fait en termes du groupe thêta, notamment on ré-utilise le type d'un point que l'on avait introduit dans le chapitre sur les 2-isogénies. On dit qu'un point est de type de Montgomery si ce dernier est un carré. Une courbe elliptique a alors un modèle de Montgomery si et seulement si elle possède un point de type de Montgomery, ce que l'on relie aussi aux conditions précédentes sur les représentations galoisiennes, cela correspond au grand Théorème 8.4.3 donnant 12 conditions équivalentes au fait qu'un point soit de type de Montgomery. Sur un corps fini k, il n'y a qu'une seule classe non-triviale dans $k^*/k^{*,2}$, on dit alors qu'un point est de type anti-Montgomery si ce n'est pas un carré, et on donne dans le Théorème 8.4.4 des conditions similaires pour qu'un point soit de type anti-Montgomery. On décrit aussi dans cette section comment apparaissent les modèles thêta twistés dans notre étude.

Enfin, on regarde dans la dernière partie si le modèle thêta peut être intéressant pour ECM, notamment si l'on peut atteindre le même niveau de performance que les courbes de Koblitz, qui ont plus de contraintes que des courbes de Montgomery avec 2-torsion rationnelle, et qui sont donc plus difficiles à trouver.

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INTRODUCTION

CONTEXT

For the past forty years, elliptic curves have been extensively used in cryptographic protocols, whether it is key exchange (ECDH), digital signature (ECDSA), pairings-based identity-based encryption (IBE), or isogeny-based key encapsulation method (e.g. CSIDH). These protocols are used in various context, whether to authenticate on a remote server via SSH, to certify data on a biometrical passport, to identify items with RFID tags, and so on. Elliptic curve cryptography has been preferred to the more traditional protocol RSA [RSA78] because the attacks on ECC are exponential, compared to sub-exponential ones for RSA, leading to bigger parameter sizes to achieve the same security levels as elliptic curves. Moreover, they can be used in post-quantum protocols via isogeny-based cryptography. In the end, elliptic curves are extensively used in cryptography and having an efficient arithmetic is therefore essential.

The most important computation to perform on an elliptic curve is the scalar product, that is computing $n \cdot P = P + \cdots + P$. Improving this operation requires among others finding new systems of coordinates. Some of them are available to all elliptic curves, like the Weierstrass form, and some are restricted to certain families of curves, like the Montgomery form [Mon87], the Edwards form [Edw07; BL07b], the twisted Edwards form [Ber+08a; His+08], the Hessian form [Sma01], the generalized Hessian form [FJ10], the theta coordinates [GL09] and theta squared coordinates [Bos+13; HR19]. We also mention a mixed system [CGF08] where the addition law takes coordinates in one form and returns coordinates in another form.

Montgomery [Mon87] explained how one can compute $x(n \cdot P)$ with only the knowledge of the x-coordinate $x(P)$ using the 'Montgomery ladder', leading to way faster formulas than when using the full coordinates of the point P. Moreover, this computation can be achieved in constant time, and can therefore be used in a cryptographic context. In fact, this approach is so efficient that it is still used nowadays. There are two important operations required for the Montgomery ladder:

- the differential addition, that is given $x(P)$, $x(Q)$ and $x(P - Q)$ with $P \neq Q$, compute $x(P + Q)$;
- the doubling, that is given $x(P)$, compute $x(2 \cdot P)$.

Using the x-only arithmetic amounts to looking at the map $x : E \to E/\pm 1$, which is an example of a Kummer line and the main object of this thesis. Other examples of Kummer lines can be derived from the theta and theta squared / twisted coordinates cited previously. These models have been known for a while, as well as differential addition and doubling formulas.

Going the other way, given a Kummer line, how does one find formulas? Is the Kummer line related to some usual model?

RESULTS

The aim of this thesis is to answer these questions, we give a brief overview of our results here. These were made available in the following publications [RS24; RS25] and the following pre-print [BRS25].

We first describe Kummer lines in a unified manner as done in Section 4.3. A Kummer line is a degree 2 covering $\pi : E \to \mathbb{P}^1$ with exactly 4 ramification points, corresponding to the 2-torsion. Since $0_E \in E(k)$ is rational, this is a marked point. The ramification points are the important one to describe the Kummer line. This description was certainly known before our work. We look at Montgomery curves, theta and theta squared coordinates through this same scope.

Our first goal was to find doubling formulas in Chapter 6. This can be done via isogenies, that is surjective homomorphisms of abelian varieties with finite kernel. When an elliptic curve E has a rational 2-torsion point E, we can look at the rational isogeny $\varphi : E \to E' := E/\langle T \rangle$ with kernel $\{0_E, T\}$. φ can be composed with the dual isogeny $\widetilde{\varphi} : E' \to E$ to recover the multiplication by 2: $\widetilde{\varphi} \circ \varphi = [2]_E$, hence we want to derive formulas for the induced map by φ on the Kummer line: $\varphi : \mathbb{P}^1 \to \mathbb{P}^1$.

Using Mumford's theta group theory, we provide a framework in Section 6.2 that, given a Kummer line with a rational 2-torsion point, enables one to find 2-isogenies formulas between the Kummer lines algorithmically. We recover old and new formulas in a unified manner (see for instance Theorems 6.3.3, 6.3.7 and 6.3.10).

With this unified point of view, we are also able to switch between various models of Kummer lines more easily. A reason to do so is that we would like to benefit from the best arithmetic in every case:

- Montgomery curves have faster differential addition formulas than theta squared coordinates;
- theta squared coordinates have faster doubling formulas than Montgomery curves.

We introduce the hybrid ladder in Section 6.4, a variant to Montgomery ladder exploiting the above fact. This hybrid ladder is always faster than the Montgomery ladder used with theta squared coordinates, and can beat Montgomery curves when the constants used in the algorithm are small enough. One caveat though is that we require a Montgomery curve with full rational 2-torsion (we show that having theta twisted coordinates is equivalent to having a Montgomery curve with full rational 2-torsion).

A natural continuation was then to look at the differential addition on E. This time, we have to study the map $F : (P, Q) \mapsto (P + Q, P - Q)$. This is an isogeny from $E \times E$ to itself, which is an abelian surface. We cannot factor this isogeny as easily as the doubling one, this leads to a new kind of

formulas. We introduce 'half differential addition' formulas in Section 7.2, that is given a 2-isogeny φ on the elliptic curve E, compute $x(P+Q)$ from the data of $x(\varphi(P))$, $x(\varphi(Q))$ and $x(P-Q)$. The idea behind these formulas is that, since we need to compute $x(\varphi(P))$ at some point to perform the doubling, we would like to re-use this computation.

By extending our framework to a product of elliptic curves as in Section 7.4, and using once again the theta group, we can provide an algorithmic way to compute differential addition and half differential addition formulas on a Kummer line given a 2-isogeny φ. For instance with theta coordinates we give the complete calculation details in Section 7.4.2, and in Section 7.5 we deal with `Curve25519`, showing the flexibility of our framework. The extension is natural because the theta group is a general construction for abelian varieties, it is not restricted to elliptic curves.

An application of these formulas is yet another variant of the Montgomery ladder, called the half ladder and introduced in Section 7.3. Without pre-computations, we achieve the same complexity as the Montgomery ladder, unconditionally whether the base point is normalized or not, given the Montgomery curve has full rational 2-torsion once again. When compared to the right-to-left Montgomery ladder (a variant with pre-computations), or pre-computation is way faster, but the main loop is slightly slower, this again depends on whether the constants involved are small or not.

Moreover, our approach generalizes well in higher dimension, giving a half ladder in dimension 2 as explained briefly in Section 7.6, whereas the right-to-left Montgomery ladder does not have an interesting generalization to the best of our knowledge. This is one of the main reason to use tools coming from the abelian variety scope such as the theta group and theta coordinates rather than ones specific to elliptic curves.

Finally, studying the arithmetic helped to have a better vision of the Kummer line object. We can give conversions between the different models as in Section 8.2, but more importantly we can provide simple and various criteria on elliptic curves to check if they have an interesting model of Kummer line or not. We give interpretations in terms of Galois representation and modular curves (Section 8.3, see Propositions 8.3.12 and 8.3.14), in terms of the theta group (Section 8.4, see Corollaries 8.4.3 and 8.4.4), and in terms of isogeny volcanoes (Section 8.5, see Table 8.2).

Having this kind of criteria helps us find curves with good arithmetic and good torsion properties, as desired in the ECM algorithm, details are provided in Section 8.6.

PLAN OF THE THESIS

This thesis is split into two parts. The first one introduces the various key concepts we need. We first discuss public key cryptography in Chapter 1 as this is the main usage of elliptic curves algorithmically, as well as details on the arithmetic by comparing the various elementary operations and introducing the Montgomery ladder in Chapter 2. We then go into the

general theory of abelian varieties in Chapter 3, before specializing to the case of elliptic curves — which are abelian varieties of dimension 1 — and introducing their specific properties in Chapter 4. In particular, we define properly Kummer lines in this section. We end this first part with Mumford's theta group theory in Chapter 5, providing the theta functions we are using throughout this thesis as well as various tools.

In the second part, we go into the improvements of the arithmetic in dimension 1 as described above. We first start by introducing our framework for elliptic curves and derive 2-isogenies formulas in Chapter 6, we give our first variant of the Montgomery ladder, called the hybrid ladder. We then extend this framework to a product of elliptic curves in Chapter 7 to find the new half differential addition formulas, and we use those to build our half ladder. Finally, we go into the classification aspects with various descriptions in Chapter 8.

Part I

USAGE OF ELLIPTIC CURVES IN CRYPTOGRAPHY

1 PUBLIC KEY CRYPTOGRAPHY

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Cryptography is everywhere in our lives and applications are numerous, whether to exchange secret information over the Internet without a third-party having access to them, to sign electronic documents, to identify items with RFID tags in warehouses and shops, to authenticate on a remote server with SSH, to certify the security of a connection with the TLS layer [Res18], and so on[1]...

Cryptography splits into two categories: symmetric and asymmetric. Symmetric cryptography is great at encrypting messages, given two (or more) participants share a common secret, known as a secret key to encrypt and decrypt. It is also used to derive hash functions. Asymmetric cryptography — also known as public key cryptography (PKC) — relies on key pairs, one is private and must be kept secret, the other one is publicly available and should not leak any information about the private key. One major application of PKC is key exchange so that two (or more) entities can share a common secret which they can then use for symmetric encryption. This family of protocols is a key encapsulation mechanism (KEM), the most renown algorithm being the Diffie-Hellman key exchange [DH76]. Another important application is signature schemes to verify and authenticate the source of a message, the most famous is ElGamal signature scheme. It was later derived into DSA and standardized by the NIST [Sta+23].

In the context of this thesis, we are interested in cryptography based on elliptic curves, which are given by equations of the form

$$y^2 = x^3 + ax + b.$$

1 An even more exhaustive list can be found at [MvV97, Table 1.1].

They have been popularized in cryptography by Koblitz and Miller [Kob91; Mil86] independently with a variant of Diffie-Hellman known as ECDH. An elliptic curve variant of DSA also exists, it is named ECDSA and is still widely used in our days, for instance in biometric passports.

As of now, the cited protocols are known as 'pre-quantum' ones. The reason is that they are secure for our traditional computers. This is no longer true if an attacker uses Shor's algorithm [Sho97a], running on a *quantum computer* as it has a polynomial time complexity. Symmetric cryptography can be attacked with Grover's algorithm [Gro96] — also running on quantum computers — but it only saves a square root in the complexity expression. It is hard to predict if a quantum computer capable of breaking current cryptographic standards will be built in the foreseeable future. Moreover, schemes have to be adapted to their use cases: RFID tags which are physical devices are less vulnerable because the operation cost of quantum computers exceeds by far the price of a warehouse item protected by a tag. Therefore, we will see both pre-quantum and post-quantum cryptography co-exist in the upcoming future.

Isogenies are morphisms between elliptic curves and are an approach to building post-quantum cryptography. Diffie-Hellman key exchange has a variant which was first introduced by Couveignes in 1997 and rediscovered independently by Rostovtsev-Stolbunov in 2006 [Cou06; RS06]. They are currently a promising option for post-quantum cryptography, with for instance the SQISign signature scheme [De +20]. Isogeny-based cryptography is an active research area, and a reason why arithmetic of elliptic curves and more generally abelian varieties are an important topic.

The cryptanalysis of most of these cryptosystems, whether it is RSA [RSA78], Diffie-Hellman key exchange or ElGamal signature scheme relies on factoring numbers with medium-sized factors, the state of the art to achieve this is Elliptic Curve Method (ECM) [Len87].

In Section 1.1, we will first introduce the discrete logarithm problem which is the foundation of several protocols, including the Diffie-Hellman key exchange and ElGamal signature scheme which are presented in Section 1.2. In Section 1.3, we will focus on the factorization of integers via elliptic curves (ECM), and then we will introduce some isogeny-based cryptography in Section 1.4. Finally, we will discuss side-channel attacks and how protocols can be made resistant to those in Section 1.5.

1.1 DISCRETE LOGARITHM PROBLEM

The discrete logarithm problem (DLP) in a cyclic group is one of the main security assumption in several PKC protocols as we will see in Section 1.2. Basics on the topic are introduced in [MvV97, § 3.6].

Definition 1.1.1 (DLP) *Let* G *be a cyclic group of order* n, g *a generator of* G, *and* $h \in G$. *The discrete logarithm of* h *in base* g *is the unique* $x \in \mathbb{Z}/n\mathbb{Z}$ *such that* $g^x = h$, *it is denoted* $\log_g(h)$.

In this context, the discrete logarithm problem (DLP) is, given g *and* h*, find* $\log_g(h)$.

In the sequel, n will denote the cardinality of G and ℓ its binary size, that is $\ell = \lceil \log_2 n \rceil$. The reason is that when we will talk about complexity, the interesting parameter will be ℓ and not n, but

The hardness of DLP depends on the group G. We will consider some examples here:

- If G is the additive group $\mathbb{Z}/n\mathbb{Z}$, 1 is a generator and for any $g \in G$, $\log_1(g) = g$, DLP is solved in constant-time.
- If one consider a generic generator g of $\mathbb{Z}/n\mathbb{Z}$ instead, then g is coprime to n, there is a Bézout relation $ug + vn = 1$ with $u, v \in \mathbb{Z}$ and therefore $\log_g(h) \equiv uh \bmod n$ for any $h \in G$. Since extended Euclidean algorithm runs in $O((\log_2 n)^2) = O(\ell^2)$ operations, this is polynomial and not secure too [MvV97, Alg. 2.107].

We will provide more details on these two cases later in this section.

- On the multiplicative group $\mathbb{F}_p^*$ where p is prime, the best known attacks are sub-exponential.
- On elliptic curves, the best known attacks are exponential.

On a generic group, the most naive algorithm is the exhaustive search, which runs in $O(n) = O(2^\ell)$ and is therefore exponential. A square root can be saved on the complexity with the baby-step giant-step algorithm, using a time-memory tradeoff. We are still in the context of $G = \langle g \rangle$ of order n; let $m = \lceil \sqrt{n} \rceil$ and assume $h = g^x$. Then x can be written as $x = im + j$ with unique $0 \leqslant i, j < m$ and $g^x = (g^m)^i g^j$. Therefore, we are looking for elements i and j such that $h\tilde{g}^i = g^j$, where $\tilde{g} = g^{-m}$:

Algorithm 1.1.2 (Baby-step giant-step) *Let* G *be a cyclic group of order* n *and generator* g, $m = \lceil \sqrt{n} \rceil$ *and* $h = g^x \in G$. *Set* $\tilde{g} = g^{-m}$:

Baby-step

1. *Compute the list* $L_1 = [(0, g^0), (1, g^1), \dots, (m-1, g^{m-1})]$.

Giant-step

2. *Compute the list* $L_2 = [(0, h\tilde{g}^0), (1, h\tilde{g}^1), \dots, (m-1, h\tilde{g}^{m-1})]$.
3. *Concatenate both list into* L*, and sort* L *by its second component.*
4. *Look for two consecutive elements with the same second component. These two elements have the form* $(i, k) \in L_1$ *and* $(j, k) \in L_2$ *for some* $k \in G$*, return* $x = im + j$.

The collision is illustrated in Fig. 1.1.

In terms of complexity, both list in steps 1 and 2 can be computed with $O(m)$ group multiplications. Sorting the list L in step 3 can be achieved in $O(m \log m)$ comparisons, which is negligible compared to $O(m)$ multiplications. The last step takes at most $O(m)$ look-up in the list L, again negligible. Therefore, the total cost is $O(m) = O(2^{\ell/2})$ group multiplications. This is still exponential, but saves a factor 2, at the cost of storing $O(m) = O(2^{\ell/2})$ elements. Pollard's rho algorithm [MvV97, § 3.6.3] achieves the same complexity time-wise, but with a negligible storage required and is generally

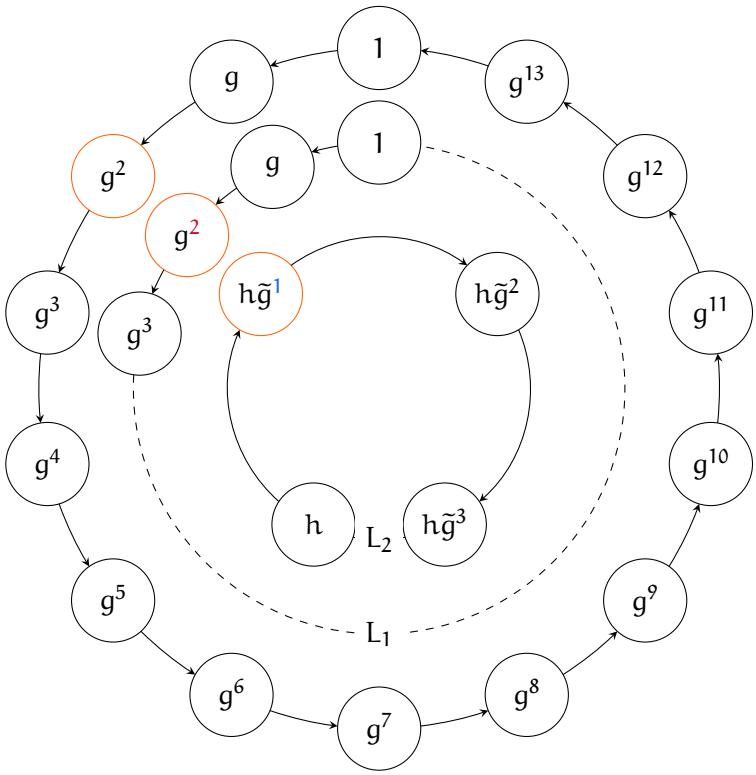

Figure 1.1: Baby-step giant-step on a group with 14 elements — here $h = g^6$, and $m = 4$, the collision occurs when the three elements are aligned. Here, $i = 2$ and $j = 1$, hence $x = i + mj = 6$.

preferred. We decided to present the baby-step giant-step approach because it is more general and applicable in other contexts: one can save a square root time-wise with a memory trade-off. For more specific groups, one could use index-calculus algorithms [MvV97, § 3.6.5].

One last method we would like to discuss is the Pohlig-Hellman algorithm [MvV97, § 3.6.4]. Assume $n = p_1^{e_1} \dots p_r^{e_r}$ is the prime decomposition of n, and $h = g^x \in \langle g \rangle$. Because of the Chinese remainder theorem, the group $\mathbb{Z}/n\mathbb{Z}$ in which lives x can be decomposed as

$$\mathbb{Z}/n\mathbb{Z} \simeq \mathbb{Z}/p_1^{e_1}\mathbb{Z} \times \dots \times \mathbb{Z}/p_r^{e_r}\mathbb{Z}.$$

The isomorphism is explicit and can recover the discrete logarithm x from the elements $x_i \equiv x \bmod p_i^{e_i}$ in $O(\ell^2)$ operations [MvV97, Alg. 2.121]. Therefore, Pohlig-Hellman algorithm reduces to computing the elements x_i which is achieved via a discrete logarithm in $\mathbb{Z}/p_i^{e_i}\mathbb{Z}$ instead. The whole algorithm requires $O\left(\sum_{i=1}^{r} e_i(\ell + \sqrt{p_i})\right)$ group multiplications [MvV97, Fact 3.65] when the decomposition of n is given. Therefore, if the group order has small prime factors, solving the DLP in this group is way simpler thanks to the Chinese remainder theorem and Pohlig-Hellman algorithm, this is why in cryptographic usage, the group G generally has order p, $2p$ or $4p$ where p is a large prime, to avoid attacks with this method, and this will be our assumption from now on.

As discussed above, the group choice is crucial, a good candidate is the multiplicative group $\mathbb{F}_p^*$. The best algorithm to solve DLP in this case is sub-exponential and uses the number field sieve [Gor93; LL93; Sch93; Sch00].

Definition 1.1.3 (Sub-exponential complexity) *Let $x > 2$ be a parameter of the algorithm, $0 \leqslant \alpha \leqslant 1$, $c > 0$. Set*

$$L_x[\alpha, c] = \exp\Big((c + o(1))(\ln x)^{\alpha}(\ln \ln x)^{1-\alpha}\Big)$$

If $0 < \alpha < 1$, an algorithm requiring $L_x[\alpha, c]$ operations is said to run in sub-exponential time in $\log_2 x$. If $\alpha = 0$, this corresponds to polynomial time and if $\alpha = 1$, exponential one.

The lower α, the faster the algorithm. In the case of solving DLP, the complexity is $L_n[1/3, (64/9)^{1/3}]$. This is clearly better for cryptographic purposes than polynomial time, but Shoup proved in 1997 [Sho97b] that a generic algorithm on a generic group must perform at least $O(\sqrt{n})$ operations, the same complexity as baby-step giant-step. The issue here is finding generic groups, without specific structure. This is the reason why elliptic curves are used in cryptography, they have a group structure but no additional one like the multiplicative group $\mathbb{F}_p^*$ where the field structure of $\mathbb{F}_p$ can be exploited.

This is of importance when choosing security parameters, say one wants to have a protocol with s bits of security. On elliptic curves, since the generic algorithm runs in $O(2^{\ell/2})$, one must have s proportional to $\ell/2$, that is ℓ. On the multiplicative group however, because NFS solves DLP in sub-exponential time $L_n[1/3, (64/9)^{1/3}] = O(2^{\ell^{1/3}})$, the number of input bits ℓ must be proportional to s^3, which can be an issue when considering hardware implementation and chip size.

Finally, DLP is not quantum resistant as explained in the introduction, thanks to Shor's algorithm [Sho97a].

1.2 PROTOCOLS

In this section, we will discuss two major protocols in asymmetric cryptography: Diffie-Hellman key exchange and ElGamal signature scheme. References to the handbook of cryptography are respectively [MvV97, § 12.6.1] and [MvV97, § 11.5.2]. Another great review about Diffie-Hellman is the following [Smi18].

1.2.1 Diffie-Hellman key exchange

In 1976, Diffie and Hellman [DH76] revolutionized cryptography with their new key establishment protocol. If Alice and Bob want to discuss over a secure channel for instance, they must first agree on a common secret key to encrypt their messages. How can they achieve that over an insecure channel without leaking their secret key? This is the question Diffie and Hellman answered. The protocol is as follows:

Algorithm 1.2.1 (Diffie-Hellman key exchange) *Let G be a cyclic group of order n, set g a generator. These data are publicly available.*

1. *Alice chooses a secret random integer $0 < a < n$ and computes $g_a = g^a$. She sends g_a to Bob over the insecure channel.*

2. *Similarly, Bob chooses a secret random integer $0 < b < n$, computes $g_b = g^b$ and sends g_b to Alice.*

3. *Alice computes $(g_b)^a = g^{ab}$ with her secret a, Bob computes $(g_a)^b = g^{ab}$ with his secret b. They now have a common secret key: g^{ab}.*

The situation is also summarized in Fig. 1.2.

$$\text{Public data: } G = \langle g \rangle$$

$$\begin{array}{ccc} \text{Alice} & & \text{Bob} \\ \text{Secret } a & & \text{Secret } b \\ g_a = g^a & \xrightarrow{\quad g_a \quad} & g_a \\ g_b & \xleftarrow{\quad g_b \quad} & g_b = g^b \\ (g_b)^a = g^{ab} & & (g_a)^b = g^{ab} \end{array}$$

Figure 1.2: Diffie-Hellman key exchange

If Eve — a passive attacker — listens to the insecure channel, then she would need to derive g^{ab} from g_a and g_b, which can be achieved by solving DLP in G. The problem is known as the computational Diffie-Hellman problem [MvV97, § 3.7]:

Definition 1.2.2 (CDHP) *Let G be a cyclic group with generator g. The computational Diffie-Hellman problem (CDHP) is, given two elements g^a, g^b, find g^{ab}.*

While CDHP and DLP are slightly different, they are under mild assumptions equivalent in terms of hardness [MW99].

The security of CDHP raises the question of choosing G accordingly; commonly used groups are $\mathbb{F}_p^*$ for some prime p or $\langle P \rangle$ where P is a point of high order on an elliptic curve E over a finite field, in that case the variant is called ECDH for Elliptic Curve Diffie-Hellman.

Choosing a secure group is not enough however: Alice and Bob should not choose small exponents a and b, the order of the group should not have too many prime factors, …

As explained in the conclusion of Section 1.1, protocols based on DLP are not quantum resistant. There is a variant of Diffie-Hellman using group actions, we will give the general protocol here and an example will come later in Section 1.4. This idea was developed first by Couveignes and then rediscovered independently by Rostovtsev-Stolbunov [Cou06; RS06].

Algorithm 1.2.3 (Diffie-Hellman key exchange with group action) *Let* G *be an abelian group acting on a set* X *simply and transitively, that is for any* $x, y \in X$*, there is a unique* $g \in G$ *such that* $g \cdot x = y$*, where* $\cdot$ *denotes the group action. Let* $x \in X$ *be a public element, as well as the group* G*.*

1. *Alice chooses a secret random element* $a \in G$ *and computes* $x_a = a \cdot x$*. She sends* x_a *to Bob over the insecure channel.*

2. *Similarly, Bob chooses a secret random element* $b \in G$*, computes* $x_b = b \cdot x$ *and sends* x_b *to Alice.*

3. *Alice computes* $a \cdot x_b = (ab) \cdot x$ *with her secret* a*. Bob on his side computes* $b \cdot x_a = (ba) \cdot x = (ab) \cdot x$ *with his secret* b*. They now have a common secret key:* $(ab) \cdot x$*.*

The security of this variant doesn't rely on DLP any more, but on something called hard homogeneous spaces:

Definition 1.2.4 (Hard homogeneous spaces) *A pair* (G, X) *of a commutative group* G *acting on a set* X *simply and transitively is called a principal homogeneous space. A hard homogeneous space is a principal homogeneous space where the action can be computed, and the two following problems are computationally hard to solve:*

- *Vectorization: given* $x, y \in X$*, find the unique* $g \in G$ *such that* $g \cdot x = y$*.*

- *Parallelization: given* $a \cdot x$ *and* $b \cdot x$ *where* $x \in X$ *and* $a, b \in G$*, compute* $(ab) \cdot x$*.*

Vectorization is analogous to DLP and parallelization to CDHP, and the variant of Diffie-Hellman with group action relies on parallelization.

Remark 1.2.5 *In fact, when considering a cyclic group* X *of order* n *and* G *the group* $\mathbb{Z}/n\mathbb{Z}$ *acting via* $a \cdot x := x^a$*, vectorization corresponds to DLP and parallelization to CDHP.*

1.2.2 ElGamal signature scheme

In the previous section, we did not discuss man-in-the-middle attacks, i.e. when the attacker is active. They could for instance spoof messages to recover additional information from Alice and Bob. A workaround to these attacks is authenticating the source of the message, which can be achieved with signature schemes. We will introduce the ElGamal one here [ElG85], which involves two persons: the signer who wants to sign a message, and the verifier who wants to check that the message is signed by the signer.

A signature scheme splits into three parts: a key generation algorithm creating a public key and private key pair for the signer, a signature generation algorithm for the signer and a verification algorithm for the verifier. We will start with the key generation on a cyclic group.

Algorithm 1.2.6 (Key generation) *Let* G *be a cyclic group of order* n *publicly available, altogether with a generator* g*. The signer should do the following:*

1. *Choose a random integer* $0 < a < n$ *and compute* $g_a = g^a$.

2. *Return the public key* g_a*, the secret key is the integer* a.

Recovering the secret key with this key generation is exactly the discrete logarithm problem in G, and therefore the same constraints on the choice of G applies: n should have at least one large prime factors to avoid Pohlig-Hellman attacks, ...

For the signature scheme, we will require a one-way hash function, which informally is a compressing function easy to compute and for which finding a pre-image is a hard problem [MvV97, Def. 9.1 and 9.3].

Definition 1.2.7 (One-way hash function) *A hash function is a map* h *which satisfies the following properties:*

- *Compression: any bit string* $x \in \{0,1\}^*$ *of arbitrary length is mapped to a bit string* $h(x) \in \{0,1\}^\ell$ *of fixed length* ℓ.

- *Ease of computation: given* h *and an input* x*, the output* $h(x)$ *is easy to compute.*

A hash function h *is furthermore said to be one-way if it satisfies the following additional properties:*

- *Pre-image resistance: given an output* y *for which an input* x *is not known, it is computationally impossible to find the input* x.

- *Second pre-image resistance: given an input* x*, it is computationally impossible to find a second input* $x' \neq x$ *such that* $h(x) = h(x')$.

The underlying assumption is that h is collision-free: it is very unlikely to find two inputs giving a same hash.

The ElGamal signature scheme is the following, it is also described in Fig. 1.3 below.

Algorithm 1.2.8 (ElGamal signature scheme) *We re-use the notations from Algorithm 1.2.6. The group* $G = \langle g \rangle$ *is publicly available and of order* n*. The public key of the signer is* $g_a = g^a$ *where* a *is their private key. Consider a one-way hash function* $h : \{0,1\}^* \to \mathbb{Z}/n\mathbb{Z}$ *and a message* m *to be signed.*

The signature generation for the signer is as follows:

1. *Select a random integer* $0 < k < n$ *with* $\gcd(k, n) = 1$.

2. *Compute* $g_0 = g^k$ *and* $e \equiv k^{-1}(h(m) - ah(g_0)) \bmod n$*, where* $h(g_0)$ *should be understood as the hash of the bit string corresponding to* $g_0 \in G$.

3. *Return the signature* (g_0, e).

A more rigorous definition would include an encoding function $f : G \to \{0,1\}^*$ *and consider* $h(f(g_0))$.

The verifier has access to the group $G = \langle g \rangle$*, the public key* g_a*, the hash function* h*, the original message* m *and the signature* (g_0, e)*. The signature verification is as follows:*

1. *Compute* $v_1 = g_a^{h(g_0)} g_0^e$.

Public data: $G = \langle g \rangle$ of order n
$h : \{0,1\}^* \to \mathbb{Z}/n\mathbb{Z}$ a one-way hash function

Signer Alice		Verifier Bob
Message m	$\xrightarrow{\text{Message } m}$	
Key generation	$\xrightarrow{\text{Public key } g_a}$	$v_1 = g^{h(m)}$
Secret a		
$g_a = g^a$		$v_2 = g_a^{h(g_0)} g_0^e$
Signature generation	$\xrightarrow{\text{Signature } (g_0, e)}$	
Secret k, $\gcd(k, n) = 1$		
$g_0 = g^k$		Accept if $v_1 = v_2$
$e \equiv_n k^{-1}(h(m) - ah(g_0))$		

Figure 1.3: ElGamal signature scheme

2. *Compute* $v_2 = g^{h(m)}$.

3. *Accept the signature if* $v_1 = v_2$*, otherwise reject.*

A signature is accepted if and only if $g^{ah(g_0)+ke} = g^{h(m)}$, that is if and only if $ek \equiv h(m) - ah(g_0) \bmod n$ because g is of order n.

The input can be of arbitrary length, this is a characteristic of a signature scheme *with appendix* as the ElGamal one, where the input message is required for verification. Another family of scheme is the one *with message recovery*, where the original message is computed during the verification.

If an attacker wanted to forge a signature, they could choose a random k as in step 1 and compute $g_0 = g^k$. They would then require to compute the exponent e, which cannot be done without the secret key a and therefore solving DLP in the group G. At best, they could pick a random $e \in \mathbb{Z}/n\mathbb{Z}$, which has probability $1/n$ to be correct and decays with the size n.

The first step in the signature generation requires a random number k and therefore the scheme is said to be randomized, as opposed to a deterministic scheme. This is necessary with this approach, as if a same k is used for two messages m_1 and m_2, then a is probably leaked. Indeed, consider e_1 and e_2 such that $e_i \equiv k^{-1}(h(m_i) - ah(g_0)) \bmod n$ for $i = 1, 2$. Then $(e_1 - e_2)k \equiv h(m_1) - h(m_2) \bmod n$, and if both $e_1 - e_2$ and $h(s)$ are invertible modulo n, the attacker can compute k and finally recover $a \equiv h(g_0)^{-1}(h(m_1) - ke_1) \bmod n$.

In both Diffie-Hellman key exchange and ElGamal signature scheme, the security assumption relies heavily on solving the DLP, hence a natural choice in these protocols is a cyclic subgroup of an elliptic curve given the hardness of DLP in this context. It is therefore essential to have an efficient arithmetic to compute the scalar product — that is the multiple of a given point on the elliptic curve — and we will see how this can be achieved in this thesis.

1.3 ELLIPTIC CURVE METHOD

So far, we have seen protocols based on DLP, which can be solved on the multiplicative group with NFS, a factorization algorithm. On top of that, many of the pre-quantum protocols relies on RSA [RSA78], where the public data is a composite integer $n = pq$ with two large primes p and q, these two being secret. Therefore, the security of RSA-based protocols relies on the hardness of factoring a large integer with large prime factors. In shorter terms, factoring integers is essential to cryptography.

The most efficient generic algorithm is NFS, running in a sub-exponential time $L_n[1/3, (64/9)^{1/3}]$. However, when looking for 'small'- and 'medium'-sized prime factors, elliptic curve method (ECM) is the state-of-the-art, and is also used in NFS first to try to reduce the input size. We will describe ECM here. Since elliptic curves are not introduced until Chapter 4, we will give some knowledge here while trying to keep those minimal, more details can be found in the corresponding chapter later.

This section is mainly inspired by [Coh93, § 10.3] and Lenstra's seminal paper [Len87].

1.3.1 Elliptic curves over a field

Let k be a field with characteristic different from 2 and 3. An elliptic curve E over k is an algebraic curve given by an equation of the form

$$E : y^2 = x^3 + ax + b,$$

where $a, b \in k$ and $4a^3 + 27b^2 \neq 0$. $E(k)$ is then the set of points $(x, y) \in k^2$ satisfying the equation of E. While looking at the curve projectively by setting $x = X/Z$ and $y = Y/Z$, the equation becomes

$$Y^2Z = X^3 + aXZ^2 + bZ^3.$$

In the projective space, the point at infinity $(0 : 1 : 0)$ is also a point on the elliptic curve. We denote it by $0_E \in E(k)$.

Proposition 1.3.1 *Let $P, Q \in E(k)$ and define $P + Q$ in the following way:*

- *if $P = 0_E$, let $P + Q := Q$. Similarly, if $Q = 0_E$, let $P + Q := P$;*
- *otherwise, $P = (x_1, y_1)$ and $Q = (x_2, y_2)$:*
  - *if $x_1 \neq x_2$, set*

    $$\lambda = \frac{y_2 - y_1}{x_2 - x_1},\ x_3 = \lambda^2 - x_1 - x_2,\ y_3 = -y_1 - \lambda(x_3 - x_1).$$

    Finally, define $P + Q := (x_3, y_3)$;
  - *if $x_1 = x_2$, because P and Q are elements of $E(k)$, $y_1 = \pm y_2$ and:*
    - *if $y_1 = -y_2$, set $P + Q := 0_E$;*

* *otherwise,* $y_1 = y_2 \neq 0$*, set*

$$\lambda = \frac{3x_1^2 + a}{2y_1},\ x_3 = \lambda^2 - 2x_1,\ y_3 = -y_1 - \lambda(x_3 - x_1).$$

Finally, define $P + Q := (x_3, y_3)$.

Then the map $(P, Q) \mapsto P + Q$ *has its image in* $E(k)$ *and defines an abelian group law on* $E(k)$.

At first glance, there does not seem to be a justification to why these formulas and not other ones. A geometric interpretation is given in Definition 4.1.3. Deriving the algebraic formulas from the geometric approach is way more natural, but since we want to keep this introduction minimal, we only give the (rather unoptimized) affine formulas for now.

1.3.2 Elliptic curves over $\mathbb{Z}/n\mathbb{Z}$

We now assume that k is not a field any more, but that $k = \mathbb{Z}/n\mathbb{Z}$ where n may be a composite integer. We want to define projectively elliptic curves over $\mathbb{Z}/n\mathbb{Z}$. Some adjustments must be made:

Definition 1.3.2 (Projective spaces over $\mathbb{Z}/n\mathbb{Z}$) *Let* $m \geqslant 0$ *and*

$$A = \{(x_0, \dots, x_m) \in (\mathbb{Z}/n\mathbb{Z})^{m+1} \mid \gcd(x_0, \dots, x_m, n) = 1\}.$$

The set A *is endowed with the equivalence relation* $x \sim y$ *if there is* $\lambda \in (\mathbb{Z}/n\mathbb{Z})^*$ *such that* $x = \lambda y$*. The* m*-th projective space is then* $\mathbb{P}^m(\mathbb{Z}/n\mathbb{Z}) := A/\sim$*, the equivalence class of* $(x_0, \dots, x_m)$ *is denoted* $(x_0 : \cdots : x_m)$.

If n is a prime number, this is the usual projective space over a finite field. $(\mathbb{Z}/n\mathbb{Z})^m$ can be identified as an affine subset of $\mathbb{P}^m(\mathbb{Z}/n\mathbb{Z})$:

$$\mathbb{A}^m(\mathbb{Z}/n\mathbb{Z}) := \{(x_0 : \cdots : x_{m-1} : 1) \mid (x_0, \dots, x_{m-1}) \in (\mathbb{Z}/n\mathbb{Z})^m\}.$$

$\mathbb{P}^{m-1}(\mathbb{Z}/n\mathbb{Z})$ can also be identified as a subset of $\mathbb{P}^m(\mathbb{Z}/n\mathbb{Z})$:

$$\{(x_0 : \cdots : x_{m-1} : 0) \mid (x_0 : \cdots : x_{m-1}) \in \mathbb{P}^{m-1}(\mathbb{Z}/n\mathbb{Z})\}.$$

If n is prime, this gives a disjoint union of $\mathbb{P}^m(\mathbb{Z}/n\mathbb{Z})$ as

$$\mathbb{P}^m(\mathbb{Z}/n\mathbb{Z}) = \mathbb{A}^m(\mathbb{Z}/n\mathbb{Z}) \sqcup \mathbb{P}^{m-1}(\mathbb{Z}/n\mathbb{Z}).$$

However, if n is composite, this decomposition does not hold any more, there is a special set to consider:

$$\mathbb{P}^m_{\text{spec}}(\mathbb{Z}/n\mathbb{Z}) := \{(x_0 : \cdots : x_m) \mid x_m \neq 0, x_m \notin (\mathbb{Z}/n\mathbb{Z})^*\}.$$

This is precisely the set of elements where $\gcd(x_m, n)$ is a non-trivial factor of n. Now the complete decomposition is

$$\mathbb{P}^m(\mathbb{Z}/n\mathbb{Z}) = \mathbb{A}^m(\mathbb{Z}/n\mathbb{Z}) \sqcup \mathbb{P}^{m-1}(\mathbb{Z}/n\mathbb{Z}) \sqcup \mathbb{P}^m_{\text{spec}}(\mathbb{Z}/n\mathbb{Z}).$$

Example 1.3.3 *Let $n = 6$, we have $(\mathbb{Z}/6\mathbb{Z})^* = \{1, 5\}$ and the elements of $\mathbb{P}^1(\mathbb{Z}/6\mathbb{Z})$ are:*

- *the affine points* $(0 : 1), (1 : 1), (2 : 1), (3 : 1), (4 : 1), (5 : 1)$;
- *the point at infinity* $(1 : 0)$;
- *the special points* $(1 : 2), (3 : 2), (5 : 2), (1 : 3), (2 : 3)$.

Consider p a prime divisor of n. The reduction map from $\mathbb{P}^m(\mathbb{Z}/n\mathbb{Z})$ to $\mathbb{P}^m(\mathbb{F}_p)$ where every coefficient is reduced modulo p is well-defined. This map is denoted red_p. The following lemma helps to classify affine points:

Lemma 1.3.4 *Let $P \in \mathbb{P}^m(\mathbb{Z}/n\mathbb{Z})$, then $P \in \mathbb{A}^m(\mathbb{Z}/n\mathbb{Z})$ if and only if for all prime p dividing n, $\mathrm{red}_p(P) \in \mathbb{A}^m(\mathbb{F}_p)$.*

Equivalently, $P \in \mathbb{P}^{m-1}(\mathbb{Z}/n\mathbb{Z}) \sqcup \mathbb{P}^m_{spec}(\mathbb{Z}/n\mathbb{Z})$ if and only if there is a prime p dividing n such that $\mathrm{red}_p(P) \in \mathbb{P}^{m-1}(\mathbb{F}_p)$.

Proof *The second statement is the opposite of the first one. Furthermore, if $P \in \mathbb{A}^m(\mathbb{Z}/n\mathbb{Z})$, it is straight-forward that $\mathrm{red}_p(P) \in \mathbb{A}^m(\mathbb{F}_p)$ for any prime p dividing n.*

We then assume that for all prime p dividing n, $\mathrm{red}_p(P) \in \mathbb{A}^m(\mathbb{F}_p)$ and we set $P = (x_0 : \cdots : x_m)$. By assumption, if p is a prime divisor of n, $x_m \equiv 1 \bmod p$. Therefore, $\gcd(x_m, n) = 1$ and x_m is invertible modulo n. Hence, $P = (x_0 x_m^{-1} : \cdots : x_{m-1} x_m^{-1} : 1) \in \mathbb{A}^m(\mathbb{Z}/n\mathbb{Z})$. □

We can now give the definition of an elliptic curve over $\mathbb{Z}/n\mathbb{Z}$:

Definition 1.3.5 (Elliptic curve over $\mathbb{Z}/n\mathbb{Z}$) *Let n be an integer coprime to 6. An elliptic curve E over $\mathbb{Z}/n\mathbb{Z}$ is a subset of $\mathbb{P}^2(\mathbb{Z}/n\mathbb{Z})$, denoted $E(\mathbb{Z}/n\mathbb{Z})$, where the points $(X : Y : Z)$ verify the projective equation*

This corresponds to the usual assumption $\mathrm{char}\, k \neq 2, 3$.

$$Y^2 Z = X^3 + aXZ^2 + bZ^3,$$

with $a, b \in \mathbb{Z}/n\mathbb{Z}$ and such that $4a^3 + 27b^2 \in (\mathbb{Z}/n\mathbb{Z})^*$.

The definition again coincides with the usual one if n is a prime and the reduction maps are compatible: $\mathrm{red}_p : E(\mathbb{Z}/n\mathbb{Z}) \to E(\mathbb{F}_p)$ for any prime divisor p of n. However, the affine group law defined in Proposition 1.3.1 does not hold any more. To overcome this, a solution would be to give the projective version of the group law and this would turn $E(\mathbb{Z}/n\mathbb{Z})$ into a group too with neutral element $0_E = (0 : 1 : 0)$. Another approach, as suggested by Lenstra, is to ignore this and to try to do computations with the affine formulas, since we only want to factor n. In fact, consider the decomposition $E(\mathbb{Z}/n\mathbb{Z}) = E_{\mathrm{std}} \sqcup E_{\mathrm{spec}}$ where

$$E_{\mathrm{std}} := \big(E(\mathbb{Z}/n\mathbb{Z}) \cap \mathbb{A}^2(\mathbb{Z}/n\mathbb{Z})\big) \cup \{0_E\}$$

and

$$E_{\mathrm{spec}} := \big((E(\mathbb{Z}/n\mathbb{Z}) \cap \mathbb{P}^1(\mathbb{Z}/n\mathbb{Z})) \cup (E(\mathbb{Z}/n\mathbb{Z}) \cap \mathbb{P}^2_{\mathrm{spec}}(\mathbb{Z}/n\mathbb{Z}))\big) \setminus \{0_E\}.$$

The special points E_{spec} are in this context the interesting ones to factor n, if $P \in E_{\mathrm{spec}}$ then:

- Either $P \in \mathbb{P}^1(\mathbb{Z}/n\mathbb{Z})$, P can be written $P = (X : Y : 0)$ and by the curve equation one has $X^3 \equiv 0 \bmod n$. Now, since $P \neq 0_E$, this means that $X \not\equiv 0 \bmod n$ and $d = \gcd(X, n)$ is a non-trivial factor of n.
- Otherwise, $P \in \mathbb{P}^2_{\text{spec}}(\mathbb{Z}/n\mathbb{Z})$ and if $P = (X : Y : Z)$ then by definition $\gcd(Z, n)$ is a non-trivial factor of n.

If two points P and Q are in E_{std}, then one can try to use formulas from Proposition 1.3.1 to compute $P + Q$. If the coefficient λ can be computed in $\mathbb{Z}/n\mathbb{Z}$, then $P + Q$ is well-defined via these formulas and $P + Q \in E_{\text{std}}$. Otherwise, the denominator of λ is non-invertible in $\mathbb{Z}/n\mathbb{Z}$ which may yield a non-trivial factor of n, and $P + Q \in E_{\text{spec}}$.

This illustrates the algorithmic nature of $E(\mathbb{Z}/n\mathbb{Z})$, we hope to land in the special subset E_{spec} by doing computations with elements from E_{std}.

With the full group structure, if p is any prime divisor of n and P, Q are two points in $E(\mathbb{Z}/n\mathbb{Z})$, then $\text{red}_p(P + Q) = \text{red}_p(P) + \text{red}_p(Q)$. One must be cautious with reduction to the origin however, if $P \in E(\mathbb{Z}/n\mathbb{Z})$ is such that for any prime divisor p of n, $\text{red}_p(P) = 0_E$, this does not always imply that $P = 0_E$ when n is not square-free.

Lemma 1.3.6 *Let $P \in E_{\text{std}}$, the following assertions are equivalent:*

1. $P = 0_E$.
2. *For all prime divisor p of n, $\text{red}_p(P) = 0_E$.*
3. *There is a prime divisor p of n such that $\text{red}_p(P) = 0_E$.*

Proof *The implications $1 \implies 2 \implies 3$ are trivial, we therefore consider a prime divisor p of n such that $\text{red}_p(P) = 0_E$. Because $P \in E_{\text{std}}$, either $P = (x : y : 1)$ for some $x, y \in \mathbb{Z}/n\mathbb{Z}$, or $P = 0_E$. But $\text{red}_p(x : y : 1) \neq 0_E$, therefore $P = 0_E$ and $3 \implies 1$.* □

One last thing we need to discuss is how to compute $m \cdot P = \underbrace{P + \cdots + P}_{m \text{ times}}$ and its cost. The naive approach would take $m - 1$ additions on the elliptic curve, which is rather suboptimal. Instead, fast exponentiation (also known as double-and-add) can achieve $O(\log_2 m)$ additions.

Algorithm 1.3.7 (Fast exponentiation) *Let G be a group, $g \in G$ and m an ℓ-bits integer. Set $g_\ell = 1$, decompose $m = (b_{\ell-1}, \ldots, b_0)_2$ in binary form and do the following for i going from $\ell - 1$ down to 0:*

- *if $b_i = 0$, compute $g_i = g_{i+1}^2$;*
- *otherwise, $b_i = 1$, compute $g_i = g_{i+1}^2 g$.*

Return $g_0 = g^m$.

One could also use addition chains, but more details on those will be given in Section 2.2. If m is a product $m = q_1 \ldots q_r$, then $m \cdot P$ can be computed as

$$m \cdot P = q_r \cdot (q_{r-1} \cdot (\cdots (q_1 \cdot P))).$$

Each step being $O(\log_2 q_i)$ additions, this method also takes $O(\log_2 m)$ additions.

1.3.3 Smooth numbers

Smooth numbers are a way to quantify the size of prime factors of a composite integer n.

Definition 1.3.8 (Smooth numbers) *Let $w \geqslant 2$, a positive integer k is said to be w-smooth if for all prime divisor p of k, $p \leqslant w$. If $v \geqslant 1$, we denote the number of w-smooth numbers less than v by $\Psi(v, w)$.*

The least common multiple of all w-smooth numbers less than v can be simply expressed in the following, and will justify some choices in ECM later:

Lemma 1.3.9 *Let $v \geqslant 1$, $w \geqslant 2$, then:*

$$\operatorname{lcm}\{k \leqslant v \mid k \text{ is } w\text{-smooth}\} = \prod_{\substack{p \leqslant w \\ p \text{ prime}}} p^{\lfloor \frac{\log_2 v}{\log_2 p} \rfloor}.$$

In particular:

$$\operatorname{lcm}(1, \dots, w) = \prod_{\substack{p \leqslant w \\ p \text{ prime}}} p^{\lfloor \frac{\log_2 w}{\log_2 p} \rfloor}.$$

Proof *Set $m = \operatorname{lcm}\{k \leqslant v \mid k \text{ is } w\text{-smooth}\}$ and m_0 to be the right-side product. Consider $k \leqslant v$ a w-smooth number, and p a prime divisor of k, set e to be the exponent of p in k. Then $p^e \leqslant v$ implies $e \leqslant \frac{\log_2 v}{\log_2 p}$. Since this holds for every prime divisor p, $k \mid m_0$, and therefore $m \mid m_0$. On the other hand, if $p \leqslant w$ is a prime and $e_p = \lfloor \frac{\log_2 v}{\log_2 p} \rfloor$, then $p^{e_p} \leqslant v$, and it is w-smooth by construction, hence $p^{e_p} \mid m$. Therefore, $m = m_0$.*

Regarding the second equality, every integer less than w is w-smooth, hence $\operatorname{lcm}(1, \dots, w) = \operatorname{lcm}\{k \leqslant w \mid k \text{ is } w\text{-smooth}\}$ and the result comes from the first equality. □

An important theorem on smooth numbers — due to de Bruijn, Canfield, Erdös and Pomerance [Bru51; CEP83] — is that we know asymptotically how many there are:

Theorem 1.3.10 *The following holds when $v \to \infty$ uniformly, for u verifying $(\ln v)^{\varepsilon} < u < (\ln v)^{1-\varepsilon}$ and for a fixed $0 < \varepsilon < 1$:*

$$\Psi(v, v^{1/u}) = v u^{-u(1+o(1))}.$$

In particular for any $c \neq 0$,

The definition of L_v still holds for $c < 0$.

$$\Psi(v, L_v[1/2, c]) = v L_v[1/2, -1/2c]$$

where the function L_v is introduced in Definition 1.1.3.

1.3.4 The ECM algorithm

Since the affine addition law is not completely defined over $\mathbb{Z}/n\mathbb{Z}$, the computation of $m \cdot P$ may fail, but that is not a problem, in fact it generally is a good news: because of the previous discussion, this mean that we can recover a non-trivial divisor of n. This is the main idea of Lenstra's ECM [Len87], inspired by the $p-1$ method (another factorization algorithm):

Algorithm 1.3.11 (Elliptic Curve Method) *Let n be a composite integer which is not a prime power, v, w and h parameters. Set*

$$m = \operatorname{lcm}\{k \leqslant v + 2\sqrt{v} + 1 \mid k \textit{ is } w\textit{-smooth}\}.$$

While no factor of n has been found and less than h curves have been tested, do the following:

1. *Choose $a, x, y \in \mathbb{Z}/n\mathbb{Z}$ at random, set $b = y^2 - x^3 - ax$, E the elliptic curve with equation $Y^2Z = X^3 + aXZ^2 + bZ^3$ and $P = (x : y : 1)$, where $P \in E_{std}$ by construction.*

2. *Try to compute $m \cdot P$:*

   a) *if there is some $\mu \mid m$ such that $\mu \cdot P \in E_{spec}$, then a divisor of n has been found;*

   b) *otherwise, $m \cdot P \in E_{std}$ and the computation ends, go back to step 1 with a new curve.*

We dismiss the case of prime powers because those can be detected more easily. The method can be refined in several ways: one could add a stage 2, choose curves with 'good properties', ... Zimmermann and Dodson did a great survey about ECM and its optimizations [ZD06].

An alternative approach is to keep going with the computations. If at some point we want to add points P and Q which are distinct modulo n and equal modulo p for some prime factor p of n, then we will use the wrong formulas in $E(\mathbb{F}_p)$, and ultimately land on '$(0 : 0 : 0)$' modulo p, which is not a projective point. p can then be recovered by computing the greatest common divisor of n and a coordinate of the resulting '$P + Q$' modulo n.

Remark 1.3.12 *Montgomery modified the random selection of the elliptic curve modulo n with the usage of a list or a family of elliptic curves E over $\mathbb{Q}$ with a non-torsion point P. When factoring n, one computes the scalar product of the reduction of P modulo n on $E(\mathbb{Z}/n\mathbb{Z})$. For instance, one can consider the curve $y^2 = x^3 + ax + 1$ with the point $P = (0, 1)$.*

Step 1 of ECM can therefore be replaced as follows:

1′ let E be the elliptic curve over $\mathbb{Z}/n\mathbb{Z}$ with equation

$$Y^2Z = X^3 + aXZ^2 + Z^3,$$

set $P = (0 : 1 : 1) \in E(\mathbb{Z}/n\mathbb{Z})$.

Another common choice for m is $m = w!$ or $m = \operatorname{lcm}(1, \dots, w)$.

We will now discuss the complexity of the algorithm and how to choose the parameters accordingly. The computation in step 2 takes $O(\log_2 m)$ group additions, let us call h the expectancy of the number of repetitions of step 2. Because

$$\log_2 m = \sum_{\substack{p \leqslant w \\ p \text{ prime}}} \left\lfloor \frac{\log_2(v + 2\sqrt{v} + 1)}{\log_2 p} \right\rfloor \log_2 p = O(w \log_2(v)),$$

ECM takes $O(hw \log_2(v))$ group operations. Since the complexity is sub-exponential and the cost of a group operation is polynomial, in the following, we will not distinguish between group and binary operations.

Remark 1.3.13 *A more precise bound is* $\log_2 m = O(\pi(w) \log_2 v)$ *where* $\pi(w)$ *is the number of primes less than* w*. Since* $\pi(w) \sim \frac{w}{\ln w}$*, this will not have any impact on the final complexity.*

Lenstra gives a sufficient condition for a curve E to find a non-trivial divisor:

Proposition 1.3.14 *We re-use the notations of Algorithm 1.3.11. Consider a curve* E *over* $\mathbb{Z}/n\mathbb{Z}$ *with a point* $P \in E_{std}$*. Let* $p \neq q$ *be two prime divisors of* n *and assume the following conditions:*

1. $p \leqslant v$.
2. $\#E(\mathbb{F}_p)$ *is* w*-smooth.*
3. *Let* r *be the order of the element* $\operatorname{red}_p P$ *in* $E(\mathbb{F}_p)$*, and consider* ℓ *the largest prime divisor of* r*, assume* $\ell \nmid \#E(\mathbb{F}_q)$.

Then E *will find a non-trivial divisor of* n.

A comment on the choice of parameters: the first assumption shows that v determines the expected size of the prime factors of n, this does not imply that the found divisor will be less than v.

Proof *The Hasse-Weil [Sil09, § V, Thm. 1.1] bound ensures* $\#E(\mathbb{F}_p) \leqslant p + 2\sqrt{p} + 1$*, and by the first assumption* $\#E(\mathbb{F}_p) \leqslant v + 2\sqrt{v} + 1$*. Assumption 2 also states that* $\#E(\mathbb{F}_p)$ *is* w*-smooth, hence by definition* $\#E(\mathbb{F}_p) \mid m$*. By Lemma 1.3.9,*

$$m = \prod_{\substack{k \leqslant w \\ k \text{ prime}}} k^{e_k}, \quad e_k = \left\lfloor \frac{\log_2(v + 2\sqrt{v} + 1)}{\log_2 k} \right\rfloor.$$

For any prime divisor k *of* $\#E(\mathbb{F}_p)$*, its exponent is less than* e_k*. In particular, if* α *is the exponent of* ℓ *in* r*, then* $\alpha \leqslant e_\ell$*, and since* $\#E(\mathbb{F}_p)$ *is* w*-smooth,* $\ell \leqslant w$*. We set*

$$m_0 = \prod_{\substack{k < \ell \\ k \text{ prime}}} k^{e_k} \times \ell^{\alpha - 1},$$

By construction $m_0, m_0\ell \mid m$ *and* $m_0 \cdot \operatorname{red}_p P \neq 0_E$, $m_0\ell \cdot \operatorname{red}_p P = 0_E$*. We will show that at least one of* $m_0 \cdot P$ *and* $m_0\ell \cdot P$ *is in* E_{spec}*:*

- *If* $m_0\ell \cdot P \in E_{std}$*, then by Lemma 1.3.6,* $m_0\ell \cdot P = 0_E$ *and by reducing modulo* q*,* $m_0\ell \cdot \mathrm{red}_q P = 0_E$*. By the third assumption,* $\ell \nmid \#E(\mathbb{F}_q)$*, hence* $m_0 \cdot \mathrm{red}_q P = 0_E$*. Then,* $m_0 \cdot P \in E_{spec}$*, otherwise this would contradict Lemma 1.3.6.*

- *Otherwise,* $m_0\ell \cdot P \in E_{spec}$*.*

Hence, there is always a $\mu \mid m$ *such that* $\mu \cdot P \in E_{spec}$*, which means that ECM recovers successfully a non-trivial divisor of* n*.* □

The end of this proof is the main idea of ECM, we are looking for points that reduce to the origin for some prime divisor but not for all of them. This is strongly related to the curve order being w-smooth.

The following proposition from Lenstra [Len87, Prop 2.8] bounds the probability of success of ECM:

Proposition 1.3.15 *Let* p *be the smallest prime divisor of* n*, and assume* $p \leqslant v$*. Let* $f_p(w)$ *be the probability that a random integer picked uniformly in* $]p+1-\sqrt{p}, p+1+\sqrt{p}[$ *is* w*-smooth. Then there is a computable constant* $c > 1$ *such that the success probability of Algorithm 1.3.11 with parameters* n, v, w, h *is at least* $1 - c^{-hf_p(w)/\log_2(v)}$*.*

We will not prove this, the important information is that h and $\frac{\log_2 v}{f_p(w)}$ should have the same order of magnitude to have a decent chance of success. With this in mind, ECM takes around $O\left(\frac{w}{f_p(w)}(\log_2 v)^2\right)$ additions. We now need a conjecture on $f_p(w)$ introduced by Lenstra and extensively studied afterwards, see for instance [Gra08].

Conjecture 1.3.16 (Smooth numbers in short intervals) *The probability of an integer chose at random uniformly in the interval* $]p+1-\sqrt{p}, p+1+\sqrt{p}[$ *to be* w*-smooth is the same as the one when picking the integer in the interval* $[1, p]$*. More formally:*

$$f_p(w) = \frac{\Psi(p, w)}{p}.$$

There are higher dimension variants of ECM such as HECM [Cos10] where a similar conjecture is proven.

Regarding the complexity, using Theorem 1.3.10 alongside the conjecture, if $w = L_p[1/2, c]$ with $c > 0$:

$$\frac{w}{f_p(w)} = \frac{L_p[1/2, c]}{\Psi(p, L_p[1/2, c])} = L_p[1/2, c + 1/2c].$$

The constant is optimal when $c = 1/\sqrt{2}$, and the number of additions involved in ECM is then $O\left(L_p[1/2, \sqrt{2}](\log_2 v)^2\right)$. Since v bounds p, the logarithm is negligible, and ECM runs in $L_p[1/2, \sqrt{2}]$. While this is not as good as NFS in general, the complexity relies on the smallest prime factor of n. If for instance this factor is of size $L_n[1/2, c_1]$, then ECM runs in $L_n[1/4, c_2]$ for some constants $c_1, c_2 > 0$, which is faster than NFS. This justifies why ECM is particularly good at finding small factors of an integer.

1.4 ISOGENY-BASED CRYPTOGRAPHY

Elliptic curve cryptography is pre-quantum because it relies on the DLP which is broken by Shor's algorithm on a quantum computer. Still, since they have an efficient arithmetic, it would be interesting to have an alternative based on elliptic curves and resistant to quantum computers. In this section, we will briefly introduce isogeny-based cryptography, which solves this problem. We will mainly be referring to the following review of Smith [Smi18].

1.4.1 Isogenies

The issue with elliptic curves comes from the group structure, and a countermeasure was to introduce hard homogenous spaces (HHS), as defined in Definition 1.2.4. We will in this section give an example of a HHS. We need some material first, which will be more detailed in the upcoming chapters (namely Chapters 3 and 4), so as before with elliptic curves we will try to be minimal.

Once we have elliptic curves, it is natural to study maps between them. We will consider that our elliptic curves are defined over a field k.

Definition 1.4.1 *An isogeny $\varphi : E \to E'$ is a surjective morphism of elliptic curves (as algebraic varieties) defined over k which is also a group homomorphism.*

Some notable properties of isogenies is that they have a finite kernel, and it is enough to check that $\varphi(0_E) = 0_{E'}$ to recover a group homomorphism.

Note that two elliptic curves could be isogenous over the closure $\overline{k}$ but not over the field of definition k.

On elliptic curves, we can define some standard form of isogenies, and derive properties from this standard form. This is a more intuitive approach than the general one on abelian varieties developed in Section 3.2, and fits perfectly our introduction here. Our main reference is a course of Andrew Sutherland given at MIT[2]. We will assume that $\operatorname{char} k \neq 2, 3$ here.

Proposition 1.4.2 *If E and E' are two elliptic curves defined over k in short Weierstrass form and $\varphi : E \to E'$ is an isogeny defined over k, then φ is given by an affine rational map*

$$\varphi : (x, y) \mapsto \left(\frac{u(x)}{v(x)}, \frac{s(x)}{t(x)} y \right)$$

where $u, v, s, t \in k[x]$ are polynomials where $\gcd(u, v) = \gcd(s, t) = 1$.

When $\varphi : E \to E'$ is in standard form, we can define its degree as $\deg \varphi := \max(\deg u, \deg v)$, and it is separable if the derivative of $\frac{u(x)}{v(x)}$ is non-zero. Otherwise, we say that φ is inseparable. The degree is multiplicative, that is if $\varphi : E \to E'$ and $\psi : E' \to E''$ are two isogenies, then $\deg(\psi \circ \varphi) = (\deg \psi)(\deg \varphi)$. If φ is a separable isogeny, then $\deg \varphi = \# \ker \varphi$.

Example 1.4.3 *For any non-zero integer n, the multiplication by n, denoted $[n]_E : E \to E$ and which maps $P \in E$ to $n \cdot P$, is an isogeny of degree n^2.*

2 https://math.mit.edu/classes/18.783/2023/index.html, in particular chapter 4.

An important proposition is the existence of a dual isogeny.

Proposition 1.4.4 *If $\varphi : E \to E'$ is an isogeny of degree n, then there is a unique isogeny $\widetilde{\varphi} : E' \to E$, called the dual isogeny, such that*

$$\varphi \circ \widetilde{\varphi} = [n]_{E'} \text{ and } \widetilde{\varphi} \circ \varphi = [n]_E.$$

Moreover, $\deg \widetilde{\varphi} = n$, $\widetilde{\widetilde{\varphi}} = \varphi$ *and* φ *is defined over* k *if and only if* $\widetilde{\varphi}$ *is defined over* k.

One could look at the relation 'being isogenous over k'. The previous proposition shows that this is an equivalence relation. Over a finite field $\mathbb{F}_q$, we have a simple criterion to test if two elliptic curves are isogenous over $\mathbb{F}_q$, due to Tate [Tat66].

Theorem 1.4.5 *Let E and E' be two elliptic curves defined over $\mathbb{F}_q$. Then E and E' are isogenous over $\mathbb{F}_q$ if and only if $\#E(\mathbb{F}_q) = \#E'(\mathbb{F}_q)$.*

The proof of this theorem is not constructive however, and in fact finding an isogeny is a hard problem which we would like to exploit for our cryptographic purposes. We also do not recover any information on the degree of such an isogeny.

A modern point of view is looking at this problem by finding a path in a graph. We introduce a few more notions before that.

Definition 1.4.6 *If $E : y^2 = x^3 + ax + b$ is an elliptic curve in short Weierstrass form defined over k, its j-invariant is given by*

$$j(E) := 1728\frac{4a^3}{4a^3 + 27b^2}.$$

This is a very important invariant, as two elliptic curves E and E' are isomorphic over $\overline{k}$ if and only if $j(E) = j(E')$. Note that the isomorphism may not be rational! For our purpose, we will need to work with isomorphisms over k.

Two isogenies $\varphi_i : E_i \to E'_i$ for $i = 1, 2$ are said to be isomorphic over k if there are isomorphisms $\psi : E_1 \xrightarrow{\sim} E_2$ and $\psi' : E'_1 \xrightarrow{\sim} E'_2$ such that $\psi \circ \varphi_1 = \varphi_2 \circ \psi'$. We finally define the graph of isogenies where:

- vertices are given by class of isomorphisms of elliptic curves over k (represented by their j-invariant);
- edges are given by isogenies between elliptic curves up to isomorphisms over k.

An example is given in Fig. 1.4[3].

3 This graph was generated using code from [Arp+23].

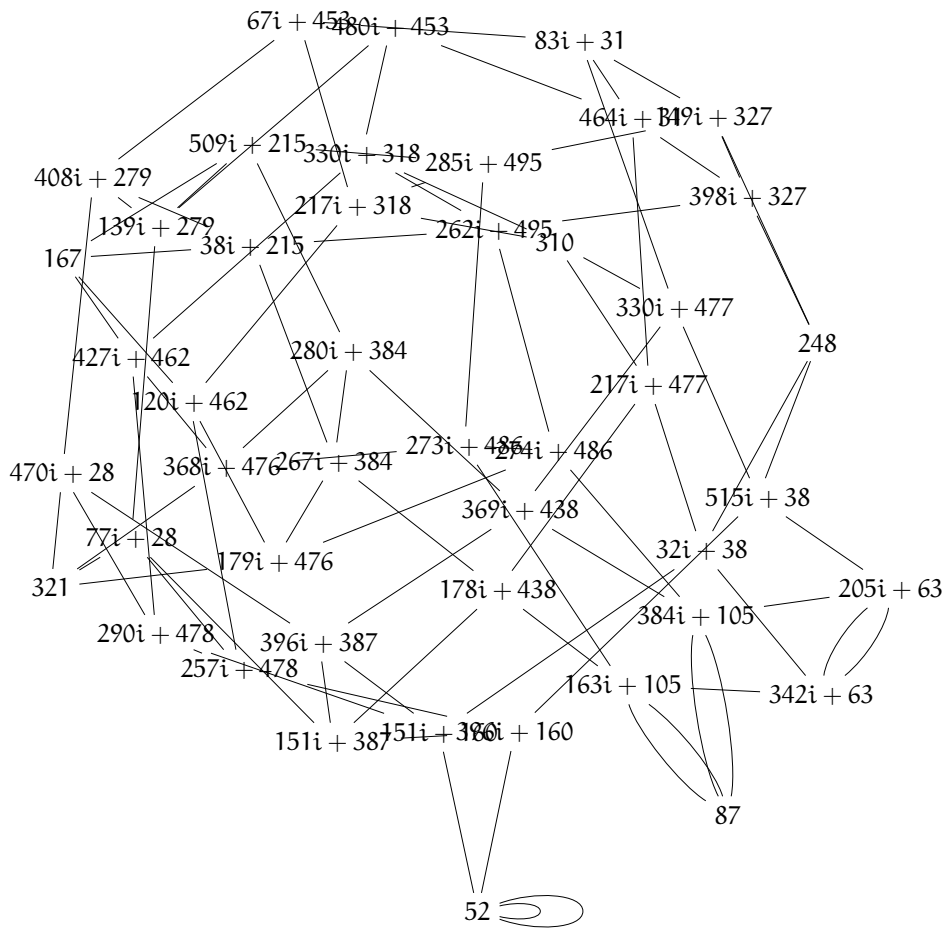

Figure 1.4: A (supersingular) isogeny graph over $\overline{\mathbb{F}_{547}}$. Only isogenies of degree 3 have been represented here, and $i^2 = -1$.

1.4.2 An example of a hard homogenous space

We will explain how to construct properly a principal homogenous space here, and in particular describe the action to walk through the isogeny graph. Given an elliptic curve E, we are looking for endomorphisms on it.

Definition 1.4.7 *If an isogeny goes from E to itself, it is said to be an endomorphism of E. Additionally, the trivial map $[0]_E$ is also an endomorphism. There set is denoted* $\operatorname{End} E$. *It has a ring structure for addition and composition.*

We have seen that multiplication by n for any $n \in \mathbb{Z}$ is an isogeny, and in fact is an endomorphism, so $\mathbb{Z} \subseteq \operatorname{End} E$. We will work over a finite field $\mathbb{F}_q$ in this section, where we have a Frobenius endomorphism.

Example 1.4.8 *The Frobenius endomorphism π_q (over $\overline{\mathbb{F}_q}$) is given by $(x, y) \mapsto (x^q, y^q)$. It verifies the equation ${\pi_q}^2 - t\pi_q + q = 0$ for some integer t verifying $|t| \leqslant 2\sqrt{q}$. t is called the trace of π_q.*

In particular, $\#E(\mathbb{F}_q) = q + 1 - t$ since the set of $\mathbb{F}_q$-rational points is exactly the set of points fixed by the Frobenius.

The graph in Fig. 1.4 was generated in the case $t = 0$.

For the sake of simplicity, we will only deal with the case $t \neq 0$, and in that case the elliptic curve is said to be ordinary, otherwise it is supersingular. The endomorphism ring is then isomorphic to an order (i.e. a $\mathbb{Z}$-lattice of rank 2) in the quadratic imaginary field $K = \mathbb{Q}(\pi_q)$, and containing $\mathbb{Z}[\pi]$, of discriminant $t^2 - 4q$. More details on this and the structure of the corresponding isogeny graph will be given in Section 8.5.

To build a HHS, we are particularly interested in the case where this order is maximal, that is when $\operatorname{End} E \simeq \mathcal{O}_K$, the ring of integers of K. Consider an ideal $\mathfrak{a}$ of $\mathcal{O}_K$. If it is principal and generated by an endomorphism φ (under the identification $\operatorname{End} E \simeq \mathcal{O}_K$), then any endomorphism $\psi \in \mathfrak{a}$ is isomorphic to φ. Since we are interested in isogenies up to isomorphism, it makes sense to look at the ideal class group of $\mathcal{O}_K$ instead, which is denoted $\operatorname{Cl} \mathcal{O}_K$. If $[\mathfrak{a}] \in \operatorname{Cl} \mathcal{O}_K$, we define

$$E[\mathfrak{a}] := \bigcap_{\varphi \in \mathfrak{a}} \ker \varphi.$$

This is by construction a finite subgroup of E, and using Theorem 3.2.8, there is a unique separable isogeny φ with kernel $E[\mathfrak{a}]$. Its image is denoted $E/E[\mathfrak{a}]$.

This is how we define our action of $G = \operatorname{Cl} \mathcal{O}_K$ on X, the set of isomorphism classes over $\mathbb{F}_q$ of elliptic curves E such that $\operatorname{End} E \simeq \mathcal{O}_K$: if $\mathfrak{a} \in \operatorname{Cl} \mathcal{O}_K$ and $E \in X$, we set $\mathfrak{a} \cdot E := E/E[\mathfrak{a}]$. Couveignes conjectured that this action defines a HHS, where vectorization and parallelization are exactly finding path in the isogeny graph. CSIDH [Cas+18], a Diffie-Hellman key exchange, is based on that[4].

Classically, the fastest algorithms solving vectorization and parallelization are generic and run in $O(q^{1/4})$, hence are exponential. On quantum computers, there is a subexponential isogeny evaluation algorithm, which helps to solve vectorization in subexponential time, more details are given in [Smi18, § 14].

4 To be more accurate, CSIDH uses supersingular curves where $\operatorname{End} E$ is a commutative order in a quadratic imaginary field, which is not always the case when the curve is supersingular.

This is where we will end our introduction on isogeny-based cryptography, it is an active research area with several, more complicated, protocols and even more technical details, for instance we did not discuss how to use effectively the isomorphism $\mathrm{End}\, E \simeq \mathrm{Cl}\, \mathcal{O}_K$.

1.5 SIDE-CHANNEL ATTACKS

So far we discussed several protocols and algorithms, and some attacks based on the mathematical foundation. In this section, we would like to give a small insight on side-channel attacks, which revolve around exploiting additional data obtained from physical measurements or by exploiting implementation flaws. A great review of side-channel attacks on elliptic curves has been realized in 2013 [Dan+13]. We will describe two of them.

We described the fast exponentiation in Algorithm 1.3.7 to perform the scalar multiplication efficiently. However (using additive notations):

- if the current bit is 0, the algorithm performs one doubling;
- if the current bit is 1, the algorithm performs one doubling and one addition.

Since the formulas for doubling a point and adding two distinct points are not the same in Proposition 1.3.1, they won't have the same footprint in terms of power consumption (illustrated in Fig. 1.5) and this can be exploited in a passive attack by analysing current, as suggested by Coron [Cor99]. A counter-measure would be to use the same formulas to add two points,

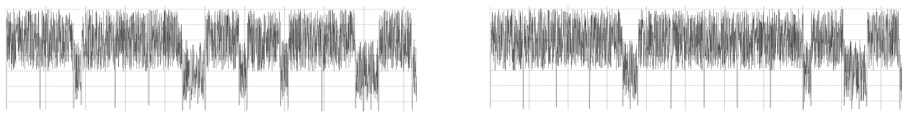

Figure 1.5: The left curve represents the power consumption of a doubling, the right curve the power consumption of an addition [Dan+13, Fig. 2]

whether they are distinct or not. These unified formulas are still prone to power analysis attacks in some cases as suggested by Walter, see [Wal04] for more details. A more common fix is using constant-time algorithms, where the operations are the same independently of the current bit, we will present the Montgomery ladder in Chapter 2 as an example.

Instead of analysing the protocol, an attacker could inject intentional errors to simplify the computation as suggested by Ciet and Joye [CJ05]. Let E be an elliptic curve with equation $y^2 = x^3 + ax + b$ and a point $P \in E$. If the coefficient a is faulty and replaced by a', then there is some constant b' such that $P \in E'$ where E' is given by $y^2 = x^3 + a'x + b'$. b' is not involved in the computation according to formulas from Proposition 1.3.1, hence the

scalar multiplication will compute $Q = k \cdot P \in E'$ without any issue. The coefficients of the curve E' can be recovered via

$$\begin{cases} y_P^2 = x_P^3 + a'x_P + b', \\ y_Q^2 = x_Q^3 + a'x_Q + b'. \end{cases}$$

Finally, if P has small order ω in E', the attacker can compute the DLP in E' to recover $k \bmod \omega$. By performing this attack several times, they can have enough equation to compute the secret k with the Chinese remainder theorem.

2 ARITHMETIC

CONTENTS

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2.3 Montgomery ladder 37

In Chapter 1, we introduced several protocols and algorithms and discussed their complexity in terms of group additions. It is therefore essential to have an efficient arithmetic on the groups involved. In the multiplicative group $\mathbb{F}_p^*$, the group law comes from the elementary and low-level operations that are additions, subtractions, multiplications and divisions modulo an integer. For our cryptographic purposes, we require a certain number of bits to have secure cryptosystems (for instance, 256), and they are generally bigger than the 64-bit precision of processors, hence we will need multi-precision arithmetic.

On elliptic curves, these operations are also involved, but to compare the various formulas, one generally counts the number of elementary operations required. For instance in Proposition 1.3.1:

- when adding two distinct points, one requires to perform 1 multiplication, 1 square, 1 inversion and 6 additions/subtractions;
- when doubling a point, one requires to perform 1 multiplication, 2 squares, 1 inversion, 4 additions/subtractions and a multiplication by 2 and 3 (which reduces to 3 more additions).

To compare accurately, one needs to know the relative costs of these operations.

This thesis focuses on the arithmetic of elliptic curves, but we first need to introduce some notions on the computational cost of elementary operations. This will be done in Section 2.1. Section 2.2 will focus on computing efficiently g^n on any abstract group, and finally Section 2.3 will introduce the most used algorithm to perform scalar multiplication on elliptic curves: the Montgomery ladder.

2.1 ELEMENTARY OPERATIONS

In this section, we will discuss the main operations: addition/subtraction, multiplication, squaring, and inversion modulo an integer m. We will describe how they can be implemented and compare their relative cost. Our references for this section will be [BZ11, § 1, § 2; MvV97, § 14.2, § 14.3].

Uniqueness is achieved by assuming $x_{n-1} \neq 0$. However, being able to pad numbers with zeroes on the left is convenient, therefore we will not assume uniqueness.

As stated in the introduction, we want to deal with multi-precision arithmetic, that is an integer x will be decomposed

$$x = x_{n-1}\beta^{n-1} + x_{n-2}\beta^{n-2} + \cdots + x_0 = \sum_{i=0}^{n-1} x_i\beta^i$$

where $0 \leqslant x_i < \beta$. β is the base, it will be fixed during this whole chapter. Common values for β are 2^w where w is the word-length. Usually $w = 32$ or 64, depending on the processor architecture. When n is the maximum integer such that $x_{n-1} \neq 0$, n is the number of limbs of x (or length if the context is not ambiguous).

Our main interest is the comparison of the order of magnitude of the different operations, we are not specialized at all in finite field arithmetic and are not intending to discuss many optimizations that can be achieved on their arithmetic (which is also very dependent on the context). Table 2.1 gives a comparison of the operations we are going to introduce in this section.

	Relative comparison to **M**
Addition and Subtraction **a**	between 0.02**M** and 0.3**M**
Multiplication **M**	1**M**
Multiplication by a small constant **c**	between 0.02**M** and 0.3**M**
Squaring **S**	between 0.67**M** and 0.8**M**
Inversion **I**	between 7**M** and 150**M**

Table 2.1: Comparison of the main arithmetic operations in prime finite fields of cardinality between 1 and 4 machine words in the $\mathtt{mp}\mathbb{F}_q$, NTL and ZEN implementations [GT07, Table 1].

2.1.1 Addition and subtraction

If the length are different, zeroes can be appended to the left of the shortest integer.

These are the most elementary operations, the algorithms are very simple. In this section, we will consider integers of the same length n, we will decompose them as

$$x = x_{n-1}\beta^{n-1} + x_{n-2}\beta^{n-2} + \cdots + x_0, \; y = y_{n-1}\beta^{n-1} + y_{n-2}\beta^{n-2} + \cdots + y_0. \quad (2.1)$$

The addition and subtraction are given in Algorithms 2.1 and 2.2.

β should be chosen such that the operation modulo β is trivial. The two algorithms cost $O(n)$ machine instructions, and as we will see this is negligible compared to multiplication.

Algorithm 2.1: Addition of integers

Input: Integers x, y of word-length n as in Eq. (2.1)
Output: $x + y$

```
def Addition(x, y):
    c ← 0;
    for i ← 0 to n − 1 do
        z_i ← x_i + y_i + c mod β;
        if x_i + y_i + c < β then c ← 0 else c ← 1;
    end
    z_n ← c;
    return z_n β^n + z_{n−1} β^{n−1} + ··· + z_0;
```

Algorithm 2.2: Subtraction of integers

Input: Integers x, y of word-length n as in Eq. (2.1)
Output: $x - y$

```
def Subtraction(x, y):
    c ← 0;
    for i ← 0 to n − 1 do
        z_i ← x_i − y_i − c mod β;
        if x_i − y_i − c ⩾ 0 then c ← 0 else c ← 1;
    end
    z ← z_{n−1} β^{n−1} + z_{n−2} β^{n−2} + ··· + z_0;
    if c = 0 then return z else return Subtraction(0, z);
```

Moreover, if m is an integer, and $x, y \in \mathbb{Z}/m\mathbb{Z}$ are represented with integers $0 \leqslant x, y < m$, computing the class of $x + y$ and $x - y$ is easy because:

- If $x + y \geqslant m$, then $0 \leqslant (x + y) - m < m$.
- If $x \geqslant y$, then $0 \leqslant x - y < m$.
- If $x < y$, then $0 \leqslant (x + m) - y < m$.

Therefore, performing addition and subtraction modulo m requires at most one additional step and is therefore still linear.

In our case, we will work in a context where inputs will have the same length and modulo a same integer m, therefore we tend to omit these from our notations.

Notation 2.1.1 *The cost of an addition or subtraction will be denoted* **a**.

2.1.2 Multiplication

The addition and multiplication of two integers fitting in a machine word have the same cost up to a small factor between 1 and 2. However, when the number of limbs increases, we will see that performing a multiplication costs much more than an addition. Having fast multiplications is a whole topic of research, here we will only cover the naive approach and the Karatsuba's

algorithm. In the addition algorithm, padding with zeroes to ensure both numbers have same word-length n is cost-free since the output size will be at least n and at most $n+1$. This will not be the case any more with multiplication, we consider two integers x and y of respective word-length n and t:

$$x = x_{n-1}\beta^{n-1} + x_{n-2}\beta^{n-2} + \cdots + x_0, \; y = y_{t-1}\beta^{t-1} + y_{t-2}\beta^{t-2} + \cdots + y_0. \quad (2.2)$$

The word-length of $x \cdot y$ is at most $n+t$, the naive approach (also known as schoolbook method) is given in Algorithm 2.3. This is exactly the way

Algorithm 2.3: Naive multiplication of integers

Input: Integers x, y of word-length respectively n and t as in Eq. (2.2)
Output: $x \cdot y$

```
def SchoolbookMultiplication(x, y):
    z ← 0;
    for j ← 0 to t − 1 do          // Compute intermediate products
        w_j ← 0;
        c ← 0;
        for i ← 0 to n − 1 do                  // Compute w_j = x · y_j
            uβ + v ← x_i y_j + c;                              // u, v < β
            w_j ← w_j + vβ^i;
            c ← u;
        end
        w_j ← w_j + cβ^n;
    end
    for j ← 0 to t − 1 do              // Add intermediate products
        z ← z + w_j β^j;
    end
    return z;
```

multiplication is taught in primary school, both main loop requires $O(nt)$ operations (the shifts in Algorithm 2.3 are done in constant time because of the representation). This cost is quadratic for two integers of similar length and therefore much more costly than an addition. The important operation to improve in this algorithm is the multiplication of an integer of word-length n with a single-precision integer, this is linear in n.

Karatsuba's algorithm [KO62] is based on the following observation: let x, y be integers of length n in base β. Set $k = \lceil n/2 \rceil$ and consider the decomposition $x = a\beta^k + b, y = c\beta^k + d$ where a, b, c, d are of length k. Then:

$$xy = (a\beta^k + b)(c\beta^k + d) = ac\beta^{2k} + (ac + bd - (a-b)(c-d))\beta^k + bd.$$

Hence, multiplying two integers of length n requires only 3 multiplications of words of length $\lceil n/2 \rceil$. By repeating this recursively, the multiplication of two integers of word-length n can be achieved asymptotically in $O(n^{\log_2 3})$, and $\log_2 3 \approx 1.585$. This is a neat gain, but one must keep in mind that this is asymptotic, and some constants are hidden in the O notation: the schoolbook algorithm will be faster for integers with small number of words. In the

cryptographic context, Karatsuba's algorithm is viable. Other algorithms faster asymptotically exist, but they require even longer integers to be worth it.

The algorithms discussed as of now concern the multiplication of integers, but multiplication modulo an integer m is a bit more involved. Let n be the word-length of m, and $x, y \in \mathbb{Z}/m\mathbb{Z}$, represented as two integers lower than m. We will assume there length is n too. Of course, the simplest approach would be the following:

- Compute xy as a product of integer.
- Reduce xy modulo m with a Euclidean division, this can be performed in $O(n(n+1))$ word operations [BZ11, Alg. 1.6].

In the end, the cost is quadratic, just like a standard multiplication. But the Euclidean division has an impact on the hidden constant in the O notation. Some integer representations such as the Montgomery reduction [MvV97, § 14.3.2] avoid doing the full division which is the limiting factor. These algorithms also perform a quadratic multiplication (or better if using Karatsuba's method), but they improve the constant involved compared to the one of the Euclidean division. Discussing all the variants is out of the scope of this thesis, the usual way to compare algorithms involving basic operations is to express everything in terms of multiplications.

One last note on this topic, a modular multiplication by a small integer (one or two words) is also linear in the size of the word, therefore we will separate these two.

Notation 2.1.2 *The cost of a multiplication will be denoted* $\mathbf{M}$. *The cost of a multiplication by a small integer will be denoted* $\mathbf{c}$.

In Table 2.1, we use the value $0.02\mathbf{M} \leqslant \mathbf{a}, \mathbf{c} \leqslant 0.3\mathbf{M}$, which is realistic in practice with integers of length 2 to 10 words.

Remark 2.1.3 *In the introduction, we reduced a multiplication by 3 to two additions. This is also linear, but certainly the constant in* $\mathbf{a}$ *is better than the one in* $\mathbf{c}$. *For small enough values, it is preferable to transform the multiplication by a sequence of additions.*

In the same vein, we mentioned that the multiplication in Algorithm 2.3 of Algorithm 2.3 can be achieved in constant time because they correspond to a word shift. Multiplication by any power of 2 can also be done in constant time with bit shifts, and this can be combined to the above remark to enhance small multiplications. For instance multiplication by 10 of an integer x *can be done with two left bit shifts and one addition using the formula* $10x =$ `(x << 3) + (x << 1)`.

2.1.3 Squaring

Modulo an integer m, computing the square of x can be achieved as the product $x^2 \equiv x \cdot x \bmod m$.

Notation 2.1.4 *The cost of a squaring will be denoted* $\mathbf{S}$.

What this shows is that $\mathbf{S} \leqslant \mathbf{M}$, but can we achieve a better bound? What is the limit? Since $4xy = (x+y)^2 - (x-y)^2$, we already know that the lowest bound is $\mathbf{S} \geqslant 0.5\mathbf{M}$.

Consider the following formula over complex numbers, where $a, b \in \mathbb{R}$:

$$(a+bi)^2 = a^2 - b^2 + 2abi = (a-b)(a+b) + 2abi.$$

Over $\mathbb{C}$, a standard multiplication is done with the following Karatsuba-like formula, with $a, b, c, d \in \mathbb{R}$:

$$(a+bi)(c+di) = ac - bd + (ac + bd - (a-b)(c-d))i.$$

Hence, a multiplication in $\mathbb{C}$ costs three real multiplications, whereas a square costs only two, in this case $\mathbf{S} = \frac{2}{3}\mathbf{M}$. These formulas can be transposed as well to finite field arithmetic. Consider a prime $p \equiv 3 \bmod 4$, then -1 is not a square modulo p, consider i a square root of -1 in a quadratic extension and use the representation $\mathbb{F}_{p^2} = \mathbb{F}_p[i]$. Using the above formulas in $\mathbb{F}_{p^2}$, $\mathbf{S} = 0.67\mathbf{M}$.

This is generally the best result that can be achieved, it is not limited to the above cases. For instance, with more involved machinery, Bernstein [Ber06] achieved $\mathbf{S} = 0.67\mathbf{M}$ over the field $\mathbb{F}_p$ with $p = 2^{255} - 19$, which is not covered by the previous case since $p \equiv 1 \bmod 4$.

One commonly used value in the literature[1] is $\mathbf{S} = 0.8\mathbf{M}$, taking into account the several optimizations that can be achieved.

2.1.4 Greatest common divisor and modular inversion

Let m be an integer, and $0 < x < m$. Computing the inverse of x modulo m (if it exists) is achieved via an extended Euclidean algorithm. If $d = \gcd(x, m)$, there are integers u and v such that $ux + vm = d$. Therefore, either $d = 1$ and $ux \equiv 1 \bmod m$, so u is the inverse of x modulo m, or $d > 1$ and x is not invertible modulo m. Just like the Euclidean division, computing the Bézout relation is quadratic, but the constant is quite huge compared to the multiplication.

Notation 2.1.5 *The cost of a modular inverse will be denoted* $\mathbf{I}$.

The explicit-formulas database uses $\mathbf{I} = 100\mathbf{M}$, the value can vary quite a lot depending on the context, the several optimizations that can be achieved if m has a certain form, ...In [SL23], the authors compute some relative costs of modular operations and find $\mathbf{I} \approx 6.5\mathbf{M}$. The important message here is that inversions should be avoided at all cost when possible, for instance by using projective coordinates in the case of elliptic curves.

If one has n inversions to perform simultaneously, Montgomery suggested an algorithm which reduces the cost to 1 inversion and $3n-3$ multiplications

1 For instance, in the explicit-formulas database for elliptic curves: https://www.hyperelliptic.org/EFD/index.html.

Lemma 2.1.6 (Simultaneous inversion) *Consider* $m > 1$, *and* $x_1, \dots, x_n \in (\mathbb{Z}/m\mathbb{Z})^*$, *then* $x_1^{-1}, \dots, x_n^{-1}$ *can be computed in* $1\mathbf{I} + (3n-3)\mathbf{M}$.

A proof is available in [SB01, Lem. 3.1] for instance. This algorithm is also called 'Montgomery's trick' sometimes. Asymptotically per bit, an inversion can therefore be reduced to 3 multiplications, which we will denote by $1\mathbf{I} \overset{\text{asy}}{=} 3\mathbf{M}$.

2.2 ADDITION CHAINS

In this section we will define addition chains, they are used to generalize the fast exponentiation introduced in Algorithm 1.3.7. We will also discuss Lucas chains, a subfamily introduced by Montgomery for elliptic curves. General addition chains are defined in [MvV97, § 14.6.2].

Definition 2.2.1 (Addition chain) *Let* n *be a positive integer, an addition chain of length* r *is a finite sequence* $(a_0, \dots, a_r)$ *such that* $a_0 = 1$, $a_r = n$, *and for every* $1 \leqslant k \leqslant r$, *there are two indices* $0 \leqslant i_k, j_k < k$ *such that* $a_k = a_{i_k} + a_{j_k}$.

An example for $n = 43$ is the chain $(1, 2, 4, 8, 9, 17, 26, 43)$ of length 7:

$$\begin{array}{llll} 2 = 1 + 1; & 4 = 2 + 2; & 8 = 4 + 4; & 9 = 8 + 1; \\ 17 = 9 + 8; & 26 = 17 + 9; & 43 = 26 + 17. & \end{array}$$

$r = 7$ is the shortest length that can be achieved for $n = 43$. Finding the shortest length is a hard problem in general.

Addition chains can be derived from the binary decomposition of n:

Algorithm 2.2.2 *Let* n *be an* ℓ*-bits integer decomposed as* $(b_{\ell-1}, b_{\ell-2}, \dots, b_0)_2$ *in base* 2, *start the chain with a* 1, *and for* i *going from* $\ell - 2$ *down to* 0, *do the following:*

- *let* a *be the last element in the sequence;*
- *if* $b_i = 0$, *append* $2a$ *to the sequence;*
- *otherwise,* $b_i = 1$, *append* $(2a, 2a + 1)$ *to the sequence.*

The constructed sequence is an addition chain for n.

$n = 43$ decomposes as $(1, 0, 1, 0, 1, 1)_2$ in base 2, yielding the addition chain $(1, 2, 4, 5, 10, 20, 21, 42, 43)$ of length $r = 8$, which is not optimal.

We already introduced the fast exponentiation in Algorithm 1.3.7, a generalization is the addition chain exponentiation:

Algorithm 2.2.3 (Addition chain exponentiation) *Let* G *be a group,* g *an element of* G, n *an integer and* $(a_0, a_1, \dots, a_r)$ *an addition chain of length* r *representing* n. *For* k *going from* 1 *to* r, *let* $0 \leqslant i_k, j_k < k$ *be the indices such that* $a_k = a_{i_k} + a_{j_k}$. *Compute* $g^{a_k} = g^{a_{i_k}} \times g^{a_{j_k}}$. *Return* $g^{a_r} = g^n$.

If r is the length of the chain representing n, the algorithm takes r group operations to compute g^n, this is why it is necessary to find short chains. We called it a generalization of the fast exponentiation because applying Algorithm 2.2.3 to the addition chain derived from the binary decomposition is exactly Algorithm 1.3.7. In the case of 43, the fast exponentiation requires 8 multiplications whereas the fastest addition chain can be completed in 7.

Finally, we would like to introduce a subfamily of addition chains, due to Montgomery [Mon83]:

Definition 2.2.4 (Lucas chain) *Let n be an integer represented by an addition chain $(a_0, a_1, \dots, a_r)$ of length r. For every $1 \leqslant k \leqslant r$, let $0 \leqslant i_k, j_k < k$ be the indices such that $a_k = a_{i_k} + a_{j_k}$. The sequence is said to be a Lucas chain if for every $1 \leqslant k \leqslant r$:*

- *either $a_{i_k} = a_{j_k}$;*
- *or otherwise, there is some index $0 \leqslant m_k < k$ such that $|a_{i_k} - a_{j_k}| = a_{m_k}$.*

For instance, any integer of the Fibonacci sequence can be represented by a Lucas chain, such as $(1, 2, 3, 5, 8, 13, 21, 34, 55)$.

We will have to do a short introduction to Kummer lines of elliptic curves to justify the use of Lucas chains. This is the main topic of this thesis and will be properly introduced in Chapter 4. Let E be an elliptic curve, $P \in E$ and $x(P)$ the x-coordinate of P. Given the equation of E and $x(P)$ it is possible to recover the y-coordinate up to a sign. We then prefer to work on the set $\mathcal{K} = \{x(P) \mid P \in E\}$ where $x(0_E) := \infty$, but what about the group law? Since $x(-P) = x(P)$, given two distinct points P and Q, it is not possible to distinguish $x(P+Q)$ from $x(P-Q)$ with the only data of $x(P)$ and $x(Q)$, as illustrated in Fig. 2.1. Fortunately, we can expect 'differential addition

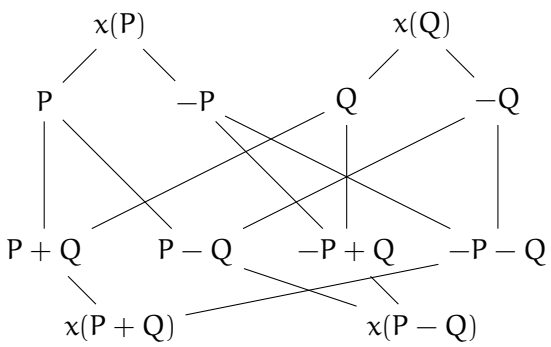

Figure 2.1: Group law is not well-defined on the Kummer line

formulas' which can compute $x(P+Q)$ from the data of $x(P)$, $x(Q)$ and $x(P-Q)$. Montgomery did so in [Mon87] on the curve

$$E : y^2 = x(x^2 + \mathcal{A}x + 1),$$

as well as formulas to compute $x(2 \cdot P)$ from $x(P)$ and some curve constants, those are given below in Eqs. (2.3) and (2.4). Given a Lucas chain representing n, Algorithm 2.2.3 can be used to compute $x(n \cdot P)$ with Montgomery's formulas because of the differential addition, benefiting from the efficiency

of the formulas compared to Proposition 1.3.1 and the speed of fast exponentiation.

Montgomery also provided a heuristic algorithm named PRAC to compute Lucas chains [Mon83, § 7]. The issue with all these methods, as already stated in Section 1.5, is that they are vulnerable to side-channel attacks. In the next section, we will present the most efficient way to do arithmetic on elliptic curves in a cryptographic context.

2.3 MONTGOMERY LADDER

The Montgomery ladder [Mon87] was introduced in 1987 along differential addition and doubling formulas. We will work in a field k, with $\mathcal{A}, \beta \in k$ and on the Montgomery curve

$$E : \beta y^2 = x(x^2 + \mathcal{A}x + 1).$$

The formulas are the following, for two distinct points $P, Q \in E$:

$$x(P+Q)x(P-Q) = \left(\frac{x(P)x(Q)-1}{x(P)-x(Q)}\right)^2, \tag{2.3}$$

$$x(2 \cdot P) = \frac{\left(x(P)^2 - 1\right)^2}{4x(P)\left(x(P)^2 + \mathcal{A}x(P) + 1\right)}. \tag{2.4}$$

Remark 2.3.1 *Note that the constant β is never involved in the computation. Indeed, β only describes what we will call later a twist of the elliptic curve, and working only with the x-coordinate / with a Kummer line does not keep track of the twist of the elliptic curve.*

We kept for now the affine point of view, but as explained earlier in this chapter, inversions are to be avoided. A solution is to use projective coordinates instead. Write $x(P) = \frac{X(P)}{Z(P)} := (X(P) : Z(P))$ and $x(0_E) := (1 : 0)$. Since $(\lambda X : \lambda Z) = (X : Z)$ for any $\lambda, X, Z \in k$, inversions can be changed into multiplications. The optimized formulas on Montgomery curves are described in Algorithms 2.4 and 2.5.

In the end:

- A differential addition costs $2\mathbf{M} + 2\mathbf{S} + 1\mathbf{m} + 6\mathbf{a}$. Here $\mathbf{m}$ is the multiplication by X_0/Z_0, we distinguish it from the others because in some context, the quotient can be pre-computed, reducing to only one multiplication $1\mathbf{M}$, and even $1\mathbf{c}$ if the quotient is small. Otherwise, with projective coordinates, if no pre-computation can be done, $\mathbf{m}$ just counts as $2\mathbf{M}$.

- A doubling costs $2\mathbf{M} + 2\mathbf{S} + 1\mathbf{m}_0 + 4\mathbf{a}$. Here $\mathbf{m}_0$ is the multiplication by d which is always a constant of the algorithm. If it is small, it counts as $1\mathbf{c}$, otherwise it is $1\mathbf{M}$.

This is way faster than using the affine formulas from Proposition 1.3.1.

Algorithm 2.4: Differential addition on a Montgomery curve

Input: $x(P) = (X_1 : Z_1)$, $x(Q) = (X_2 : Z_2)$ and $x(P - Q) = (X_0 : Z_0)$
Output: $x(P + Q) = (X : Z)$

```
def DiffAdd(x(P), x(Q), x(P − Q)):
    u ← (X1 + Z1)(X2 − Z2);
    v ← (X1 − Z1)(X2 + Z2);
    w ← (u + v)^2;
    t ← (u − v)^2;
    X ← w;
    Z ← (X0/Z0) t;
    return (X : Z);
```

Algorithm 2.5: Doubling on a Montgomery curve

Input: $x(P) = (X_1 : Z_1)$
Output: $x(2 \cdot P) = (X : Z)$
Data: $d = \frac{A+2}{4}$

```
def Doubling(x(P)):
    u ← (X1 + Z1)^2;
    v ← (X1 − Z1)^2;
    t ← u − v;
    X ← uv;
    Z ← t(v + dt);
    return (X : Z);
```

Remark 2.3.2 *The differential addition formulas do not hold if* $x(P - Q) = (0 : 1)$. *In this case,* $P - Q = T$ *is a 2-torsion point and* $x(P + Q) = x(2 \cdot Q + T)$. *If* $x(2 \cdot Q) = (X : Z)$, *then* $x(2 \cdot Q + T) = (Z : X)$ *according to Eq.* (6.2).

Notation 2.3.3 *As above, in the context of elliptic curves,* $\mathbf{m}_0$ *denotes a multiplication by some curve constant. In the context of Montgomery ladder,* $\mathbf{m}$ *denotes the multiplication by the difference in differential addition.*

Moreover, Montgomery generalized the fast exponentiation from Algorithm 1.3.7 to compute efficiently the scalar multiplication on Kummer lines, taking into account the differential addition. The algorithm is described in Algorithm 2.6 and illustrated in Fig. 2.2.

Each step is exactly one differential addition and one doubling, whether the bit is zero or one, and can therefore be made resistant to side-channel attacks. The general cost of a step is $4\mathbf{M} + 4\mathbf{S} + 1\mathbf{m} + 1\mathbf{m}_0 + 10\mathbf{a}$[2]. Moreover, in the differential addition, the difference is always the base point. Therefore, depending on the protocol, the base point can be chosen with a small x-coordinate such that $1\mathbf{m}$ counts as $1\mathbf{c}$. In the end, assuming the constant d is small ($1\mathbf{m}_0 = 1\mathbf{c}$), the range of the cost of one ladder step on a Montgomery curve is given in Table 2.2.

2 Two additions could be saved by grouping differential addition and doubling and re-using partial sums.

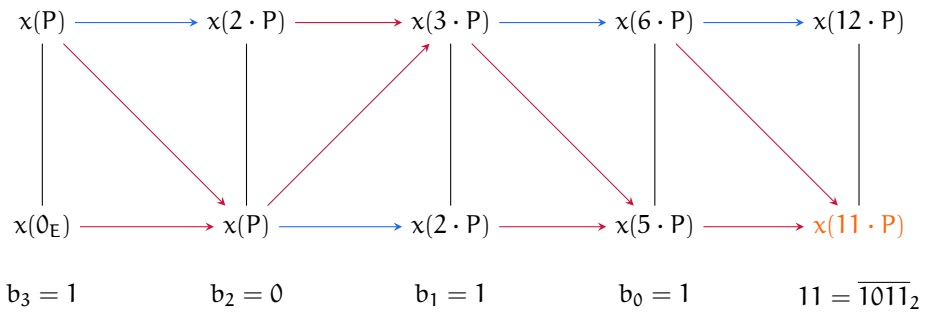

Figure 2.2: Montgomery ladder to compute $x(11 \cdot P)$. Doublings are represented with blue arrows, and differential additions with red ones.

Algorithm 2.6: Scalar multiplication with Montgomery ladder

Input: $n = (b_{\ell-1}, b_{\ell-2}, \dots, b_0)_2$ an ℓ-bits integer, $x(P)$ the x-coordinate of a point $P \in E$, written projectively

Output: $x(n \cdot P)$

```
def MontgomeryLadder(n, x(P)):
    x(R) ← x(0_E);
    x(S) ← x(P);
    for i ← ℓ − 1 down to 0 do
        if b_i = 1 then
            x(R) ← DiffAdd(x(R), x(S), x(P));
            x(S) ← Doubling(x(S));
        else if b_i = 0 then
            x(S) ← DiffAdd(x(R), x(S), x(P));
            x(R) ← Doubling(x(R));
        end
    end
    return x(R);
```

Base point	Cost
Not normalized	$6\mathbf{M} + 4\mathbf{S} + 1\mathbf{c}$
Normalized	$5\mathbf{M} + 4\mathbf{S} + 1\mathbf{c}$
Normalized, small value	$4\mathbf{M} + 4\mathbf{S} + 2\mathbf{c}$

Table 2.2: Possible ladder step cost on a Montgomery curve

3 ABELIAN VARIETIES

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This chapter aims to be an introduction to abelian varieties and their most important properties. There are several reasons for such a chapter:

- Elliptic curves are a particular case of abelian varieties;
- The theta group theory developed in Chapter 5 holds on any abelian variety, and provides theta coordinates, which are available in every dimension;
- While this thesis focuses on dimension 1, some of the ideas can be generalized in higher dimension, and we would like to have these openings.

This is probably the most technical part of this thesis, it introduces properly isogenies in a general context and justifies some important results on divisors that we will use later on. Chapter 4 will specialize to the case of elliptic curves and intend to introduce them using their specificities.

Since the modern theta group theory is mainly due to Mumford, we will naturally refer to his book on abelian varieties too [Mum74]. Another great reference are the lecture notes from Milne [Mil08], as well as the unfinished book of van der Geer and Moonen [GM]. We will assume several results from algebraic geometry throughout this chapter. We will try to provide most results over a perfect field k, its algebraic closure is denoted $\overline{k}$.

In Section 3.1, we first introduce the definition of an abelian variety, as well as important properties. We also give some details on the correspondence between divisors and line bundles as we will use the language of divisors in this thesis. We then introduce an important class of homomorphisms, the isogenies in Section 3.2. We will particularly study the polarization associated to a divisor and give some basic results on the dual of an abelian variety in Section 3.3.

3.1 GENERALITIES

In this whole chapter, we should make the distinction between a *morphism* of algebraic varieties (which is a regular map between varieties) and a *homomorphism* of abelian varieties (which will be a morphism of varieties which respects the group structure). Since there is no consensus on the definition of an algebraic variety, we should specify it here.

Definition 3.1.1 (Affine variety) *An affine variety over* k *is a geometrically reduced* k*-ringed space* $(X, \mathcal{O}_X)$ *which is isomorphic to an algebraic subset of some* k^n *endowed with its structural sheaf.*

Definition 3.1.2 (Algebraic variety) *An algebraic variety over* k *is a geometrically reduced* k*-ringed space* $(X, \mathcal{O}_X)$ *with a finite open covering* $X = \cup_i U_i$ *such that:*

- X *is connected;*
- *For every* i, $(U_i, \mathcal{O}_X(U_i))$ *is an affine variety of finite type;*
- *Separation axiom: for every affine variety* Y *and for every pair of morphisms* $f, g : Y \to X$, *the set* $\{y \in Y \mid f(y) = g(y)\}$ *is closed in* Y.

This corresponds to Mumford definition in his red book [Mum88]. Another commonly used definition omits the connected assumption, such as Milne or Serre. Abelian varieties are connected though and since our intent is not to give a complete treatment of algebraic varieties but rather an introduction to abelian ones, we will go with Mumford's definition.

Remark 3.1.3 *The above definitions can be rephrased from the scheme point of view:*

- *an affine variety is an affine geometrically reduced* k*-scheme of finite type;*
- *an algebraic variety is a geometrically reduced connected separated* k*-scheme of finite type.*

We will use the following usual convention: if X is a variety defined over k, the set of k-rational points is denoted $X(k)$, and an element $x \in X$ is considered over the algebraic closure, that is $x \in X(\overline{k})$. For instance, $\mathbb{P}^n$ is the projective space of dimension n, defined over k, and

$$\mathbb{P}^n = \{(x_0 : \cdots : x_n) \mid x_i \in \overline{k}\},\ \mathbb{P}^n(k) = \{(x_0 : \cdots : x_n) \mid x_i \in k\}.$$

3.1.1 Definition and first properties

With this convention set, we should now define an abelian variety, this is also the last time we will explicitly mention the redundant connected assumption.

Definition 3.1.4 (Abelian variety) *An abelian variety over* k *is a connected complete algebraic variety* X *with a group law* $m : X \times X \to X$ *such that* m *and the inverse map are both morphisms of varieties, and the neutral element* 0_X *is a* k*-rational point. The* k*-rational points are denoted* $X(k)$*, whereas* $X = X(\overline{k})$ *denotes the geometric points.*

From now on, if we consider a variety X, we mean an algebraic variety with no further assumption: an abelian variety will always be called that way. We will also note the group law on an abelian variety additively as we will see in Corollary 3.1.8 that it is commutative. Another property we want to derive is the following.

Proposition 3.1.5 *An abelian variety* X *defined over* k *is smooth.*

This is a corollary of [GM, Prop. 1.5].

We will heavily use the following theorem.

Theorem 3.1.6 (Rigidity lemma) *Consider* X *a complete algebraic variety,* Y *and* Z *two algebraic varieties, and* f *a morphism* $f : X \times Y \to Z$ *such that there are some* $y_0 \in Y(k)$, $z_0 \in Z(k)$ *verifying* $f(X \times \{y_0\}) = \{z_0\}$*. Then* f *factors through the projection* $p_Y : X \times Y \to Y$*: there is a morphism* $g : Y \to Z$ *such that* $f = g \circ p_Y$*.*

Proof *We may assume* $k = \overline{k}$*. Let* $x_0 \in X$*, and set* $g : y \in Y \mapsto f(x_0, y)$*. By separation axiom, we only need to show that* f *and* $g \circ p_Y$ *coincides on an open subset of* $X \times Y$*. Let* U *be an affine open neighbourhood of* z_0 *in* Z *and set* $F = Z \setminus U$*. It is closed, so* $f^{-1}(F)$ *is too in* $X \times Y$ *and since* X *is complete, by definition* $G = p_Y(f^{-1}(F))$ *is closed in* Y*.*

Moreover, $y \in G$ *if and only if there is some* $x \in X$ *such that* $f(x, y) \notin U$*, so* y_0 *is not in* G *by assumption. Let* $V = Y \setminus G$ *be an open subset of* Y*, we have* $y_0 \in V$ *and more generally, for every* $y \in V$*,* $f(X \times \{y\}) \subseteq U$*. Therefore, since the complete variety* $X \times \{y\}$ *is embedded into the affine variety* U *via* f*, it reduces to a singleton* $\{f(x_0, y)\}$*. Finally, on the open set* $X \times V$*:*

$$\forall x \in X, y \in V,\ f(x, y) = f(x_0, y) = g \circ p_Y(x, y). \qquad \square$$

The rigidity lemma actually puts some constraints on abelian varieties.

Corollary 3.1.7 *Let* X *and* Y *be two abelian varieties, and consider a morphism of algebraic varieties* $f : X \to Y$*. Then* $f = g + f(0_X)$ *for some* $g : X \to Y$ *a homomorphism of abelian varieties.*

Proof *Assume we know the result when* $f(0_X) = 0_Y$*. Then, by applying it to* $g = f - f(0_X)$*, we recover that* g *is a homomorphism and* $f = g + f(0_X)$*.*

We now focus on the case $f(0_X) = 0_Y$*, set* $\varphi : X \times X \to Y$ *given by:*

$$\forall x, y \in X,\ \varphi(x, y) = f(x + y) - f(x) - f(y).$$

Since $\varphi(X \times \{0_X\}) = \{0_Y\}$ *and* X *is complete by definition, we get by Theorem 3.1.6 that* $\varphi(x, y) = \varphi(0_X, y) = 0_Y$ *for any* $x, y \in X$*. Hence,* f *is a homomorphism.* $\square$

Note that the above corollary does not use the commutativity, hence we can re-use it to prove the following.

Corollary 3.1.8 *The group law on an abelian variety X is commutative.*

Proof *Consider $i : X \to X$ the inverse map. It is a morphism by definition of an abelian variety, and since $i(0_X) = 0_X$, it is a homomorphism by Corollary 3.1.7. Therefore, for any $x, y \in X$, $i(x+y) = i(x) + i(y) = -x - y$ on the one hand, and $i(x+y) = -y - x$ on the other hand, hence $x + y = y + x$.* □

3.1.2 Divisors and line bundles

So far we only discussed the structure of an abelian variety. Divisors are an important tool in the understanding of an abelian variety, and we will use those extensively in Chapter 5. They also are important in regard to the polarization. In this section, we should give some reminders on divisors and line bundles. We will refer to [Har77, § II.6].

We are not aiming at giving a complete description in the broadest context possible in algebraic geometry, but rather in the one of abelian varieties. Hence, we will consider an abelian variety X and benefit from some of its inherent properties. In particular, it is a normal and irreducible smooth variety.

Divisors

We will only describe Weil divisors here, consider an abelian variety X.

We only require a normal and irreducible smooth variety actually.

Definition 3.1.9 (Weil divisor) *A prime divisor on X is an irreducible subvariety of codimension one. The group of (Weil) divisors is the free abelian group generated by prime divisors, and is denoted $\operatorname{Div} X$. We will write $D \in \operatorname{Div} X$ as $D = \sum_i n_i Y_i$ where $n_i \in \mathbb{Z}$ with finitely many $n_i \neq 0$, and Y_i prime divisors on X.*

A divisor $D = \sum_i n_i Y_i$ is said to be effective if for all i, $n_i \geqslant 0$, we will write $D \geqslant 0$ in that case, and $D \geqslant D'$ if $D - D' \geqslant 0$ for two divisors $D, D' \in \operatorname{Div} X$.

If X is a curve, a divisor is then a sum of formal points. Some divisors we are interested in are the principal ones.

Definition 3.1.10 (Principal divisor) *Let Y be a prime divisor on X, then it has an associated valuation v_Y. Let $n \geqslant 0$ be an integer. For any $f \in k(X)$ non-zero, f has a zero of order n along Y if $v_Y(f) = n$ and a pole of order n if $-v_Y(f) = n$.*

For any $f \in k(X)$ which is non-zero, we define $\operatorname{div} f = \sum_Y v_Y(f) Y$ where the sum runs over all prime divisors of X. We call $\operatorname{div} f$ a principal divisor, and we denote there set $\operatorname{Princ} X$.

This is well-defined because $v_Y(f) = 0$ for all except finitely many prime divisors Y [Har77, § II, Lem. 6.1]. Therefore, $\operatorname{Princ} X \subseteq \operatorname{Div} X$. Moreover, it is a subgroup of $\operatorname{Div} X$ since, for any $f, g \in k(X)$ non-zero, $\operatorname{div} \frac{f}{g} = \operatorname{div} f - \operatorname{div} g$.

Classifying divisors is done up to linear equivalence, which we define here.

Definition 3.1.11 (Class group) *Two divisors $D, D' \in \operatorname{Div} X$ are linearly equivalent if $D - D' = \operatorname{div} f$ for some $f \in k(X)$, that is if $D - D' \in \operatorname{Princ} X$. We denote it by $D \sim D'$, it is an equivalence relation. We define the class group of X via $\operatorname{Cl} X = \operatorname{Div} X / \operatorname{Princ} X$. The class of $D \in \operatorname{Div} X$ in $\operatorname{Cl} X$ is denoted $[D]$.*

Line bundles

Divisors share similarities with line bundles. We will stay under the assumption that X is an abelian variety, although this is not necessary.

Definition 3.1.12 *A locally free $\mathcal{O}_X$-module of rank 1 is called an invertible sheaf. A line bundle on X is a vector bundle of rank 1.*

[Har77, § II, Ex. 5.18] actually shows that there is a bijection between isomorphism classes of vector bundles of rank n and isomorphism classes of locally free sheaves of rank n. In our context, there should be no confusion and we will use the names 'line bundles' and 'invertible sheaves' interchangeably.

Proposition 3.1.13 *If $\mathcal{L}$ and $\mathcal{M}$ are line bundles on X, then $\mathcal{L} \otimes \mathcal{M}$ is a line bundle. Moreover, if $\mathcal{L}$ is a line bundle on X, there is a line bundle $\mathcal{M}$, such that $\mathcal{L} \otimes \mathcal{M} \simeq \mathcal{O}_X$. We denote it by $\mathcal{L}^{-1} := \mathcal{M}$.*

This corresponds to [Har77, § II, Prop. 6.12]. On isomorphism classes, there is a natural group structure.

Definition 3.1.14 (Picard group) *The Picard group of X, denoted $\operatorname{Pic} X$ is the group of isomorphism classes of line bundles on X, with the induced operation by $\otimes$. The isomorphism class of a line bundle $\mathcal{L}$ is denoted $[\mathcal{L}]$.*

We are especially interested in the correspondence with divisors, which we should describe now.

Definition 3.1.15 *Consider a divisor $D \in \operatorname{Div} X$. For any open subset $U \subseteq X$, define the following space:*

$$\Gamma(D)(U) := \{f \in k(X)^* \mid \operatorname{div} f_{|U} + D_{|U} \geqslant 0\} \cup \{0\},$$

where $f_{|U}$ is the natural restriction to U, and if $D = \sum_Y n_Y Y$, $D_{|U} = \sum_Y n_Y Y \cap U$ where the sum runs on prime divisors Y such that $Y \cap U \neq \emptyset$.

We will also denote by $\Gamma(D) := \Gamma(D)(X)$. Those are finite dimensional vector spaces, and if $U \subseteq V$ are two open subsets of X, then $\Gamma(D)(V) \subseteq \Gamma(D)(U)$, this defines a sheaf $\mathcal{O}_X(D)$ with the natural restriction maps, called the sheaf of sections above D.

Remark 3.1.16 *The line bundle $\mathcal{O}_X(D)$ is also often denoted $\mathcal{L}(D)$ to refer to the Riemann-Roch space. Another usual notation for $\Gamma(D)(U)$ is also $\Gamma(U, D)$.*

When working from the line bundle point of view, if $\mathcal{L}$ is a line, the corresponding space of (global) sections above $\mathcal{L}$ is $\Gamma(X, \mathcal{L})$, or $H^0(X, \mathcal{M})$ from the cohomological point of view.

This sheaf is in fact a line bundle, and relates having isomorphic line bundles with having linearly equivalent divisors.

Proposition 3.1.17 *Let $D, D' \in \operatorname{Div} X$, then $\mathcal{O}_X(D - D') \simeq \mathcal{O}_X(D) \otimes \mathcal{O}_X(D')^{-1}$ and $D \sim D'$ if and only if $\mathcal{O}_X(D) \simeq \mathcal{O}_X(D')$ as line bundles.*

Moreover, if $D \geqslant D'$, then $\Gamma(D') \subseteq \Gamma(D)$.

This is [Har77, § II, Prop. 6.13]. In the context of an abelian variety, we have the important following correspondence:

Theorem 3.1.18 *On an abelian variety* X*, the map* $[D] \mapsto [\mathcal{O}_X(D)]$ *is an isomorphism between* $\operatorname{Cl} X$ *and* $\operatorname{Pic} X$*. In particular principal divisors correspond to trivial line bundles.*

This is part of a more general result [Har77, § II, Cor. 6.14–6.16]. Note that we haven't stated yet that an abelian variety is smooth and projective, which induce the required properties to show this result. Since we will be using the language of divisors later in this thesis, and we require line bundles for the next theorems, it was natural to introduce both at this point, including the correspondence. One could check that the arguments are not circular.

3.1.3 Theorems of the cube and the square

Before beginning this section, we give the definition of the pullback of a morphism. If $f : X \to Y$ is a morphism of algebraic varieties, then it induces $f^* : k(Y) \to k(X)$ via $s \mapsto s \circ f$.

In the next section, we will define isogenies. In this context, the pullback of a divisor can be expressed in the following manner. Consider Z a prime divisor in Y, then $f^*Z = \sum_T T$ where the sum goes over the prime divisors T of X such that $T \subseteq f^{-1}(Z)$. We then extend this definition to $\operatorname{Div} Y$ by linearity, setting a map $f^* : \operatorname{Div} Y \to \operatorname{Div} X$.

The reason we have to restrict to isogenies is that they are flat morphisms.

Remark 3.1.19 *From the line bundle point of view, if* f *is a morphism of algebraic varieties, then the pullback of a line bundle is still a line bundle. We do not require the restriction to an isogeny.*

The theorems of the cube and the square are essential results to study polarization later on.

Theorem 3.1.20 (Theorem of the cube) *Let* X, Y *be complete varieties,* Z *a variety and* $(x_0, y_0, z_0) \in X(k) \times Y(k) \times Z(k)$ *base points. If* $\mathcal{L}$ *is a line bundle on* $X \times Y \times Z$ *such that its restrictions to*

$$X \times Y \times \{z_0\},\ X \times \{y_0\} \times Z,\ \{x_0\} \times Y \times Z$$

are trivial, then $\mathcal{L}$ *is trivial.*

A proof by Mumford is available at [Mum74, § 6, pp. 55–58]. Intuitively, being trivial on the three axes ensures triviality in the whole space. Several corollaries come from this.

Corollary 3.1.21 *Let* X *be a variety, and* Y *an abelian one. For any line bundle* $\mathcal{L}$ *on* Y *and morphisms* $f, g, h : X \to Y$*, the following isomorphism holds:*

$$(f+g+h)^*\mathcal{L} \otimes f^*\mathcal{L} \otimes g^*\mathcal{L} \otimes h^*\mathcal{L} \simeq (f+g)^*\mathcal{L} \otimes (f+h)^*\mathcal{L} \otimes (g+h)^*\mathcal{L}.$$

Proof *Let* $p_i : Y \times Y \times Y \to Y$ *be the projection to the* i*-th component,* $m_{ij} = p_i + p_j$ *and* $m = p_1 + p_2 + p_3$*. We consider the following line bundle on* $Y \times Y \times Y$*:*

$$\mathcal{M} = m^*\mathcal{L} \otimes m_{12}^*\mathcal{L}^{-1} \otimes m_{13}^*\mathcal{L}^{-1} \otimes m_{23}^*\mathcal{L}^{-1} \otimes p_1^*\mathcal{L} \otimes p_2^*\mathcal{L} \otimes p_3^*\mathcal{L}.$$

When restricted to $\{0_Y\} \times Y \times Y$*, the map* m *and* m_{23} *are the same, while* m_{12} *is the same as* p_2 *and* m_{13} *as* p_3*. Finally,* p_1 *reduces to the zero map. Therefore, every term has its inverse appearing when* $\mathcal{M}$ *is restricted to* $\{0_Y\} \times Y \times Y$*, and* $\mathcal{M}$ *is trivial on this product. Similarly, it is trivial on* $Y \times \{0_Y\} \times Y$ *and* $Y \times Y \times \{0_Y\}$*. By Theorem 3.1.20,* $\mathcal{M}$ *is trivial.*

Let $F = (f, g, h) : Y \times Y \times Y \to Y \times Y \times Y$*, we then have* $f + g + h = m \circ F$*,* $f + g = m_{12} \circ F$*,* $f + h = m_{13} \circ F$*,* $g + h = m_{23} \circ F$*,* $f = p_1 \circ F$*,* $g = p_2 \circ F$ *and* $h = p_3 \circ F$*. Therefore, the pullback of* $\mathcal{M} \simeq \mathcal{O}_X$ *via* F *gives the result.* □

In particular, we recover the theorem of the square from this.

Corollary 3.1.22 (Theorem of the square) *Consider* X *an abelian variety,* $\mathcal{L}$ *a line bundle on* X*. For any* $a \in X$*, denote by* t_a *the translation by* a*. Then for any* $x, y \in X(k)$*,*

$$t_{x+y}^*\mathcal{L} \otimes \mathcal{L} \simeq t_x^*\mathcal{L} \otimes t_y^*\mathcal{L}.$$

Proof *Since for any* $a \in X(k)$*,* $t_a = \mathrm{id}_X + a$*, we recover the result from Corollary 3.1.21 with* f *and* g *constant maps equal to* x *and* y *respectively, and* $h = \mathrm{id}_X$*.* □

The theorem of the square can be rephrased on divisors. For any divisor $D \in \operatorname{Div} X$ and $x, y \in X(k)$:

$$t_{x+y}^*D + D \sim t_x^*D + t_y^*D.$$

3.1.4 Abelian varieties are projective

Among the properties of abelian varieties, projective is a rather important one as it provides a natural system of coordinates. Mumford and Milne both prove it using linear systems. We will omit the proof and refer to [Mum74, § 6, App. 1; Mil08, Thm. 6.4]. From an algorithmic point of view, we are more interested in the consequences of this result.

Theorem 3.1.23 *Every abelian variety* X *is projective, that is there is a closed immersion* $X \to \mathbb{P}^n$ *for some* $n > 0$*.*

First, this result enables us to use the language of divisors from now on since $\operatorname{Cl} X \simeq \operatorname{Pic} X$ for smooth projective varieties. By abuse of notation, we will consider divisor classes $[D] \in \operatorname{Pic} X$. More importantly, now that we have this immersion, is there some efficient way to express the group law of the abelian variety in the projective space? Chapter 5 will go into details of building canonical coordinates, and we will see in the second part of this thesis how this can be exploited to study the arithmetic of elliptic curves, with a methodology that can be generalized to higher dimension.

Definition 3.1.24 (Ample and very ample divisor) *Let* D *be a divisor such that* $\Gamma(D) \neq \{0\}$*, and consider a basis* $(f_0, \dots, f_n)$ *of* $\Gamma(D)$*.* D *is said to be very ample if the map*

$$x \in X \mapsto (f_0(x) : \dots : f_n(x)) \in \mathbb{P}^n$$

is a closed immersion. D *is said to be ample if* mD *is very ample for some* $m > 0$*.*

Proof of Theorem 3.1.23 (which we omitted) shows that there are ample divisors on an abelian variety. This is how we will be constructing our system of coordinates in Chapter 5 with a natural basis. Here are some properties of these.

Proposition 3.1.25 *Consider two divisors* D *and* D'.

1. *If* D *and* D' *are ample, then* $D + D'$ *is ample too.*

2. *If* $D \sim D'$ *and* D' *is ample, then* D *is ample.*

3. *If* D *is ample, then* $D_{|Y}$ *is ample on* Y *for any closed subvariety* $Y \subseteq X$.

Proof *Consider two ample divisors* D *and* D', *for a convenient choice of* m, *both* mD *and* mD' *are very ample. Because* mD' *is very ample,* $\Gamma(mD')$ *is not trivial, and there is an effective divisor* $D_0 \geqslant 0$ *such that* $mD' \sim D_0$. *Hence,* $mD + D_0 \geqslant mD$, *and* $\Gamma(mD) \subseteq \Gamma(mD + D_0)$. *Since* mD *is very ample,* $mD + D_0$ *is very ample too by completing a basis of* $\Gamma(mD)$.

Finally, $mD + D_0 \sim m(D + D')$, *so* $\Gamma(mD + D_0) \simeq \Gamma(m(D + D'))$ *and the isomorphism is given by* $f \mapsto fs$ *where* $\operatorname{div} s = D_0 - mD'$, *therefore, for any* $f_0, \dots, f_r \in \Gamma(mD + D_0)$:

$$(f_0(x) : \cdots : f_r(x)) = (f_0 s(x) : \cdots : f_r s(x)).$$

The closed immersion is the same independently of this isomorphism, this proves (1).

(2) comes from a similar reasoning, consider m *such that* mD' *is very ample. Then* $\Gamma(mD) \simeq \Gamma(mD')$ *and by a similar argument* D *is also ample.*

Finally, if m *is such that* mD *is very ample, consider a basis* $\Gamma(mD) = \langle f_0, \dots, f_r \rangle$. *A basis of* $\Gamma(D)(Y)$ *is then* $\langle f_{0|Y}, \dots, f_{r|Y} \rangle$, *and* $\Gamma(D)(Y) = \Gamma(D_{|Y})$ *is the vector space of sections above* $D_{|Y}$ *in* Y *because of the sheaf structure. This proves (3).* □

One last remark is the following complicated theorem [Mum74, § 17].

Theorem 3.1.26 *For any ample divisor* $D \in \operatorname{Div} X$, mD *is very ample if* $m \geqslant 3$.

3.2 ISOGENIES

We now look into homomorphisms of abelian varieties, especially isogenies.

Definition 3.2.1 (Isogeny) *A homomorphism of abelian varieties* $f : X \to Y$ *is an isogeny if it has finite kernel and is surjective.*

We could use the following equivalent definition.

Proposition 3.2.2 *Let* $f : X \to Y$ *be a homomorphism of abelian varieties, the following assertions are equivalent:*

1. f *is an isogeny;*

2. $\dim X = \dim Y$ *and* f *is surjective;*

Dimension in the sense of varieties.

3. $\dim X = \dim Y$ *and* f *has finite kernel.*

Proof *This corresponds to Milne proof [Mil08, Prop. 7.1]. If* $y = f(x)$ *for some* $x \in X$*, then the fibre* $f^{-1}(y)$ *is isomorphic to* $\ker f$ *via the translation* $t_x : x_0 \mapsto x_0 + x$*. The dimension of a fibre verifies the following inequality [Har77, § II, Ex. 3.22]*

$$\dim f^{-1}(y) \geqslant \dim X - \dim f(X),$$

and this is an equality on a dense open subset $U \subseteq f(X)$*. Since* X *is complete,* $f(X)$ *is a closed subvariety of* Y*. Combining this with the previous result, for every* $y \in f(X)$*,* $\dim f^{-1}(y) = \dim X - \dim f(X)$*.*

If we assume that f *is an isogeny, then* $f(X) = Y$ *and* $\dim \ker f = 0$ *since it is finite. This gives* $\dim X = \dim Y$*.*

If $\dim X = \dim Y$ *and* f *is surjective, then* $\dim \ker f = 0$*, hence it is finite.*

Finally, if $\dim X = \dim Y$ *and* f *has finite kernel, then* $\dim f(X) = \dim Y$*, hence* $f(X) = Y$ *and* f *is surjective.* □

If $f : X \to Y$ is an isogeny, X is called the domain and Y the co-domain. This terminology comes from the categorical point view, where f is an arrow in the category of abelian varieties.

An isogeny $f : X \to Y$ induces $f^* : k(Y) \to k(X)$, it is injective and therefore a finite extension of function fields.

Definition 3.2.3 *An isogeny* $f : X \to Y$ *is said to be separable (resp. (purely) inseparable) if the extension of function fields* $k(X)/k(Y)$ *is separable (resp. (purely) inseparable).*

We define the degree of f *(resp. separable degree, inseparable degree) to be the degree (resp. separable degree, inseparable degree) of the extension* $[k(X) : f^*k(Y)]$ *(resp.* $[k(X) : f^*k(Y)]_s$*,* $[k(X) : f^*k(Y)]_i$*). We denote the degree (resp. separable degree, inseparable degree) of* f *by* $\deg f$ *(resp.* $\deg_s f$*,* $\deg_i f$*).*

Since the degree of field extension is multiplicative, the degree of an isogeny is too. We have $\# \ker f = \deg_s f$ (see [Mum74, Eq. (*)]). We are particularly interested in separable isogenies in the sequel, since then $\# \ker f = \deg f$. Here is one common example of a purely inseparable isogeny.

Example 3.2.4 (Frobenius endomorphism over a finite field) *Assume* k *is a finite field* $\mathbb{F}_q$*, and* X *is an abelian variety defined over* k*. The map* $x \in \overline{k} \mapsto x^q$ *extends to* $\mathbb{P}^n \to \mathbb{P}^n$ *naturally, and since* X *is projective and defined over* k*, it yields an endomorphism* $\pi_q : X \to X$*. It is a purely inseparable isogeny, of degree* q^g*.*

We will note it $[n]$ *when the context is not ambiguous.*

We will look into a particular isogeny, the multiplication by a scalar. Let $n \in \mathbb{Z}$ be an integer and consider $[n]_X : X \to X$ the map given by $x \mapsto nx$. We denote its kernel $X[n]$, the subgroup of n-torsion points. We first require a technical lemma.

Lemma 3.2.5 *Let* $D \in \operatorname{Div} X$*, then for any* $n \geqslant 1$

$$[n]^*D \sim \frac{n(n+1)}{2} D + \frac{n(n-1)}{2} [-1]^*D.$$

Proof *By applying Corollary 3.1.21 to* $f = [n]$, $g = [1]$ *and* $h = [-1]$, *we get*

$$[n+1]^*D \sim D + [-1]^*D + 2[n]^*D - [n-1]^*D.$$

We want to prove the result by induction. If $n = 1$, *it is trivial. Let* $n \geqslant 2$, *we now assume it for any* $1 \leqslant k \leqslant n$ *and want to prove it at rank* $n+1$. *If we replace* $[n]^*D$ *and* $[n-1]^*D$ *by the assumption, we have*

$$[n+1]^*D \sim \left(1 + 2\frac{n(n+1)}{2} - \frac{n(n-1)}{2}\right)D + \left(1 + 2\frac{n(n-1)}{2} - \frac{(n-1)(n-2)}{2}\right)[-1]^*D.$$

Since

$$\begin{cases} 1 + 2\frac{n(n+1)}{2} - \frac{n(n-1)}{2} = \frac{(n+1)(n+2)}{2}, \\ 1 + 2\frac{n(n-1)}{2} - \frac{(n-1)(n-2)}{2} = \frac{n(n+1)}{2}, \end{cases}$$

we recover the expected result by induction. □

Proposition 3.2.6 $[n]_X$ *is an isogeny, that is* $X[n]$ *is a finite subgroup of* X, *or equivalently* X *is a divisible group, i.e. for any* $n \geqslant 1$, $nX = X$. *Moreover:*

Note that this is on the closure and not the rational points, which is generally false.

- $\deg [n]_X = n^{2g}$ *where* $g = \dim X$.
- *Assume* $\operatorname{char} k = p > 0$, $[n]_X$ *is separable if and only if* $p \nmid n$.
- *If* $p \nmid n$, $X[n] \simeq (\mathbb{Z}/n\mathbb{Z})^{2g}$.
- *There is an integer* $0 \leqslant i \leqslant g$ *such that for all* $m \geqslant 1$, $X[p^m] \simeq (\mathbb{Z}/p^m\mathbb{Z})^i$.

Proof *We consider* $n \geqslant 1$, *the result for negative integers comes from* $[-n] = [n] \circ [-1]$. $[n]$ *is clearly a homomorphism and since* $\dim X = \dim X$, *by Proposition 3.2.2, being an isogeny is equivalent to* $X[n]$ *being finite or to* $[n]$ *being surjective, which itself is equivalent to* X *being divisible. We will prove that* $X[n]$ *is finite. Consider an ample divisor* D *on* X, *which exists by Theorem 3.1.23.*

Recall from Lemma 3.2.5 that

$$[n]^*D \sim \frac{n(n+1)}{2}D + \frac{n(n-1)}{2}[-1]^*D.$$

Since $[-1]$ *is an automorphism,* $[-1]^*D$ *is also ample. Moreover,* $n(n+1)/2 > 0$, $n(n-1)/2 \geqslant 0$, *so* $[n]^*D$ *is ample by Proposition 3.1.25, as well as its restriction to* $X[n]$. *If the trivial divisor is very ample, since* $\Gamma(0) = k$, *this means we have a closed immersion* $X[n] \to \mathbb{P}^0$, *hence* $\dim X[n] = 0$, *or equivalently* $X[n]$ *is finite.*

The second part of the proof corresponds to [Mum74, § 6, App. 3]. □

If $f : X \to Y$ is an isogeny with kernel K and degree d, then $K \subseteq X[d]$ (see [GM, Ex. 4.4]), which is of rank $2g$. Hence, K is of type at most $2g$. This leads to the following definition.

Definition 3.2.7 *Let* $g = \dim X$, *consider integers* $n_{2g} \mid n_{2g-1} \mid \cdots \mid n_1$, *which may be equal to* 1. *A* $(n_{2g}, \dots, n_1)$*-isogeny is an isogeny with kernel* K *such that* $(n_{2g}, \dots, n_1)$ *are the elementary divisors of* K.

Going the other way, given a finite subgroup K of X, can we find an abelian variety Y and an isogeny $f : X \to Y$ such that $\ker f = K$? The answer is yes, as the following theorem shows.

Theorem 3.2.8 *Given a finite subgroup K of X, there is an abelian variety Y and a separable isogeny $f : X \to Y$ such that, as a topological space (Y, f) is the quotient of X where K acts by translation. The pair (Y, f) is unique up to isomorphism, Y is denoted as X/K.*

Moreover, there is a bijection between the set of finite subgroups $K \subseteq X$ and the set of separable isogenies $f : X \to Y$, modulo the equivalence relation where two isogenies $f_1 : X \to Y_1$, $f_2 : X \to Y_2$ are related if there is an isomorphism $g : Y_1 \to Y_2$ such that $f_2 = g \circ f_1$. The isogeny f verifies $K = \ker f$ and $Y = X/K$.

This is a particular case of [Mum74, § 7, Thm. p. 66, Thm. 4].

Since we are dealing with quotients, we are interested in a factorization theorem.

Theorem 3.2.9 *If $f : X \to Y$ and $g : X \to Z$ are two isogenies such that $\ker f \subseteq \ker g$, then there is a unique isogeny $h : Y \to Z$ such that $h \circ f = g$. Moreover, if f and g are rational, then h is rational too.*

Remark 3.2.10 *The theorem holds for inseparable isogenies when the kernel inclusion is seen through the scope of finite schemes.*

This is a particular case of [Mum74, § 12, Thm. 1]. An important corollary is the following:

Proposition 3.2.11 *Consider an isogeny $f : X \to Y$ of degree d, then there is a unique $g : Y \to X$ such that*

$$g \circ f = [d]_X \text{ and } f \circ g = [d]_Y.$$

g is called the contragedient isogeny and is denoted $\tilde{f}$. It has degree d^{2g-1}.

Proof *Since f is of degree d, we have $\ker f \subseteq X[d]$, and $X[d]$ is the kernel of $[d]_X$, hence by Theorem 3.2.9 there is a unique $g : Y \to X$ such that $g \circ f = [d]_X$. Because $[d]_Y \circ f = f \circ [d]_X = f \circ g \circ f$, we have $([d]_Y - f \circ g) \circ f = 0$. Since f is surjective, this means that $f \circ g = [d]_Y$.*

In particular, because the degree is multiplicative, we immediately get that $\deg g = d^{2g-1}$ since $\deg [d]_X = d^{2g}$ and $\deg f = d$. □

Another consequence is the decomposition of an isogeny into the product of isogenies of prime degree, at least on the algebraic closure.

Proposition 3.2.12 *Consider an isogeny $f : X \to Y$ of degree $d = \ell_1 \dots \ell_r$ where the ℓ_i are prime numbers for $1 \leqslant i \leqslant r$. Then there are isogenies $f_i : X_i \to X_{i+1}$ of degree ℓ_i for $1 \leqslant i \leqslant r$ such that $f = f_r \circ \dots \circ f_1$ (in particular, $X_1 = X$ and $X_{r+1} = Y$).*

The inseparable part of the degree can be dealt with using the Frobenius isogeny induced by $x \mapsto x^p$.

Proof *Assume f is separable, this means that none of the primes ℓ_i is equal to $p = \operatorname{char} k$ if $p > 0$.. Since $\ell_1 \mid d = \# \ker f$, consider a subgroup K_1 of order ℓ_1 contained in the kernel of f. This induces an isogeny $f_1 : X \to X_2 := X/K_1$ of degree ℓ_1 by Theorem 3.2.8. Theorem 3.2.9 ensures that f factors as $f = g \circ f_1$ where $g : X_2 \to Y$. g is then separable of degree $\ell_2 \dots \ell_r$, and the process can be repeated by induction.* □

3.3 DUALITY

Here we introduce the dual of an abelian variety. We first need to define the polarization. We will then discuss the consequences on isogenies, and briefly introduce principal polarizations. Consider an abelian variety X through this section.

3.3.1 Polarization

Recall from Definition 3.1.14 that the Picard group $\operatorname{Pic} X$ is the group of equivalence classes of linearly equivalent divisors (under the correspondence Theorem 3.1.18).

Definition 3.3.1 (Polarization) *Let $D \in \operatorname{Div} X$ be a divisor, and set the map $\Lambda(D) : X \to \operatorname{Pic} X$ to be $x \mapsto [t_x^* D - D]$. This is the polarization associated to the divisor D.*

Remark 3.3.2 *On an arbitrary (perfect) field k, a polarization is a rational isogeny λ such that its extension to $\overline{k}$ is of the form $\Lambda(D)$.*

Because of Corollary 3.1.22, this is a group homomorphism. We denote its kernel by $H(D)$.

Proposition 3.3.3 *Let $D, D' \in \operatorname{Div} X$ be two divisors.*

1. $\Lambda(D + D') = \Lambda(D) + \Lambda(D')$;
2. *For any $f \in k(X)$, $\Lambda(\operatorname{div} f) = 0$, in particular if $D \sim D'$, then $\Lambda(D) = \Lambda(D')$;*
3. *For any $x \in X$, $\Lambda(t_x^* D) = \Lambda(D)$;*
4. *$H(D)$ is a Zariski-closed subgroup of X;*
5. *Assume D is effective. Then $H(D)$ is finite if and only if D is ample.*

The first three items are straight-forward computations, elements (4) and (5) are part of [Mum74, § 6, App. 1].

Definition 3.3.4 *Two divisors D and D' are said to be algebraically equivalent if $\Lambda(D) = \Lambda(D')$.*

We have seen that linearly equivalent implies algebraically equivalent, but the converse does not hold in general.

Our goal is to define an abelian variety relatively 'close' to X in some sense. We will see that $\operatorname{Pic}^0 X$ is a good fit.

Definition 3.3.5 *$\operatorname{Pic}^0 X$ is the subgroup of $\operatorname{Pic} X$ consisting of classes of divisors $[D]$ such that $\Lambda(D) = 0$.*

This definition makes sense because of Proposition 3.3.3 (2). In particular, because of item (3), we have that $\operatorname{im} \Lambda(D) \subseteq \operatorname{Pic}^0 X$ for any divisor D, and the following sequence is exact:

$$0 \to \operatorname{Pic}^0 X \to \operatorname{Pic} X \to \operatorname{Hom}(X, \operatorname{Pic}^0 X),$$

where the last map is given by $[D] \mapsto \Lambda(D)$. Here are some other characterization of $\operatorname{Pic}^0 X$:

Proposition 3.3.6 *The following assertions are equivalent:*

1. $[D] \in \operatorname{Pic}^0 X$;
2. *For every* $x \in X$, $t_x^* D \sim D$;
3. $m^* D \sim p_1^* D + p_2^* D$ *where the maps* $m, p_1, p_2 : X \times X \to X$ *are respectively the addition law, and the projection on the first and second component.*

Proof *(2) is equivalent to* $\Lambda(D)(x) = 0$ *for every* $x \in X$, *that is* $\Lambda(D) = 0$, *which is the definition of* $[D] \in \operatorname{Pic}^0 X$.

Using the see-saw principle, $m^* D - p_1^* D - p_2^* D$ *is trivial if and only if it is trivial on* $\{0_X\} \times X$ *and* $X \times \{x\}$ *for any* $x \in X$ *(see for instance [Mum74, § 5, Cor. 6]). It is always trivial on* $\{0_X\} \times X$, *and on* $X \times \{x\}$ *it restricts to* $t_x^* D - D$. *Therefore, (3) is equivalent to (2) and (1).* □

Consider $f, g : Y \to X$ morphisms of varieties (Y is not assumed to be abelian). $f + g$ can be written as $m \circ (f, g)$, and if $[D] \in \operatorname{Pic}^0 X$, by composing with (f, g) in the above proposition (3), we get

$$(f + g)^* D \sim f^* D + g^* D.$$

In particular, for n a positive integer, we get by induction $[n]_X^* D \sim nD$. Using Lemma 3.2.5 for $n = 2$:

$$2D \sim [2]^* D \sim 3D + [-1]^* D,$$

so $[-1]^* D \sim -D$, we say that D is antisymmetric in that case. Similarly, if $[-1]^* D \sim D$, we say that D is symmetric. We will redefine those later.

The last thing to mention is that the polarization is surjective for ample divisors.

Theorem 3.3.7 *Consider an ample divisor* D *and let* $[D'] \in \operatorname{Pic}^0 X$. *Then, for some* $x \in X$, $D' \sim t_x^* D - D$, *that is* $\Lambda(D)(x) = [D']$ *and* $\operatorname{im} \Lambda(D) = \operatorname{Pic}^0 X$.

Mumford proves it in [Mum74, § 8, Thm. 1]. One usage is the following:

Proposition 3.3.8 *A divisor* D *and an ample divisor* D' *are algebraically equivalent if and only if there is some* $x \in X$ *such that* $D \sim t_x^* D'$.

Proof *First assume that the divisors are algebraically equivalent, that is* $\Lambda(D) = \Lambda(D')$. *Then* $\Lambda(D - D') = 0$, *and* $[D - D'] \in \operatorname{Pic}^0 X$. *Because* $\Lambda(D')$ *is surjective, there is an* $x \in X$ *such that* $t_x^* D' - D' \sim D - D'$, *which implies* $D \sim t_x^* D'$.

Conversely, we have the equalities $\Lambda(D) = \Lambda(t_x^* D') = \Lambda(D')$ *when* $D \sim t_x^* D'$. □

Finally, if D is an ample divisor, let d be the dimension of $\Gamma(D)$, the space of sections above D. The degree of the associated polarization is then [Mum66, (IV) p. 289]

$$\deg \Lambda(D) = d^2.$$

Definition 3.3.9 D *is said to be of separable type if* $p \nmid d$ *when* $p = \operatorname{char} k$ *is non-zero.*

If D is ample of separable type, then the associated polarization is a separable isogeny.

3.3.2 Dual abelian variety

Constructing the dual variety is a hard task, especially in positive characteristic and is out of the scope of this thesis. We are more interested in the consequences.

As we have seen, $\operatorname{Pic} X$ and $\operatorname{Pic}^0 X$ are interesting groups related to X. $\operatorname{Pic} X$ is in fact a proper group scheme, and the connected component of $[0]$ is $\operatorname{Pic}^0 X$, it has an abelian variety structure.

Definition 3.3.10 (Dual abelian variety) *The dual variety of X, denoted $\widehat{X}$, is $\operatorname{Pic}^0 X$ endowed with its abelian variety structure.*

The difficulty comes from that one would like to define the dual as $X/H(D)$ for a well-chosen divisor D. However, in positive characteristic this construction requires to consider a scheme quotient rather than a variety one, and is certainly more involved.

With this settled, for any ample effective divisor D, $\Lambda(D) : X \to \widehat{X}$ is an isogeny since it has finite kernel (Proposition 3.3.3 (5)) and is surjective (Theorem 3.3.7).

Consider now an isogeny $f : X \to Y$, it induces via the pullback of divisors a morphism $f^* : \operatorname{Pic} Y \to \operatorname{Pic} X$.

Theorem 3.3.11 (Dual isogeny) *We have $f^*(\operatorname{Pic}^0 Y) \subseteq \operatorname{Pic}^0 X$ and it induces an isogeny $\widehat{f} : \widehat{Y} \to \widehat{X}$. If K is the kernel of f, then the kernel of $\widehat{f}$ is isomorphic canonically to $\widehat{K} = \operatorname{Hom}(K, \mathbb{G}_m)$, where $\mathbb{G}_m$ is the multiplicative group (as a group variety). $\widehat{f}$ is unique (up to an isomorphism) and is called the dual isogeny.*

Moreover, if f is a rational isogeny, then $\widehat{f}$ is too.

Proof *Since f is a homomorphism, we have for any $x \in X$, $f \circ t_x = t_{f(x)} \circ f$. Let $[D] \in \widehat{Y}$, in particular, for any $x \in X$,*

$$\Lambda(f^*D)(x) = [t_x^* f^* D - f^* D] = [f^*(t_{f(x)}^* D - D)] = f^*(\Lambda(D)(f(x))).$$

Since $[D] \in \widehat{Y}$, $\Lambda(D) = 0$ and therefore $[f^*D] \in \widehat{X}$.

When the isogeny is separable, Mumford shows in [Mum74, § 7, Cor. 2] that $\ker f^*$ and $\ker f$ are canonically dual. Furthermore, if $[D] \in \operatorname{Pic} Y$ is such that $[f^*D] = 0$, then $[D]$ has finite order n, and $0 = \Lambda(nD) = n\Lambda(D) = \Lambda(D)(n\cdot)$. Since $X = nX$, we recover that $\Lambda(D) = 0$, or $[D] \in \operatorname{Pic}^0 Y$. Hence, $\ker f^* = \ker \widehat{f}$.

Finally, $\Lambda(D)$ being an isogeny for any effective ample divisor D (which exists) ensure $\dim X = \dim \widehat{X}$, therefore $\widehat{f}$ is an isogeny by Proposition 3.2.2. Because its kernel must be $\ker f^*$, it is unique.

If f is rational, its kernel K is Galois invariant. Now $\widehat{K}(\overline{k}) \simeq \operatorname{Hom}(K(\overline{k}), \overline{k}^*)$, it is Galois invariant since K is, hence it is rational. □

We restricted the proof to separable isogenies, but the result holds more generally, see for instance [Mum74, § 15, Thm. 1]. A corollary is that f and $\widehat{f}$ have same degree. We also proved that for any divisor D over Y and isogeny $f : X \to Y$,

$$\Lambda(f^*D) = \widehat{f} \circ \Lambda(D) \circ f. \tag{3.1}$$

Remark 3.3.12 *When working with isogenies* $f : E \to E'$ *on elliptic curves which are abelian varieties of dimension* 1*, we often consider their dual* $\widehat{f} : E' \to E$*. This is an abuse of notation, as we should see in Section 4.2.2, where the contragedient* $\widetilde{f}$ *and the dual* $\widehat{f}$ *are the same up to isomorphisms.*

In higher dimension g*, this comparison does not make sense any more: first the dual has degree* d *if* f *has degree* d*, whereas the contragedient has degree* d^{2g-1}*, and most importantly a generic abelian variety* X *has no reason to be isomorphic to its dual* $\widehat{X}$*.*

However, one usual property of duality is an isomorphism with the bidual, as stated below.

Theorem 3.3.13 *There is a canonical isomorphism between* X *and* $\widehat{\widehat{X}}$*.*

This corresponds to [Mum74, § 13, Cor. p. 132].

3.3.3 Principal polarization

In general, polarizations are not isomorphisms since they are not injective. By combining several results about the cohomology of line bundles (namely, the Riemann-Roch theorem, the vanishing theorem and the index of a line bundle theorem, see [Mum74, § 16]), it is possible to state the following result.

Proposition 3.3.14 *If* D *is an ample effective divisor, set* $d = \dim \Gamma(D) > 0$*, then* $\deg \Lambda(D) = d^2$*.*

If we further assume that $\operatorname{char} k \nmid \dim \Gamma(D)$ when the characteristic is positive, then $\Lambda(D)$ is a separable isogeny.

We stay with the assumption that D is an ample effective divisor, then $\Lambda(D)$ is an isomorphism if and only if $\dim \Gamma(D) = 1$.

Abrreviated p.p.a.v.

Definition 3.3.15 (Principal polarization) *An abelian variety is called principally polarized if there is an ample effective divisor* D *such that* $\Lambda(D)$ *is an isomorphism. The associated polarization is said to be principal, and we define the theta divisor to be* $\operatorname{div} s + D$ *where* $\Gamma(D) = \langle s \rangle$*, it is unique up to translation by Proposition 3.3.8.*

We can refine the notion of isogeny when we are dealing with p.p.a.v. X and Y. If $f : (X, D_X) \to (Y, D_Y)$ is an isogeny where D_X and D_Y are divisors on X and Y respectively, then f is said to be polarized if f^*D_Y is algebraically equivalent to D_X.

Furthermore, denote the principal polarization on X by λ. If the polarization $\Lambda(D_X)$ factors as $\Lambda(D_X) = \lambda \circ [m]_X$ for some integer m, we say that f is an m-isogeny. In that case, for a symmetric theta divisor Θ_X, we have that D_X and $m\Theta_X$ are algebraically equivalent.

If $f : (X, \Theta_X) \to (Y, \Theta_Y)$ is an m-isogeny and Θ_X, Θ_Y are theta divisors on X and Y respectively associated to their principal polarization, then f preserves the principal polarization if $f^*\Theta_Y$ is algebraically equivalent to $n\Theta_X$.

In this context, if f is an ℓ-isogeny for some prime ℓ, the kernel of f is of rank exactly g. The reason is that it is maximal isotropic for some pairing derived from the commutator pairing defined later in Definition 5.2.1.

If f is an m-isogeny of p.p.a.v. and $m = \ell_1 \dots \ell_r$ for some primes ℓ_i, then f can be decomposed as a product of ℓ_i-isogenies of p.p.a.v. on the algebraic closure, refining Proposition 3.2.12.

4 DIMENSION 1: ELLIPTIC CURVES

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In the previous chapter, we introduced the general theory of abelian varieties. We will now look into the case of dimension 1. We will see that they exactly correspond to elliptic curves, which we introduced briefly in the context of cryptography.

While we want to specify some results of the previous chapter, we do not require the whole machinery of abelian varieties to introduce elliptic curves, in fact they have a very natural construction. Following Silverman lines [Sil09], we construct this chapter by first introducing elliptic curves with their geometric group structure and then make connections with the previous chapter. Since we will be working over curves, we can easily define the degree of a divisor and use it to simplify some previous results.

Throughout this chapter, we consider a perfect field k and denote its algebraic closure by $\overline{k}$.

In Section 4.1, we first introduce Weierstrass equations, derive a group law on them, and then relate this construction to the usual definition of an elliptic curve. In Section 4.2, we prove that elliptic curves are abelian varieties, and we study some properties on polarizations and isogenies on these. Finally, we define Kummer lines in Section 4.3 as those are closely related to elliptic curves and are the main object considered in this thesis.

4.1 DEFINITION

There are two approaches to define elliptic curves, we could first define them as smooth projective curves of genus 1 with a rational point. Or, since they are curves, we could define them via a Weierstrass equation and prove they have nice properties. The second point of view is natural, therefore we will

start with Weierstrass equations, before proving that any smooth projective curve of genus 1 is given by a Weierstrass equation.

4.1.1 Weierstrass equations

In Section 1.3, we defined elliptic curves as curves with a specific kind of equation, we should study these before defining properly an elliptic curve in Definition 4.1.10.

Definition 4.1.1 (Weierstrass equation) *A Weierstrass equation is an equation in $\mathbb{P}^2$ of the form*

$$Y^2Z + a_1XYZ + a_3YZ^2 = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3$$

where $a_1, \dots, a_6 \in \overline{k}$.

The projective point $(0:1:0)$ is a solution to this equation. Given a Weierstrass equation, we will denote by E the projective curve in $\mathbb{P}^2$ verifying the equation. E is defined over k if $a_1, \dots, a_6 \in k$ and the set of rational points in $\mathbb{P}^2(k)$ is denoted $E(k)$. The point at infinity is written $0_E = (0:1:0)$.

It is generally more convenient to write the equation in affine form by setting $x = X/Z$ and $y = Y/Z$:

$$E : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6.$$

If $\operatorname{char} k \neq 2$, we can complete the square by setting $y' = y + \frac{a_1x+a_3}{2}$:

$$E : y'^2 = x^3 + b_2x^2 + b_4x + b_6$$

where $b_2 = a_2 + \frac{a_1^2}{4}$, $b_4 = a_4 + \frac{a_1a_3}{2}$ and $b_6 = a_6 + \frac{a_3^2}{4}$. If we further assume that $\operatorname{char} k \neq 2, 3$, then the change of variable $x' = x + \frac{b_2}{3}$ gives

$$E : y'^2 = x'^3 + a_4'x' + a_6'$$

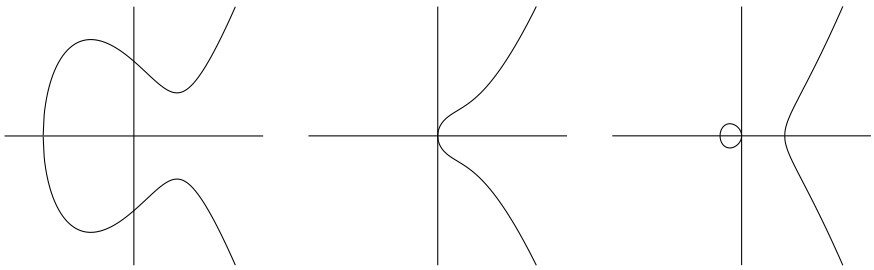
with $a_4' = b_4 - \frac{b_2^2}{3}$ and $a_6' = b_6 + \frac{2b_2^3}{27} - \frac{b_2b_4}{3}$. This last form is what we call a *short Weierstrass equation*. In the context of this thesis, we will often assume that the characteristic is not 2 or 3, and therefore we will work with short Weierstrass equations. The results in this chapter can be generalized to the general case as pointed out by Silverman [Sil09, § III, Rmk. 1.3], but the computations are much more tedious.

Consider the following short Weierstrass equation with $a, b \in k$

$$E : y^2 = x^3 + ax + b.$$

We are particularly interested in curves where this equation has no singular point, which is equivalent to the discriminant $\Delta(E) := -16(4a^3 + 27b^2)$ being non-zero ([Sil09, § III, Prop. 1.4a]). We introduce one last important notion tied to Weierstrass equations, which will prove useful later.

Definition 4.1.2 (j-invariant) *For a curve E with a short Weierstrass equation $y^2 = x^3 + ax + b$, we define its j-invariant as $j(E) := -1728\frac{(4a)^3}{\Delta(E)} = 1728\frac{4a^3}{4a^3+27b^2}$.*



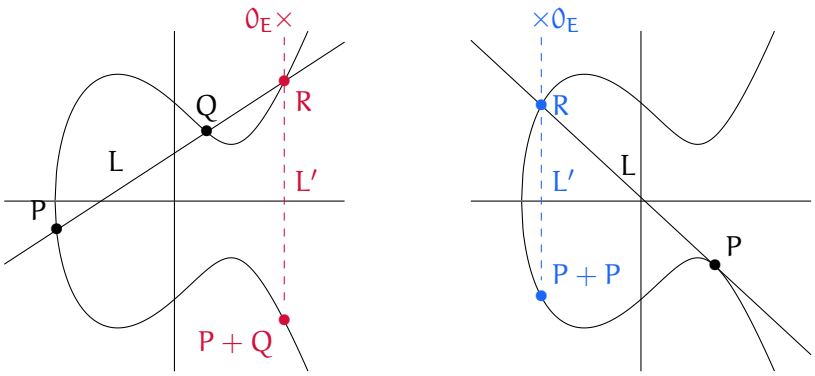
(a) $y^2 = x^3 - 3x + 3$ (b) $y^2 = x(x^2 - x + 1)$ (c) $y^2 = x(x-1)(x+0.5)$

Figure 4.1: Some real representations of Weierstrass equations

We are interested in building geometrically a group law on such curves. Since a Weierstrass equation has degree 3, it must intersect any line $L \subseteq \mathbb{P}^2$ in exactly three points (counted with multiplicities) thanks to Bézout's theorem, see [Har77, § I, Cor. 7.8].

Definition 4.1.3 *Consider a curve* E *with a Weierstrass equation, and let* $P, Q \in E$. *Consider the line* $L \subseteq \mathbb{P}^2$ *connecting* P *and* Q *(or the tangent to* E *at* P *if* $P = Q$*). Denote by* R *the third intersection point of* L *with* E*, and* L' *the line connecting* R *and* 0_E*. Then* $P + Q$ *is defined as the third intersection point of* L' *and* E.

This defines a commutative group law on E and $E(k)$ is a subgroup of E ([Sil09, § III, Prop. 2.2]). The geometric interpretation is illustrated in Fig. 4.2. In particular, the opposite of a point is its symmetric to the x-axis, and a point verifies $y = 0$ if and only if it is a non-trivial 2-torsion point.



(a) $P + Q$ when $P \neq Q$ (b) $P + P = 2 \cdot P$

Figure 4.2: Group law on E

We can find an expression for the coordinates of $P + Q$ as a rational function of the coordinates of two points $P, Q \in E$, we recall them from Proposition 1.3.1 (see also [Sil09, § III, Alg. 2.3] for the complete expression in long Weierstrass form).

Proposition 4.1.4 *Consider a curve* E *defined over* k *with Weierstrass equation* $y^2 = x^3 + ax + b$. *Let* $P, Q \in E(k)$*, then* $P + Q$ *coordinates can be computed in the following way:*

- *if* $P = 0_E$, *then* $P + Q = Q$. *Similarly, if* $Q = 0_E$, *then* $P + Q = P$;
- *otherwise,* $P = (x_1, y_1)$ *and* $Q = (x_2, y_2)$:
  - *if* $x_1 \neq x_2$, *set*

$$\lambda = \frac{y_2 - y_1}{x_2 - x_1},\ x_3 = \lambda^2 - x_1 - x_2,\ y_3 = -y_1 - \lambda(x_3 - x_1).$$

    Then $P + Q = (x_3, y_3)$;
  - *if* $x_1 = x_2$, *because* P *and* Q *are elements of* $E(k)$, $y_1 = \pm y_2$ *and:*
    * *if* $y_1 = -y_2$, *then* $P + Q = 0_E$;
    * *otherwise,* $y_1 = y_2 \neq 0$, *set*

$$\lambda = \frac{3x_1^2 + a}{2y_1},\ x_3 = \lambda^2 - 2x_1,\ y_3 = -y_1 - \lambda(x_3 - x_1).$$

    Then $P + Q = (x_3, y_3)$.

4.1.2 Divisors on curves

In this section we provide more details to Section 3.1.2 in the context of curves. In fact, an algebraic curve is of dimension 1, hence subvarieties of codimension 1 are points, and divisors are formal sums of points according to Definition 3.1.9.

Consider a curve C, for a point $P \in C$, we should consider the associated divisor $(P) \in \operatorname{Div} C$.

Definition 4.1.5 *Consider a divisor* $D = \sum_{P \in C} n_P(P)$ *where* $n_P = 0$ *for all but finitely many* $P \in C$. *We define the degree of the divisor* D *as*

$$\deg D = \sum_{P \in C} n_P.$$

The degree defines a group homomorphism $\deg : \operatorname{Div} C \to \mathbb{Z}$. Its kernel is denoted $\operatorname{Div}^0 C$. Every principal divisor has degree 0 ([Sil09, § II, Prop. 3.1b]), that is a rational function $f \in k(C)$ has as many zeroes than poles. Here is a useful lemma showing the strength of the degree.

Lemma 4.1.6 *Let* $D \in \operatorname{Div} C$ *with* $\deg D < 0$, *then* $\Gamma(D) = \{0\}$.

Proof *Let* $f \in \Gamma(D)$, *assume it is non-zero. Then* $\operatorname{div} f + D \geqslant 0$, *which implies that* $\deg(\operatorname{div} f) + \deg D = \deg D \geqslant 0$. *This is a contradiction, hence* $\Gamma(D) = \{0\}$. □

Recall that $\operatorname{Pic} C$ is the group of divisors modulo linear equivalence, and $\operatorname{Pic}^0 C = \operatorname{Div}^0 C / \operatorname{Princ} C$ is a subgroup of $\operatorname{Pic} C$, which is of particular interest when looking at abelian varieties.

We haven't made any connection with abelian varieties yet, we are solely introducing material for later on.

We do not want to introduce the differentials' formalism here as it will not be very useful in the remaining of this thesis, the main result is the existence of a canonical divisor class, any divisor K in this equivalence class is called a canonical divisor and satisfies the following theorem.

Theorem 4.1.7 (Riemann-Roch) *Let* C *be a curve of genus* g, $D \in \operatorname{Div} C$ *and* K *a canonical divisor. Then the following equality holds:*

$$\dim \Gamma(D) - \dim \Gamma(K - D) = \deg D + 1 - g.$$

This is proven in [Har77, § IV, Thm. 1.3] for instance. We can get some information on the canonical divisor from that, and how to get rid of it in the equality.

Corollary 4.1.8 *Let* K *be a canonical divisor on the curve* C *of genus* g, *and consider* $D \in \operatorname{Div} C$, *then:*

1. $\deg K = 2g - 2$;
2. $\dim \Gamma(K) = g$;
3. *if* $\deg D > 2g - 2$, *then* $\dim \Gamma(D) = \deg D + 1 - g$.

Proof *We use Riemann-Roch theorem for* $D = 0$ *and* $D = K$, *this gives the two equalities*

$$\dim \Gamma(0) - \dim \Gamma(K) = 1 - g, \ \dim \Gamma(K) - \dim \Gamma(0) = \deg K + 1 - g.$$

By summing them, we get $\deg K = 2g - 2$.

If $f \in \Gamma(0)$, *then* $\operatorname{div} f \geqslant 0$, *so it has no poles, and therefore have no zeroes either, so* $f \in k$. *Obviously,* $k \subseteq \Gamma(0)$, *hence* $\Gamma(0) = k$. *Using this in the second equality, we get*

$$\dim \Gamma(K) - 1 = (2g - 2) + 1 - g,$$

which gives $\dim \Gamma(K) = g$.

Finally, if $\deg D > 2g - 2$, *we have* $\deg(K - D) < 0$, *so* $\Gamma(K - D) = \{0\}$ *by Lemma 4.1.6 and the last equality holds.* □

The last result on divisors is the Hurwitz formula. A morphism of curves $f : C \to C'$ is either a constant map, or a surjective morphism ([Sil09, § II, Thm. 2.3]). Assume f is non-constant, the induced map $f^* : k(C') \to k(C)$ gives a field extension, which is finite ([Sil09, § II, Thm. 2.4]). Therefore, the various definitions of degree and separable / inseparable maps made in Definition 3.2.3 also make sense in this context. In particular, if $D \in \operatorname{Div} C'$, $\deg(f^*D) = (\deg f)(\deg D)$.

The ramification index of f at $P \in C$ is the valuation at P of $f^*(u)$ where $u \in k(C')$ is an uniformizer at $f(P)$. It is denoted $e_f(P)$, and verifies $e_f(P) \geqslant 1$.

Theorem 4.1.9 (Hurwitz formula) *Let* $f : C \to C'$ *be a separable morphism of smooth curves, where* C *is of genus* g *and* C' *of genus* g'. *Then*

$$2g - 2 \geqslant (\deg f)(2g' - 2) + \sum_{P \in C} (e_f(P) - 1).$$

Moreover, equality holds if $\operatorname{char} k = 0$ *or* $\operatorname{char} k = p > 0$ *and* p *does not divide* $e_f(P)$ *for all* $P \in C$.

Details can be found in [Sil09, § II, Thm. 5.9].

4.1.3 Elliptic curves

We will now properly define an elliptic curve.

Definition 4.1.10 *An elliptic curve* E *is a smooth projective curve of genus* 1 *along with a point* $0_E \in E$*. It is defined over* k*, and denoted* E/k*, if* E *is defined over* k *as a curve and* $0_E \in E(k)$*.*

Since we want to re-use previously introduced material on Weierstrass equations and the group, we should prove the following theorem:

Theorem 4.1.11 *Let* E *be an elliptic curve defined over* k*. There are functions* $x, y \in k(E)$ *such that* $\Phi : E \to \mathbb{P}^2(k), P \mapsto (x(P) : y(P) : 1), 0_E \mapsto (0 : 1 : 0)$ *gives an isomorphism of* E/k *with a curve* C *given by a Weierstrass equation*

$$C : y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6,$$

with $a_1, \dots, a_6 \in k$*.*

Conversely, any smooth curve C *given by a Weierstrass equation is an elliptic curve defined over* k *with* $0_E = (0 : 1 : 0)$*.*

Proof *Using Theorem 4.1.7 and its corollary, since* E *is of genus* 1*, for any divisor* $D \in \operatorname{Div} E$ *with* $\deg D > 0$*,* $\dim \Gamma(D) = \deg(D)$*. Consider* $D = (0_E)$*, we are interested in the following sequence, thanks to Proposition 3.1.17:*

$$\Gamma(D) \subsetneq \Gamma(2D) \subsetneq \Gamma(3D) \subsetneq \cdots \subsetneq \Gamma(6D).$$

We have $\dim \Gamma(nD) = n$*, and* $\Gamma(D) = k$ *because it clearly contains constants. We complete the basis of* $\Gamma(D)$ *in the following way:*

$$\Gamma(D) = \langle 1 \rangle,\ \Gamma(2D) = \langle 1, x_0 \rangle,\ \Gamma(3D) = \langle 1, x_0, y_0 \rangle.$$

Since $x_0 \in \Gamma(2D) \setminus \Gamma(D)$*,* x_0 *has exactly one pole of order* 2 *at* 0_E*. Similarly,* y_0 *has exactly one pole of order* 3 *at* 0_E*. Moreover,* $x_0, y_0 \in k(E)$ *since* E *is defined over* k *and* D *is rational [Sil09, § II, Prop. 5.2].*

Now, the seven following elements are in $\Gamma(6D)$*, which is of dimension 6:*

$$1, x_0, x_0^2, x_0^3, y_0, y_0^2, x_0y_0.$$

Therefore, there must be a non-trivial relation with coefficients in k *such that*

$$\alpha_0 + \alpha_1x_0 + \alpha_2x_0^2 + \alpha_3x_0^3 + \beta_1y_0 + \beta_2y_0^2 + \gamma x_0y_0 = 0.$$

If $\alpha_3 = 0$ *or* $\beta_2 = 0$*, every term has a different pole order at* 0_E*, which would imply that every coefficient is* 0*, a contradiction. We now set* $x = -\frac{x_0}{\alpha_3\beta_2}$ *and* $y = \frac{y_0}{\alpha_3^2\beta_2}$*; by dividing by* $\alpha_3^4\beta_2^2$ *we define a morphism* $\Phi : E \to \mathbb{P}^2(k)$ *given by* $\Phi(P) = (x(P) : y(P) : 1)$ *when* $P \neq 0_E$*, and since* $v_{0_E}(y) < v_{0_E}(x)$*,* $\Phi(0_E) = (0 : 1 : 0)$*. The image is a curve described by a Weierstrass equation, and* $\Phi : E \to C$ *is surjective.*

Using a result on ramification, since x *and* y *only have a pole at* 0_E*, we deduce that the map* $P \mapsto (x(P) : 1)$ *and* $P \mapsto (y(P) : 1)$ *have degree respectively* 2 *and* 3*, see [Sil09, § II, Prop. 2.6a]. Therefore,* $[k(E) : k(x)] = 2$ *and* $[k(E) : k(y)] = 3$*. Because* $[k(E) : k(x, y)]$ *divides both* 2 *and* 3*, we must have* $[k(E) : k(x, y)] = 1$*:* Φ *is of degree* 1 *and* $k(E) = k(x, y)$*.*

Details on the smoothness of C *can be found in [Sil09, § III, Prop. 3.1].*

Conversely, if C *is given by a Weierstrass equation, let* $F(X, Y, Z)$ *be the associated homogenous polynomial. We set* $x = X/Y$ *and* $z = Z/Y$*, as well as* $f(x, z)$ *the normalized polynomial. We also define* $\Phi : C \to \mathbb{P}^1(k), P \mapsto (x(P) : 1)$*. Since* $\overline{k}(C) = k[x, z]/(f)$ *and* $\overline{k}(\mathbb{P}^1) = \overline{k}(x)$*, we get that* $\deg \Phi = 3$ *since* f *is irreducible of degree 3. Moreover,* Φ *ramifies at* $P \in C$ *if and only if* $F(P) = 0$ *and* $\partial_Y F(P) = 0$*. By Bézout theorem [Har77, § I, Cor. 7.8], these two curves intersect in exactly* 2×3 *points (on* $\overline{k}$*, counted with multiplicities) since* F *is of degree 3 and* $\partial_Y F$ *of degree 2. Hence, by Theorem 4.1.9, if* g *is the genus of* C*:*

$$2g - 2 = 3(0 - 2) + 3 \times 2,$$

hence $g = 1$ *and* C *is an elliptic curve with the rational point* $0_E = (0 : 1 : 0) \in C(k)$*.* □

The functions x and y we constructed are called Weierstrass coordinates, we will provide more details on these in Section 4.3.1. We proved in the previous theorem that $k(E) = k(x, y)$ and $[k(E) : k(x)] = 2$. In particular, we look into possible changes of Weierstrass coordinates.

Proposition 4.1.12 *If* (x, y) *and* (x', y') *are two pairs of Weierstrass coordinates on an elliptic curve* E/k*, then there are some elements* $u, r, s, t \in k$ *with* $u \neq 0$ *such that*

$$x' = u^2 x + r, \ y' = u^3 y + u^2 s x + t.$$

Proof *We will assume* $\operatorname{char} k \neq 2, 3$ *and that* E *has a short Weierstrass equation for both pair of coordinates to ease the computations. Let* $a, b, a', b' \in k$ *such that*

$$E : y'^2 = x'^3 + a'x' + b', \ E : y^2 = x^3 + ax + b.$$

Because $\Gamma(2(0_E)) = \langle 1, x \rangle = \langle 1, x' \rangle$ *and* $\Gamma(3(0_E)) = \langle 1, x, y \rangle = \langle 1, x', y' \rangle$*, there are some elements* $\alpha, \beta, \gamma, \lambda, \mu \in k$ *such that*

$$x' = \lambda x + \mu, \ y' = \alpha y + \beta x + \gamma.$$

By substituting x' *and* y' *in the equation of* E *and expanding, we get the following equalities by identifying coefficients:*

$$\beta = \gamma = \mu = 0, \ \lambda^3 = \alpha^2, \ a'\lambda = a\alpha^2, \ b' = b\alpha^2.$$

Neither λ *nor* α *can be zero, we therefore set* $u = \frac{\alpha}{\lambda}$*, then* $\lambda = u^2$ *and* $\alpha = u^3$*. Hence,* $x' = u^2 x$*,* $y' = u^3 y$*,* $a' = u^4 a$*,* $b' = u^6 b$*.* □

For more detailed relations on the coefficients for long Weierstrass equations, see [Sil09, § III, Tab, 1.2].

We now look into conditions for two elliptic curves to be isomorphic, and how to find elliptic curves given a j-invariant.

Proposition 4.1.13 *Two elliptic curves* E *and* E' *are isomorphic over* $\overline{k}$ *if and only if* $j(E) = j(E')$*. Moreover, if* $j_0 \in \overline{k}$*, there is an elliptic curve* E *defined over* $k(j_0)$ *such that* $j(E) = j_0$*.*

Proof *We will assume* $\operatorname{char} k \neq 2, 3$ *here.*

Consider the short Weierstrass equations $E : y^2 = x^3 + ax + b$ *and* $E' : y'^2 = x'^3 + a'x' + b'$ *with* $a, b, a', b' \in \overline{k}$. *Assume first that* E *and* E' *are isomorphic. By Proposition 4.1.12, we must have an element* $u \in \overline{k}^*$ *such that* $x' = u^2x$, $y' = u^3y$, $a' = u^4a$, $b' = u^6b$. *Since* $j(E) = 1728\frac{4a^3}{4a^3+27b^2}$, *we get* $j(E) = j(E')$.

Conversely, if $j(E) = j(E')$, *the equation simplifies to* $b'^2a^3 = b^2a'^3$. *We are looking for* $u \in \overline{k}^*$ *such that* $x' = u^2x$, $y' = u^3y$, $a' = u^4a$ *and* $b' = u^6b$. *We look into three cases:*

- *If* $a = 0$, *then* $b \neq 0$ *since* $\Delta(E) \neq 0$, *hence* $a' = 0$ *and setting* $u = (b/b')^{1/6}$ *gives an isomorphism;* $j(E) = 0$
- *If* $b = 0$, *then* $a \neq 0$ *since* $\Delta(E) \neq 0$, *hence* $b' = 0$ *and setting* $u = (a/a')^{1/4}$ *gives an isomorphism;* $j(E) = 1728$
- *If* $a, b \neq 0$, *then* $a', b' \neq 0$. *Setting* $u = (a/a')^{1/4} = (b/b')^{1/6}$ *gives an isomorphism.* $j(E) \neq 0, 1728$

Hence, E *and* E' *are isomorphic.*

If $j_0 \in \overline{k}$, *first assume that* $j_0 \neq 0, 1728$ *and consider the curve* E *with equation* $y^2 = x^3 + a(j_0)x + b(j_0)$ *where*

$$a(j_0) := \frac{-36}{j_0 - 1728} - \frac{1}{48}, \quad b(j_0) := \frac{2}{j_0 - 1728} + \frac{1}{864}.$$

One computes $\Delta(E) = \frac{j_0^2}{(j_0-1728)^3}$ *and* $j(E) = j_0$. *It is non-singular since* $\Delta(E) \neq 0$, *hence* E *is an elliptic curve with* j*-invariant* j_0, *defined over* $k(j_0)$.

If $j_0 = 0$ *or* $j_0 = 1728$, *we have the two following curves:*

$$E : y^2 = x^3 + 1, \ \Delta(E) = -432, \ j(E) = 0;$$
$$E : y^2 = x^3 + x, \ \Delta(E) = -64, \ j(E) = 1728. \qquad \square$$

Remark 4.1.14 *In characteristic 2, the following curve has* j*-invariant* 0 *and is non-singular:*

$$E : y^2 + y = x^3, \ \Delta(E) = -27, \ j(E) = 0.$$

For the remaining cases, one can check [Sil09, § A, Prop. 1.2].

The j-invariant only tests if two curves are isomorphic over the algebraic closure, and yields no information whether they are isomorphic on the base field k, even if $j(E) \in k$!

Example 4.1.15 *Assume* k *is a finite field* $\mathbb{F}_p$ *with* $p > 3$ *prime, and consider* $\varepsilon \in \mathbb{F}_p$ *non-square. Set* $\mathbb{F}_{p^2} = \mathbb{F}_p[u]$ *where* $u^2 = \varepsilon$. *The curve* $E : y^2 = x^3 + ax + b$ *with* $a, b \in \mathbb{F}_p$ *is isomorphic over* $\mathbb{F}_{p^2}$ *to* $E' : y'^2 = x'^3 + \varepsilon^2ax' + \varepsilon^3b$ *via* $(x, y) \mapsto (x', y') := (\varepsilon x, \varepsilon^{3/2}y)$ *according to Proposition 4.1.12. Clearly,* E *and* E' *are not isomorphic over* $\mathbb{F}_p$, *but still* $j(E) = j(E') \in \mathbb{F}_p$.

This leads to the following definition:

Definition 4.1.16 *Let* E *and* E' *be elliptic curves defined over* k, E' *is said to be a twist of* E *if* E *and* E' *are isomorphic over* $\overline{k}$ *but not over* k.

Over a finite field, the previous example is what we call the quadratic twist of E, it is unique up to isomorphism because of the choice of a non-square.

In this chapter we have only discussed of elliptic curves with short Weierstrass equations so far, but recall that in Section 2.3 we introduced Montgomery curves which have the following shape for some $\mathcal{A}, \beta \in k$:

$$M : \beta y^2 = x(x^2 + \mathcal{A}x + 1).$$

These are due to Montgomery [Mon87]. One reason to use these algorithmically instead of the generic short Weierstrass equation is their fast arithmetic, especially when using only the x-coordinate as we did in Section 2.3. Not every elliptic curve has a Montgomery model on a field k not algebraically closed: Montgomery curves always have a rational 2-torsion point since $(0, 0) \in M$, whereas the polynomial $x^3 + ax + b$ may not have a root in k.

The conversion between a Montgomery curve and a short Weierstrass one is as follows.

Proposition 4.1.17 *Consider a Montgomery curve* $M : \beta y^2 = x(x^2 + \mathcal{A}x + 1)$ *with* $\mathcal{A}, \beta \in k$. *Then it is isomorphic to* $E : v^2 = u^3 + au + b$, *with*

$$a = \frac{1}{\beta^2}\left(1 - \frac{\mathcal{A}}{3}\right), \ b = \frac{\mathcal{A}}{3\beta^3}\left(\frac{2\mathcal{A}^2}{9} - 1\right);$$

the isomorphism $M \xrightarrow{\sim} E$ *is given by*

$$(x, y) \mapsto (u, v) := \left(\frac{1}{\beta}\left(x + \frac{\mathcal{A}}{3}\right), \frac{y}{\beta}\right), \ \left(\beta u - \frac{\mathcal{A}}{3}, \beta v\right) \leftarrow\!\shortmid (u, v).$$

Conversely, E *is isomorphic to* M *iff there is* $\alpha \in k$ *such that* $\alpha^3 + a\alpha + b = 0$ *and* $3\alpha^2 + a = \beta^{-2}$, *in that case* $\mathcal{A} = 3\alpha\beta$.

Proof *The isomorphism can be checked through a direct computation. The condition for* E *to be isomorphic to* M *corresponds to [OKS00, Prop. 1]* □

Finally, we refer to [CS18] for a survey on Montgomery curves.

Another example of curves used frequently in cryptography are Edwards [Edw07] and twisted Edwards curves [Ber+08a], with equation

$$E_{a,d} : ax^2 + y^2 = 1 + dx^2y^2$$

where $a, d \in k^*$ and $a \neq d$ (Edwards originally introduced the case $a = 1$). The name twisted Edwards comes from the fact that $E_{a,d}$ is isomorphic to $E_{1,d/a}$ over $k(\sqrt{a})$, hence it is a quadratic twist of an Edwards curve. Their arithmetic is also quite fast and has been quite optimized [BL07a; BL07b; His+08]. This is no coincidence since a twisted Edwards curve is always isomorphic to a Montgomery curve.

Proposition 4.1.18 *Consider a Montgomery curve* $M : \beta y^2 = x(x^2 + \mathcal{A}x + 1)$ *with* $\mathcal{A}, \beta \in k$. *Then it is isomorphic to* $E : au^2 + v^2 = 1 + du^2v^2$, *with*

$$a = \frac{\mathcal{A} + 2}{\beta}, \ b = \frac{\mathcal{A} - 2}{\beta};$$

the isomorphism $M \xrightarrow{\sim} E$ *is given by*

$$(x, y) \mapsto (u, v) := \left(\frac{x}{y}, \frac{x-1}{x+1}\right), \left(\frac{1+v}{1-v}, \frac{1+v}{(1-v)u}\right) \leftarrow\!\shortmid (u, v).$$

Conversely, a twisted Edwards curve is isomorphic to a Montgomery curve via the above isomorphism, where

$$\mathcal{A} = 2\frac{a+d}{a-d}, \ \beta = \frac{4}{a-d}.$$

There are many more models of elliptic curves available on the explicit-formulas database[1]. In this thesis, we are mainly interested in Montgomery curves and models derived from them since they have been used for the past forty years.

4.2 ELLIPTIC CURVES AS ABELIAN VARIETIES

In this section we should look at abelian varieties of dimension 1, and specify some results from Chapter 3 in this context. An important theorem is the following.

Theorem 4.2.1 *An abelian variety* X *of dimension* 1 *is an elliptic curve.*

Proof *If* X *is of dimension* 1*, it is a smooth curve, and it is projective by Theorem 3.1.23. We denote by* g *its genus. Consider a canonical divisor* K *of* X*. By Corollary 4.1.8,* $\deg K = 2g - 2$*. On the other hand, Mumford proves [Mum74, § 4, Q. 4.(iv)] that the tangent bundle is trivial, this implies that the class of* K *is trivial, and* $\deg K = 0$*, therefore* $g = 1$*. This is the definition of an elliptic curve.*□

For the rest of this section E will be an elliptic curve.

4.2.1 Polarization

We first want to study the dual of an elliptic curve $\widehat{E} = \mathrm{Pic}^0 E$, which was introduced in Definitions 3.3.5 and 3.3.10. Since the divisor $D = (0_E)$ is of degree 1, we have already checked that $\dim \Gamma((0_E)) = 1$ using Riemann-Roch theorem. D is obviously effective, and ample since $3D$ gives the projective embedding via Weierstrass coordinates.

Therefore, the polarization $\Lambda((0_E)) : P \mapsto [(-P) - (0_E)]$ is an isomorphism between E and its dual, and an elliptic curve is always principally polarized with theta divisor (0_E). This is one of the many specificities of dimension 1. We will denote this principal polarization λ in this section.

Remark 4.2.2 *On elliptic curves, we generally consider the isomorphism* $P \mapsto [(P) - (0_E)]$ *which is simply* $\lambda \circ [-1]$*. However, this does not match the usual sign convention for polarization on abelian varieties. This is why we prefer to work with the polarization* λ *here.*

1 https://www.hyperelliptic.org/EFD/index.html

We were also very interested in ample and very ample divisors in the previous chapter. The degree helps to classify them. Up to now, we have seen that

- If $\deg D < 0$, then $\Gamma(D) = \{0\}$;
- If $\deg D > 0$, then $\dim \Gamma(D) = \deg D$.

In the second case, since $\Gamma(D)$ is non-trivial, there is an element $f \in \Gamma(D)$ such that $D' = \operatorname{div} f + D \geqslant 0$, and so $D \sim D'$ with $D' \geqslant 0$. Hence, we can focus ourselves on effective divisors when the degree is positive. [Har77, § IV, Ex. 3.3.3] shows that on an elliptic curve, D is very ample if and only if $\deg D \geqslant 3$, and therefore D is ample if and only if $\deg D > 0$. It remains to study the case when the degree is 0, we will link this to $\operatorname{Pic}^0 E$ in this section.

One matter we should discuss is the group law on E. On the one hand, there is the one coming from the abelian variety and also induced by λ^{-1}. On the other hand, there is the geometric group law from Definition 4.1.3. We should check that those are indeed the same.

Proposition 4.2.3 *The group law induced by* $\operatorname{Pic}^0 E$ *is the same as the geometric group law from Definition 4.1.3.*

Proof *We want to show that* $\lambda(P+Q) = \lambda(P) + \lambda(Q)$ *for every* $P, Q \in E$ *where the left addition is via the geometric group law and the right one is the addition of divisor classes.*

Consider $P, Q, R \in E$ *such that* $P + Q + R = 0_E$. *By construction, the line in* $\mathbb{P}^2$ *going through* P, Q *and* R *intersects* E *exactly at those three points. The equation of the line is given by a homogenous polynomial* $F(X, Y, Z)$ *of degree* 1. *We also consider the line going through* $P + Q$, R *and* 0_E, *we denote its equation by* $G(X, Y, Z)$.

Since $3 \cdot 0_E = 0_E$, *the line with equation* $Z = 0$ *intersects* E *in* 0_E *with multiplicity* 3. *Therefore:*

$$\operatorname{div}(F/Z) = (P) + (Q) + (R) - 3(0_E), \ \operatorname{div}(G/Z) = (P+Q) + (R) - 2(0_E).$$

This gives $\operatorname{div}(G/F) = ((P+Q) - (0_E)) - ((P) - (0_E)) - ((Q) - (0_E))$, *which is principal, hence* $\lambda(-P-Q) = \lambda(-P) + \lambda(-Q)$. *Since this is true for any* $P, Q \in E$, *the equality* $\lambda(P+Q) = \lambda(P) + \lambda(Q)$ *also holds.* □

The first addition is in $\operatorname{Div} E$, *the second one in* $\mathbb{Z}$ *and the third one in* E.

Proposition 4.2.4 *A divisor* $D = n_1(P_1) + \cdots + n_r(P_r)$ *with points* $P_1, \ldots, P_r \in E$ *is principal if and only if* $n_1 + \cdots + n_r = 0$ *and* $n_1 \cdot P_1 + \cdots n_r \cdot P_r = 0_E$.

Proof *We have* $\lambda\left(-\sum_i n_i \cdot P_i\right) = \sum_i n_i \lambda(-P_i)$, *hence*

$$\begin{aligned}\lambda(-n_1 \cdot P_1 - \cdots - n_r \cdot P_r) &= [n_1(P_1) + \cdots + n_r(P_r) - (n_1 + \cdots + n_r)(0_E)] \\ &= [D - (\deg D)(0_E)].\end{aligned}$$

If D *is principal then its degree is* 0 *and* $[D] = 0$, *therefore*

$$n_1 + \cdots + n_r = 0 \text{ and } n_1 \cdot P_1 + \cdots + n_r \cdots P_r = 0_E.$$

Conversely, if the two equalities above hold then $[D] = 0$, *that is* D *is principal.* □

This helps to prove that on the algebraic closure, $\operatorname{Pic}^0 E$ is the subgroup of divisors of degree 0 up to linear equivalence.

Proposition 4.2.5 $\operatorname{Pic}^0 E$ *is equal to* $\operatorname{Div}^0 E/\operatorname{Princ} E$ *on the algebraic closure.*

Proof *Consider* $[D] \in \operatorname{Pic}^0 E$, *we have seen in Proposition 3.3.6 that this is equivalent to* $t_P^* D \sim D$ *for all* $P \in E$.

If $D = \sum_i n_i(P_i)$, *then* $t_P^* D = \sum_i n_i(P_i - P)$ *and by Proposition 4.2.4, these are linearly equivalent if and only if they have same degree (which is the case) and*

$$\sum_i n_i \cdot P_i = \sum_i n_i \cdot (P_i - P) \text{ iff } \deg D \cdot P = 0_E.$$

Therefore, $[D] \in \operatorname{Pic}^0 E$ *if and only if for all* $P \in E$, $\deg D \cdot P = 0_E$. *On an infinite field, this is equivalent to* $\deg D = 0$ *and* $\operatorname{Pic}^0 E = \operatorname{Div}^0 E/\operatorname{Princ} E$. □

We end this part on polarization with a last characterization of algebraically equivalent divisors.

Proposition 4.2.6 *Two divisors* $D, D' \in \operatorname{Div} E$ *on an elliptic curve are algebraically equivalent on the algebraic closure if and only if* $\deg D = \deg D'$.

Proof *If* $\deg D = 0$, *then* $[D] \in \operatorname{Pic}^0 E$ *thanks to Proposition 4.2.5, hence* $\Lambda(D) = \Lambda(D')$ *is equivalent to* $\Lambda(D') = 0$, *or alternatively to* $[D'] \in \operatorname{Pic}^0 E$, *that is to* $\deg D' = 0 = \deg D$.

If $\deg D < 0$, *since* $\Lambda(0) = 0$, $\Lambda(-D) = -\Lambda(D)$ *and we can reduce to the case where* $\deg D > 0$. *Assume first that* D *and* D' *are algebraically equivalent. In this situation,* D *is ample, so we can use Proposition 3.3.8 to show that there is some* $P \in E$ *such that* $D \sim t_P^* D'$. *Hence,* $\deg D = \deg D'$.

Conversely, if $\deg D = \deg D'$, *we want to show that there is some* $P \in E$ *such that* $D \sim t_P^* D'$ *since* D *is ample. If we write* $D = \sum_i n_i(P_i)$ *and* $D' = \sum_j m_j(Q_j)$, *by Proposition 4.2.4 this is equivalent to*

$$\sum_i n_i \cdot P_i - \sum_j m_j Q_j = -\deg D' \cdot P.$$

On the algebraic closure, such a $P \in E$ *exists because the multiplication* $[\deg D']_E$ *is surjective.* □

4.2.2 Isogenies

Regarding isogenies, we have a few interesting results. First, if K is a finite subgroup of an elliptic curve E of order d, then there is a unique separable isogeny $f : E \to E'$ with kernel $\ker f = K$. Assume f is not the multiplication by $\sqrt{d}$ (or equivalently $K \neq E[\sqrt{d}]$), and that $\operatorname{char} k \nmid d$ if $\operatorname{char} k > 0$. Then K is of rank 1 because of the discussion in Section 3.3.3, that is K a cyclic subgroup of E, generated by an element of d-torsion.

From such an isogeny, we can derive two more

- the dual isogeny $\widehat{f} : \widehat{E'} \to \widehat{E}$ from Theorem 3.3.11;

- the contragedient isogeny $\tilde{f} : E' \to E$ such that

$$\tilde{f} \circ f = [d]_E \text{ and } f \circ \tilde{f} = [d]_{E'}$$

from Proposition 3.2.11.

Denote by λ and λ' the principal polarization on E and E' respectively. Since E and E' are principally polarized, we want the isogeny f to preserve the polarizations. We set $f_0 = \lambda^{-1} \circ \hat{f} \circ \lambda' : E' \to E$. It is an isogeny of degree $\deg \hat{f} = \deg f = d$ since the degree is multiplicative, and λ, λ' are isomorphisms. Hence, $\ker f_0 \subseteq E[d]$; consider $P \in E[d]$, we have

$$f_0 \circ f(P) = \lambda^{-1} \circ \Lambda(f^*(0_{E'}))(P)$$

by Eq. (3.1). Moreover,

$$f^*(0_{E'}) = \sum_{T \in K} (T) \implies \Lambda(f^*(0_{E'}))(P) = \left[\sum_{T \in K} ((T - P) - (T)) \right].$$

The inner divisor is principal by Proposition 4.2.4: it has degree 0 and on the elliptic curve

$$\sum_{T \in K} (T - P - T) = -\#K \cdot P = -d \cdot P = 0_E$$

by assumption, hence $\ker f_0 \circ f = E[d]$, and similarly $\ker f \circ f_0 = E'[d]$. By uniqueness, we have $f_0 = \tilde{f}$, so the dual and contragedient isogenies are the same up to the canonical isomorphisms between the elliptic curve and its dual. This is very specific to elliptic curves, as we have already stressed out earlier that there is a simple obstruction with the degree in higher dimension.

4.3 KUMMER LINES

In this section, we introduce the notion of Kummer line which will be central in this thesis. Although we described this framework in [RS24; RS25; BRS25], these facts are general knowledge on elliptic curves, and it felt more natural to put it in the preliminaries part. We assume that $\operatorname{char} k \neq 2$.

4.3.1 Weierstrass coordinates

Recall that we used Weierstrass coordinates while proving Theorem 4.1.11.

Definition 4.3.1 (Weierstrass coordinates) *Weierstrass coordinates for an elliptic curve E defined over k are two sections x and y such that*

$$\Gamma(2(0_E)) = \langle 1, x \rangle, \ \Gamma(3(0_E)) = \langle 1, x, y \rangle.$$

They verify the long Weierstrass equation

$$E : y^2 = x^3 + a_2 x^2 + a_4 x + a_6$$

for some coefficients $a_2, \dots, a_6 \in k$.

Consider some Weierstrass coordinates x, y. By Theorem 4.1.11 they provide an embedding $E \to \mathbb{P}^2(k)$. However, since x is even ($x(P) = x(-P)$ for any $P \in E$), the map $x : E \to \mathbb{P}^1, P \mapsto (x(P) : 1), 0_E \mapsto (1 : 0)$ factors through $E/\pm 1$. We will come back to it in the next section on models of Kummer lines.

It is more convenient to work with projective coordinates to avoid divisions; the elliptic curve has a projective equation $Y^2Z = X^3 + a_2X^2Z + a_4XZ^2 + a_6Z^3$ in $\mathbb{P}^2$. Working in projective coordinates means we pass from the divisor point of view to the line bundle point of view, with the additional structure coming from restrictions. Set

$$\Gamma((0_E)) = \langle Z_0 \rangle,\ \Gamma(2(0_E)) = \langle X_0, Z_0^2 \rangle,\ \Gamma(3(0_E)) = \langle X := X_0Z_0, Y, Z := Z_0^3 \rangle.$$

The affine Weierstrass coordinates are $x = X/Z = X_0/Z_0^2$ and $y = Y/Z$.

Since we are mainly interested in models of Kummer lines in this thesis, we will change notations and denote $\Gamma(2(0_E)) = \langle X, Z \rangle$, where $Z = Z_0^2$. With these notations, the full projective Weierstrass coordinates are XZ_0, Y, ZZ_0.

Another reason to use projective coordinates is that we want to deal with every model of Kummer lines, in particular those where the neutral element is not sent to infinity. If $x \in \Gamma(2(0_E))$ is an affine Weierstrass coordinate (resp. $(X : Z)$ are projective Weierstrass coordinates) this amounts to allow working with the affine coordinate $x' = \frac{ax+b}{cx+d}$ with $c \neq 0$ (resp. $(X' : Z') = (aX + bZ : cX + dZ)$). Notice that the divisor of poles of x' is not equal to $2(0_E)$ when $c \neq 0$, but to a linearly equivalent divisor, so x' is not a section of $2(0_E)$, whereas X' and Z' still are.

4.3.2 Models of Kummer lines

Let E/k be an elliptic curve, and $\pi : E \to E/\pm 1$ be the projection, it is a degree 2 cover. This map ramifies at the 2-torsion points $E[2]$. Since there are 4 of them on $\overline{k}$, $E/\pm 1$ is a marked curve of genus 0, the marked point being $\pi(0_E)$, hence is isomorphic to $\mathbb{P}^1$ over k. In fact, taking an affine Weierstrass model $y^2 = h(x)$ with $h(x)$ of degree 3 and the neutral point 0_E being at infinity, since $-P = (x_P, -y_P)$, it is classical that the quotient $\pi : E \to E/\pm 1$ is given by $(X : Y : Z) \mapsto (X : Z)$; this is equivalent to checking that the field $k(x)$ is the subfield of $k(E)$ invariant by the involution $[-1]$.

Definition 4.3.2 *A Kummer line is a map $\pi : E \to \mathbb{P}^1$ defined over k through which $[-1]$ factorizes, modulo the identification of π with $\pi' : E' \to \mathbb{P}^1$ whenever there is an isomorphism (of elliptic curves) $\varphi : E \xrightarrow{\sim} E'$ defined over $\overline{k}$ such that $\pi' = \pi \circ \varphi$.*

An isomorphism of Kummer lines $\pi : E \to \mathbb{P}^1$ and $\pi' : E' \to \mathbb{P}^1$ is any isomorphism $\varphi : \mathbb{P}^1 \to \mathbb{P}^1$ defined over k, such that φ extends to an isomorphism $\psi : E \to E'$ over $\overline{k}$ such that $\pi' \circ \psi = \varphi \circ \pi$.

Remark 4.3.3 *Note that in Definition 4.3.2, the morphisms between elliptic curves are defined over the algebraic closure $\overline{k}$, whereas the maps to the projective line are defined over k.*

For instance let us consider an elliptic curve E defined over a finite field k and its quadratic twist E'. Then what the definition says is that we identify the Kummer

$$\begin{tikzcd} E \arrow[rr, "\varphi/\bar{k}"] \arrow[dr, "\pi/k"'] & & E' \arrow[dl, "\pi'/k"] \\ & \mathbb{P}^1 & \end{tikzcd} \qquad \begin{tikzcd} E \arrow[r, "\psi/\bar{k}"] \arrow[d, "\pi/k"'] & E' \arrow[d, "\pi'/k"] \\ \mathbb{P}^1 \arrow[r, "\varphi/k"'] & \mathbb{P}^1 \end{tikzcd}$$

Figure 4.3: Commutative diagrams for Definition 4.3.2

lines of E *and* E' *even if* E *and* E' *are only isomorphic on the algebraic closure but not on* k *itself.*

Although the definition given above might seem ad hoc, it is quite natural if we consider the Kummer line as a stack quotient rather than a scheme quotient, we refer to Section 8.7 for more details.

Not every degree 2 cover $E \to \mathbb{P}^1$ factorizes through $[-1]$, hence defines a Kummer line. Indeed, degree 2 covers are given by degree 2 divisors, as the pullback of (∞) on $\mathbb{P}^1$ (by Riemann-Roch a degree 2 divisor has two global sections, hence defines a morphism to $\mathbb{P}^1$), and such a divisor D is, up to linear equivalence, of the form $D \sim (P) + (0_E) \sim 2(P')$. It corresponds to the cover $R \mapsto x(R - P')$, which factorizes through $[-1]$ if and only if P' is a point of 2-torsion, that is if D is linearly equivalent to $2(0_E)$. Here is an alternative definition.

Definition 4.3.4 *A Kummer line of an elliptic curve* E *is the datum of a degree* 2 *cover* $\pi : E \to \mathbb{P}^1$ *with* 4 *distinct ramification points, one of which is rational and marked:*

$$\exists 0_E \in E(k), \exists T_1, T_2, T_3 \in E \text{ with } \#\pi^{-1}(\pi(P)) = \begin{cases} 1 & \text{if } P \in \{0_E, T_1, T_2, T_3\}, \\ 2 & \text{otherwise.} \end{cases}$$

The following proposition shows that these definitions are indeed equivalent.

Proposition 4.3.5 *Let* E *be an elliptic curve and* $\pi : E \to \mathbb{P}^1$ *a map, both defined over* k*. The following assertions are equivalent:*

1. π *is a Kummer line in the sense of Definition 4.3.2.*
2. π *is a Kummer line in the sense of Definition 4.3.4.*

Moreover, the fibres are given by $\pi^{-1}(\pi(P)) = \{\pm P\}$ *for* $P \in E$ *and the* 4 *ramification points in Definition 4.3.4 are exactly the* 2*-torsion points.*

Proof *Assume first that* π *is a Kummer line in the sense of Definition 4.3.2. Let* x, y *be Weierstrass coordinates on* E*, we consider* $p : E \to \mathbb{P}^1$ *with* $p(P) = (x(P) : 1)$ *if* P *is an affine point and* $p(0_E) = (1 : 0)$*.* p *is a degree* 2 *cover which factors by* $[-1]$*, hence we can identify with* π *by the defining universal property. If* $P \in E$*, the fibres are* $p^{-1}(p(P)) = \{\pm P\}$*, so ramification points must verify* $P = -P$*, which is equivalent to* $P \in E[2]$*. This gives exactly* 4 *ramification points over* $\bar{k}$*, and since* 0_E *is rational,* p *and* π *are Kummer lines in the sense of Definition 4.3.4.*

Conversely, assume π is a Kummer line in the sense of Definition 4.3.4. This induces a field extension $\pi^* : k(\mathbb{P}^1) \to k(E)$ of degree 2. Since $\operatorname{char} k \neq 2$, it can be written as $k(\mathbb{P}^1) = k(x)$ and $k(E) = k(x, y)$ with $y^2 = h(x)$, where h is a Weierstrass equation in x. Consider σ a Galois automorphism, we must have $\sigma(x) = x$, and since $y^2 \in k(x)$, $\sigma(y) = \pm y$. If σ is non-trivial, this means that $\sigma(y) = -y$, which corresponds to the involution $[-1] : P \mapsto P$ on the elliptic curve. The equation $\sigma \circ \pi^* = \pi^*$ becomes $\pi \circ [-1] = \pi$ on the elliptic curve E, hence π factors through $[-1]$, and it is a Kummer line in the sense of Definition 4.3.2. □

Remark 4.3.6 *By Riemann-Hurwitz formula (Theorem 4.1.9), if we suppose that C is a smooth projective curve and $\pi : C \to \mathbb{P}^1$ is a degree 2 cover defined over k with 4 ramification points over $\overline{k}$, then C is an elliptic curve (over $\overline{k}$, and over k if one of the ramification points is rational).*

We will from now on prefer the definition from Definition 4.3.4. Since $\pi : E \to \mathbb{P}^1$ is a morphism of curves, it is surjective, therefore the important data is stored in the ramification points, which is how we will describe Kummer line in the sequel. Here is an example.

Example 4.3.7 *The marked point is denoted with a $*$. Consider a Kummer line with the following ramification points:*

$$(1:0)^*, \quad (\alpha_1 : 1), \quad (\alpha_2 : 1), \quad (\alpha_3 : 1),$$

where the α_i are potentially defined over an extension of k. Then the corresponding elliptic curve has equation, with some $\beta \in k$:

$$E : \beta y^2 = (x - \alpha_1)(x - \alpha_2)(x - \alpha_3). \tag{4.1}$$

Conversely, starting from Eq. (4.1), if the point at infinity is denoted 0_E, then the following map is a degree 2 covering with 4 ramification points which correspond to the 2-torsion:

$$\begin{aligned} E &\xrightarrow{\pi} \mathbb{P}^1 \\ (x, y) &\mapsto (x : 1), \\ 0_E &\mapsto (1 : 0). \end{aligned}$$

As stated earlier, a Kummer line cannot distinguish between an elliptic curve E and a quadratic twist E'. Over a finite field k, this amounts in the previous example to a choice of $\beta \in k^*/k^{*,2}$. Denote $\pi' : E' \to \mathbb{P}^1$ a Kummer line associated to the quadratic twist. If $p \in \mathbb{P}^1$ is not a ramification point of π and π', then there are two points $P \in E$ and $P' \in E'$ of order $n > 2$ such that $\pi(P) = \pi'(P') = p$, and either $P \in E(k)$ or $P' \in E'(k)$. Thus pushing the rational point along a 2-isogeny allows keeping track of the twist on the codomain even while working on the Kummer lines.

We will denote Kummer lines by $\mathcal{K}$, where $\mathcal{K} \simeq \mathbb{P}^1$. We will also forget about the π notation when it is not ambiguous and write $[P] = \pi(P)$, or even $P \in \mathcal{K}$ when the context is not ambiguous.

As already detailed in Sections 2.2 and 2.3, the addition law does not hold any more on the Kummer line, however there exists differential addition formulas that, given $[P]$, $[Q]$ and $[P - Q]$, return $[P + Q]$.

One last important lemma is the following, given a rational 2-torsion point, it will be essential to compute the map $P \mapsto P + T$ on the Kummer line in Chapters 6 and 7 to use the theta group theory.

Lemma 4.3.8 (Translation by a 2-torsion point) *Consider an elliptic curve* E *with a rational 2-torsion point* $T \in E[2](k)$ *and* $\pi : E \to \mathbb{P}^1$ *a Kummer line. Then the translation by* T *map* $t_T : \pi(P) \mapsto \pi(P + T)$ *is well-defined and is a homography on* $\mathbb{P}^1$.

Proof *Since* $T = -T$, *if* $P, Q \in E$ *with* $P = \pm Q$ *then* $\pi(P + T) = \pi(Q + T)$, *hence the map* $P \mapsto \pi(P + T)$ *factors through* π *and* t_T *is well-defined and is a morphism of* $\mathbb{P}^1$ *because the translation on* E *is a morphism of algebraic curves. It is bijective because the translation on* E *is surjective and if* $\pi(P + T) = \pi(Q + T)$ *then* $\pi(P) = \pi(Q)$ *since* T *is a 2-torsion point. Therefore,* t_T *is a homography.* $\square$

5 THE THETA GROUP

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In this last preliminary chapter, we will introduce the theta group theory developed by Mumford in [Mum66; Mum67a; Mum67b] for the algebraic theory and [Mum83; Mum84; Mum91] for a more analytic approach. While there is a large material on the subject, we will focus on introducing the basics of the theta group, theta structures and theta functions; these are mainly detailed in [Mum66; Mum83]. This chapter also refers to more detailed results in [Rob10, Chap. 3].

The theta group is a helpful tool to study various properties of abelian varieties, in particular it provides a universal system of coordinates for any abelian variety in any dimension. We will use this theory in Chapters 6 to 8 to derive isogenies and addition formulas, as well as to classify different models of elliptic curves.

Remark 5.0.1 (Convention) *While Mumford uses the language of ample line bundles, for convenience we will use the one of ample divisors here, as we already did in Section 3.1.2. If we have two line bundles $\mathcal{L} \simeq \mathcal{O}_X(D)$ and $\mathcal{M} \simeq \mathcal{O}_X(D')$ where D and D' are divisors on the abelian variety X, then $\mathcal{L} \simeq \mathcal{M}$ iff D and D' are linearly equivalent (Proposition 3.1.17). Let us set a direction for the isomorphism $\varphi : \mathcal{L} \xrightarrow{\sim} \mathcal{M}$, we associate to it an element $s \in k(X)$ such that $\operatorname{div} s = D' - D$. We make this choice because the natural isomorphism $\Gamma(D') \xrightarrow{\sim} \Gamma(D)$ is then the multiplication by s, which is an important property we will require in the theory later.*

In this chapter, we will work over an algebraically closed field $k = \overline{k}$ except in Section 5.4, X will denote an abelian variety of dimension g over k. We will only consider ample divisors of separable type in this chapter (see Definition 3.3.9).

In Section 5.1, we will introduce the theta group with some basic properties, before relating it to Heisenberg groups in Section 5.2. In Section 5.3 we will define theta structures and see how we can derive the system of coordinates of theta functions from them. The last two sections are more specialized to the case of dimension 1, we first discuss in Section 5.4 how to find twisted theta coordinates from the theta functions when the base field is not algebraically closed, and we then discuss the analytic aspect of theta functions in Section 5.5.

5.1 DEFINITION AND FIRST PROPERTIES

Recall from Chapter 3 that two divisors D and D' are linearly equivalent if there is an element $s \in k(X)$ such that $\operatorname{div} s = D - D'$, this is an equivalence relation denoted $\sim$ and the set of linear equivalence classes is the Picard group $\operatorname{Pic} X$. The subgroup $\operatorname{Pic}^0 X$ has an abelian variety structure, it is called the dual abelian variety of X and denoted $\widehat{X}$.

Let D be an ample divisor of separable type on X, consider the induced polarization

$$\Lambda(D) : X \to \widehat{X}$$

which maps an element $x \in X$ to the element $[t_x^* D - D]$ in $\widehat{X} = \operatorname{Pic}^0(X)$, and where $t_x : X \to X$ is the translation by x. We denote its kernel $H(D)$.

If $x \in H(D)$, then there is an element $s \in k(X)$ such that $\operatorname{div} s = t_x^* D - D$, this is how we construct the theta group:

We should define elements as tuples (x, g_x), but then the notations would become too cumbersome.

Definition 5.1.1 *Let D be a divisor on X, we set*

$$G(D) = \{g_x \in k(X) \mid \exists x \in H(D), \operatorname{div} g_x = t_x^* D - D\}.$$

Let $g_x, g_y \in G(D)$ for some $x, y \in H(D)$, we set

$$g_x \cdot g_y = g_x \times t_x^* g_y : z \mapsto g_x(z) g_y(z + x).$$

If $g_x \in G(D)$ for some $x \in H(D)$ and $s \in \Gamma(D)$, we define

$$g_x \cdot s = g_x \times t_x^* s : z \mapsto g_x(z) s(z + x).$$

We have $\operatorname{div} g_x \cdot g_y = \operatorname{div} g_x + \operatorname{div}(t_x^* g_y)$, so by construction

$$\operatorname{div} g_x \cdot g_y = t_x^* D - D + t_x^*(t_y^* D - D) = t_{x+y}^* D - D.$$

Hence, $g_x \cdot g_y$ is in $G(D)$ associated to $x + y \in H(D)$, this defines a group law on $G(D)$. Similarly, one can check that for any $s \in \Gamma(D)$, $g_x \in G(D)$, $g_x \cdot s \in \Gamma(D)$; in fact, $G(D)$ acts linearly on $\Gamma(D)$, we will come back to the study of this representation in Section 5.2.

Remark 5.1.2 *The neutral element in $G(D)$ is 1 associated to $0 \in H(D)$ and the inverse of $g_x \in G(D)$ for $x \in H(D)$ is given by $g_x^{-1} = 1/t_{-x}^* g_x$, associated to $-x \in H(D)$.*

Let us consider the natural projection $\pi_D : g_x \in G(D) \mapsto x \in H(D)$.

Proposition 5.1.3 *The following sequence is exact*

$$1 \to k^* \xrightarrow{i_D} G(D) \xrightarrow{\pi_D} H(D) \to 0$$

where, for any $\lambda \in k$, $i_D(\lambda)$ *is the constant function equals to* λ *associated to* $0 \in H(D)$ *and the map* $\pi_D : G(D) \to H(D)$ *is the one described above.*

Proof *The first map* i_D *is the natural injection of* k^* *into* $G(D)$ *and is therefore clearly injective. By definition of* $G(D)$*, the last map* π_D *is also surjective. Let* $g_0 \in G(D)$ *associated to* $0 \in H(D)$*; then* $\operatorname{div} g_0 = 0$*, hence* $g_0 = \lambda \in k^*$ *is a constant map. The sequence is therefore exact.* □

We are particularly interested in subgroups of $H(D)$ preserving this sequence, they are defined in [Mum66, § 1, Def. p. 291].

Definition 5.1.4 (Level subgroups) *A level subgroup* $\widetilde{K} \subseteq G(D)$ *is a subgroup such that* $k^* \cap \widetilde{K} = \{1\}$*, that is* $\widetilde{K}$ *is isomorphic to its image* $K \subseteq H(D)$*. If* $K \subseteq H(D)$*, any level subgroup* $\widetilde{K} \subseteq G(D)$ *such that* $\widetilde{K} \simeq K$ *is called a lift of* K*.*

In particular, a level subgroup is necessarily abelian; in Section 5.2.1 we will give a condition on the existence of level subgroups based on the commutator pairing.

In the second part of this thesis, we will be particularly interested in symmetric elements of the theta group, here is a definition:

Definition 5.1.5 (Symmetric divisors and elements) *Let* $\iota : x \in X \mapsto -x$*, this is an involution of* X*. If* D *is a divisor,* D *is symmetric if* $\iota^* D \sim D$*. For such a divisor, an element* $g_x \in G(D)$ *is said to be symmetric if* $\iota^* g_x = g_x^{-1}$ *(up to the isomorphism* $G(D) \simeq G(\iota^* D)$*).*

Let $f : X \to Y$ be an isogeny between two abelian varieties. Deriving explicit formulas for f is not trivial, and studying sections above certain divisors can help us to do so. We must however look for certain constraints between the considered divisors:

Definition 5.1.6 (Descent of a divisor) *Let* $f : X \to Y$ *be an isogeny and* D' *a divisor on* Y*.* D' *is a descent of a divisor* D *on* X *relative to* f *if* $f^* D'$ *is linearly equivalent to* D*.*

In particular in dimension 1*,* $(\deg f)(\deg D') = \deg D$*.*

These descent data have been introduced by Grothendieck and Mumford explains in his article how the theta group encodes those data with respect to an isogeny. In particular, we know how to describe lifts of the kernel of an isogeny and how it acts on the space of section; here is a condensed theorem of various Mumford results:

Theorem 5.1.7 (Mumford) *Let* D *be a divisor on* X*,* K *a subgroup of* $H(D)$*, and* $f : X \to X/K$ *the isogeny with kernel* K*. Let* D' *be a descent of* D *via* f*, that is* $f^* D' \sim D$*.*

1. *Let* $\alpha \in k(X)$ *such that* $\operatorname{div} \alpha = D - f^* D'$ *and set* $\widetilde{K} = \{\frac{t_x^* \alpha}{\alpha} \mid x \in K\}$*. The set* $\widetilde{K}$ *does not depend on the choice of* α*, it is a lift of* K *and the map* $[D'] \in \operatorname{Pic}(X/K) \mapsto \widetilde{K}$ *is a bijection between the set of descents of* D *via* f *and the set of lifts of* K*.*

2. $f^{-1}(H(D')) \subseteq H(D)$ *and, if* $C_{\widetilde{K}}(D)$ *is the centralizer of* $\widetilde{K}$ *in* $G(D)$*, then*

$$C_{\widetilde{K}}(D) = \{g_x \in G(D) \mid x \in H(D) \text{ and } f(x) \in H(D')\}.$$

Moreover, $C_{\widetilde{K}}(D)/\widetilde{K} \simeq G(D')$ *canonically.*

3. *We have* $\Gamma(D)^{\widetilde{K}} \simeq \Gamma(D')$*, where* $\Gamma(D)^{\widetilde{K}}$ *is the set of sections of* $\Gamma(D)$ *invariant by the action of* $\widetilde{K} \subseteq G(D)$*.*

4. *Assume* D *is symmetric. Then* D' *is symmetric if and only if the elements of* $\widetilde{K}$ *are symmetric.*

Proof *(1) corresponds to [Mum66, § 1, Prop. 1].*

(2) is [Mum66, § 1, Prop. 2], the isomorphism comes from the map from $C_{\widetilde{K}}(D)$ *to* $G(D')$ *that to some* g_x *associates the only* g_y *such that* $g_x = \frac{t_x^*\alpha}{\alpha} f^* g_y$*, where* $y = f(x)$ *and* α *is as in (1). It is a surjective morphism with kernel* $\widetilde{K}$*, hence by factorization there is an isomorphism* $C_{\widetilde{K}}(D)/\widetilde{K} \simeq G(D')$*.*

To prove (3), we set $\varphi : s' \in \Gamma(D') \mapsto \frac{f^* s'}{\alpha}$ *where* α *is as in (1). We have*

$$\operatorname{div} \varphi(s') + D = f^*(\operatorname{div} s' + D') \geqslant 0,$$

so $\varphi(s') \in \Gamma(D)$*, and if* $g_x = \frac{t_x^*\alpha}{\alpha} \in \widetilde{K}$ *for some* $x \in K$*, we have*

$$g_x \cdot \varphi(s') = g_x t_x^* \left(\frac{f^* s'}{\alpha} \right) = \frac{(t_x^*\alpha)(t_x^* f^* s')}{\alpha(t_x^*\alpha)} = \frac{f^* s'}{\alpha} = \varphi(s').$$

The last equality holds because $x \in K$ *so* $f \circ t_{-x} = f$*, hence* $\operatorname{im} \varphi \subseteq \Gamma(D)^{\widetilde{K}}$*.* φ *is clearly a morphism, if* $\varphi(s') = 0$ *then* $f^* s' = 0$ *and because* f *is surjective,* f^* *is injective so* $s' = 0$*:* φ *is injective. Let* $s \in \Gamma(D)^{\widetilde{K}}$*, we set* $s'' = \alpha s$*, this is an element of* $\Gamma(f^* D')$*. Moreover, if* $x \in K$ *and* $g_x \in \widetilde{K}$ *is the associated element, we have* $g_x \cdot s = s$ *which is equivalent to* $t_x^*(\alpha s) = \alpha s$*. Hence, for any* $x \in K$*,* $t_x^* s'' = s''$*, it is invariant by* K *and can then be factored by* f*: there is* $s' \in k(X/K)$ *such that* $s'' = s' \circ f = f^* s'$*, so* $\varphi(s') = s$ *and* φ *is surjective.*

We now prove (4). We have the following linear equivalences:

$$D \sim \iota^* D \sim \iota^* f^* D' \sim f^* \iota^* D'.$$

Hence, $\iota^* D'$ *also descends to* D*, the corresponding kernel being* $\iota^* \widetilde{K}$*. So* D' *being symmetric is equivalent to* $[\iota^* D'] = [D']$ *and by (1), this is the same as* $\iota^* \widetilde{K} = \widetilde{K}$*. Assume first that* D' *is symmetric. If* $g_x \in \widetilde{K}$ *is above* $x \in K$*, then* $\iota^* g_x \in \widetilde{K}$ *is above* $-x$*, but the only element above* $-x$ *in* $\widetilde{K}$ *is* g_x^{-1}*, hence* $\iota^* g_x = g_x^{-1}$ *and the elements of* $\widetilde{K}$ *are symmetric. Conversely, if* $\widetilde{K}$ *consists of symmetric elements, it is clear that* $\iota^* \widetilde{K} = \widetilde{K}$ *and therefore that* D' *is symmetric.* □

5.2 HEISENBERG GROUPS

In the first section we saw how the theta group helps to understand the possible descents of a divisor via an isogeny. We now want to make these data more explicit by deriving canonical coordinates on abelian varieties to derive formulas for our isogenies. In this section we introduce the Heisenberg groups, whose behaviour is similar to the one of theta groups.

5.2.1 The commutator pairing

Because of the exactness of $1 \to k^* \to G(D) \to H(D) \to 0$, the following definition makes sense:

Definition 5.2.1 (Commutator pairing) *Let $x, y \in H(D)$ and choose an element $g_x \in G(D)$ (resp. g_y) associated to x (resp. y). We set*

$$e_D(x, y) = g_x g_y g_x^{-1} g_y^{-1}.$$

Since $\operatorname{div} e_D(x, y) = 0$, we can see it as an element of k^*. Because k^* is the kernel of the projection, the value does not depend on the choice of g_x and g_y. In fact e_D is a skew-symmetric bilinear pairing on $H(D)$:

Proposition 5.2.2 $e_D : H(D) \times H(D) \to k^*$ *verifies the following properties:*

- $e_D(x_1 + x_2, y) = e_D(x_1, y)e_D(x_2, y)$ *for any* $x_1, x_2, y \in H(D)$,
- $e_D(x, y_1 + y_2) = e_D(x, y_1)e_D(x, y_2)$ *for any* $x, y_1, y_2 \in H(D)$,
- $e_D(x, x) = 1$ *for any* $x \in H(D)$,
- $e_D(y, x) = e_D(x, y)^{-1}$ *for any* $x, y \in H(D)$.

Proof *Let $g_x, g_{x_1}, g_{x_2}, g_y \in G(D)$ associated to $x, x_1, x_2, y \in H(D)$. Then $g_{x_1} g_{x_2}$ is associated to $x_1 + x_2$ and*

$$e_D(x_1 + x_2, y) = g_{x_1} g_{x_2} g_y (g_{x_1} g_{x_2})^{-1} g_y^{-1} = g_{x_1} e_D(x_2, y) g_y g_{x_1}^{-1} g_y^{-1}.$$

Because $e_D(x_2, y) \in k^*$, it commutes with elements of $G(D)$ and therefore $e_D(x_1 + x_2, y) = e_D(x_1, y)e_D(x_2, y)$. The same holds for bilinearity on the right side.

It is immediate that $e_D(x, x) = g_x g_x g_x^{-1} g_x^{-1} = 1$. By bilinearity:

$$e_D(x + y, x + y) = e_D(x, x)e_D(x, y)e_D(y, x)e_D(y, y) = e_D(x, y)e_D(y, x).$$

Hence, $e_D(x, y)e_D(y, x) = 1$. □

Remark 5.2.3 *In Section 8.3.1 we will define the Weil pairing on an elliptic curve. It can be defined more generally on an elliptic curve, and is — up to some sign convention — equal to the commutator pairing associated to some divisor D, see for instance [GM, § 11] or the following blog note from Martin Orr[1].*

The commutator pairing gives a nice criterion for the existence of level subgroups:

Proposition 5.2.4 *Let K be a subgroup of $H(D)$, then there is a lift of K denoted $\widetilde{K} \subseteq G(D)$ if and only if $e_D \equiv 1$ when restricted to elements of K.*

K is said to be isotropic for e_D.

Proof *If $\widetilde{K}$ is a lift of K, then $\widetilde{K}$ is abelian and therefore $e_D \equiv 1$ on K. Conversely, assume K is isotropic for e_D and consider $x \in K$, $\widetilde{g}_x \in G(D)$ associated to x. Denote by ℓ the order of $x \in K$, then $\widetilde{g}_x^\ell$ is above $\ell x = 0$ in $G(D)$: there is $\lambda \in k^*$ such that $\widetilde{g}_x^\ell = \lambda$.*

1 http://www.martinorr.name/blog/2011/08/29/weil-pairings/

Because k *is algebraically closed, we consider* $\mu \in k^*$ *an* ℓ*-th root of* λ*, we then set* $g_x = \mu^{-1}\tilde{g}_x$*. By construction* $g_x^\ell = 1$ *and* g_x *must be of order* ℓ*.*

By the structure theorem of finite abelian groups, decompose

$$K = \langle x_1, \dots, x_r \rangle$$

where the order of x_i *is* ℓ_i*. We construct* $g_i \in G(D)$ *above* x_i *of order* ℓ_i *as earlier and we set*

$$\widetilde{K} = \langle g_1, \dots, g_r \rangle.$$

Let $1 \leqslant i, j \leqslant r$*. By assumption,* K *is isotropic for* e_D*, i.e.* $e_D(x_i, x_j) = 1$ *which implies* $g_i g_j = g_j g_i$*, therefore* $\widetilde{K}$ *is abelian, finite and by construction* $\widetilde{K} \simeq K$ *via the projection.* □

Given our skew-symmetric pairing, we are interested in having a symplectic decomposition compatible with this pairing, we are looking into the following equivalent conditions:

(i) The centre of $G(D)$ — denoted $Z(G(D))$ — is k^*.

(ii) e_D is non-degenerate, that is

$$\forall x \in H(D) \setminus \{0\},\ \exists y \in H(D),\ e_D(x, y) \neq 1.$$

(iii) There are two subgroups K_1, K_2 of $H(D)$ such that $H(D) = K_1 \oplus K_2$, both K_1 and K_2 are isotropic for e_D and e_D is a non-degenerate pairing of K_1 and K_2. In particular, $K_2 \simeq \mathrm{Hom}(K_1, k^*) =: \widehat{K_1}$.

$\widehat{G} = \mathrm{Hom}(G, k^*) \simeq G$ if G is a finite abelian group by duality.

Indeed, if we consider $x \in H(D)$ with $g_x \in G(D)$ a lift, then $e_D(x, y) = 1$ for all $y \in H(D)$ is equivalent to $g_x \in Z(G(D))$. Therefore, (ii) is equivalent to ($x = 0$ or $g_x \notin Z(G(D))$). But $x = 0$ iff $g_x \in k^*$ and because $k^* \subseteq Z(G(D))$, (i) and (ii) are equivalent.

If we assume (iii), we consider $x \neq 0$ in $H(D)$, decomposed as $x_1 + x_2$ in $K_1 \oplus K_2$. Assume $x_1 \neq 0$ by symmetry. Because e_D is non-degenerate, there exists $y \in K_2$ such that $e_D(x_1, y) \neq 1$. Since $e_D(x_2, y) = 1$, we therefore have $e_D(x, y) \neq 1$, hence e_D is non-degenerate. The converse is obtained via a usual symplectic decomposition (or Lagrangian decomposition) of skew-symmetric pairing, see for instance [Dav10, Lemma 5.2].

The construction is similar to the one on vector spaces with a skew-symmetric bilinear form.

Remark 5.2.5 *Another condition equivalent to (ii) is:*

(ii′) The map $\omega : H(D) \to \widehat{H(D)} = \mathrm{Hom}(H(D), k^*)$ *is an isomorphism:*

$$\omega : x \mapsto (y \mapsto e_D(x, y)).$$

(ii) is equivalent to $\ker \omega = \{0\}$ *and because in our case* $H(D)$ *and* $\widehat{H(D)}$ *have same cardinal, this is equivalent to* ω *being an isomorphism.*

What matters more importantly is the following theorem:

Theorem 5.2.6 $G(D)$ *satisfies conditions (i) – (iii).*

Proof *Mumford proves it in [Mum66, § 1, Thm. 1].* □

We will denote a symplectic decomposition satisfying (iii) by

$$H(D) = K_1(D) \oplus K_2(D)$$

in the sequel, with $K_2(D) \simeq \widehat{K_1(D)}$ via e_D.

The commutator pairing behaves well on the theta group and gives a convenient decomposition of $H(D)$. In particular, because $K_1(D) \simeq \widehat{K_2(D)} \simeq K_2(D)$, $K_1(D)$ and $K_2(D)$ have the same elementary divisors, and so they occur in pair in $H(D)$. Let $(\delta_1, \dots, \delta_r)$ be the elementary divisors of $K_1(D)$, and set $d = \operatorname{lcm}(\delta_1, \dots, \delta_r)$. Then for every $x \in H(D)$, $dx = 0$, hence $H(D) \subseteq X[d] \simeq (\mathbb{Z}/d\mathbb{Z})^{2g}$ if g is the dimension of X, but $H(D)$ is of rank $2r$, so $r \leqslant g$.

Definition 5.2.7 (Type of a divisor) *Let $\delta_g \mid \delta_{g-1} \mid \cdots \mid \delta_1$ be integers with $\delta_i \geqslant 1$ such that $(\delta_1, \delta_2, \dots, \delta_g)$ are the elementary divisors of $K_1(D)$. The type (or level) of D is the tuple $\delta = (\delta_1, \dots, \delta_g)$.*

We may pad with 1 the sequence to always have g elements.

Note that, because D is assumed to be of separable type (see Definition 3.3.9), $\operatorname{char} k \nmid \delta_1$.

Example 5.2.8 *In this chapter, we will consider a long example over an elliptic curve $X = E$, with the divisor $D = 2(0_E)$. This is the interesting case in Chapters 6 and 7. D is an ample divisor, the polarization $\Lambda(D)$ is the map $P \in E \mapsto [2(-P) - 2(0_E)]$. Therefore, $\Lambda(D)(P) = 0$ if and only if $2 \cdot P = 0_E$ using Proposition 4.2.4, that is $H(D) = E[2]$. We know that $E[2] = \langle T_1, T_2 \rangle$ where T_1, T_2 are 2-torsion points with $e_D(T_1, T_2) = -1$. This gives a natural symplectic decomposition with $K_i(D) = \langle T_i \rangle$ for $i = 1, 2$, and the type of D is $\delta = 2$.*

5.2.2 Heisenberg groups

We now want to go backwards, the theta group has a behaviour very close to the one of Heisenberg groups. In the sequel we set $\delta = (\delta_1, \dots, \delta_g)$ as in Definition 5.2.7.

Definition 5.2.9 (Heisenberg group) *We set the following groups:*

$$K(\delta) = \mathbb{Z}/\delta_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/\delta_r\mathbb{Z}, \ \widehat{K(\delta)} = \operatorname{Hom}(K(\delta), k^*), \ H(\delta) = K(\delta) \times \widehat{K(\delta)}.$$

Let $G(\delta) = k^* \times K(\delta) \times \widehat{K(\delta)} = k^* \times H(\delta)$ as a set and define the following group law on $G(\delta)$:

$$(\alpha, u, \chi) \cdot (\alpha', u', \chi') = (\alpha\alpha'\chi'(u), u + u', \chi\chi').$$

$G(\delta)$ is called the Heisenberg group.

Checking that this indeed defines a group law is a straight-forward computation. Given the notations, we want to show a connection between the Heisenberg group $G(\delta)$ and the theta group $G(D)$. First, the sequence

$$1 \to k^* \xrightarrow{i_\delta} G(\delta) \xrightarrow{\pi_\delta} H(\delta) \to 0$$

is exact, where i_δ given by $i_\delta(\alpha) = (\alpha, 0, 1)$ is the natural injection of k^* and π_δ defined as $\pi_\delta(\alpha, x, \chi) = (x, \chi)$ is the projection to $H(\delta)$. We may identify k^* with its image and simply write the element $(\alpha, 0, 1) \in G(\delta)$ as α.

Remark 5.2.10 *The neutral element on* $G(\delta)$ *is* $(1, 0, 1)$*. The inverse of an element is given by* $(\alpha, u, \chi)^{-1} = (\chi(u)\alpha^{-1}, -u, \chi^{-1})$*. Moreover, every element* (α, u, χ) *can be decomposed as* $(\alpha, u, \chi) = \alpha \cdot (1, 0, \chi) \cdot (1, u, 1) = \frac{\alpha}{\chi(u)} \cdot (1, u, 1) \cdot (1, 0, \chi)$*.*

$H(\delta)$ also comes with a natural pairing $e_\delta : H(\delta) \times H(\delta) \to k^*$ given by (see for instance [Lys22, Eq. 2])

$$e_\delta((u, \chi), (u', \chi')) = \frac{\chi'(u)}{\chi(u')}.$$

It is bilinear, skew-symmetric, and non-degenerate, so it admits a symplectic decomposition. If we identify $K(\delta)$ with $K(\delta) \times \{1\}$ and $\widehat{K(\delta)}$ with $\{0\} \times \widehat{K(\delta)}$, we have $H(\delta) = K_1(\delta) \oplus K_2(\delta)$ with $K_1(\delta) = K(\delta)$ and $K_2(\delta) = \widehat{K(\delta)}$. To go further with this identification, we will simplify the notations like $e_\delta((u, 1), (0, \chi)) = \chi(u)$ to $e_\delta(u, \chi)$ for $u \in K(\delta)$ and $\chi \in \widehat{K(\delta)}$.

Remark 5.2.11 *One can check that, if* $g = (\alpha, u, \chi), h = (\alpha', u', \chi') \in G(\delta)$*, then the commutator* $ghg^{-1}h^{-1}$ *is in* k^* *and is equal to* $\frac{\chi'(u)}{\chi(u')} = e_\delta((u, \chi), (u', \chi'))$*.*

By construction, if D is a divisor of type δ, then $H(D)$ and $H(\delta)$ are isomorphic, what we are interested in are isomorphisms preserving the symplectic structure.

Definition 5.2.12 *A symplectic isomorphism* $\varphi : (H(D), e_D) \xrightarrow{\sim} (H(\delta), e_\delta)$ *is a group isomorphism* $\varphi : H(D) \xrightarrow{\sim} H(\delta)$ *such that:*

$$\forall x, y \in H(D),\ e_D(x, y) = e_\delta(\varphi(x), \varphi(y)).$$

The following lemma helps us to classify the symplectic isomorphisms [Rob10, Prop. 3.3.1].

Lemma 5.2.13 *There is a bijection between the set of symplectic isomorphisms* $\varphi : (H(D), e_D) \xrightarrow{\sim} (H(\delta), e_\delta)$ *such that* $\varphi(K_i(D)) \subseteq K_i(\delta)$ *for* $i = 1, 2$ *and the set of group isomorphisms* $\lambda : K_1(D) \xrightarrow{\sim} K(\delta)$*.*

Proof *For any isomorphism* $\lambda : K_1(D) \xrightarrow{\sim} K(\delta)$*, we define* $\widehat{\lambda} : K_2(D) \to \widehat{K(\delta)}$ *by*

$$\widehat{\lambda} : y \mapsto e_D(\lambda^{-1}(\cdot), y).$$

It is clearly an injective morphism because e_D *is non-degenerate, and by cardinality it is an isomorphism.*

Then, for any $x = x_1 + x_2 \in H(D)$ *where* $x_i \in K_i(D)$ *for* $i = 1, 2$*, we set* $\varphi : x \mapsto (\lambda(x_1), \widehat{\lambda}(x_2))$*.* φ *is an isomorphism because* λ *and* $\widehat{\lambda}$ *are, and* $\varphi(K_i(D)) \subseteq K_i(\delta)$*. Moreover, if* $x = x_1 + x_2 \in H(D)$ *and* $y = y_1 + y_2 \in H(D)$ *with* $x_i, y_i \in K_i(D)$ *for* $i = 1, 2$*, then:*

$$e_D(x, y) = e_D(x_1, y_2)e_D(x_2, y_1).$$

We will show that for any $x \in K_1(D)$*,* $y \in K_2(D)$*,* $e_D(x, y) = e_\delta(\varphi(x), \varphi(y))$*, which is enough to prove that* φ *is symplectic by the previous formula. Let* $u = \lambda(x)$ *and* $\chi = \widehat{\lambda}(y)$*:*

$$e_D(x, y) = e_D(\lambda^{-1}(u), y) = \widehat{\lambda}(y)(u) = \chi(u).$$

Therefore, $e_D(x,y) = e_\delta(u,\chi)$*, which is also* $e_\delta(\lambda(x),\widehat{\lambda}(y)) = e_\delta(\varphi(x),\varphi(y))$ *for any* $x \in K_1(D)$*,* $y \in K_2(D)$*, and* φ *is a symplectic isomorphism.*

Conversely, if $\varphi : (H(D), e_D) \xrightarrow{\sim} (H(\delta), e_\delta)$ *is a symplectic isomorphism such that* $\varphi(K_i(D)) \subseteq K_i(\delta)$ *for* $i = 1,2$*, we let* λ *be the restriction of* φ *to* $K_1(D)$*, and* μ *its restriction to* $K_2(D)$*. Because* φ *is an isomorphism,* λ *(resp.* μ*) is still injective; its image is a subgroup of* $K_1(\delta)$ *(resp.* $K_2(\delta)$*) by assumption and by cardinality they must be equal. So* λ *(resp.* μ*) is an isomorphism from* $K_1(D)$ *to* $K(\delta)$ *(resp. from* $K_2(D)$ *to* $\widehat{K(\delta)}$*).*

Let $x \in K_1(D)$ *and* $y \in K_2(D)$*, because* φ *is symplectic:*

$$e_D(x,y) = e_\delta(\varphi(x),\varphi(y)) = e_\delta(\lambda(x),\mu(y)) = \mu(y)(\lambda(x)).$$

Therefore, $\mu(y) = e_D(\lambda^{-1}(\cdot), y)$ *for any* $y \in K_2(D)$*, that is* $\mu = \widehat{\lambda}$ *and the lemma follows.* □

We can now prove our correspondence between the theta group and the corresponding Heisenberg group.

Corollary 5.2.14 (Theorem 5.2.6) *If* δ *is the type of* D*, then the following sequences are isomorphic:*

$$\begin{tikzcd}
1 \arrow[r] & k^* \arrow[r, "i_D"] \arrow[d, "\|"] & G(D) \arrow[r, "\pi_D"] \arrow[d, "\wr"] & H(D) \arrow[r] \arrow[d, "\wr"] & 0 \\
1 \arrow[r] & k^* \arrow[r, "i_\delta"] & G(\delta) \arrow[r, "\pi_\delta"] & H(\delta) \arrow[r] & 0
\end{tikzcd}$$

Proof *Our goal is to construct isomorphisms* $G(D) \simeq G(\delta)$ *and* $H(D) \simeq H(\delta)$ *such that the diagram commutes. Let* $H(D) = K_1(D) \oplus K_2(D)$ *be a symplectic decomposition, we can consider lifts to* $G(D)$ *by Proposition 5.2.4 because* $K_i(D)$ *is isotropic for* $i = 1,2$*, we denote those lifts* $\widetilde{K}_1 \simeq K_1(D)$ *and* $\widetilde{K}_2 \simeq K_2(D)$*.*

Because $K_1(D)$ *and* $K(\delta)$ *have the same elementary divisors, we can consider an isomorphism* $\lambda : K_1(D) \xrightarrow{\sim} K(\delta)$*, to which corresponds a symplectic isomorphism* $\varphi : (H(D), e_D) \xrightarrow{\sim} (H(\delta), e_\delta)$ *with* $\varphi(K_i(D)) \subseteq K_i(\delta)$ *for* $i = 1,2$ *by Lemma 5.2.13. Now, let* $u \in K(\delta)$ *and* $\chi \in \widehat{K(\delta)}$*, there is a unique* $g_u \in \widetilde{K}_1$ *above* $\varphi^{-1}(u)$*. Similarly, we consider* $g_\chi \in \widetilde{K}_2$ *the unique element above* $\varphi^{-1}(\chi)$*. If* $\alpha \in k^*$*, we then set* $j : G(\delta) \to G(D)$ *via*

$$j : (\alpha, u, \chi) \mapsto \alpha g_\chi g_u.$$

We first show that this is a morphism. By uniqueness, if $u, u' \in K(\delta)$*, then* $g_{u+u'} = g_u g_{u'}$ *and if* $\chi, \chi' \in \widehat{K(\delta)}$*, then* $g_{\chi\chi'} = g_\chi g_{\chi'}$*. Therefore:*

$$j((\alpha,u,\chi)\cdot(\alpha',u',\chi')) = j(\alpha\alpha'\chi'(u), u+u', \chi\chi') = \alpha\alpha'\chi'(u) g_\chi g_{\chi'} g_u g_{u'}.$$

Since φ *is symplectic,* $\chi'(u) = e_\delta(u,\chi') = e_D(\varphi^{-1}(u), \varphi^{-1}(\chi'))$*. By definition of the commutator pairing, we then have:*

$$g_u g_{\chi'} = e_D(\varphi^{-1}(u), \varphi^{-1}(\chi')) g_{\chi'} g_u = \chi'(u) g_{\chi'} g_u.$$

Hence,

$$j((\alpha,u,\chi)\cdot(\alpha',u',\chi')) = \alpha\alpha' g_\chi g_u g_{\chi'} g_{u'} = j(\alpha,u,\chi)\cdot j(\alpha',u',\chi').$$

So j is a morphism, which clearly commutes with the injections and projections involved in the diagram by construction. If (α, u, χ) is in the kernel of j, then $\alpha g_\chi g_u = 1$, so $\varphi^{-1}(u, \chi) = 0$, which implies $(u, \chi) = (0, 1)$ and $g_u = g_\chi = 1$. Finally, $\alpha = 1$ and j is injective. If $g_x \in G(D)$ is above $x \in H(D)$, we set $(u, \chi) = \varphi(x)$, then $g_x = \alpha g_\chi g_u$ for some $\alpha \in k^*$, and $j(\alpha, u, \chi) = g_x$: j is surjective, and it is an isomorphism. □

As of now, we have mainly studied the convenient group structure of $G(D)$ but as already seen in Theorem 5.1.7 (3), the action of the theta group on sections of a divisor helps to derive sections of a descent canonically, which will be our main concern in Sections 6.2.1 and 7.4. Because of the isomorphism of sequences in the previous corollary, the action of the theta group on sections has an equivalent in the corresponding Heisenberg group. It turns out this last one has a nice description.

The definition of a level subgroup in $G(\delta)$ is analogous to the one in $G(D)$.

Proposition 5.2.15 *$G(\delta)$ admits a unique (up to isomorphism) irreducible representation where k^* acts by multiplication, we denote it $V(\delta)$. If V is a representation of $G(\delta)$ on which k^* acts by multiplication, then V is isomorphic to the direct sum of r copies of $V(\delta)$, and $r = \dim(V^{\widetilde{K}})$ for any maximal level subgroup $\widetilde{K} \subseteq G(\delta)$, where $V^{\widetilde{K}}$ is the subspace of V of elements $\widetilde{K}$-invariant.*

Proof *This corresponds to [Mum66, § 1, Prop. 3].* □

Therefore, there is also a unique irreducible representation of $G(D)$ where k^* acts by multiplication, it turns out we have already defined it:

Theorem 5.2.16 *If D is an ample divisor of separable type, then $\Gamma(D)$ is the unique irreducible representation of $G(D)$ on which k^* acts by multiplication.*

Proof *The fact that it is unique has already been explained, and the action of k^* can be easily verified on the Definition 5.1.1. For irreducibility, see [Mum66, § 1, Thm. 2].* □

We will set a few notations for the sequel.

Definition 5.2.17 *Let $V(\delta)$ be the vector space of functions on $K(\delta)$ with values in k. The irreducible representation associated is given by*

$$\rho_\delta : G(\delta) \to GL(V(\delta)),\ (\alpha, u, \chi) \mapsto (f \mapsto \alpha \times \chi \times t_u^* f).$$

Similarly, the irreducible representation $\Gamma(D)$ derived from the action in Definition 5.1.1 will be denoted

$$\rho_D : G(D) \to GL(\Gamma(D)),\ g_x \mapsto (s \mapsto g_x \times t_x^* s).$$

5.3 THETA STRUCTURES AND FUNCTIONS

So far we have been introducing tools which have nice connections, our goal here will be to define a canonical system of coordinates on any abelian variety associated to some divisor. We will set in this section an ample divisor D of separable type δ.

5.3.1 Theta structures

While proving Corollary 5.2.14, we constructed an isomorphism between $G(D)$ and $G(\delta)$ which was the identity on k^*. This isomorphism was derived from a symplectic isomorphism $\varphi : (H(D), e_D) \xrightarrow{\sim} (H(\delta), e_\delta)$, this is what we will call a theta structure.

Definition 5.3.1 (Theta structure) *A theta structure of* $G(D)$ *is the data of an isomorphism* $\Theta_D : G(D) \xrightarrow{\sim} G(\delta)$ *which is the identity on* k^*.

Let $\Theta_D : G(D) \xrightarrow{\sim} G(\delta)$ be a theta structure, by factorization we can recover an isomorphism $\overline{\Theta_D} : H(D) \xrightarrow{\sim} H(\delta)$ which makes the diagram of Corollary 5.2.14 commutative. Because Θ_D is the identity on k^*, if $g, h \in G(D)$ and $u = \Theta_D(g), v = \Theta_D(h) \in G(\delta)$, we have

$$e_D(\pi_D(g), \pi_D(h)) = ghg^{-1}h^{-1} = \Theta_D(ghg^{-1}h^{-1}) = uvu^{-1}v^{-1}.$$

According to Remark 5.2.11, $uvu^{-1}v^{-1}$ is exactly $e_\delta(\pi_\delta(u), \pi_\delta(v))$, and because our diagram is commutative:

$$e_D(\pi_D(g), \pi_D(h)) = e_\delta(\overline{\Theta_D}(\pi_D(g)), \overline{\Theta_D}(\pi_D(h))).$$

Hence, $\overline{\Theta_D}$ is a symplectic isomorphism.

Having a symplectic isomorphism is not enough to recover a theta structure though, we required a symplectic decomposition of $H(D)$ compatible with the natural one on $H(\delta)$ while proving Corollary 5.2.14. Fortunately, it is easy to define one; consider the natural section of $H(\delta)$ to $G(\delta)$ given by $\sigma_\delta(u, \chi) = (1, u, \chi)$ for $(u, \chi) \in H(\delta)$. This induces a section $\sigma_D : H(D) \to G(D)$ via $\sigma_D = \Theta_D^{-1} \circ \sigma_\delta \circ \overline{\Theta_D}$. σ_δ is not a group morphism, however its restriction to $K_i(\delta)$ for $i = 1, 2$ is, we will denote it σ_δ^i. If we set $K_i(D) := \overline{\Theta_D}^{-1}(K_i(\delta))$, then we have a symplectic decomposition of $H(D)$ and the restriction of σ_D to $K_i(D)$ — denoted σ_D^i — is exactly $\Theta_D^{-1} \circ \sigma_\delta^i \circ \overline{\Theta_D}$. In the end, we have a correspondence between theta structures and symplectic isomorphisms with additional data [Rob10, Prop. 4.3.3], the situation is also summarized in Fig. 5.1.

More generally, the restriction to any isotropic subgroup K *is a group morphism since it has a lift* $\widetilde{K}$ *such that* $\widetilde{K} \cap k^* = \{1\}$ *by Proposition 5.2.4.*

$$\begin{tikzcd}
1 \arrow[r] & k^* \arrow[r, "i_D"] \arrow[d, "\|"] & G(D) \arrow[r, shift left, "\pi_D"] \arrow[d, "\Theta_D"] & H(D) \arrow[l, dashed, shift left, "\sigma_D"] \arrow[r] \arrow[d, "\overline{\Theta_D}"] & 0 \\
1 \arrow[r] & k^* \arrow[r, "i_\delta"] & G(\delta) \arrow[r, shift left, "\pi_\delta"] & H(\delta) \arrow[l, dashed, shift left, "\sigma_\delta"] \arrow[r] & 0
\end{tikzcd}$$

Figure 5.1: Theta structure and induced applications

Proposition 5.3.2 *There is a bijection between theta structures* $\Theta_D : G(D) \xrightarrow{\sim} G(\delta)$ *and triplets* $(\overline{\Theta_D}, \sigma_D^1, \sigma_D^2)$ *where* $\overline{\Theta_D} : H(D) \xrightarrow{\sim} H(\delta)$ *is a symplectic isomorphism,* $K_i(D) := \overline{\Theta_D}^{-1}(K_i(\delta))$ *and* $\sigma_i^D : K_i(D) \to G(D)$ *is a section of* $G(D)$ *for* $i = 1, 2$ *which is also a group morphism.*

Proof *Let* $\Theta_D : G(D) \xrightarrow{\sim} G(\delta)$ *be a theta structure, we associate to it the triplet* $(\overline{\Theta_D}, \sigma_D^1, \sigma_D^2)$ *as constructed in the discussion preceding the property. The surjectivity of the defined map has been treated in the proof of Corollary 5.2.14. If two theta structures* Θ_D *and* Θ'_D *ends on the same triplet* $(\overline{\Theta_D}, \sigma_D^1, \sigma_D^2)$*, then*

$$\Theta_D \circ \sigma_D^i = \sigma_\delta^i = \Theta'_D \circ \sigma_D^i$$

for $i = 1, 2$*. Because they are the identity on* k^**, we must have* $\Theta_D = \Theta'_D$*, so it is indeed bijective.* □

What this tells us is that any symplectic isomorphism $\overline{\Theta_D} : H(D) \xrightarrow{\sim} H(\delta)$ always corresponds to a unique theta structure $\Theta_D : G(D) \xrightarrow{\sim} G(\delta)$ since we know how to construct sections σ_D^i for $i = 1, 2$. From now on, we will always consider we are in the situation where the symplectic decomposition of $H(D) = K_1(D) \oplus K_2(D)$ is compatible with $\overline{\Theta_D}$ via $K_i(D) = \overline{\Theta_D}^{-1}(K_i(\delta))$ for $i = 1, 2$. Another information is that, since there is a finite number of symplectic isomorphisms, there is a finite number of theta structures on a theta group.

Example 5.3.3 *Continuing Example 5.2.8 with* $D = 2(0_E)$*, the Heisenberg group* $G(2)$ *is just* $k^* \times \mathbb{Z}/2\mathbb{Z} \times \widehat{\mathbb{Z}/2\mathbb{Z}}$*. There is only one choice for a symplectic isomorphism* $\overline{\Theta_D} : H(2) \xrightarrow{\sim} \mathbb{Z}/2\mathbb{Z}$*, that is* $T_1 \mapsto 1$*. The non-trivial element in* $\widehat{\mathbb{Z}/2\mathbb{Z}}$ *is given by* $1 \mapsto -1$*. We will see later that this gives* 4 *possible theta structures.*

One last thing we need to introduce is the level structure. While proving Corollary 5.2.14 we implicitly used the canonical level subgroups

$$\widetilde{K}_1(\delta) := \{(1, u, 1) \mid u \in K(\delta)\},\ \widetilde{K}_2(\delta) := \{(1, 0, \chi) \mid \chi \in \widehat{K(\delta)}\},$$

they verify $\sigma_\delta^i(K_i(\delta)) = \widetilde{K}_i(\delta)$ for $i = 1, 2$. For $(u, \chi) \in H(\delta)$, we constructed elements g_u and g_χ above $\overline{\Theta_D}^{-1}(u) \in K_1(D)$ and $\overline{\Theta_D}^{-1}(\chi) \in K_2(D)$ respectively. Since the diagram is commutative,

$$\Theta_D(g_u) = (1, u, 1),\ \Theta_D(g_\chi) = (1, 0, \chi).$$

Remark 5.3.4 *As stated in Remark 5.2.10, an element* $(\alpha, u, \chi) \in G(\delta)$ *can be decomposed as* $\alpha \cdot (1, 0, \chi) \cdot (1, u, 1)$*. Therefore,* $\Theta_D^{-1}(\alpha, u, \chi) = \alpha g_\chi g_u$ *and not* $\alpha g_u g_\chi$ *as one could expect.*

We then have canonical level subgroups in $G(D)$ too given by

$$\widetilde{K}_1(D) := \Theta_D^{-1}(\widetilde{K}_1(\delta)),\ \widetilde{K}_2(D) := \Theta_D^{-1}(\widetilde{K}_2(\delta)).$$

$\widetilde{K}_1(D)$ is the group of elements g_x and $\widetilde{K}_2(D)$ the group of elements g_χ. They define what we call a level structure.

Definition 5.3.5 (Level structure) *A level structure on* $G(D)$ *is the data of two level subgroups* $\widetilde{K}_1, \widetilde{K}_2 \subseteq G(D)$ *such that* $\pi_D(\widetilde{K}_1)$ *and* $\pi_D(\widetilde{K}_2)$ *give a symplectic decomposition of* $H(D)$*.*

Our canonical level structure has the advantage that $\widetilde{K}_i(D)$ descends to $K_i(\delta)$ when composing morphisms, for $i = 1, 2$. Giving a level structure is equivalent to giving the two sections σ_D^1 and σ_D^2.

To summarize, a theta structure is the same as a symplectic isomorphism and two sections by Proposition 5.3.2 and:

- Lemma 5.2.13 states that the data of the symplectic isomorphism corresponds to indexing $K_1(D)$ via $K(\delta)$;
- the above discussion shows that the data of the two sections corresponds to a level structure on $G(D)$.

5.3.2 Theta functions

In this section we assume that the divisor D is *very ample*. Recall that $m = \deg D = \dim \Gamma(D)$, we set a basis of $\Gamma(D) = \langle s_1, \dots, s_m \rangle$. D being very ample means that there is an embedding $I : X \hookrightarrow \mathbb{P}^{m-1}$ explicitly given by $I(x) = (s_1(x) : \cdots : s_m(x))$, it is unique up to a projective change of basis, that is up to an element of $\mathrm{PGL}_m(k)$. How does one choose a natural basis? Theta structures enable us to do so, let $\Theta_D : G(D) \xrightarrow{\sim} G(\delta)$ be a theta structure.

1. Since ρ_D (resp. ρ_δ) is the unique irreducible representation of $G(D)$ (resp. $G(\delta)$) where k^* acts by multiplication, we have an isomorphism of representations $\beta : \Gamma(D) \xrightarrow{\sim} V(\delta)$, that is
$$\forall g \in G(D), s \in \Gamma(D),\ \beta(\rho_D(g)s) = \rho_\delta(\Theta_D(g))\beta(s). \tag{5.1}$$
2. Since $K(\delta)$ is finite, a natural basis $(X_u)_{u \in K(\delta)}$ of $V(\delta)$ is given by, for $u, v \in K(\delta)$: $\#K(\delta) = m$
$$X_u(v) := \begin{cases} 1 & \text{if } u = v, \\ 0 & \text{otherwise.} \end{cases}$$
Using β, we derive a basis of $\Gamma(D)$ via $\theta_u = \beta^{-1}(X_u)$ for any $u \in K(\delta)$.
3. By Schur's lemma, β is unique up to some scalar, therefore the embedding $I : X \hookrightarrow \mathbb{P}^{m-1}, x \mapsto (\theta_u(x))_{u \in K(\delta)}$ does not depend on the choice of β.

Definition 5.3.6 (Theta functions) *The elements $(\theta_u)_{u \in K(\delta)}$ are called the theta functions. The element $I(0_X)$ from the above construction is called the theta-null point.*

Because of Lemma 5.2.13, indexing via $K(\delta)$ is the same as indexing via $K_1(D)$; therefore, depending on the context, we may use the basis $(\theta_x)_{x \in K_1(D)}$ where $\theta_x := \theta_{\overline{\Theta_D(x)}}$. We will also drop the notation I and simply write $x = (\theta_y(x))_{y \in K_1(D)}$.

We will now explain how one can compute the family of theta functions. We first look at the action of an element (α, u, χ) on X_v, where $v \in K(\delta)$:
$$\rho_\delta(\alpha, u, \chi)X_v = \alpha\chi t_u^* X_v = \alpha\chi X_{v-u} = \alpha\chi(v-u)X_{v-u}. \tag{5.2}$$

In particular, $\rho_\delta(1,0,\chi)X_0 = X_0$ for any $\chi \in K_2(\delta)$, that is $X_0 \in V(\delta)^{\widetilde{K}_2(\delta)}$. By Proposition 5.2.15, since $\widetilde{K}_2(\delta)$ is a maximal level subgroup and $V(\delta)$ is irreducible, we have that $V(\delta)^{\widetilde{K}_2(\delta)}$ is of dimension 1, and X_0 is a generator. This space is isomorphic to $\Gamma(D)^{\widetilde{K}_2(D)}$ via β, which is therefore generated by θ_0.

Example 5.3.7 *From the unique symplectic isomorphism of Example 5.3.3, we recover a level structure $\widetilde{K}_i(D) = \langle g_i \rangle$ where g_i is an element of order 2 of the theta group above T_i for $i = 1, 2$. If g is another element of order 2 above T_i, then $g^2 = 1 = g_i^2$, so $g = \pm g_i$: there are only two possible choices above each point, that is 4 possible theta structures. Our sections are then given by $\sigma_D^i(T_i) = g_i$ for $i = 1, 2$.*

Here D is not very ample, we only have a degree 2 covering $E \to E/\pm 1 \to \mathbb{P}^1$. We can still use it to build theta functions, but they have to factor through $[-1]$.

The construction of the theta basis shows that there is a theta function θ_0 which is invariant by g_2, and the second function is given by $\theta_1 = \rho_D(g_1)\theta_0$. We can also index the function with 0_E and T_1 as $\theta_{0_E} := \theta_0$ and $\theta_{T_1} := \theta_1$.

If we now consider the isogeny $f : X \to X/K_2(D)$ with kernel $K_2(D)$, there is a divisor D' which is a descent of D associated to $\widetilde{K}_2(D)$ via Theorem 5.1.7 (1). By (3), there is an explicit isomorphism between $\Gamma(D')$ and $\Gamma(D)^{\widetilde{K}_2(D)}$: consider $\alpha \in k(X)$ with divisor $D - f^*D'$, and s a non-trivial element of $\Gamma(D')$. While proving (3) of Theorem 5.1.7, we gave the isomorphism: θ_0 is equal to $\lambda \frac{f^*s}{\alpha}$ for some $\lambda \in k^*$. Since we will consider these functions projectively later on, the constant will not matter.

The computation from Eq. (5.2) can be transposed on theta functions via the isomorphism β. Let $(\alpha, u, \chi) \in G(\delta)$, and consider $x_1 + x_2 = \overline{\Theta_D}^{-1}(u, \chi)$ a symplectic decomposition. Then $\Theta_D^{-1}(\alpha, u, \chi) = \alpha\sigma_D^2(x_2)\sigma_D^1(x_1)$. Since $\rho_\delta(\alpha, u, \chi)X_v = \alpha e_\delta(v - u, \chi)X_{v-u}$ for any $v \in K(\delta)$, if $y = \overline{\Theta_D}^{-1}(v)$, we have by Eq. (5.1):

$$\rho_D(\alpha\sigma_D^2(x_2)\sigma_D^1(x_1))\theta_y = \alpha e_D(y - x_1, x_2)\theta_{y-x_1}. \tag{5.3}$$

In particular, for any $x \in K_1(D)$, $\rho_D(\sigma_D^1(-x))\theta_0 = \theta_x$, so any theta function can be recovered from θ_0.

One last thing is the importance of the theta null point, its coordinates determine the whole theta structure. Consider $x \in X$ and $y \in H(D)$, as well as an element $g_y \in G(D)$ above y. We have $x + y = (\theta_z(x + y))_{z \in K_1(D)}$, that is $x + y = (t_y^*\theta_z(x))_z$. Because these are projective points, we can multiply by g_y every coordinate, and because $\rho_D(g_y)\theta_z = g_y \times t_y^*\theta_z$ by definition, $x + y = (\rho_D(g_y)\theta_z(x))_z$. Note that this does not depend on the choice of g_y. Therefore, if $y = y_1 + y_2$ with $y_i \in K_i(D)$ for $i = 1, 2$, we consider $g_y = \sigma_D^2(y_2)\sigma_D^1(y_1)$. Using Eq. (5.3):

$$x + y = \left(e_D(z - y_1, y_2)\theta_{z-y_1}(x)\right)_{z \in K_1(D)} \tag{5.4}$$

In particular when $x = 0_X$, we can compute the coordinates of y for any $y \in H(D)$ from the coordinates of 0_X, this completely determines $\overline{\Theta_D}$.

Example 5.3.8 *Continuing Example 5.3.7 where the 2-torsion is* $E[2] = \langle T_1, T_2 \rangle$ *and* $e_D(T_1, T_2) = -1$*, we have a basis of theta functions* $\theta_0 =: \theta_{0_E}$ *and* $\theta_1 =: \theta_{T_1}$*. Write the coordinates of* $0_E = (\theta_0(0_E) : \theta_1(0_E)) = (a : b)$*. We can determine the remaining coordinates of the 2-torsion from that using Eq. (5.4). Recall that* $T_i \in K_i(D)$ *for* $i = 1, 2$ *and the last 2-torsion point symplectic decomposition is* $T_3 = T_1 + T_2$*:*

- $T_1 = (e_D(0_E - T_1, 0_E)\theta_{0_E - T_1}(0_E) : e_D(T_1 - T_1, 0_E)\theta_{T_1 - T_1}(0_E))$*, so*
$$T_1 = (\theta_{-T_1}(0_E) : \theta_{0_E}(0_E)) = (b : a).$$
- $T_2 = (e_D(0_E - 0_E, T_2)\theta_{0_E - 0_E}(0_E) : e_D(T_1 - 0_E, T_2)\theta_{T_1 - 0_E}(0_E))$*, so*
$$T_2 = (\theta_{0_E}(0_E) : -\theta_{T_1}(0_E)) = (-a : b).$$
- $T_3 = (e_D(0_E - T_1, T_2)\theta_{0_E - T_1}(0_E) : e_D(T_1 - T_1, T_2)\theta_{T_1 - T_1}(0_E))$*, so*
$$T_3 = (-\theta_{-T_1}(0_E) : \theta_{0_E}(0_E)) = (-b : a).$$

This is precisely the model of Kummer line we will introduce in Example 7.1.2 and that we will study in the second part of this thesis, among others.

Moreover, the section σ_D can also be derived from the theta null point, let $x \in H(D)$ and decompose $x = x_1 + x_2$ with $x_i \in K_i(D)$. Consider also $y \in K_1(D)$. From Eq. (5.3) and the definition of the action:

$$\sigma_D(x) \times t_x^* \theta_y = e_D(y - x_1, x_2)\theta_{y - x_1}.$$

At least one coordinate of the theta null point is not zero because it is a projective point, we choose y such that $\theta_{y - x_1}(0_X) \neq 0$. Therefore, $\sigma_D(x)$ is the unique section of $k(X)$ such that

$$\begin{cases} \operatorname{div} \sigma_D(x) = t_x^* D - D; \\ \sigma_D(x)(0) = \frac{e_D(y - x_1, x_2)\theta_{y - x_1}(0_X)}{\theta_y(x)}. \end{cases}$$

Proposition 5.3.2 then shows that the theta null point completely determines the theta structure Θ_D.

5.3.3 Dual theta functions and Hadamard change of basis

We consider the special case of a divisor D of type $\delta = (2, 2, \dots, 2)$, therefore $H(D) = X[2]$. We consider a symplectic decomposition $K_1(D) = \langle T_1, \dots, T_g \rangle$ and $K_2(D) = \langle R_1, \dots, R_g \rangle$ where T_i and R_j are points of 2-torsion, as well as elements $e_1, \dots, e_g$ and $f_1, \dots, f_g$ of order 2 in the theta group such that $\widetilde{K}_1(D) = \langle e_1, \dots, e_g \rangle$ and $\widetilde{K}_2(D) = \langle f_1, \dots, f_g \rangle$ are the level subgroup associated to the symplectic decomposition. The element e_i is above T_i and f_i above R_i.

We will be numbering functions with $K(\delta) = (\mathbb{Z}/2\mathbb{Z})^g$, an element $n \in K(\delta)$ will be indexed as $n = (n_1, \dots, n_g)$. We first recover a theta function θ_0, invariant by $\widetilde{K}_2(D)$. The other elements come from Eq. (5.3) with:

$$\forall n \in K(\delta),\ \theta_n := \rho_D(e_1^{n_1} \cdots e_g^{n_g})\theta_0.$$

There is a natural map when we flip the symplectic decomposition. Let $K_1'(D) = \langle T_1', \dots, T_g' \rangle$ and $K_2'(D) = \langle R_1', \dots, R_g' \rangle$ where $T_i' = R_i$ and $R_i' = T_i$ for $1 \leqslant i \leqslant g$. Therefore, $K_1'(D) = K_2(D)$ and vice-versa. We also re-use the theta group elements with $e_i' := f_i$ above T_i' and $f_i' := e_i$ above R_i'. With this symplectic basis, we get a new set of theta functions with θ_0' invariant by $K_2'(D) = K_1(D)$ and

$$\forall n \in K(\delta),\ \theta_n' := \rho_D(f_1^{n_1} \cdots f_g^{n_g})\theta_0'.$$

To simplify notations, we will write $e^n = e_1^{n_1} \dots e_g^{n_g}$ and $f^n = f_1^{n_1} \dots f_g^{n_g}$ for any $n \in K(\delta)$. By construction of the symplectic basis, $e^n f^m = \varepsilon_{mn}^{(g)} f^m e^n$ for any $m, n \in K(\delta)$ where $\varepsilon_{mn}^{(g)} = (-1)^{\langle m,n \rangle}$ and $\langle \cdot, \cdot \rangle$ denotes the natural scalar product on $\mathbb{Z}^g$. Construct the matrix $H_g = \left(\varepsilon_{mn}^{(g)}\right)_{mn}$ of size $2^g \times 2^g$, this is the Hadamard matrix.

Consider $m, n \in K(\delta)$, and set $\overline{m} = (m_1, \dots, m_{g-1})$, $\overline{n} = (n_1, \dots, n_{g-1})$. When looking at the values of m_g and n_g, we have four possible combinations leading to:

$$\varepsilon_{mn}^{(g)} = \begin{cases} \varepsilon_{\overline{m}\overline{n}}^{(g-1)} & \text{if } m_g = 0 \text{ or } n_g = 0; \\ -\varepsilon_{\overline{m}\overline{n}}^{(g-1)} & \text{if } m_g = 1 \text{ and } n_g = 1. \end{cases}$$

Therefore, our Hadamard matrix has a nice recursive structure:

$$H_g = \begin{pmatrix} H_{g-1} & H_{g-1} \\ H_{g-1} & -H_{g-1} \end{pmatrix},\ H_0 = (1),$$

where the first column correspond to $n_g = 0$, the second one to $n_g = 1$, the first line to $m_g = 0$ and the second one to $m_g = 1$.

The reason to introduce this matrix is that it permutes the theta functions $(\theta_m)_m$ and the dual theta functions $(\theta_m')_m$. Indeed, consider the element

$$\sum_{n \in K(\delta)} \theta_n',$$

it is by construction invariant by $K_2(D)$, thus it is colinear to θ_0:

$$\sum_{n \in K(\delta)} \theta_n' = \lambda \theta_0,\ \lambda \in k^*.$$

Now, by using the action of e^m on θ_0 for any $m \in K(\delta)$, we get

$$\lambda \theta_m = \sum_{n \in K(\delta)} \rho_D(e^m)\theta_n'.$$

Since $\theta_n' = \rho_D(f^n)\theta_0'$, we have

$$\rho_D(e^m)\theta_n' = \rho_D(e^m f^n)\theta_0' = \varepsilon_{mn}^{(g)} \rho_D(f^n e^m)\theta_0' = \varepsilon_{mn}^{(g)} \rho_D(f^n)\theta_0' = \varepsilon_{mn}^{(g)} \theta_n'.$$

In the end, $H_g(\theta_m')_m = \lambda(\theta_m)_m$. Since we are looking at these coordinates projectively, the factor λ does not matter, and H_g is involutive because H_g^{-1} is proportional to H_g.

Example 5.3.9 *On an elliptic curve $X = E$ with the divisor $D = 2(0_E)$, the dual theta functions of the ones defined in Example 5.3.7 verify*

$$(\theta_0' : \theta_1') = (\theta_0 + \theta_1 : \theta_0 - \theta_1).$$

5.4 RATIONAL THETA FUNCTIONS

In this section only, we will not assume that k is algebraically closed. The preceding construction shows that for any ample divisor D, there are theta structures $\Theta_D : G(D) \xrightarrow{\sim} G(\delta)$ over the closure $\overline{k}$ and therefore theta functions if D is very ample[2], but these may not be rational, that is elements of $k(X)$. We will describe how we can twist those to recover rational functions.

We will denote by $\mathfrak{G}_k = \mathrm{Gal}(\overline{k}/k)$ the absolute Galois group of k. The action of $\mathfrak{G}_k$ on $\overline{k}$ can be extended naturally to X and $\overline{k}(X)$: for any $x \in X$ and $s \in \overline{k}(X)$, an element $\sigma \in \mathfrak{G}_k$ acts on x coordinate-wise and on s coefficient-wise. We have that $(s(x))^\sigma = s^\sigma(x^\sigma)$ where the action is denoted with a superscript σ.

Moreover, if $D = \sum_i n_i Y_i$ is a divisor, $\sigma \in \mathfrak{G}_k$ also acts on $\operatorname{Div} X$ via $D^\sigma = \sum_i n_i \sigma(Y_i)$. In particular, if $x \in X$ and $D \in \operatorname{Div} X$, $(t_x^* D)^\sigma = t_{\sigma(x)}^* D^\sigma$.

Assume that D is a rational divisor, that is for all $\sigma \in \mathfrak{G}_k$, $D^\sigma = D$. Then the theta group $G(D)$ is stable under the action of $\mathfrak{G}_k$: if $g_x \in G(D)$ is above $x \in H(D)$, then g_x^σ is an element of $G(D)$ above $\sigma(x)$. A quick computation shows that, for any $\sigma \in \mathfrak{G}_k$, $g_x \in G(D)$ and $s \in \Gamma(D)$:

$$(\rho_D(g_x)s)^\sigma = \rho_D(g_x^\sigma)s^\sigma.$$

Finding linear combinations of rational theta functions can be tricky as this highly depends on the structure of $\mathfrak{G}_k$ and the number of coordinates. We will elaborate on the methodology through the example of an elliptic curve over a finite field, as a sequel to Example 5.3.8. From our basis $(\theta_i)_i$, we first want to compute the family $(\theta_i^\sigma)_i$ for every $\sigma \in \mathfrak{G}_k$.

Example 5.4.1 *This is a continuation of previous examples; recall that we are working over an elliptic curve $X = E$, with ample divisor $D = 2(0_E)$. The kernel of the polarization is $H(D) = E[2] = \langle T_1, T_2 \rangle$ with $e_D(T_1, T_2) = -1$, and $K_j(D) = \langle T_j \rangle$ is the symplectic decomposition for $j = 1, 2$. We will work over a finite field $k = \mathbb{F}_q$ and $\operatorname{char} k = p$, the absolute Galois group is then cyclic and generated by the Frobenius: $\mathfrak{G}_k = \langle \pi_q \rangle$.*

We re-use the theta structure of Example 5.3.7, where $\widetilde{K}_j(D) = \langle g_j \rangle$ for $j = 1, 2$. First, we need to compute an element of order 2 above $T_3 = T_1 + T_2$. We already have that $g_1 g_2$ is an element above T_3, and using the commutator pairing, $(g_1 g_2)^2 = -1$, we will then set $g_3 = i g_2 g_1$ where i is a square root of -1. We will assume that $i \in \mathbb{F}_q$ here for the sake of simplicity.

Recall that the theta basis in the previous example was based on a function θ_0 invariant by g_2, and the second function is given by $\theta_1 = g_1 \cdot \theta_0$.

In this example, we will denote the linear representation ρ_D with a dot.

We want to compute $\theta_j^{\pi_q}$ for $j = 1, 2$. These depend on two factors:

- *How does π_q permute the 2-torsion? Since $\pi_q(0_E) = 0_E$, this gives 6 possibilities;*
- *Given a permutation of the 2-torsion, how does π_q acts on g_1 and g_2? As explained above, there are exactly 2 choices for each g_j, therefore we have $2^2 = 4$ possibilities.*

2 or when we go through the Kummer variety $X \to X/\pm 1$, as we did in Example 5.3.7.

We will not explain the computations for the 24 examples, only for a few of them. The complete list is given in Table 5.1.

We first deal with the case where the 2-torsion is rational, that is $\pi_q(T_j) = T_j$ *for* $1 \leqslant j \leqslant 3$. *That means that* $g_1^{\pi_q}$ *must be above* $\pi_q(T_1) = T_1$ *and similarly for* T_2.

1. *Assume* $g_1^{\pi_q} = g_1$ *and* $g_2^{\pi_q} = g_2$. *Then* $\theta_0^{\pi_q} = (g_2 \cdot \theta_0)^{\pi_q} = g_2^{\pi_q} \cdot \theta_0^{\pi_q}$. *Therefore,* $\theta_0^{\pi_q} = g_2 \cdot \theta_0^{\pi_q}$, *it is* g_2*-invariant, so colinear to* θ_0: $\theta_0^{\pi_q} = \lambda\theta_0$ *for some* $\lambda \in \overline{k}^*$.

   Using the definition of θ_1 *and the assumption, we get* $\theta_1^{\pi_q} = g_1 \cdot \theta_0^{\pi_q} = \lambda\theta_1$. *Projectively,* $(\theta_0^{\pi_q} : \theta_1^{\pi_q}) = (\theta_0 : \theta_1)$, *the theta functions are already rational, which is coherent with the assumption where every element is rational.*

2. *Assume now* $g_1^{\pi_q} = -g_1$ *and* $g_2^{\pi_q} = g_2$. *Then* $\theta_0^{\pi_q} = g_2 \cdot \theta_0^{\pi_q}$ *is again* g_2*-invariant, and* $\theta_1^{\pi_q} = -g_1 \cdot \theta_0^{\pi_q}$. *As before, by taking into account the minus sign, we get projectively* $(\theta_0^{\pi_q} : \theta_1^{\pi_q}) = (-\theta_0 : \theta_1)$.

3. *When* $g_1^{\pi_q} = g_1$ *and* $g_2^{\pi_q} = -g_2$, *we get* $\theta_0^{\pi_q} = -g_2 \cdot \theta_0^{\pi_q}$ *and* $\theta_1^{\pi_q} = g_1 \cdot \theta_0^{\pi_q}$. $\theta_0^{\pi_q}$ *is not* g_2*-invariant this time. Because of the pairing,* $g_2 g_1 = -g_1 g_2$, *we compute* $g_2 \cdot \theta_1^{\pi_q} = -g_1 g_2 \cdot \theta_0^{\pi_q} = \theta_1^{\pi_q}$: *we have a* g_2*-invariant element,* $\theta_1^{\pi_q}$ *is colinear to* θ_0.

   Since $\theta_0^{\pi_q} = g_1 \cdot \theta_1^{\pi_q}$ *and* $\theta_1 = g_1 \cdot \theta_0$, *we get* $(\theta_0^{\pi_q} : \theta_1^{\pi_q}) = (\theta_1 : \theta_0)$.

4. *Finally, if* $g_1^{\pi_q} = -g_1$ *and* $g_2^{\pi_q} = -g_2$, *we have a composition of the previous cases, and we recover* $(\theta_0^{\pi_q} : \theta_1^{\pi_q}) = (-\theta_1 : \theta_0)$.

Here, finding a g_2*-invariant function was easy, we will look at the case where* π_q *transposes* T_1 *and* T_2. *In this case* T_3 *is therefore rational.* $g_1^{\pi_q}$ *must lie above* T_2 *and* $g_2^{\pi_q}$ *above* T_1.

1. *Assume first that* $g_1^{\pi_q} = g_2$ *and* $g_2^{\pi_q} = g_1$. *As done previously, we compute* $\theta_0^{\pi_q} = g_2^{\pi_q} \cdot \theta_0^{\pi_q} = g_1 \cdot \theta_0^{\pi_q}$. *Similarly,* $\theta_1^{\pi_q} = g_2 \cdot \theta_0^{\pi_q}$. *Because* g_2 *is of order 2, the element* $\theta_0^{\pi_q} + g_2 \cdot \theta_0^{\pi_q} = \theta_0^{\pi_q} + \theta_1^{\pi_q}$ *is* g_2*-invariant, therefore colinear with* θ_0: *there is* $\lambda \in \overline{k}^*$ *such that* $\lambda\theta_0 = \theta_0^{\pi_q} + \theta_1^{\pi_q}$.

   Now, using the definition of $\theta_1 = g_1 \cdot \theta_0$, *we get* $\lambda\theta_1 = g_1 \cdot \theta_0^{\pi_q} + g_1 \cdot \theta_1^{\pi_q}$. *The first term is* $\theta_0^{\pi_q}$, *and the second comes from* $g_1 \cdot \theta_1^{\pi_q} = g_1 g_2 \cdot \theta_0^{\pi_q} = -\theta_1^{\pi_q}$. *Therefore,* $(\theta_0 : \theta_1) = (\theta_0^{\pi_q} + \theta_1^{\pi_q} : \theta_0^{\pi_q} - \theta_1^{\pi_q})$.

   We want to compute the inverse of the homography $(X : Z) \mapsto (X+Z : X-Z)$, *which is involutive, so finally* $(\theta_0^{\pi_q} : \theta_1^{\pi_q}) = (\theta_0 + \theta_1 : \theta_0 - \theta_1)$.

2. *When* $g_1^{\pi_q} = -g_2$ *and* $g_2^{\pi_q} = g_1$, *we find that* $\theta_0^{\pi_q} - \theta_1^{\pi_q}$ *is* g_2*-invariant, and* $(\theta_0^{\pi_q} : \theta_1^{\pi_q}) = (\theta_0 + \theta_1 : -\theta_0 + \theta_1)$.

3. *Assume* $g_1^{\pi_q} = g_2$ *and* $g_2^{\pi_q} = -g_1$. *A* g_2*-invariant is again* $\theta_0^{\pi_q} + \theta_1^{\pi_q}$, *and* $(\theta_0^{\pi_q} : \theta_1^{\pi_q}) = (\theta_0 - \theta_1 : \theta_0 + \theta_1)$.

4. *If* $g_1^{\pi_q} = -g_1$ *and* $g_2^{\pi_q} = -g_2$, *we get* $(\theta_0^{\pi_q} : \theta_1^{\pi_q}) = (-\theta_0 + \theta_1 : \theta_0 + \theta_1)$ *by composition.*

	$g_1^{\pi_q}$	$g_2^{\pi_q}$	g_2-invariant	$g_1 \cdot \theta_0^{\pi_q}$	$g_1 \cdot \theta_1^{\pi_q}$	$(\theta_0^{\pi_q} : \theta_1^{\pi_q})$
			(A) $\pi_q(T_1) = T_1$, $\pi_q(T_2) = T_2$, $\pi_q(T_3) = T_3$			
(1)	g_1	g_2	$\theta_0^{\pi_q}$	$\theta_1^{\pi_q}$	—	$(\theta_0 : \theta_1)$
(2)	$-g_1$	g_2	$\theta_0^{\pi_q}$	$-\theta_1^{\pi_q}$	—	$(-\theta_0 : \theta_1)$
(3)	g_1	$-g_2$	$\theta_1^{\pi_q}$	—	$\theta_0^{\pi_q}$	$(\theta_1 : \theta_0)$
(4)	$-g_1$	$-g_2$	$\theta_1^{\pi_q}$	—	$-\theta_0^{\pi_q}$	$(-\theta_1 : \theta_0)$
			(B) $\pi_q(T_1) = T_2$, $\pi_q(T_2) = T_1$, $\pi_q(T_3) = T_3$			
(1)	g_2	g_1	$\theta_0^{\pi_q} + \theta_1^{\pi_q}$	$\theta_0^{\pi_q}$	$-\theta_1^{\pi_q}$	$(\theta_0 + \theta_1 : \theta_0 - \theta_1)$
(2)	$-g_2$	g_1	$\theta_0^{\pi_q} - \theta_1^{\pi_q}$	$\theta_0^{\pi_q}$	$-\theta_1^{\pi_q}$	$(\theta_0 + \theta_1 : -\theta_0 + \theta_1)$
(3)	g_2	$-g_1$	$\theta_0^{\pi_q} + \theta_1^{\pi_q}$	$-\theta_0^{\pi_q}$	$\theta_1^{\pi_q}$	$(\theta_0 - \theta_1 : \theta_0 + \theta_1)$
(4)	$-g_2$	$-g_1$	$\theta_0^{\pi_q} - \theta_1^{\pi_q}$	$-\theta_0^{\pi_q}$	$\theta_1^{\pi_q}$	$(-\theta_0 + \theta_1 : \theta_0 + \theta_1)$
			(C) $\pi_q(T_1) = T_3$, $\pi_q(T_2) = T_2$, $\pi_q(T_3) = T_1$			
(1)	g_3	g_2	$\theta_0^{\pi_q}$	$i\theta_1^{\pi_q}$	—	$(-\theta_0 : i\theta_1)$
(2)	$-g_3$	g_2	$\theta_0^{\pi_q}$	$-i\theta_1^{\pi_q}$	—	$(\theta_0 : i\theta_1)$
(3)	g_3	$-g_2$	$\theta_1^{\pi_q}$	—	$-i\theta_0^{\pi_q}$	$(-i\theta_1 : \theta_0)$
(4)	$-g_3$	$-g_2$	$\theta_1^{\pi_q}$	—	$i\theta_0^{\pi_q}$	$(i\theta_1 : \theta_0)$
			(D) $\pi_q(T_1) = T_1$, $\pi_q(T_2) = T_3$, $\pi_q(T_3) = T_2$			
(1)	g_1	g_3	$\theta_0^{\pi_q} + i\theta_1^{\pi_q}$	$\theta_1^{\pi_q}$	$\theta_0^{\pi_q}$	$(\theta_0 - i\theta_1 : -i\theta_0 + \theta_1)$
(2)	$-g_1$	g_3	$\theta_0^{\pi_q} - i\theta_1^{\pi_q}$	$-\theta_1^{\pi_q}$	$-\theta_0^{\pi_q}$	$(\theta_0 - i\theta_1 : i\theta_0 - \theta_1)$
(3)	g_1	$-g_3$	$\theta_0^{\pi_q} - i\theta_1^{\pi_q}$	$\theta_1^{\pi_q}$	$\theta_0^{\pi_q}$	$(\theta_0 + i\theta_1 : i\theta_0 + \theta_1)$
(4)	$-g_1$	$-g_3$	$\theta_0^{\pi_q} + i\theta_1^{\pi_q}$	$-\theta_1^{\pi_q}$	$-\theta_0^{\pi_q}$	$(-\theta_0 - i\theta_1 : i\theta_0 + \theta_1)$
			(E) $\pi_q(T_1) = T_2$, $\pi_q(T_2) = T_3$, $\pi_q(T_3) = T_1$			
(1)	g_2	g_3	$\theta_0^{\pi_q} + \theta_1^{\pi_q}$	$-i\theta_1^{\pi_q}$	$i\theta_0^{\pi_q}$	$(i\theta_0 + \theta_1 : i\theta_0 - \theta_1)$
(2)	$-g_2$	g_3	$\theta_0^{\pi_q} - \theta_1^{\pi_q}$	$i\theta_1^{\pi_q}$	$i\theta_0^{\pi_q}$	$(i\theta_0 + \theta_1 : -i\theta_0 + \theta_1)$
(3)	g_2	$-g_3$	$\theta_0^{\pi_q} + \theta_1^{\pi_q}$	$i\theta_1^{\pi_q}$	$-i\theta_0^{\pi_q}$	$(i\theta_0 - \theta_1 : i\theta_0 + \theta_1)$
(4)	$-g_2$	$-g_3$	$\theta_0^{\pi_q} - \theta_1^{\pi_q}$	$-i\theta_1^{\pi_q}$	$-i\theta_0^{\pi_q}$	$(-i\theta_0 + \theta_1 : i\theta_0 + \theta_1)$
			(F) $\pi_q(T_1) = T_3$, $\pi_q(T_2) = T_1$, $\pi_q(T_3) = T_2$			
(1)	g_3	g_1	$\theta_0^{\pi_q} - i\theta_1^{\pi_q}$	$\theta_0^{\pi_q}$	$-\theta_1^{\pi_q}$	$(i(\theta_0 + \theta_1) : -\theta_0 + \theta_1)$
(2)	$-g_3$	g_1	$\theta_0^{\pi_q} + i\theta_1^{\pi_q}$	$\theta_0^{\pi_q}$	$-\theta_1^{\pi_q}$	$(i(\theta_0 + \theta_1) : \theta_0 - \theta_1)$
(3)	g_3	$-g_1$	$\theta_0^{\pi_q} + i\theta_1^{\pi_q}$	$-\theta_0^{\pi_q}$	$\theta_1^{\pi_q}$	$(i(\theta_0 - \theta_1) : \theta_0 + \theta_1)$
(4)	$-g_3$	$-g_1$	$\theta_0^{\pi_q} - i\theta_1^{\pi_q}$	$-\theta_0^{\pi_q}$	$\theta_1^{\pi_q}$	$(-i(\theta_0 - \theta_1) : \theta_0 + \theta_1)$

Table 5.1: Action of π_q on $(\theta_0 : \theta_1)$ in every possible cases

Once we know the action of $\mathfrak{G}_k$ on the theta functions, we would like to determine linear combinations of these functions which are rational. The new functions are twisted theta functions: we are still putting a system of coordinates on the original abelian variety X, however we have twisted the theta structure by an automorphism of $G(\delta)$.

Example 5.4.2 *This is the sequel of Example 5.4.1. Assume first that the 2-torsion of the elliptic curve is rational, this corresponds to situation (A) in Table 5.1. Additionally, assume that there is a rational element g_2 of order 2 above T_2. This reduces to the cases (A.1, A.2).*

If $g_1^{\pi_q} = \varepsilon g_1$ with $\varepsilon^2 = 1$, then $(\theta_0^{\pi_q} : \theta_1^{\pi_q}) = (\varepsilon\theta_0 : \theta_1)$. Because the conjugation by π_q only introduces a change of sign, we are looking for rational functions (θ'_0, θ'_1) such that $(\theta'_0 : \theta'_1) = (\alpha\theta_0 : \beta\theta_1)$.

We then get $({\theta'_0}^{\pi_q} : {\theta'_1}^{\pi_q}) = (\alpha^{\pi_q}\varepsilon\theta_0 : \beta^{\pi_q}\theta_1)$, and we are aiming for

$$\frac{\alpha^{\pi_q}}{\beta^{\pi_q}}\varepsilon = \frac{\alpha}{\beta}.$$

We also have that $(\theta_0(0_E)^{\pi_q} : \theta_1(0_E)^{\pi_q}) = (\varepsilon\theta_0(0_E) : \theta_1(0_E))$, which is a convenient choice of α/β. By setting $(\alpha : \beta) = (\theta_0(0_E) : \theta_1(0_E))$, we get the following twisted theta functions:

$$(\theta'_0 : \theta'_1) = (\theta_0(0_E)\theta_0 : \theta_1(0_E)\theta_1).$$

This situation will be more detailed in Example 8.4.10 and is what we accordingly call a theta twisted model.

Another important example is when there is at least one rational 2-torsion point with an associated rational element of order 2 in the theta group. To simplify, we will consider that this 2-torsion point is T_2. This situation corresponds to either (A.1, A.2) or (C.1, C.2) in Table 5.1. We already discussed the first case above, so we will focus on the second one. We will assume that $i \in \mathbb{F}_q$ is a square root of -1. In this case, $(\theta_0^{\pi_q} : \theta_1^{\pi_q}) = (\varepsilon\theta_0 : i\theta_1)$ when $g_1^{\pi_q} = \varepsilon g_3$. We once again are looking for elements $(\theta'_0 : \theta'_1) = (\alpha\theta_0 : \beta\theta_1)$.

In the setting of (A), the rationality leads to the same equation as before, when $g_1^{\pi_q} = \varepsilon g_1$:

$$\frac{\alpha^{\pi_q}}{\beta^{\pi_q}}\varepsilon = \frac{\alpha}{\beta}.$$

We already checked earlier that $(\theta_0(0_E)^{\pi_q} : \theta_1(0_E)^{\pi_q}) = (\varepsilon\theta_0(0_E) : \theta_1(0_E))$, or equivalently

$$\frac{\theta_1(0_E)^{\pi_q}}{\theta_0(0_E)^{\pi_q}}\varepsilon = \frac{\theta_1(0_E)}{\theta_0(0_E)}.$$

In the setting of (C) where $g_1^{\pi_q} = \varepsilon g_3$, the equation is

$$\frac{\alpha^{\pi_q}}{\beta^{\pi_q}}\varepsilon = i\frac{\alpha}{\beta}.$$

This time, $(\theta_0(0_E)^{\pi_q} : \theta_1(0_E)^{\pi_q}) = (\varepsilon\theta_0(0_E) : i\theta_1(0_E))$, so

$$\frac{\theta_1(0_E)^{\pi_q}}{\theta_0(0_E)^{\pi_q}}\varepsilon = i\frac{\theta_1(0_E)}{\theta_0(0_E)}.$$

This is our choice of α and β, and the twisted theta functions are

$$(\theta_0' : \theta_1') = (\theta_1(0_E)\theta_0 : \theta_0(0_E)\theta_1) = \left(\frac{\theta_0}{\theta_0(0_E)} : \frac{\theta_1}{\theta_1(0_E)}\right).$$

The above situation with a rational 2-torsion point and an associated rational element of order 2 in the theta group corresponds to a Montgomery curve, as we will detail in Example 8.4.10.

5.5 THETA FUNCTIONS IN DIMENSION 1 OVER $\mathbb{C}$

In this last section, we describe the theta functions analytically over $\mathbb{C}$ as done by Mumford in [Mum83; Mum84; Mum91]. This section is independent of the algebraic theory developed earlier in this chapter and is taken from [BRS25, § 3.2]. We will restrict to the case of dimension 1 in this section for the sake of simplicity, but a similar approach can be considered in dimension g by replacing elements τ from the Poincaré upper half-plane $\mathbb{H}$ with matrices Ω of size $g \times g$ of the Siegel space of dimension g.

In the first part of this chapter, we constructed theta functions algebraically. But those can be naturally expressed as analytic functions over $\mathbb{C}$, and it is an interesting approach to find formulas for the arithmetic on complex numbers. We do not intend to explicit the connection with the algebraic construction.

Let $\mathbb{H}$ be the Poincaré half-plane $\mathbb{H} = \{\tau \in \mathbb{C} \mid \operatorname{im}\tau > 0\}$ and consider $\tau \in \mathbb{H}$. We can define a lattice $\Lambda_\tau = \mathbb{Z} + \tau\mathbb{Z}$ and an elliptic curve on $\mathbb{C}$ can be seen as a torus $E_\tau = \mathbb{C}/\Lambda_\tau$. Our first objective is to define some mappings $E_\tau \to \mathbb{P}^{\ell-1}$ with simple algebraic addition law. Theta functions will help us to do so, more details can be found in [Mum83, § II].

First, for $\tau \in \mathbb{H}$, the Jacobi theta function is defined as follows for every $z \in \mathbb{C}$:

$$\vartheta(z;\tau) = \sum_{n\in\mathbb{Z}} \exp(i\pi n^2\tau + 2i\pi nz). \tag{5.5}$$

This function is well-defined for every $z \in \mathbb{C}$ (see [Mum83, § II.1, Prop. 1.1]).

If $a, b \in \mathbb{Z}$, $\ell \in \mathbb{N}^*$, we also define theta functions with characteristics (we use the notations of Gaudry and Lubicz [GL09]):

$$\vartheta_\ell\,[a,b]\,(z;\tau) = \sum_{n\in\mathbb{Z}} \exp\left(i\pi\left(n+\frac{a}{\ell}\right)^2\tau + 2i\pi\left(n+\frac{a}{\ell}\right)\left(z+\frac{b}{\ell}\right)\right). \tag{5.6}$$

Let us list a series of classical properties of the theta functions:

Proposition 5.5.1 *Let $z \in \mathbb{C}$, $\ell \in \mathbb{N}^*$, $a, b \in \mathbb{Z}$, $m \in \mathbb{Z}$, then:*

$$\begin{aligned}
\vartheta_\ell\,[a,b]\,(z;\tau) &= \exp(i\pi a^2\tau/\ell^2 + 2i\pi a(z+b/\ell)/\ell)\vartheta(z+a\tau/\ell+b/\ell;\tau),\\
\vartheta_\ell\,[a,b]\,(z+m;\tau) &= \exp(2i\pi am/\ell)\vartheta_\ell\,[a,b]\,(z;\tau),\\
\vartheta_\ell\,[a,b]\,(z+m\tau;\tau) &= \exp(-2i\pi bm/\ell)\exp(-i\pi m^2\tau - 2i\pi mz)\vartheta_\ell\,[a,b]\,(z;\tau).
\end{aligned}$$

These functions are still well-defined if $a, b \in \mathbb{R}$, but we generally restrict to the case where $a, b \in \mathbb{Z}$.

Theta functions with characteristic are a particular case of functions of level ℓ.

Definition 5.5.2 *Let $\tau \in \mathbb{H}$ and $\ell \in \mathbb{N}^*$. An entire function $f : \mathbb{C} \to \mathbb{C}$ is said to be of level ℓ if for all $z \in \mathbb{C}$, $m \in \mathbb{Z}$:*

$$f(z+m) = f(z) \text{ and } f(z+m\tau) = \exp(-i\pi\ell m^2\tau - 2i\pi\ell mz)f(z)$$

We call R_ℓ^τ the vector space of functions of level ℓ.

The point of defining such functions is that if we take a family $f_1, \dots, f_r \in R_\ell^\tau$ and $\lambda \in \Lambda_\tau$, then there is a function $c_\ell : \mathbb{C} \to \mathbb{C}$ such that for all $1 \leqslant i \leqslant r$, for all $z \in \mathbb{C}$, $f_i(z+\lambda) = c_\ell(z)f_i(z)$, hence in the projective space, $(f_1 : \cdots : f_r)$ is invariant by the lattice Λ_τ.

We will now relate theta functions with characteristics with functions of level ℓ. Because of the definition in Eq. (5.6), at a set ℓ and with integer characteristics, it is clear that we can define for a given τ exactly ℓ^2 functions. The following theorem shows that in fact we only require ℓ of them.

Theorem 5.5.3 ([Mum83], § II.1, Prop. 1.3) *For every $\tau \in \mathbb{H}$ and $\ell \in \mathbb{N}^*$, $\dim R_\ell^\tau = \ell$.*

Mumford's proof builds explicit bases via theta functions with characteristics.

From now on, we will focus on level $\ell = 2$ as this is the interesting one regarding Kummer lines. We will now give some of the families used in Mumford's proof:

Definition 5.5.4 *Let $\tau \in \mathbb{H}$, for every $z \in \mathbb{C}$, we define the following theta functions:*

$$\theta_0^\tau(z) = \vartheta_2[0,0](z;\tau/2), \quad \theta_1^\tau(z) = \vartheta_2[0,1](z;\tau/2).$$

Proposition 5.5.5 ([Mum83], § II.1, Prop. 1.3) *$\left(\theta_0^\tau, \theta_1^\tau\right)$ is a basis of R_2^τ.*

These functions correspond to the one from Example 5.3.7!

For every elliptic curve E on $\mathbb{C}$, there exists τ such that $E = E_\tau$, to which we associate the previously defined theta functions, giving a degree 2 cover thanks to the following theorem by Lefschetz proven in [Mum83, § II.1, Thm. 1.3] and [Mum74, § I.3]:

Theorem 5.5.6 (Lefschetz) *Let $\tau \in \mathbb{H}$. If E_τ is an elliptic curve on $\mathbb{C}$, denote by $\mathcal{K}_\tau = E_\tau/\pm 1$, and $\bar{z} \in \mathcal{K}_\tau$ the class of $z \in E_\tau$. If (f, g) is a basis of R_2^τ, then the following map $\Phi : \mathcal{K}_\tau \to \mathbb{P}^1$ is well-defined and is an embedding:*

$$\Phi : \bar{z} \mapsto (f(z) : g(z)).$$

Equivalently, $\pi : E_\tau \to \mathbb{P}^1, z \mapsto (f(z) : g(z))$ is a Kummer line as defined in Definition 4.3.2.

The more general version of the theorem also states that if the level ℓ is greater than 3, then we directly have an embedding $E_\tau \to \mathbb{P}^{\ell-1}$.

As for the arithmetic, we will need the following formulas relating theta functions with different characteristics and τ, we will only give them in level 2, however more general formulas can be found in [Igu72, § IV.1, Thm. 2]:

Proposition 5.5.7 (Duplication formulas) *Let* $a_1, b_1, a_2, b_2 \in \mathbb{Z}$, $z_1, z_2 \in \mathbb{C}$ *and* $\tau \in \mathbb{H}$. *Let* $2w_1 = z_1 + z_2$, $2w_2 = z_1 - z_2$, $2u_1 = a_1 + a_2$, $2u_2 = a_1 - a_2$, $2v_1 = b_1 + b_2$, $2v_2 = b_1 - b_2$. *Then:*

$$2\vartheta_2[a_1, b_1](z_1;\tau)\vartheta_2[a_2, b_2](z_2;\tau) = \sum_{t\in\{0,1\}} e^{-i\pi a_1 t}\vartheta_2[2u_1, v_1 + t](w_1;\tau/2)\cdot\vartheta_2[2u_2, v_2 + t](w_2;\tau/2),$$

$$\vartheta_2[a_1, b_1](z_1;\tau)\vartheta_2[a_2, b_2](z_2;\tau) = \sum_{t\in\{0,1\}} \vartheta_2[u_1 + t, 2v_1](2w_1;2\tau)\cdot\vartheta_2[u_2 + t, 2v_2](2w_2;2\tau).$$

Proof *The first formula is [Igu72, § IV.1, Thm. 2]. The second one can then be derived from the first one, let* A *be the quantity*

$$\sum_{t\in\{0,1\}} \vartheta_2\left[\frac{a_1 + a_2}{2} + t, b_1 + b_2\right](2w_1;2\tau)\cdot\vartheta_2\left[\frac{a_1 - a_2}{2} + t, b_1 - b_2\right](2w_2;2\tau).$$

Also set, for $t \in \{0, 1\}$*:*

$$a_1' = \frac{a_1 + a_2}{2} + t, \quad a_2' = \frac{a_1 - a_2}{2} + t, \quad b_1' = b_1 + b_2, \quad b_2' = b_1 - b_2.$$

A quick computation yields:

$$a_1' + a_2' = a_1 + 2t, \quad a_1' - a_2' = a_2, \quad b_1' + b_2' = 2b_1, \quad b_1' - b_2' = 2b_2.$$

By applying the first formula inside the sum of A*, since* $2t = 0 \bmod 2$*, we get:*

$$A = \frac{1}{2}\sum_{t,r\in\{0,1\}} e^{-i\pi a_1' r}\vartheta_2[a_1, b_1 + r](z_1;\tau)\cdot\vartheta_2[a_2, b_2 + r](z_2;\tau).$$

We then split when $r = 0$ *and* $r = 1$*. If* $r = 0$*, then there is no dependency on* t*, and we recover* $2\vartheta_2[a_1, b_1](z_1;\tau)\vartheta_2[a_2, b_2](z_2;\tau)$ *in the sum. If* $r = 1$*, the theta functions do not depend on* t *and the exponential is* $\exp(-i\pi a_1' r) = \exp(-i\pi\frac{a_1+a_2}{2})\exp(-i\pi t)$*. Because* $1 + \exp(-i\pi) = 0$*, we finally get:*

$$A = \vartheta_2[a_1, b_1](z_1;\tau)\vartheta_2[a_2, b_2](z_2;\tau). \qquad \square$$

Part II

IMPROVEMENTS OF THE ARITHMETIC IN DIMENSION 1

6 FINDING 2-ISOGENIES FORMULAS BETWEEN KUMMER LINES

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This chapter corresponds to [RS24] published in IACR Communications in Cryptology. As we have seen in Section 2.3, having efficient differential addition and doubling formulas on a Kummer line enables us to compute quickly the scalar multiplication via the Montgomery ladder. A way to improve doubling formulas is by splitting the doubling into the composition of a 2-isogeny and its contragedient (which is the dual on an elliptic curve) as explained in Proposition 3.2.11. The goal of this chapter is to study extensively the 2-isogenies between any Kummer lines, in order to derive old and new doubling formulas, as these are already used in the Montgomery model [Mon87] and the theta model [GL09]. Our main result is to provide a methodology to find 2-isogenies formulas.

Our main objective was to better understand the isogeny formulas from isogeny-based cryptography, and in particular why the Montgomery model has fast 2-isogenies, in the hope to extend these formulas to higher dimension. For dimension 1, as we will see in Section 6.6, our translated 2-isogeny formulas are faster than the usual ones. However, as shown in [CH17], it is faster to decompose a 2^n-isogeny as a product of 4-isogenies rather than a product of 2-isogenies.

To count operations, we will use notations introduced in Sections 2.1 and 2.3.

Methodology

We proceed to a systematic study of 2-isogenies between Kummer lines by combining two tools:

1. First, we use that a Kummer model is completely determined by its four ramification points as explained in Section 4.3.2. Keeping track of the ramification along the isogeny allows us to keep track of the model, without resorting to formal groups as in Vélu's formulas;

2. Secondly we make a systematic use of the theta groups and their action on sections to find invariant sections, using theory of Chapter 5.

This allows us to recover the efficient 2-isogenies formulas already known in the literature in a unified manner, showing the flexibility of our framework [FJP14; CH17; Ren18].

In Chapter 7, we extend this strategy to study differential additions in dimension one on Kummer lines, using the fact that differential additions can be described by a 2-isogeny in dimension two, on a product of two Kummer lines.

We mainly focus in this chapter on the study of 2-isogenies between models of Kummer lines which have a specific Galois action on their 2^n-torsion (like the Legendre model, the Montgomery model and the theta model). This is indeed the most interesting situation: the isogeny interacts with the Galois action. So either we lose some of the Galois information on the codomain, which means that we can only describe another type of model on the codomain (like Theta to Montgomery, Montgomery to Legendre, or Legendre to Montgomery), or we need to assume that we are given a supplementary input. For instance, for 2-isogeny formulas from theta to theta, we need a 8-torsion point above the kernel to find a theta model of the codomain.

By contrast, a model requiring, say, a rational 3-torsion point T would not have this problem with a 2-isogeny formula, since the image of T by the isogeny would immediately give a model of the codomain. In a similar vein, handling the case of odd degree isogenies in the models mentioned above (Montgomery, Legendre, Theta), is in some sense easier since the Galois structure on the 2^n-torsion is respected through the isogeny, see Section 6.7 for formulas.

Results

On the Montgomery model with full rational 2-torsion, we can evaluate a 2-isogeny, translated by a suitable 2-torsion point in $2\mathbf{S} + 1\mathbf{m}_0$, compared to $2\mathbf{M} + 1\mathbf{m}_0$ for the non-translated image. Composing with the translated dual isogeny, we obtain a translated doubling formula in $4\mathbf{S} + 2\mathbf{m}_0$, compared to $2\mathbf{M} + 2\mathbf{S} + 1\mathbf{m}_0$ for a standard doubling.

Using the translated doubling in the Montgomery ladder, and keeping track of the translation by the 2-torsion point, we obtain a hybrid ladder arithmetic which costs $3\mathbf{M}+6\mathbf{S}+2\mathbf{m}_0$ per bit, compared to $5\mathbf{M}+4\mathbf{S}+1\mathbf{m}_0$ for the standard ladder for the Montgomery model. Thus, if $\mathbf{m}_0$ is sufficiently small, we obtain a more efficient scalar multiplication (while retaining the standard side channel protection of the Montgomery ladder). We remark also that the ladder for the theta model costs $3\mathbf{M} + 6\mathbf{S} + 3\mathbf{m}_0$, hence our hybrid ladder is always faster than the theta ladder.

Similar work

In [Mor+22], the authors introduce the generalized Montgomery coordinate h on an elliptic curve E, which can be seen as the composition $h = x \circ \varphi$ of an isogeny $\varphi : E \to E'$ to an elliptic curve in Montgomery form, with the x-coordinate on E' [Mor+22, Thm. 13]. They then give formulas for isogenies and scalar multiplication in the generalized Montgomery coordinate.

Our work is in an orthogonal direction. If the isogeny φ is of degree n, a generalized Montgomery coordinate $h = x \circ \varphi$ is a section of a divisor of degree $2n$ on E. The work of [Mor+22] may thus be seen as specifying a special model associated to a section of $2n(0_E)$ and developing the arithmetic and isogenies on this model.

By contrast, we focus only on sections of $2(0_E)$ to describe models of the Kummer line, but we don't impose conditions on the model; our framework allows us to derive efficient isogeny formulas between different Kummer models, including models where the neutral point is not at infinity.

In [KS20], the authors use the theta squared coordinates on the Kummer line. One can see that their doubling formulas in their Table 2 are indeed the one we recover up to a translation by a 2-torsion point in Algorithm 6.1. Our ladder is however slightly faster because we are using Montgomery differential addition instead of the theta squared one, a comparison is available in Table 6.2.

Roadmap

We first recall Vélu's affine approach in Section 6.1, before switching to the projective one in the sequel. In Section 6.2 we specify the theory of Chapter 5 to divisors $2(0_E)$ and $4(0_E)$ to develop our isogeny framework. We use it in Section 6.3 to study 2-isogenies between Montgomery models, and we apply those in Section 6.4 where we introduce the hybrid ladder. In Section 6.5, we provide more examples of our technique with different ramification structure to show its flexibility. We briefly discuss applications to fast evaluations of 2^n-isogenies in Section 6.6. Finally, in Section 6.7, we explain how to deal with odd degree isogenies.

Sections 6.5 to 6.7 were part of an appendix, the core result being the framework and the hybrid ladder.

In this chapter, we work over a perfect field k of characteristic different from 2.

6.1 AN ANALOGY WITH VÉLU'S FORMULAS

In this section, we first consider an affine point of view. This is an approach based on Vélu's work [Vél71].

Let E/k be an elliptic curve, $T \in E[2](k)$ a rational 2-torsion point, and $K = \{0_E, T\}$ the kernel generated by T. Set $\varphi : E \to E' := E/K$ to be the corresponding isogeny. Neutral elements on E and E' are denoted 0_E and $0_{E'}$ respectively.

Starting from a Kummer model of E, with affine coordinate $x \neq 1$, we want to build a Kummer model of E' with affine coordinate $x' \neq 1$. For simplicity here, we will take x, x' to be Weierstrass coordinates as described in Section 4.3.1.

Our goal is to express x' as a rational function of x. The coordinate x' is a section of the divisor $2(0_{E'})$, its pullback by φ is $\varphi^*(2(0_{E'})) = 2(0_E) + 2(T)$. In particular, the coordinate $x' \circ \varphi$ on E is a section of $2(0_E) + 2(T)$ which is invariant by translation by T. Conversely, any coordinate $u \in \Gamma(2(0_E) + 2(T))$ which is invariant by translation by T can be decomposed as $x' \circ \varphi$ for some $x' \in \Gamma(2(0_{E'}))$, by the universal property of the quotient.

In particular, since $x \in \Gamma(2(0_E))$, the trace $u = x(\cdot) + x(\cdot + T)$ is an element of $\Gamma(2(0_E) + 2(T))$ and is certainly invariant by translation, so is of the form $x' \circ \varphi$.

This is exactly like the first step of Vélu's formulas (where Vélu takes traces under the kernel of the isogeny of the coordinates x and y respectively, because he works with sections of $3(0_E)$). In the second step of Vélu's formulas, Vélu then uses the formal group of E to recover equations for E'.

In our case, we only need to recover a model of the Kummer line $E'/\pm 1$; by Section 4.3.2 it is completely determined by the image by u of the four ramification points of the Kummer line $E/\pm 1$. Since we are working with Weierstrass coordinates in our example, one of these ramification points is at infinity and is sent to infinity.

In the rest of this chapter, rather than working with the affine coordinates x, x', we will work directly with the projective coordinates X, Z, X', Z' (in other words, their numerators and denominators).

6.2 A GENERAL FRAMEWORK FOR 2-ISOGENIES ON A KUMMER LINE

In this section, we will redo Section 6.1 but with projective coordinates directly. Although conceptually more abstract (we need to use Mumford's theory of the theta group from Chapter 5), the same techniques apply to higher dimensional abelian varieties, by contrast to Vélu's method, as explained in Remark 6.2.5 below.

Throughout this section, we will have some examples based on the following setting:

Main example 6.2.1 (Example 4.3.7) *We will work on an elliptic curve E with a rational 2-torsion point. Without loss of generality, we can assume the origin to be*

at infinity and the rational 2-torsion to correspond to $x = 0$ by a change of variable. Its equation is then

$$E : \beta y^2 = x(x^2 + \mathcal{A}x + \gamma)$$

for some $\mathcal{A}, \beta, \gamma \in k$. Denote by $\alpha \in \overline{k}$ a root of $x^2 + \mathcal{A}x + \gamma$. The corresponding Kummer line is given by the following ramification, with marked point denoted by $*$:

$$0_E = (1 : 0)^*, \quad T_1 = (0 : 1), \quad T_2 = (\alpha : 1), \quad T_3 = (\gamma : \alpha).$$

We also have the additional equation $\mathcal{A} = -\alpha - \frac{\gamma}{\alpha}$.

6.2.1 Descents of divisors

We reuse the notations of Section 6.1. We have the two projective coordinates X, Z associated to the divisor $2(0_E)$, which is a shortcut to say they are sections in $\Gamma(2(0_E))$. Then X^2, XZ, Z^2 are sections of $4(0_E)$, and we would like to find, like in Section 6.1, a basis of linear combinations $aX^2 + bXZ + cZ^2$ which descends to projective coordinates X', Z' on E', associated to the divisor $2(0_{E'})$.

Remark 6.2.2 (Even coordinates) *A section s is said to be even if it is invariant by the action of $[-1]$: $[-1]^*s = s$.*

Let $D_n = n(0_E)$. The multiplication map $\Gamma(D_n) \otimes \Gamma(D_m) \to \Gamma(D_{n+m})$ is surjective when $n, m \geqslant 2, n + m \geqslant 5$, but $\Gamma(D_2) \otimes \Gamma(D_2) \to \Gamma(D_4)$ only surjects onto even global sections: all global sections of D_2 are even, so their products have to be even.

In particular, since $\Gamma(4(0_E))$ is of dimension 4, there is a fourth linearly independent projective coordinate, which is generated by an odd coordinate, because the space of even coordinates is of dimension 3.

On the other hand, the sections s we want to construct on E' are sections of $2(0_{E'})$. Hence, they are even, and their pullbacks f^*s too. Therefore, it is enough to look at the even sections $\Gamma^+(D_4) = \langle X^2, XZ, Z^2 \rangle$.

A first difficulty that arises is that, although $D_4 = 4(0_E)$ is very convenient to use to get a nice basis of the even sections subspace, it is not invariant by translation by T, so it does not descend to a divisor on E'.

We first need to find a divisor D_4'' linearly equivalent to D_4 which is invariant by translation by T. Then sections of D_4'' invariant by translation descends to coordinates on E'; we then need to express this translation invariance for the sections of D_4'' in terms of the sections of D_4.

Mumford's theory of the theta group introduced in Chapter 5 answers these questions, in the general case of an abelian variety. For simplicity, we restrict to the case of a divisor D on an elliptic curve E here.

Before looking into specific descents, we state a lemma that will prove useful in the sequel.

Lemma 6.2.3 *Consider the divisor $D = 2n(0_E)$ for some integer n, and P a non-trivial 2-torsion point. An element $g_P \in G(D)$ above P is of order 2 if and only if it is symmetric.*

$H(D)$ *is the kernel of the polarization.*

Proof *Because* D *is of even degree,* $E[2] \subset E[2n] = H(D)$*, so considering an element* g_P *in the theta group* $G(D)$ *makes sense. A result of Mumford [Mum66, § 2, Prop. 2-3, pp. 307-309] then states that* $[-1]^* g_P = g_P$*. Recall that* g_P *is symmetric if and only if* $[-1]^* g_P = g_P^{-1}$*, thus, given the previous equality, the result holds.* □

Descents of $2(0_E)$

Recall that T is a rational 2-torsion point on E, $K = \{0_E, T\}$ is the associated kernel and $\varphi : E \to E'$ the corresponding isogeny. Denote the 2-torsion on E by $E[2] = \{0_E, T_1, T_2, T_3\}$ with $T_1 = T$. Set also $\overline{T_1}$ and $\underline{T_1}$ the two 4-torsion points above T_1, which may not be rational, and finally set $T_1' = \varphi(T_2) = \varphi(T_3)$, $T_2' = \varphi(\overline{T_1})$ and $T_3' = \varphi(\underline{T_1})$. T_1' is always rational on E', even if T_2 and T_3 are not.

We start by studying the descents of $D_2 = 2(0_E)$. If D' is a descent of D_2, there is a lift $\widetilde{K}$ of K to $G(D_2)$ by Theorem 5.1.7 (1). This lift must be generated by an element g_T above T, and g_T is of order 2, that is symmetric by Lemma 6.2.3. So $\widetilde{K}$ is composed of symmetric elements and D' must be symmetric by Theorem 5.1.7 (4).

We also have $\varphi^* D'$ linearly equivalent to D_2, so $\deg \varphi^* D'$ must be 2, which implies $\deg D' = 1$ because $\deg \varphi^* D' = (\deg \varphi)(\deg D')$. Finally, because this is up to linear equivalence, we can look at $D' \geqslant 0$. These two conditions force $D' \sim (P)$ for some $P \in E'$, and since D' is symmetric, $[-1]^* D' \sim D'$ i.e. $(-P) \sim (P)$, which happens if and only $P \in E'[2]$. We then have four possible descents for D_2:

- $\varphi^*(0_{E'}) = (0_E) + (T_1) \nsim D_2$ because $T_1 \neq 0_E$;
- $\varphi^*(T_1') = (T_2) + (T_3) \nsim D_2$ because $T_2 + T_3 = T_1 \neq 0_E$;
- $\varphi^*(T_2') = (\overline{T_1}) + (\overline{T_1} + T_1) \sim D_2$ because $2 \cdot \overline{T_1} + T_1 = 2 \cdot T_1 = 0_E$;
- $\varphi^*(T_3') = (\underline{T_1}) + (\underline{T_1} + T_1) \sim D_2$ because $2 \cdot \underline{T_1} + T_1 = 2 \cdot T_1 = 0_E$.

In the end, there is at most two descents of D_2, and those are distinct because $(T_2') \nsim (T_3')$ since $T_2' \neq T_3'$. We now try to compute a symmetric element $\tilde{g}_T \in G(D_2)$ above T.

Let $g_T \in G(D_2)$ be above T, with no further assumption. This exists because we assumed that T is rational. Because T is a 2-torsion point, we must have $\operatorname{div} g_T^2 = 0$, hence there is $\lambda_T \in k^*$ such that $g_T^2 = \lambda_T$. Therefore, the element $\tilde{g}_T = \frac{g_T}{\sqrt{\lambda_T}}$ does not depend on the choice of g_T (up to a sign) and is symmetric, but may not be rational. This is the symmetric element we were looking for, as well as $-\tilde{g}_T$, they are the two elements giving the descents of D_2.

Since taking another lift g_T changes the element λ_T by a square, it is well-defined in $k^*/k^{*,2}$.

Definition 6.2.4 (Type of a point) *The element* $[\lambda_T] \in k^*/k^{*,2}$ *defined above is called the type of* T*. We say that* T *is of Montgomery type if* λ_T *is a square in* k*.*

From the discussion above, D_2 descends to a rational symmetric divisor on E' if and only if T is of Montgomery type. But these divisors are rational if

and only if $\{\overline{T_1}, \overline{T_1} + T_1 = -\overline{T_1}\}$ is invariant, if and only if the cyclic degree 4 subgroup generated by $\overline{T_1}$ is rational (the same holds for $\underline{T_1}$).

In Section 8.3.1, we will introduce the Tate pairing $e_{T,2}$ and check that $e_{T,2}(T,T) = g_T(R+T)/g_T(R)$ modulo a square for any $R \in E$ (see Theorem 8.4.1). On the other hand, by construction $\lambda_T = g_T(R+T)g_T(R)$ for any $R \in E$, which is in the same class as the self Tate pairing $e_{T,2}(T,T)$.

This justifies the terminology 'Montgomery type': T is of Montgomery type if and only if T can be sent to the point $(0,0)$ on a Montgomery model because of Corollary 8.4.3. In particular, there can be an asymmetry: D_2 may descend to a symmetric divisor on E' via φ, while $2(0_{E'})$ may not descend to a symmetric divisor on E via the dual isogeny $\widetilde{\varphi}$.

Descents of $4(0_E)$

The situation becomes much simpler when looking at the possible descents of $D_4 = 4(0_E)$ to a degree 2 divisor on E', which is all we need to construct a Kummer line model for E'.

If D' is a descent of D_4 with respect to φ, let $\widetilde{K}$ be the associated lift of the kernel to $G(D_4)$. By Lemma 6.2.3, $\widetilde{K}$ consists of symmetric elements, so we must look for a symmetric D'. Furthermore, by the degree we must have $\deg D' = 2$ and since we are looking for divisors up to linear equivalence, we can look at $D' \geqslant 0$. We can therefore restrict to $D' \sim 2(P)$ or $D' \sim (P) + (Q)$ for $P, Q \in E'$. But $2(P) \sim (2 \cdot P) + (0_{E'})$ and $(P) + (Q) \sim (P - Q) + (0_{E'})$, we then are looking at $D' \sim (P) + (0_{E'})$ for some $P \in E'$. This last divisor is symmetric if and only $P \in E'[2]$, we have once again four potential choices for descents of D_4:

- $\varphi^*(2(0_{E'})) = 2(0_E) + 2(T_1) \sim D_4$ because $2 \cdot T_1 = 0_E$;
- $\varphi^*((T_1') + (0_{E'})) = (T_2) + (T_3) + (0_E) + (T_1) \sim D_4$ because
$$T_1 + T_2 + T_3 = 2 \cdot T_1 = 0_E;$$
- $\varphi^*((T_2') + (0_{E'})) = (\overline{T_1}) + (\overline{T_1} + T_1) + (0_E) + (T_1) \nsim D_4$ because
$$2 \cdot \overline{T_1} + 2 \cdot T_1 = T_1 \neq 0_E;$$
- $\varphi^*((T_3') + (0_{E'})) = (\underline{T_1}) + (\underline{T_1} + T_1) + (0_E) + (T_1) \nsim D_4$ because
$$2 \cdot \underline{T_1} + 2 \cdot T_1 = T_1 \neq 0_E.$$

There are only two descents and we can give the corresponding elements of $G(D_4)$ generating the lifts. If $\tilde{g}_T$ is a symmetric element in $G(D_2)$ above T, we consider $\tilde{g}_T^{\otimes 2}$ where the tensor product here denotes the usual scalar multiplication of sections, not the product in $G(D_2)$. This is a symmetric element of $G(D_4)$ above T, so it is of order 2 by Lemma 6.2.3 and generates a lift $\widetilde{K}$. On top of that, if $\tilde{g}_T = \frac{g_T}{\sqrt{\lambda_T}}$, then $\tilde{g}_T^{\otimes 2} = \frac{g_T^{\otimes 2}}{\lambda_T}$, which is always rational. This $\widetilde{g_T}^{\otimes 2}$ is canonical because it does not depend on the choice of sign in $\widetilde{g_T}$ and the lift $\widetilde{K}$ corresponds to the descent of D_4 to $D_2' \sim 2(T_2') \sim 2(T_3')$, because

of the shape of the kernels in Theorem 5.1.7 (1), and (3) gives that the global sections $\Gamma(D_2')$ are precisely the global sections in $\Gamma(D_4)$ invariant under the action by $\tilde{g}_T^{\otimes 2}$.

The other descent of D_4 to $(T_1') + (0_{E'})$ is given by $-\widetilde{g_T}^{\otimes 2}$.

Methodology

We can now sketch our algorithm to compute 2-isogenies between Kummer lines:

1. compute the action of the symmetric element $\tilde{g}_T^{\otimes 2}$ on X^2, XZ, Z^2;
2. find a basis X', Z' of invariant functions;
3. recover the Kummer model of E' embedded by X', Z'.

We detail these steps in the next sections.

Remark 6.2.5 (Vélu's formula) *By Section 6.1, we certainly do not need Theorem 5.1.7 to find a divisor linearly equivalent to D_4 and invariant by T: simply take $D_4'' = 2(0_E) + 2(T)$. The pullback of $D_2' = 2(0_{E'})$ by the isogeny φ is the divisor $D' = 2(T) + 2(0_E)$. The usual strategy to construct isogenies on elliptic curves is to use Vélu's formula, which provide (via a trace) sections of D' invariant under translation by T. Notably, the affine sections of D_2 are $1, x$, and 1 is already invariant. In Vélu's formula, we compute the trace $x'(P) = x(P) + x(P+T)$, the function x' has poles of order 2 at T and 0_E hence is a section of D', and is invariant by translation by T by construction.*

The link with Vélu's formula and the theta group is as follows: since D_4'' is invariant by translation by T, the constant function 1 provides a canonical symmetric element $\widetilde{h}_T''$ above T in $G(D_4'')$, which corresponds to the canonical descent of D_4'' to $2(0_{E'})$. Sections of D_4'' invariant by T thus corresponds by Theorem 5.1.7 (3) to sections of $2(0_{E'})$ on E'.

We will see below in Main example 6.2.9 that as expected, on E' these sections are exactly the same as the ones we obtain through sections of D_4 invariant by $\widetilde{h}_T = \tilde{g}_T^{\otimes 2}$.

So Vélu's formula can be seen as a special case of the more general framework of descending sections and divisors through theta groups actions. The reason we work directly with theta groups is that it provides a more flexible framework to study isogenies. In particular, it is slightly more convenient to work with the divisor $D_2 = 2(0_E)$ than with $(0_E) + (T)$.

More importantly, in higher dimensions, we don't have analogues of Vélu's formula for an ℓ-isogeny. Namely, if we start with an ample divisor Θ of degree 1 associated to a principal polarization, then the traces of Θ under the points of K, a maximal isotropic subgroup of $A[\ell]$ will be an invariant divisor of degree ℓ^{g^2}, hence descends to a divisor on $B = A/K$ of degree $\ell^{g(g-1)}$, so is associated to a principal polarization if and only if $g = 1$. Therefore, taking traces of principal polarization does not work to build invariant divisors of the correct degree on A, and we need the full power of the theta group framework as developed by Mumford.

In this chapter, we study 2-isogenies between Kummer lines, from which we can deduce doubling formulas (by composing with the dual isogeny). In Chapter 7, we will extend this to differential addition formulas. This amounts to studying the

dimension 2 isogeny given by $E \times E \to E \times E, (P, Q) \mapsto (P + Q, P - Q)$. *The action of the theta group* $G(D_2)$ *on the global sections* X, Z *that we study in this chapter will be crucial to extend the doubling formulas to differential additions.*

6.2.2 Computing 2-isogenies

We want to compute a 2-isogeny generated by a 2-torsion point T. We will describe in this section how to build degree 2 maps which are invariant under a translation by T on Kummer lines, and how to recover 2-isogenies from that.

Remark 6.2.6 *The automorphism* $t_T : P \mapsto P + T$ *on the elliptic curve can be pushed to* $\mathbb{P}^1$ *via* π *because* T *is of 2-torsion. It is an involutive map, therefore it is an automorphism of* $\mathbb{P}^1$*, i.e. it is a homography.*

First, consider the matrix $[M_T] \in \mathrm{PGL}_2(k)$ associated to the homography t_T.

Main example 6.2.7 (cont. Main example 6.2.1) *With* $T = T_1 = (0 : 1)$*, the homography* t_T *is given by* $t_T(0_E) = T_1$, $t_T(T_1) = 0_E$ *and* $t_T(T_2) = T_3$. *We are looking for* $a, b, c, d \in k$ *such that* $t_T(X : Z) = (aX + bZ : cX + dZ)$*:*

- $t_T(1 : 0) = (a : c) = (0 : 1)$, *i.e.* $a = 0$.
- $t_T(0 : 1) = (b : d) = (1 : 0)$, *i.e.* $d = 0$.
- $t_T(\alpha : 1) = (b : c\alpha) = (\gamma : \alpha)$, *i.e.* $b = c\gamma$.

So $t_T(X : Z) = (bZ : cX) = (\gamma Z : X)$ *and the associated matrix in* $\mathrm{PGL}_2(k)$ *is*

$$[M_T] = \left[\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right] = \left[\begin{pmatrix} 0 & \gamma \\ 1 & 0 \end{pmatrix}\right].$$

We lift this matrix to $M_T \in \mathrm{GL}_2(k)$. By definition, since T is a 2-torsion point, $[M_T^2] = [I_2]$, so $M_T^2 = \lambda_T I_2$. This lift is associated to an explicit element g_T in the theta group $G(D_2)$. Indeed, we have a projective action of $E[2]$ on $\Gamma(D_2) \simeq k^2$ given by translation by a 2-torsion point on projective coordinates. The canonical action defined in Definition 5.1.1 lifts this to an affine group action of $G(D_2)$ on $\Gamma(D_2)$. Since this group action is faithful, we can represent an element $g \in G(D_2)$ by the corresponding action matrix. This also comes from the discussion to express $P + T$ coordinates from coordinates of P between Eqs. (5.3) and (5.4).

In particular, the element λ_T associated to M_T is exactly the type associated to g_T in Definition 6.2.4. Similarly, λ_T depends on the chosen lift M_T, whereas $\frac{1}{\sqrt{\lambda_T}} M_T$ does not (up to a sign). This is the invariant matrix of interest, it is of order 2, and corresponds to the action of a symmetric lift $\tilde{g}_T$.

We want to build quadratic forms in (X, Z) invariant by $\frac{1}{\sqrt{\lambda_T}} M_T$, which will be said to be T-invariants. Because we are working on quadratic forms, the choice of sign does not matter any more. We will look at the action of this matrix on X^2, Z^2 and XZ.

Remark 6.2.8 *If q is a quadratic form and $M = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, the action of M over q is given by $M \cdot q(X, Z) = q(aX + bZ, cX + dZ)$. Here, $\sqrt{\lambda_T}$ may be in some quadratic extension of k, but we can work around that, if $M_T = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$:*

$$\left(\frac{1}{\sqrt{\lambda_T}} M_T\right) \cdot q(X, Z) = q\left(\frac{aX + bZ}{\sqrt{\lambda_T}}, \frac{cX + dZ}{\sqrt{\lambda_T}}\right) = \frac{1}{\lambda_T}\left(M_T \cdot q(X, Z)\right).$$

Main example 6.2.9 (cont. Main example 6.2.7) *From previous example, we consider the lift $M_T = \left(\begin{smallmatrix} 0 & \gamma \\ 1 & 0 \end{smallmatrix}\right)$, then $M_T^2 = \gamma I_2$. The type of T is $[\gamma]$. We first compute the action of M_T on X^2, Z^2 and XZ, and then divide by γ:*

$$\frac{1}{\gamma}\left(M_T \cdot X^2\right) = \gamma Z^2; \quad \frac{1}{\gamma}\left(M_T \cdot Z^2\right) = \frac{1}{\gamma} X^2; \quad \frac{1}{\gamma}\left(M_T \cdot XZ\right) = XZ.$$

We notice that XZ is already invariant, to build another one we can consider a trace for the matrix action on quadratic forms, for instance:

$$q_1 = X^2 + \frac{1}{\gamma}\left(M_T \cdot X^2\right) = X^2 + \gamma Z^2, \textit{then } \frac{1}{\gamma}\left(M_T \cdot q_1\right) = q_1.$$

It appears that we recover the same invariant projective sections using Vélu's formula. By Remark 6.2.5, Vélu's formula build the invariant affine section:

$$x'(P) = x(P) + x(P + T) = x(P) + \frac{\gamma}{x(P)} = \frac{X(P)}{Z(P)} + \frac{\gamma Z(P)}{X(P)} = \frac{X^2(P) + \gamma Z^2(P)}{X(P)Z(P)}.$$

The numerator and denominators of this function are precisely the above invariant sections.

Say we have two linearly independent quadratic forms q_1 and q_2 which are T-invariant where T is a 2-torsion point on the Kummer line $\mathcal{K} = E/\pm 1$, and consider a basis $u, v \in \mathrm{Span}(q_1, q_2)$. Set $M_T = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ the matrix of t_T and $[\lambda_T]$ the type of T, where $t_T : \mathcal{K} \to \mathcal{K}, P \mapsto P + T$.

If $\mathcal{K}' = E'/\pm 1$, set the following degree 2 map, which is well-defined by the properties of quadratic forms:

$$\begin{aligned} f : \mathcal{K} &\to \mathcal{K}' \\ (X : Z) &\mapsto (u(X, Z) : v(X, Z)). \end{aligned}$$

Since u and v are T-invariant, we get $f(P+T) = f(t_T(P)) = f(P)$ by construction. What remains to do is determining the codomain $\mathcal{K}'$ using the extra 2-torsion points we have.

Main example 6.2.10 (cont. Main example 6.2.9) *We use the quadratic forms $q_1(X, Z) = X^2 + \gamma Z^2$ and $q_2(X, Z) = XZ$ from the previous example, and consider for instance $u = q_1$ and $v = q_2$ (or any linear combination of q_1 and q_2). Then $f : (X : Z) \mapsto (X^2 + \gamma Z^2 : XZ)$ can be computed in $1\mathbf{M} + 2\mathbf{S} + 1\mathbf{m}_0$. We have by construction of f that $f(0_E) = f(T_1)$, and since $T_3 = T_2 + T_1$, we also have $f(T_2) = f(T_3)$. A quick computation yields:*

$$f(0_E) = (1 : 1) \textit{ and } f(T_2) = (\alpha^2 + \gamma : \alpha) = (-\mathcal{A} : 1).$$

We are trying to build an isogeny with kernel T_1, so $f(0_E)$ should be sent to the origin $0_{E'}$, and $f(T_2)$ is a 2-torsion point on $\mathcal{K}'$. Note that $f(T_2)$ is always rational even if T_2 is not, this is a more general fact coming from the following lemma.

Lemma 6.2.11 $f(T_2)$ *is rational.*

Proof *Let σ be a Galois element on the field k. If T_2 is invariant by σ, so is T_3 (because $T_3 = T_2 + T_1$) and the image by f is obviously invariant by σ too. However, if T_2 is not invariant by σ, then $\sigma(T_2) \neq T_2$, so we must have $\sigma(T_2) = T_3$ because T_1 is rational. But then, since σ commutes with f:*

$$\sigma(f(T_2)) = f(\sigma(T_2)) = f(T_3) = f(T_2). \qquad \square$$

To recover the remaining 2-torsion, consider a 4-torsion point $\overline{T_1}$ above T_1 which may not be rational. On the Kummer line, since $P = -P$ for any P and $3 \cdot \overline{T_1} = -\overline{T_1}$, the coordinates of $\overline{T_1}$ can be found by solving $\overline{T_1} + T_1 = \overline{T_1}$. If $\overline{T_1} = (X : Z)$, using the translation t_{T_1}, this leads to:

$$(\gamma Z : X) = (X : Z) \text{ i.e. } \frac{X}{Z} = \pm\sqrt{\gamma}.$$

Then $\overline{T_1} = (\sqrt{\gamma} : 1)$ and $\underline{T_1} = (-\sqrt{\gamma} : 1)$ are the 4-torsion points above T_1. The remaining 2-torsion points on $\mathcal{K}'$ are $f(\overline{T_1})$, $f(\underline{T_1})$.

An optional step is to put $\mathcal{K}'$ in a nice shape by a homography, but this is not mandatory and can gain some operations. We will give more details in the next section.

6.3 2-ISOGENIES ON MONTGOMERY CURVES

We will focus on Montgomery curves in this section, which corresponds to the case $\gamma = 1$ in Main example 6.2.1. Recall that we have computed the type of $(0 : 1)$ in Main example 6.2.9, it is $[1]$, hence $(0 : 1)$ is of Montgomery type. The equation of the corresponding curve is

$$E : \beta y^2 = x(x^2 + \mathcal{A}x + 1).$$

The Montgomery Kummer line is denoted $\pi : E \to \mathcal{K} \simeq \mathbb{P}^1$ over k with constant $\mathcal{A} \in k$. The ramification of our Kummer line is then:

$$0_E = (1 : 0)^*, \quad T_1 = (0 : 1), \quad T_2 = (A : B), \quad T_3 = (B : A) = T_2 + T_1.$$

Thus, $\mathcal{A} = -\frac{(A-B)^2}{AB}$ and

$$d := \frac{\mathcal{A} + 2}{4} = \frac{-(A-B)^2}{4AB} = \frac{(A-B)^2}{(A-B)^2 - (A+B)^2}. \tag{6.1}$$

Instead of computing $\mathcal{A}$ to determine the curve (up to a quadratic twist), we will compute d because it holds the same information, and it is the natural constant used in doubling formulas on Montgomery curves, see Algorithm 2.5. When the codomain of an isogeny is a Montgomery curve, we want to compute d efficiently to get all the necessary information about the codomain.

We computed in Main example 6.2.9 two T_1-invariant quadratic forms:

$$q_1(X, Z) = X^2 + Z^2, \quad q_2(X, Z) = XZ.$$

We also have the translation by T_1 denoted $t_{T_1} : \mathcal{K} \to \mathcal{K}$ computed in Main example 6.2.7:

$$t_{T_1} : (X : Z) \mapsto (Z : X). \tag{6.2}$$

We will need the translation by T_2 later too, which is given by:

$$t_{T_2} : (X : Z) \mapsto (AX - BZ : BX - AZ). \tag{6.3}$$

Remark 6.3.1 *In Section 6.4, when the curve E is fixed for scalar multiplication, the multiplication $\mathbf{m}_0$ by d or $(A : B)$, will count as one arithmetic multiplication since we can assume that $B = 1$. For the computation of a 2^n-isogeny chain in Section 6.6, they will be given as quotients, in which case we will count them as two multiplications because of the projective coordinates.*

Remark 6.3.2 *The ramification of the Montgomery Kummer line is invariant under the involution $(X : Z) \mapsto (Z : X)$, corresponding to translation by T_1. If we apply the Hadamard transform $H(X : Z) = (X + Z : X - Z) = (X' : Z')$, we obtain a new model $H\mathcal{K}$ where the ramification becomes*

$$(1 : 1), \quad (-1 : 1), \quad (A + B : A - B), \quad (A + B : B - A),$$

which is invariant by the involution $(X' : Z') \mapsto (-X' : Z')$.

The quadratic forms invariant by the canonical affine lift of this involution are $q_1' = X'^2$ and $q_2' = Z'^2$. On $\mathcal{K}$, these quadratic forms correspond to $(X + Z)^2$ and $(X - Z)^2$, which indeed span the same vector spaces as q_1, q_2 above.

6.3.1 Isogeny with kernel T_1

Assume in this section that $\frac{A}{B} \in k$, so we have the full 2-torsion on our curve. Recall we have these independent 4-torsion points above T_1:

$$\overline{T_1} = (1 : 1), \quad \underline{T_1} = (-1 : 1) = \overline{T_1} + T_2.$$

We will use the following invariant quadratic forms from Remark 6.3.2, using the notations of Main example 6.2.9:

$$\begin{cases} u(X, Z) = (X + Z)^2 = q_1(X, Z) + 2q_2(X, Z), \\ v(X, Z) = (X - Z)^2 = q_1(X, Z) - 2q_2(X, Z). \end{cases}$$

Set $f_0 : (X : Z) \mapsto ((X + Z)^2 : (X - Z)^2)$, it is invariant by t_{T_1} by construction. Set $A' = A + B$ and $B' = A - B$, the image of the ramification is the following:

$$\begin{aligned} f_0(0_E) &= (1 : 1)^* = f_0(T_1), & f_0(T_2) &= (A'^2 : B'^2) = f_0(T_3), \\ f_0(\overline{T_1}) &= (1 : 0), & f_0(\underline{T_1}) &= (0 : 1). \end{aligned}$$

To get a Montgomery shaped ramification, we will multiply by $C : (X : Z) \mapsto (B'X : A'Z)$, set $f = C \circ f_0$, then:

$$\begin{aligned} 0_{E'} &= f(\overline{T_1}) = (1 : 0), & T_1' &= f(\underline{T_1}) = (0 : 1), \\ T_2' &= f(R_1) = (A_2 : B_2), & T_3' &= f(0_E) = (B_2 : A_2)^*. \end{aligned}$$

Up to a translation by T_3' we recover the 2-isogeny with kernel T_1 and the image is Montgomery shaped.

Theorem 6.3.3 (Translated 2-isogeny with kernel T_1) *Let $\varphi : \mathcal{K} \to \mathcal{K}'$ be the 2-isogeny with kernel T_1 on the Montgomery Kummer line $\mathcal{K}$ with extra 2-torsion $(A : B)$, and assume $\frac{A}{B} \in k$. Set $(A' : B') = (A + B : A - B)$, then $\varphi = f + T_3'$ where:*

$$f : (X : Z) \mapsto \left(B'(X+Z)^2 : A'(X-Z)^2\right).$$

f can be computed in $2\mathbf{S} + 1\mathbf{m}_0 + 2\mathbf{a}$, the codomain $\mathcal{K}'$ is a Montgomery Kummer line and the corresponding curve constant d' can be computed in $2\mathbf{S} + 1\mathbf{a}$ with:

$$d' = \frac{B^2}{B^2 - A^2}.$$

Proof *The fact that φ is the 2-isogeny with kernel T_1 and that the image is a Montgomery Kummer line is straight-forward from the reasoning above. The curve constant d' comes from Eq. (6.1) and the definition of $(A' : B')$:*

$$d' = \frac{(A' - B')^2}{(A' - B')^2 - (A' + B')^2} = \frac{B^2}{B^2 - A^2}.$$

□

Proposition 6.3.4 (Translated dual isogeny) *Using notation of Theorem 6.3.3, the dual isogeny of φ is given by $\widetilde{\varphi} = \widetilde{f} + T_3$ where:*

$$\widetilde{f} : (X : Z) \mapsto \left(B(X+Z)^2 : A(X-Z)^2\right).$$

Then, for any $P \in \mathcal{K}$, $\widetilde{f} \circ f(P) = 2 \cdot P + T_2$ can be computed in $4\mathbf{S} + 2\mathbf{m}_0 + 4\mathbf{a}$ as in Algorithm 6.1.

Proof *Because the Hadamard transform is an involution, the codomain of $\widetilde{f}$ is $\mathcal{K}$. We can then set $\varphi_0 = \widetilde{f} + T_3'$, which is the 2-isogeny with kernel T_1' thanks to Theorem 6.3.3. Let's check that $\varphi_0 \circ \varphi = [2]$ on the Kummer line. We will use the following formula:*

$[2]$ is the multiplication by 2.

$$\begin{aligned}
\varphi_0(\varphi(P)) &= \varphi_0(f(P) + T_3') \\
&= \varphi_0(f(P)) + \varphi_0(T_3') \\
&= \widetilde{f}(f(P)) + \widetilde{f}(T_3') + 2 \cdot T_3 \\
\varphi_0(\varphi(P)) + T_2 &= \widetilde{f}(f(P)).
\end{aligned}$$

We then study $\varphi_0 \circ \varphi$ on the 2-torsion:

$$\begin{aligned}
&\varphi_0(\varphi(0_E)) = 0_E, && \varphi_0(\varphi(T_1)) = \widetilde{f}(T_3') + T_2 = 2 \cdot T_2 = 0_E, \\
&\varphi_0(\varphi(T_2)) = \widetilde{f}(T_2') + T_2 = 0_E, && \varphi_0(\varphi(T_3)) = \widetilde{f}(T_2') + T_2 = 0_E.
\end{aligned}$$

$\varphi_0 \circ \varphi \neq [0]$ (for instance, $\varphi_0(\varphi(\overline{T_1})) = T_1$), so we must have $\varphi_0 \circ \varphi = [2]$. Similarly, we prove $\varphi \circ \varphi_0 = [2]$. By uniqueness, $\varphi_0 = \widetilde{\varphi}$. The first formula then yields $\widetilde{f}(f(P)) = 2 \cdot P + T_2$. □

We will be using the translated doubling of Proposition 6.3.4 in Section 6.4 to build a 'hybrid ladder'.

Algorithm 6.1: Doubling up to a 2-torsion point in Montgomery XZ-coordinates

Input: $\pi(P) = (X_1 : Z_1)$
Output: $\pi(2 \cdot P + T_2) = (X : Z)$
Data: On $\mathcal{K}$ with extra 2-torsion $T_2 = (A : B)$,
$(A' : B') = (A + B : A - B)$

```
def DoublingTranslation(π(P)):
    u ← (X_1 + Z_1)^2;
    v ← (A'/B')(X_1 − Z_1)^2;
    X ← (u + v)^2;
    Z ← (A/B)(u − v)^2;
    return (X : Z);
```

Remark 6.3.5 *In ECC, if one always works on the same curve, it is possible to control the associated constants and make them small. That way, a multiplication by a curve constant costs way less than a multiplication by a generic number. The hybrid ladder will rely on this fact a lot. Because the translated point is T_3' and not T_1' in Theorem 6.3.3, it is not convenient to use these formulas to chain isogenies.*

Remark 6.3.6 *The formulas detailed in Algorithm 6.1 are exactly the same as the standard doubling formulas in squared theta coordinates*[1]*. This is not a coincidence, as remarked in [HR19], the Montgomery Kummer line and the squared theta Kummer line differ by translation by a 2-torsion point (one way to directly recover this result is to look at the associated ramification points from Examples 7.1.1 and 7.1.3), so using the squared theta doubling formula in Montgomery coordinates gives a translated doubling. We will come back to this in Section 8.2.*

6.3.2 Isogeny with kernel T_2

Assume once again that $\frac{A}{B} \in k$, so we have full 2-torsion on our curve. Recall that the ramification is:

$$0_E = (1 : 0)^*, \quad T_1 = (0 : 1), \quad T_2 = (A : B), \quad T_3 = (B : A) = T_2 + T_1.$$

In this section, we further assume that there is a 4-torsion point $\overline{T_2} = (a' : b')$ above T_2. Another independent 4-torsion point above T_2 is then $\underline{T_2} = \overline{T_2} + T_1$ which can be computed easily as $\underline{T_2} = (b' : a')$ thanks to Eq. (6.2). Finally, set $(a : b) = (a' + b' : a' - b')$. A useful relation that we will be using is the following:

$$(A : B) = (a^2 + b^2 : a^2 - b^2) \iff (A + B : A - B) = (a^2 : b^2). \tag{6.4}$$

This comes for instance from the doubling formulas in Proposition 6.3.4, because $2 \cdot \overline{T_2} + T_2 = 0_E$:

$$(A - B)a^2 - (A + B)b^2 = 0 \iff (A + B : A - B) = (a^2 : b^2).$$

1 https//hyperelliptic.org/EFD/g1p/auto-edwards-yzsquared.html

We will now apply the algorithm to find invariant maps by T_2. A matrix associated to t_{T_2} is:

$$M_{T_2} = \begin{pmatrix} A & -B \\ B & -A \end{pmatrix}.$$

Since $M_{T_2}^2 = (A^2 - B^2)I_2$, the type is $\lambda_{T_2} = A^2 - B^2 = 4a^2b^2$. This is a square, so T_2 is of Montgomery type.

Set $M = \frac{M_{T_2}}{\sqrt{\lambda_{T_2}}}$, we will be looking at the action of M on the following basis of quadratic forms: $(X+Z)^2$, $(X-Z)^2$ and $(X-Z)(X+Z)$:

$$M \cdot (X+Z)^2 = \frac{a^2}{b^2}(X-Z)^2 \quad M \cdot (X-Z)^2 = \frac{b^2}{a^2}(X+Z)^2.$$
$$M \cdot (X-Z)(X+Z) = (X-Z)(X+Z).$$

The invariant quadratic forms we will be using are then:

$$q_1(X,Z) = b^2(X+Z)^2 + a^2(X-Z)^2, \quad q_2(X,Z) = ab(X+Z)(X-Z). \tag{6.5}$$

By doing linear combination of q_1 and q_2, we end up with the following formulas:

Theorem 6.3.7 (Translated 2-isogeny with kernel T_2) *Let $\varphi : \mathcal{K} \to \mathcal{K}'$ be the 2-isogeny with kernel T_2 on the Montgomery Kummer line $\mathcal{K}$ with extra 2-torsion $T_2 = (A : B)$. Assume there is a 4-torsion point $\overline{T_2} = (a' : b')$ above T_2. Set $(a : b) = (a' + b' : a' - b')$ such that Eq. (6.4) holds. Finally, set f to be the following map:*

$$f : (X : Z) \mapsto \Big((b(X+Z) + a(X-Z))^2 : (b(X+Z) - a(X-Z))^2\Big).$$

Then the ramification on the image of f is:

$$0_{E'} = (1:0), \quad T_1' = (0:1), \quad T_2' = (a'^2 : b'^2)^*, \quad T_3' = (b'^2 : a'^2) = T_2' + T_1'.$$

We have $\varphi = f + T_2'$. f can be computed in $2\mathbf{S} + 1\mathbf{m}_0 + 4\mathbf{a}$, the codomain $\mathcal{K}'$ is a Montgomery Kummer line and the curve constant d' can be computed in $2\mathbf{S} + 1\mathbf{a}$ with:

$$d' = \frac{B^2 - A^2}{B^2}.$$

Proof *We have $f(X:Z) = (q_1(X,Z) + 2q_2(X,Z) : q_1(X,Z) - 2q_2(X,Z))$ where q_1 and q_2 are defined in Eq. (6.5) and are T_2-invariant. So $f(\cdot + T_2) = f$. It is straight-forward to compute:*

$$f(0_E) = (a'^2 : b'^2)^* = f(T_2), \qquad f(T_1) = (b'^2 : a'^2) = f(T_3),$$
$$f(\overline{T_2}) = (1:0), \qquad f(\underline{T_2}) = (0:1).$$

So the image is a Montgomery Kummer line and $\varphi = f + R_2$ with notations from the theorem.

The codomain is given by d' using Eq. (6.1):

$$d' = \frac{(a'^2 - b'^2)^2}{(a'^2 - b'^2)^2 - (a'^2 + b'^2)^2} \tag{6.6}$$

Thanks to Eq. (6.4), we can simplify the expression of d':

$$\Big((a'^2 - b'^2)^2 : (a'^2 + b'^2)^2\Big) = \Big(4a^2b^2 : (a^2 + b^2)^2\Big) = (A^2 - B^2 : A^2). \qquad \square$$

If we compute $\varphi = f + T_2'$ using the translation $t_{T_2'}$ given in Eq. (6.3), we find back the formulas for 2-isogenies given by Renes in [Ren18, Prop. 2]. We can also recover alternative shifted doubling formulas instead of Algorithm 6.1, which only differ by the number of additions.

Remark 6.3.8 *Unlike in Theorem 6.3.3, we have a translated isogeny by T_2', and the kernel was initially T_2. We can therefore chain such isogenies to compute 2^n-isogenies, more details are given in Section 6.6.*

Since the computations only involves the 4-torsion point $\overline{T_2}$ above T_2, one could keep track only of the 4-torsion points. The codomain would then be given by Eq. (6.6), which costs $4\mathbf{S} + 3\mathbf{a}$.

Proposition 6.3.9 (Dual isogeny) *Using notation of Theorem 6.3.7, the dual isogeny of φ is given by $\widetilde{\varphi}$ where:*

$$\widetilde{\varphi} : (X : Z) \mapsto \left(B(X+Z)^2 : 4AXZ\right).$$

Then $\widetilde{\varphi} \circ f(P) = 2 \cdot P + T_2$ where $P \in \mathcal{K}$ can be computed in $4\mathbf{S} + 2\mathbf{m}_0 + 7\mathbf{a}$ as in Algorithm 6.2 (using $4XZ = (X+Z)^2 - (X-Z)^2$).

This is a slight abuse of notation where we actually mean the 2-torsion on the curve.

Proof *We know that the kernel of $\widetilde{\varphi}$ is $\varphi(\mathcal{K}[2]) = \langle \varphi(T_1) \rangle = \langle T_1' \rangle$. We also have computed two T_1'-invariant quadratic forms earlier, let $\varphi_0(X : Z) = ((X+Z)^2 : XZ)$. Then $\varphi_0(\cdot + T_1') = \varphi_0$. The output ramification is:*

$$\varphi_0(0_{E'}) = (1:0)^* = \varphi_0(T_1'), \qquad \varphi_0(T_2') = \left((a'^2 + b'^2)^2 : a'^2 b'^2\right) = \varphi_0(T_3'),$$
$$\varphi_0(\overline{T_2'}) = (4:1), \qquad \varphi_0(\underline{T_2'}) = (0:1).$$

We have $\varphi_0(T_2') = (4A^2 : B^2)$ thanks to computations done in the proof of Theorem 6.3.7.

We then consider a homography $h : (X : Z) \mapsto (aX + bZ : cX + dZ)$ such that:

- $h(1:0) = (1:0)$, *setting* $c = 0$;
- $h(0:1) = (0:1)$, *setting* $b = 0$;
- $h(4:1) = (B:A)$, *giving the equation* $(4a : d) = (B : A)$.

Then, $h(4A^2 : B^2) = (4aA^2 : dB^2) = (A : B)$. Finally, the map $h \circ \varphi_0$ is the 2-isogeny with kernel T_2 and codomain $\mathcal{K}$, hence $\widetilde{\varphi} = h \circ \varphi_0$.

Since, $\widetilde{\varphi}(\varphi(P)) = 2 \cdot P = \widetilde{\varphi}(f(P)) + \widetilde{\varphi}(T_2')$, we get the last formula from the equality $\widetilde{\varphi}(T_2') = T_2$. □

6.3.3 Alternative formulas for the isogeny with kernel T_1

In this last section, we do not assume $\frac{A}{B} \in k$ any more and we will not use the additional 2-torsion. The counterpart is that we add a hypothesis about a 8-torsion point $\widetilde{T_1} = (r : s)$ above $\overline{T_1} = (1:1)$ (which itself is above $T_1 = (0:1)$). That way, we ensure that there will still be a rational 4-torsion point on the image, giving a point of Montgomery type as discussed earlier.

Algorithm 6.2: Alternative doubling up to a 2-torsion point in Montgomery XZ-coordinates

Input: $\pi(P) = (X_1 : Z_1)$
Output: $\pi(2 \cdot P + T_2) = (X : Z)$
Data: On $\mathcal{K}$ with extra 2-torsion $T_2 = (A : B)$ and 4-torsion $\overline{T_2} = (a' : b'), (a : b) = (a' + b' : a' - b')$

```
def DoublingTranslation(π(P)):
    u ← (X_1 + Z_1);
    v ← (a/b)(X_1 − Z_1);
    w ← (u + v)^2;
    t ← (u − v)^2;
    u ← (w + t)^2;
    v ← (w − t)^2;
    X ← u;
    Z ← (A/B)(u − v);
    return (X : Z);
```

We set $(\gamma : \delta) = (4rs : (r-s)^2)$, and because $2 \cdot \widetilde{T_1} = \overline{T_1} = (1 : 1)$, using Algorithm 2.5:

$$((\gamma + \delta)\delta : \gamma(\delta + d\gamma)) = (1 : 1) \iff d = \frac{\delta^2}{\gamma^2}.$$

But we have another expression for d given in Eq. (6.1), therefore:

$$(\delta^2 : \gamma^2) = (-(A - B)^2 : 4AB).$$

We are looking for a 2-isogeny with kernel T_1 without the knowledge of $\frac{A}{B}$. The result will be in a similar shape to the one in Proposition 6.3.9. As before, we start by computing invariants by the matrix $M = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, which we already did in Main example 6.2.9. We will consider:

$$M \cdot (X - Z)^2 = (X - Z)^2, \quad M \cdot XZ = XZ.$$

If $f_0 : (X : Z) \mapsto ((X - Z)^2 : XZ)$, then $f_0(\cdot + T_1) = f_0$. The codomain ramification is:

$$f_0(0_E) = (1 : 0)^* = f_0(T_1), \quad f(T_2) = ((A - B)^2 : AB) = (-4\delta^2 : \gamma^2) = f(T_3),$$
$$f(\overline{T_1}) = (0 : 1), \qquad f(\underline{T_1}) = (-4 : 1).$$

We consider a homography $h : (X : Z) \mapsto (aX + bZ : cX + dZ)$ to put it in a convenient shape. We want $h(1 : 0) = (1 : 0)$ and $h(0 : 1) = (0 : 1)$, which forces $b = 0$ and $c = 0$. Then, we naturally want to send $f_0(\widetilde{T_1}) = ((r - s)^2 : rs)$ onto $(1 : 1)$:

$$h((r - s)^2 : rs) = (1 : 1) \iff (a : d) = (\gamma : 4\delta).$$

That way, we get:

$$h(-4\delta^2 : \gamma^2) = (-4\gamma\delta^2 : 4\delta\gamma^2) = (-\delta : \gamma), \quad h(-4 : 1) = (-\gamma : \delta).$$

By setting $f = h \circ f_0$, we find formulas already known in [FJP14, Eq. (19)]:

Theorem 6.3.10 (2-isogeny with kernel T_1) *Let $\varphi : \mathcal{K} \to \mathcal{K}'$ be the 2-isogeny with kernel T_1 on the Montgomery Kummer line $\mathcal{K}$. Assume there is a rational 8-torsion point $\widetilde{T_1} = (r : s)$ above $\overline{T_1} = (1 : 1)$. Set $(\gamma : \delta) = (4rs : (r - s)^2)$, then φ is given by:*

$$\varphi : (X : Z) \mapsto \left(\gamma(X - Z)^2 : 4\delta XZ\right).$$

φ can be computed in $2\mathbf{S} + 1\mathbf{m}_0 + 3\mathbf{a}$ ($4XZ = (X + Z)^2 - (X - Z)^2$), the codomain $\mathcal{K}'$ is a Montgomery Kummer line and the curve constant d' can be computed in $4\mathbf{S} + 6\mathbf{a}$ with:

$$d' = \frac{(\gamma + \delta)^2}{(\gamma + \delta)^2 - (\gamma - \delta)^2}.$$

The computation of d' is straight-forward from Eq. (6.1). For completeness, we also provide the dual isogeny formula.

Proposition 6.3.11 (Dual isogeny) *Using notation of Theorem 6.3.10, the dual isogeny of φ is given by $\widetilde{\varphi}$ where:*

$$\begin{aligned}
\widetilde{\varphi} : (X : Z) &\mapsto (u(X, Z) + 2\delta v(X, Z) : u(X, Z) - 2\delta v(X, Z)),\\
u(X, Z) &= (\gamma + \delta)(X + Z)^2 - (\gamma - \delta)(X - Z)^2,\\
v(X, Z) &= (X + Z)(X - Z).
\end{aligned}$$

$\widetilde{\varphi}$ can be computed in $1\mathbf{M} + 2\mathbf{S} + 2\mathbf{m}_0 + 3\mathbf{a}$.

Proof *We have that $\ker \widetilde{\varphi} = \langle T_2' \rangle$, we can't apply results from Theorem 6.3.7 since we don't know the 4-torsion above T_2'. We have already computed the matrix M associated to $t_{T_2'}$ and the type $\lambda = \delta^2 - \gamma^2$:*

$$M = \begin{pmatrix} -\delta & -\gamma \\ \gamma & \delta \end{pmatrix}.$$

Furthermore, we have seen that $\frac{1}{\lambda}(M \cdot (X + Z)(X - Z)) = (X + Z)(X - Z)$, so we are looking for one other invariant:

$$\frac{1}{\lambda}\left(M \cdot (X + Z)^2\right) = \frac{(\gamma - \delta)^2}{\delta^2 - \gamma^2}(X - Z)^2 = -\frac{\gamma - \delta}{\gamma + \delta}(X - Z)^2.$$

Hence, the two invariant quadratic forms we will consider are:

$$\begin{aligned}
u(X, Z) &= (\gamma + \delta)\left((X + Z)^2 + \frac{1}{\lambda}\left(M \cdot (X + Z)^2\right)\right)\\
&= (\gamma + \delta)(X + Z)^2 - (\gamma - \delta)(X - Z)^2,\\
v(X, Z) &= (X + Z)(X - Z).
\end{aligned}$$

We then set $\varphi_0(X : Z) = (au(X, Z) + bv(X, Z) : cu(X, Z) + dv(X, Z))$, which by construction verifies $\varphi_0(\cdot + T_2') = \varphi_0$. Since we want it to be the dual of φ, we are looking for the following equations:

- $\varphi_0(\varphi(0_E)) = \varphi_0(0_{E'}) = 0_E$, *which implies* $2\delta c + d = 0$.
- $\varphi_0(\varphi(\overline{T_1})) = \varphi_0(T_1') = T_1$, *which implies* $2\delta a - b = 0$.

- $\varphi_0(\varphi(\widetilde{T_1})) = \varphi_0(\overline{T_1'}) = \overline{T_1}$, *which implies* $a = c$.

We factor by a *in* φ_0 *and get the following expression:*

$$\varphi_0 : (X : Z) \mapsto (u(X, Z) + 2\delta v(X, Z) : u(X, Z) - 2\delta v(X, Z)).$$

We then check that it behaves correctly on the remaining 2-torsion:

- $\varphi_0(\varphi(T_2)) = \varphi_0(T_2') = (4\delta\lambda : 0) = 0_E$.
- $\varphi_0(\varphi(\underline{T_1})) = \varphi_0(T_3') = (0 : 4\delta\lambda) = T_1$.

Finally, $\varphi_0 = \widetilde{\varphi}$. □

6.4 HYBRID LADDER

Let $\pi : E \to \mathbb{P}^1 \simeq \mathcal{K}$ be a Montgomery Kummer line. Recall that if one knows $\pi(P)$, $\pi(Q)$ and $\pi(P - Q)$, then it is possible to recover $\pi(P + Q)$, and using the Montgomery ladder from Algorithm 2.6, one can compute $\pi(n \cdot P)$ on the Kummer line. The key of the ladder is that, at each step, we have $\pi(R - S) = \pi(P)$, where R and S are the two points we keep track of.

6.4.1 Algorithm

Since the cost of a scalar multiplication depends linearly on the cost of one differential addition and one doubling, we are looking into improving these individually. In this chapter, we focus on the doubling. If we look at the computational cost of the multiplication by 2 in Algorithm 2.5, we get $2\mathbf{M} + 2\mathbf{S} + 1\mathbf{m}_0$.

On the other hand, the computational cost of a doubling up to a 2-torsion point in Proposition 6.3.4 and Algorithm 6.1 is $4\mathbf{S} + 2\mathbf{m}_0$. As explained in Section 2.1, a square tends to be faster than a multiplication. Here are some comparisons in Table 6.1 using $\mathbf{S} = 0.67\mathbf{M}$ and $\mathbf{c} = 0.2\mathbf{M}$.

	Doubling	Doubling up to a 2-torsion point
Detailed cost	$2\mathbf{M} + 2\mathbf{S} + 1\mathbf{m}_0$	$4\mathbf{S} + 2\mathbf{m}_0$
$\mathbf{m}_0 = \mathbf{M}$	4.33**M**	4.67**M**
$\mathbf{m}_0 = \mathbf{c}$	3.53**M**	3.07**M**

Table 6.1: Comparison of doubling formulas computational cost

For parameters where the multiplication by a curve constant is way faster than a generic multiplication, then our translated doubling is faster. By adapting the Montgomery ladder to take into account the additional 2-torsion point, we can build a new hybrid ladder in Algorithm 6.3. The major change is that, instead of having $\pi(R - S) = \pi(P)$, we allow $\pi(R - S) \in \{\pi(P), \pi(P + T_2)\}$. The correctness of our scalar multiplication is explained below, the formula for the correction step that may occur uses Eq. (6.3).

Algorithm 6.3: Scalar multiplication with hybrid ladder

Input: $n = (b_{\ell-1}, b_{\ell-2}, \dots, b_0)_2$ an ℓ-bits integer, $\pi(P)$ a point on $\mathcal{K}$
Output: $\pi(n \cdot P)$
Data: On $\mathcal{K}$, $\pi(Q) = \pi(P + T_2)$ using Eq. (6.3)

```
def HybridLadder(n, π(P)):
    π(R) ← π(0_E);
    π(S) ← π(P);
    for i ← ℓ − 1 down to 0 do
        π(D) ← π(R − S);                    // Either π(P) or π(Q)
        if b_i = 1 then
            π(R) ← DiffAdd(π(R), π(S), π(D));
            π(S) ← DoublingTranslation(π(S));
        else if b_i = 0 then
            π(S) ← DiffAdd(π(R), π(S), π(D));
            π(R) ← DoublingTranslation(π(R));
        end
    end
    if ℓ ≡ 0 mod 2 or b_0 = 0 then      // π(R) contains π(n · P + T_2)
        return π(R + T_2)
    end
    return π(R);
```

6.4.2 Correctness

Fig. 6.1 at the end of the chapter shows two steps of Algorithm 6.3 and explains why we are cycling between 1 and 2 translated points. We re-use the notations of the algorithm. Since $b_{\ell-1} = 1$, after the first step of the loop, we have computed $\pi(R) = \pi(P)$ and $\pi(S) = \pi(2 \cdot P + T_2)$. Assume we are at a step where we have computed $\pi(R_0) = \pi(m \cdot P)$ and $\pi(S_0) = \pi((m+1) \cdot P + T_2)$, that is one of the two points is translated. As depicted in Fig. 6.1, after one step, both points are shifted by T_2, and after two steps, only one point is translated by T_2.

The case where $\pi(R_0) = \pi(m \cdot P + T_2)$ and $\pi(S_0) = \pi((m+1) \cdot P)$ is analogous.

In the end, we compute correctly $\pi(n \cdot P)$ or $\pi(n \cdot P + T_2)$ by the principle of fast exponentiation. In the second case, we must add a correction step. This case occurs if:

- The loop runs an even number of times, that is when ℓ is even;
- Or, if ℓ is odd, this only happens when the last bit is 0.

6.4.3 Proof of Concept

Now, in the context of ECC, if we are working on a set curve like in ECDSA, we can choose a convenient one such that the associated constants are small.

- In the doubling (Algorithm 2.5), we directly choose d to be small, so $1\mathbf{m}_0 = 1\mathbf{c}$ for this one.

- With the additional 2-torsion point in Algorithm 6.1, the curve constants are $\frac{A}{B}$ and $\frac{A'}{B'}$. We can choose $B = 1$, and that's it because the others are tied, we end up with three constants then: A, A' and B'. In this algorithm, $2\mathbf{m}_0 = 3\mathbf{c}$.

Instead of dealing with the low level libraries' implementation of multiplication to take into account the small constants, we provide a proof of concept as well as verification scripts on GitLab[2]. The context is the following:

- We work over $\mathbb{F}_{p^{10}} = \mathbb{F}_{p^5}[i]$ where $i^2 = -1$ and $\mathbb{F}_{p^5} = \mathbb{F}_p[u]$ where $u^5 = 2$. The extension $\mathbb{F}_{p^{10}}/\mathbb{F}_{p^5}$ is to ensure that $3\mathbf{S} = 2\mathbf{M}$, and the extension $\mathbb{F}_{p^5}/\mathbb{F}_p$ is to have a large extension with trivial multiplication by u and i. A small constant corresponds to an element of $\mathbb{F}_p$. The construction obviously puts some constraints on p ($p \equiv 3 \bmod 4$ and $p \equiv 1 \bmod 5$).
- We choose $A = 1 + \mu i$ and $d = \nu + i$ for some $\mu, \nu \in \mathbb{F}_p$, that way $A' = 2 + \mu i$ and $B' = \mu i$. Multiplication by these constants are faster to deal with than multiplication by generic number over $\mathbb{F}_{p^{10}}$.
- We repeated 100 times 100 random scalar multiplications.

The chosen parameters are the following:

- $p = 14859749208866121031$.
- $\mu = 1141088753069104366$ such that $A' = 1 + \mu i$.
- $\nu = 400659849698428527$ such that $d = \nu + i$.

The results are in Table 6.2 and show that we achieve a 6.2% gain over the Montgomery ladder. For completeness, we also add a comparison with the theta ladder over which we achieve a 4.4%, the doubling formulas correspond to the ones from Algorithm 8.4, and the differential addition can be found in Algorithm 8.3.

	Montgomery ladder	Theta ladder	Hybrid ladder
Average (s)	2.400 ± 0.049	2.344 ± 0.009	2.241 ± 0.005

Table 6.2: Timings on Intel Core i5-1145G7 @ 2.60GHz

Remark 6.4.1 *Since in our application $\pi(R - S)$ varies between two constants, we could add a constraint for this one to be small too and that improves the whole time saved. However, this is not necessary for our comparison as the differential addition is the same in both ladders.*

6.5 MORE EXAMPLES OF 2-ISOGENIES

In this section, we will look at two other classical Kummer lines models: the Legendre and theta ones.

2 https://gitlab.inria.fr/nsarkis/poc-scalar-multiplication-kummer-lines

6.5.1 Legendre model

An elliptic curve is said to be in Legendre form if it has full rational 2-torsion and is then put in the following shape:

$$E : By^2 = x(x-1)(x-\gamma), \gamma \in k.$$

In terms of Kummer lines, this is a particular case of Main example 6.2.1. The ramification is as follows:

$$0_E = (1:0)^*, \quad T_1 = (0:1), \quad T_2 = (1:1), \quad T_3 = (\gamma:1).$$

We will focus on 2-isogenies with kernel T_1. Our goal is to recover the Montgomery to Legendre and Legendre to Montgomery isogeny formulas from [Ber+08b, Theorem 5.1].

Example 6.5.1 (Montgomery to Legendre) *Suppose our initial Kummer line $\mathcal{K}$ is a Montgomery one with the following ramification:*

$$0_E = (1:0)^*, \quad T_1 = (0:1), \quad T_2 = (A:B), \quad T_3 = (B:A).$$

$\frac{A}{B}$ may not be rational. We also know about the 4-torsion points above T_1, which are $\overline{T_1} = (1:1)$ and $\underline{T_1} = (-1:1)$.

We can use the invariants computed in Section 6.3.1. Set

$$f : (X:Z) \mapsto \left((X+Z)^2 : (X-Z)^2\right),$$

it is T_1-invariant and the ramification on the codomain is:

- $f(0_E) = f(T_1) = (1:1)^*$,
- $f(T_2) = f(T_3) = ((A+B)^2 : (A-B)^2)$,
- $f(\overline{T_1}) = (1:0)$,
- $f(\underline{T_1}) = (0:1)$.

We notice that this is exactly a Legendre model with $\gamma' = \frac{(A+B)^2}{(A-B)^2}$ up to translation by a 2-torsion point. As already explained, $f(T_2) = f(T_3)$ is always rational. The ramification of the codomain is:

$$0_{E'} = (1:0), \quad T'_1 = (0:1), \quad T'_2 = (1:1)^*, \quad T'_3 = (\gamma':1).$$

The 2-isogeny is then $\varphi = f + T'_2$ and can be computed in $2\mathbf{S} + 2\mathbf{a}$, γ' can be computed in $2\mathbf{S} + 2\mathbf{a}$.

Another idea is to use invariants $(X+Z)^2$ and XZ instead. Set

$$\varphi_0 : (X:Z) \mapsto ((X+Z)^2 : XZ),$$

the ramification on the codomain this time is:

- $\varphi_0(0_E) = \varphi_0(T_1) = (1:0)^*$,
- $\varphi_0(T_2) = \varphi_0(T_3) = ((A+B)^2 : AB)$,

- $\varphi_0(\overline{T_1}) = (4:1)$,
- $\varphi_0(\underline{T_1}) = (0:1)$.

We have to change the shape of our ramification, we are looking for a homography h *such that* $h(1:0) = (1:0)$, $h(4:1) = (1:1)$ *and* $h(0:1) = (0:1)$. *We find that* $h(X:Z) = (X:4Z)$ *satisfies these conditions. We set* $\varphi = h \circ \varphi_0$ *and*

$$\gamma' = h((A+B)^2 : AB) = \frac{(A+B)^2}{4AB}.$$

The ramification on the codomain is:

$$0_{E'} = (1:0)^*, \quad T_1' = (0:1), \quad T_2' = (1:1), \quad T_3' = (\gamma':1).$$

This can be computed in $2\mathbf{S} + 3\mathbf{a}$ *using* $4XZ = (X+Z)^2 - (X-Z)^2$. γ' *is also computed in* $2\mathbf{S} + 3\mathbf{a}$.

Example 6.5.2 (Legendre to Montgomery) *On the other hand, it is also possible to go from a Legendre model to a Montgomery one via a 2-isogeny. Suppose our initial Kummer line* $\mathcal{K}$ *is a Legendre one with the following ramification:*

$$0_E = (1:0)^*, \quad T_1 = (0:1), \quad T_2 = (1:1), \quad T_3 = (\gamma:1).$$

In Main example 6.2.9, we computed the following quadratic forms that are T_1*-invariant:*

$$u(X,Z) = X^2 + \gamma Z^2, \quad v(X,Z) = XZ.$$

In Main example 6.2.10, we also computed the 4-torsion above T_1*:*

$$\overline{T_1} = (\sqrt{\gamma}:1), \quad \underline{T_1} = (-\sqrt{\gamma}:1).$$

First, set $\varphi_0 : (X:Z) \mapsto (X^2 + \gamma Z^2 : XZ)$, *it is* T_1*-invariant and the ramification on the codomain is:*

- $\varphi_0(0_E) = \varphi_0(T_1) = (1:0)^*$,
- $\varphi_0(T_2) = \varphi_0(T_3) = (1+\gamma:1)$,
- $\varphi_0(\overline{T_1}) = (2\sqrt{\gamma}:1)$,
- $\varphi_0(\underline{T_1}) = (-2\sqrt{\gamma}:1)$.

To recover a Montgomery Kummer line, we also need a 4-torsion point. Set $\overline{T_2} = (r:s)$ *to be a 4-torsion point above* T_2. *Let* σ *be an element of the Galois group of our field* k. *Then either* $\sigma(\overline{T_2}) = \overline{T_2}$, *or* $\sigma(\overline{T_2}) = \underline{T_2}$ *is another 4-torsion point above* T_2 *because* $2 \cdot \sigma(\overline{T_2}) = \sigma(T_2) = T_2$ *(the equalities are on the Kummer line). On the Kummer line, since* $\overline{T_2} + T_2$ *and* $\overline{T_2}$ *are equal, we must have* $\underline{T_2} = \overline{T_2} + T_1 = \overline{T_2} + T_3$. *Hence,* $\sigma(\varphi_0(\overline{T_2})) = \varphi_0(\sigma(\overline{T_2})) = \varphi_0(\overline{T_2} + T_1) = \varphi_0(\overline{T_2})$. *In all cases,* $\varphi_0(\overline{T_2})$ *is invariant by Galois.*

The translation by T_2 *is given by* $t_{T_2} : (X:Z) \mapsto (X - \gamma Z : X - Z)$. *Because* $\overline{T_2} + T_2 = \overline{T_2}$, *we find using* t_{T_2} *that* $r^2 + \gamma s^2 = 2rs$. *Then* $\varphi_0(\overline{T_2}) = (2:1)$.

We now compute a homography $h : (X:Z) \mapsto (aX + bZ : cX + dZ)$ *to go back to a Montgomery model. It must verify:*

- $h(1:0) = (1:0)$, *i.e.* $c = 0$;
- $h(1+\gamma:1) = (0:1)$, *i.e.* $b = -a(1+\gamma)$;
- $h(2:1) = (1:1)$, *i.e.* $d = 2a + b = a(1-\gamma)$.

This yields $h(X:Z) = (X-(1+\gamma)Z : (1-\gamma)Z)$. *One can check that:*

$$h(2\sqrt{\gamma}:1) = (\sqrt{\gamma}-1 : \sqrt{\gamma}+1), \quad h(-2\sqrt{\gamma}:1) = (\sqrt{\gamma}+1 : \sqrt{\gamma}-1).$$

We end up on the following Montgomery model with the 2-isogeny $\varphi = h \circ \varphi_0$*:*

$$0_{E'} = (1:0)^*, \quad T'_1 = (0:1), \quad T'_2 = (\sqrt{\gamma}-1 : \sqrt{\gamma}+1), \quad T'_3 = (\sqrt{\gamma}-1 : \sqrt{\gamma}+1).$$

6.5.2 Theta model

In this section we look at another model where the neutral point is not at infinity this time. Let $a, b \in k$ be two constants that will define our ramification:

$$0_E = (a:b)^*, \quad T_1 = (-a:b), \quad T_2 = (b:a), \quad T_3 = (-b:a).$$

This is called the theta model with theta constants $(a:b)$, we will again focus on 2-isogenies with kernel T_1. It is derived from theta functions form Chapter 5, the ramification has been computed in Example 5.3.8.

We first want to compute potential 4-torsion points, we have:

$$t_{T_1}(X:Z) \mapsto (-X:Z), \quad t_{T_2} : (X:Z) \mapsto (Z:X), \quad t_{T_3} : (X:Z) \mapsto (-Z:X).$$

If $\overline{T_1} = (X:Z)$ is a 4-torsion point above T_1, we want to solve $\overline{T_1} + T_1 = \overline{T_1}$. With a similar approach, these are the 4-torsion points on this model:

- Above T_1: $\overline{T_1} = (1:0)$ and $\underline{T_1} = (0:1)$.
- Above T_2: $\overline{T_2} = (1:1)$ and $\underline{T_2} = (-1:1)$.
- Above T_3: $\overline{T_3} = (i:1)$ and $\underline{T_3} = (-i:1)$ with $i^2 = -1$.

Aside $\overline{T_3}$ and $\underline{T_3}$ which may not be rational, there are always two rational independent 4-torsion points on this model: $\overline{T_1}$ and $\overline{T_2}$. This is one more occurrence of a Montgomery model where this time two points of 2-torsion are required to be of Montgomery type. In the theta model, the ramification is invariant by $(X:Z) \mapsto (Z:X)$ as in the Montgomery model, but also by $(X:Z) \mapsto (-X:Z)$. In particular the full 2-torsion is always rational in the theta model.

The matrix associated to t_{T_1} is $M = \left(\begin{smallmatrix} -1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and $M^2 = I_2$, so the type is 1 as expected and M acts as:

$$M \cdot X^2 = X^2, \quad M \cdot Z^2 = Z^2, \quad M \cdot XZ = -XZ.$$

Example 6.5.3 (Theta to Montgomery) *We will use* X^2 *and* Z^2 *as the invariants, set* $f : (X:Z) \mapsto (X^2 : Z^2)$. *A quick computation yields:*

- $f(0_E) = f(T_1) = (a^2 : b^2)^*$,
- $f(T_2) = f(T_3) = (b^2 : a^2)$,
- $f(\overline{T_1}) = (1 : 0)$,
- $f(\underline{T_1}) = (0 : 1)$.

We also have $f(\overline{T_2}) = (1 : 1)$ *and* $f(\overline{T_3}) = (-1 : 1)$, *the 4-torsion above* $(1 : 0)$. *The ramification is Montgomery shaped up to a translation, the 2-isogeny is then* $\varphi = f + T_2'$ *with:*

$$0_{E'} = (1 : 0), \quad T_1' = (0 : 1), \quad T_2' = (a^2 : b^2)^*, \quad T_3' = (b^2 : a^2).$$

f *can be computed in* $2\mathbf{S}$, *the codomain in* $2\mathbf{S}$ *too.*

This can be used to find doubling formulas on the theta model. If $\widetilde{\varphi}$ *is the dual of* φ, *we have that* $\ker \widetilde{\varphi} = \langle T_1' \rangle$. *We use the same invariants as in Theorem 6.3.3 because we start on a Montgomery model, and we set* $\widetilde{\varphi}_0 : (X : Z) \mapsto ((X + Z)^2 : (X - Z)^2)$. *One computes, with* $(A^2 : B^2) = (a^2 + b^2 : a^2 - b^2)$:

- $\widetilde{\varphi}_0(0_{E'}) = \widetilde{\varphi}_0(T_1') = (1 : 1)^*$,
- $\widetilde{\varphi}_0(T_2') = \widetilde{\varphi}_0(T_3') = (A^4 : B^4)$,
- $\widetilde{\varphi}_0(\overline{T_1'}) = (1 : 0)$,
- $\widetilde{\varphi}_0(\underline{T_1'}) = (0 : 1)$.

Because we want to compute the dual, we are aiming for the following equations:

- $\widetilde{\varphi}(\varphi(0_E)) = 0_E$ *i.e.* $\widetilde{\varphi}(0_{E'}) = (a : b)$;
- $\widetilde{\varphi}(\varphi(\overline{T_3})) = T_3$ *i.e.* $\widetilde{\varphi}(\overline{T_1'}) = (-b : a)$;
- $\widetilde{\varphi}(\varphi(\overline{T_2})) = T_2$ *i.e.* $\widetilde{\varphi}(\underline{T_1'}) = (b : a)$;
- $\widetilde{\varphi}(\varphi(\overline{T_1})) = T_1$ *i.e.* $\widetilde{\varphi}(T_2) = (-a : b)$.

As usual, we look for a homography h *such that* $\widetilde{\varphi} = h \circ \widetilde{\varphi}_0$. *Using the first three equations, we find that:*

$$h : (X : Z) \mapsto (b(B^2X + A^2Z) : a(-B^2X + A^2Z)).$$

We finally check that indeed $h(A^4 : B^4) = (-a : b)$, *and therefore we have* $\widetilde{\varphi} = h \circ \widetilde{\varphi}_0$. *Because* $\widetilde{\varphi}(\varphi(P)) = 2 \cdot P$ *and* $\widetilde{\varphi}(T_2') = T_1$, *we can compute* $2 \cdot P + T_1 = \widetilde{\varphi} \circ f(P)$ *in* $4\mathbf{S} + 2\mathbf{m}_0 + 2\mathbf{a}$, *which is essentially* $2 \cdot P$ *because* $t_{T_1}(X : Z) = (-X : Z)$.

Example 6.5.4 (Theta to theta) *We recall the theta model here:*

$$0_E = (a : b)^*, \quad T_1 = (-a : b), \quad T_2 = (b : a), \quad T_3 = (-b : a).$$

Assume in this example that we have a 8-torsion point $\widetilde{T_1} = (r : s)$ *above* $\overline{T_1}$. *Using invariants* X^2 *and* Z^2, *we set* $\varphi_0(X : Z) = (X^2 + Z^2 : X^2 - Z^2)$ *and once again* $(A^2 : B^2) = (a^2 + b^2 : a^2 - b^2)$. *One computes:*

- $\varphi_0(0_E) = \varphi_0(T_1) = (A^2 : B^2)^*$,

- $\varphi_0(T_2) = \varphi_0(T_3) = (-A^2 : B^2)$,
- $\varphi_0(\overline{T_1}) = (1 : 1)$,
- $\varphi_0(\underline{T_1}) = (-1 : 1)$.

The 4-torsion is $\varphi_0(\widetilde{T_1}) = (r^2 + s^2 : r^2 - s^2) := (u : v)$, $\varphi_0(\overline{T_2}) = (1 : 0)$ *and* $\varphi_0(\overline{T_3}) = (0 : 1)$.

By setting $h : (X : Z) \mapsto (BX : AZ)$, *the ramification is put in the correct shape. It remains to check that* $(A : B)$ *is indeed rational in this context. To do so, we will look at the 4-torsion on the intermediate model, set:*

$$0_{E_0} = (A^2 : B^2)^*, \quad T_1^0 = (-A^2 : B^2), \quad T_2^0 = (1 : 1), \quad T_3^0 = (-1 : 1).$$

We then have $\overline{T_1^0} = (1 : 0)$, $\underline{T_1^0} = (0 : 1)$ *and* $\overline{T_2^0} = (u : v)$ *by the 2-isogeny. On this model, the translation by* $\overline{T_2}$ *is* $t_{T_2} : (X : Z) \mapsto (A^2Z : B^2X)$. *The 4-torsion verifies* $\overline{T_2^0} + T_2^0 = \overline{T_2^0}$, *therefore:*

$$(A^2v : B^2u) = (u : v) \iff \frac{u}{v} = \pm\frac{A}{B}.$$

Hence, $(A : B)$ *is rational, so is* h *and* $\varphi = h \circ \varphi_0$ *which gives the following theta model:*

$$0_{E'} = (A : B)^*, \quad T_1' = (-A : B), \quad T_2' = (B : A), \quad T_3' = (-B : A).$$

The 4-torsion is given by $\overline{T_1'} = \varphi(\overline{T_2}) = (1 : 0)$, $\underline{T_1'} = \varphi(\overline{T_3}) = (0 : 1)$ *and* $\overline{T_2'} = \varphi(\widetilde{T_1}) = (1 : 1)$ *when we choose* $(A : B) = (u : v)$.

We recover the usual duplication formula on theta coordinates. Since the codomain is in the theta model, we can also easily compute the dual isogeny by swapping $(a : b)$ *and* $(A : B)$ *in the formulas, and recover the doubling formulas from [GL09].*

6.6 COMPUTING 2^n-ISOGENIES

As explained in Section 3.3.3, 2^n-isogenies can be computed by decomposing them into a product of 2-isogenies, which is often required in isogeny-based cryptographic protocols, see for instance SIDH [FJP14].

Consider a point P_0 of 2^n-torsion on an elliptic curve E_0, and say one wants to compute the 2^n-isogeny with kernel generated by P_0. To do so by chaining, one wants to divide the order by 2 at each step, which can be achieved by:

- either computing $2 \cdot P_0$, in which case we stay on the curve E_0;
- or computing the image of P_0 via the 2-isogeny of kernel $2^{n-1} \cdot P_0$, in which case we end up on some curve E_1.

We then have two important operations:

- doubling;
- image by a 2-isogeny.

One thing to keep in mind is that we have to do operations in the correct order, it is not possible to compute an image without the kernel, and it is not possible to compute a doubling without the curve constant.

As a first step, one always needs to compute every $2^i \cdot P_0$. Then a naive approach would be to compute the 2-isogeny with kernel $2^{n-1} \cdot P_0$, compute every image, and repeat this process on the new curve. One could also only compute the image of P_0, compute every doubling of the new point P_1 on the new curve and repeat the process. It is convenient to represent such strategies as trees, like in Figs. 6.2a and 6.2b at the end of the chapter. The images are represented vertically, and the doublings horizontally. The leaves are 2-torsion points on the corresponding curve.

This is obviously not optimal however, too many useless points are computed, and we end up with $O(n^2)$ operations for a 2^n-isogeny. In their paper [FJP14, § 4.2.2], De Feo, Jao and Plut explain how to find optimal strategies, taking into account the relative cost of a doubling compared to an image. An example is given in Fig. 6.2c, where using a binary tree gives $O(n \log n)$ operations for a 2^n-isogeny.

In this section, we focus on 2^n-isogenies where the intermediate Kummer lines are given by Montgomery models. We will denote them as $\mathcal{K}_i$ with $i \geqslant 0$, so $\mathcal{K}_0$ is the initial Kummer line, and the ramification will be denoted as follows:

$$0_{E_i} = (1:0)^*, \quad T_1^i = (0:1), \quad T_2^i = (A_i : B_i), \quad T_3^i = (B_i : A_i) = T_2^i + T_1^i.$$

As shown in Section 6.3.2, the point $T_2^i = (A_i : B_i)$ can be used for the translated doubling formulas on $\mathcal{K}_i$. It can also be used to recover the curve constant for standard doubling, it is always rational, even if T_2^i and T_3^i are not:

$$d_i = \frac{(A_i - B_i)^2}{(A_i - B_i)^2 - (A_i + B_i)^2}.$$

We also denoted earlier $\overline{T_2^i} = (a_i' : b_i')$ the 4-torsion point above T_2^i and $(a_i : b_i) = (a_i' + b_i' : a_i' - b_i')$.

We will focus on the case where P_0 is above the 2-torsion point T_2^0, as we want to compare it to Renes formulas provided in [Ren18, Prop. 4.2], and for simplicity we neglect the cost of the additions. Recall the formulas from Theorem 6.3.7, let φ_i be the 2-isogeny with kernel T_2^i. Then the shifted isogeny is $f^i = \varphi_i + T_2^{i+1}$ where $T_2^{i+1} = (a_i'^2 : b_i'^2)$. The formula for $\varphi_i : \mathcal{K}_i \to \mathcal{K}_{i+1}$ is

$$(X:Z) \mapsto \left((b_i(X+Z) + a_i(X-Z))^2 : (b_i(X+Z) - a_i(X-Z))^2\right).$$

The codomain $\mathcal{K}_{i+1}$ is represented by T_2^{i+1} and can be computed in $2\mathbf{S}$, and the translated image costs $2\mathbf{M} + 2\mathbf{S}$.

Since $T_2^{i+1} = f^i(\overline{T_2^i})$ is the kernel of the next isogeny f^{i+1}, computing translated images by T_2^{i+1} does not matter, except at the very last step where we can use the standard non translated formula instead. Thus, similarly to what was done in Section 6.4, we can build a hybrid algorithm which combines Montgomery doubling (standard or translated by T_2^i) and our translated image formula.

Remark 6.6.1 *As we can see, our image formula only involves the 4-torsion point $\overline{T_2^i}$ above T_2^i (we can also recover from it the constant d_i), so the leaves in our strategy tree will be the 4-torsion points instead of the 2-torsion ones. This also implies that if we want to compute a 2^n-isogeny, we have to assume we are given a 2^{n+1}-torsion point, or we do the last step with Renes formulas which doesn't have this constraint.*

What matters now is to compare the standard costs operations from [Ren18, Prop. 4.2] with the ones provided in Section 6.3.2, this is done in Table 6.3. Unlike in Section 6.4, after the first 2-isogeny, we don't have control on curve constants any more, so $\mathbf{m}_0$ must be counted as two generic multiplications because of the numerator and the denominator.

Operation	Doubling		Image	
	[Mon87]	Proposition 6.3.4	[Ren18]	Theorem 6.3.7
Cost	$2\mathbf{M} + 2\mathbf{S} + 1\mathbf{m}_0$	$4\mathbf{S} + 2\mathbf{m}_0$	$2\mathbf{M} + 1\mathbf{m}_0$	$2\mathbf{S} + 1\mathbf{m}_0$
Cost ($\mathbf{m}_0 = 2\mathbf{M}$)	$4\mathbf{M} + 2\mathbf{S}$	$4\mathbf{M} + 4\mathbf{S}$	$4\mathbf{M}$	$2\mathbf{M} + 2\mathbf{S}$
Constants used	d_i	$(A_i : B_i), (A_{i+1} : B_{i+1})$	$(A_i : B_i)$	$(a_i : b_i)$

Table 6.3: Comparison of operations on Kummer lines to compute 2^n-isogenies

We see that our formulas for images should be faster than the ones by Renes. The cost of the codomain is more tricky: Renes formulas directly give d_{i+1} in $2\mathbf{S}$. Our formulas give T_2^{i+1} in $2\mathbf{S}$; but from T_2^{i+1} we can only use our translated doubling formulas, which are in this context more expensive than the standard doubling formulas. Thus, we need to compute d_{i+1} from T_2^{i+1} which costs $2\mathbf{S}$ by the formula

$$d_{i+1} = \frac{(a_i'^2 - b_i'^2)^2}{(a_i'^2 - b_i'^2)^2 - (a_i'^2 + b_i'^2)^2},$$

for a total codomain cost of $4\mathbf{S}$.

Asymptotically, since there are exactly n codomains to compute, they are negligible compared to images and doublings which are in $O(n \log n)$. An implementation to compute a 2^n-isogeny using this hybrid method with SIKEp434 parameters is available in the same GitLab repository as earlier and shows that we do end on the same curve as the one with Renes formulas. For these parameters, our implementation shows that our hybrid method is slower, because the n considered is not large enough that the faster images compensate the slower codomains.

Another reason why this would not be viable anyway is that there exists efficient 4-isogeny formulas [CH17, § A], which saves half of the steps while having competitive costs for multiplication by 4 and images, hence are much faster. Indeed, the 4-isogeny codomain costs $4\mathbf{S}$ (computing d_{i+2} from d_i and $\overline{T_2^i}$), and a 4-isogeny image costs $6\mathbf{M} + 2\mathbf{S}$. We remark that this is the same cost as combining our translated 2-isogeny image with the standard 2-isogeny image. In particular, by composing our translated 2-isogeny image twice, there is also a translated (by a 2-torsion point) 4-isogeny image in only $4\mathbf{M} + 4\mathbf{S}$. However, to be able to use these translated images, our codomain formula would be slower.

Only in some hypothetical context where we would need to compute a lot of images — assuming we already know the codomains — the hybrid approach would be faster.

6.7 ODD DEGREE ISOGENIES ON KUMMER LINES

In this section, we extend the work of [Ren18] to build isogenies of odd degrees on any model of a Kummer line.

Let E be an elliptic curve, and K be a cyclic kernel of odd degree ℓ, and $f : E \to E'$ the corresponding isogeny. To build a model of the Kummer line associated to $E' = E/K$, we need to build sections of $2(0_{E'})$, hence invariant sections of $f^*(2(0_{E'})) = \sum_{T \in K} 2(T)$ on E.

If s is an invariant section, its associated divisor $\operatorname{div} s$ is invariant. The converse is not true, there is an obstruction coming from the Weil-Cartier pairing.

Lemma 6.7.1 *Consider a principal divisor D invariant by K associated to a section s_D and decompose it as*

$$D = \operatorname{div} s_D := \sum_{i \in I} a_i \sum_{T \in K} (P_i + T)$$

for a finite set of points of $\{P_i \in E \mid i \in I\}$. Set $Q := \sum_{i \in I} a_i P_i$. Then s_D is invariant by translation if and only if $Q \in K$.

Proof *Since $\operatorname{div} s_D$ is invariant by K, if $T \in K$, the function $s_D(\cdot + T)$ has the same divisor as s_D, hence differ by a constant. By definition of the Weil-Cartier pairing e_f, this constant is precisely equal to $e_f(T, f(Q))$. So s_D is invariant by K if and only if $Q \in E[\ell]$ is orthogonal to K, if and only if $Q \in K$, if and only if $f(Q) = 0_{E'}$.*

Another equivalent proof is to remark that s_D is invariant by translation if and only if D descends to a divisor $D' = \sum_{i \in I} a_i \sum_{T \in K} (f(P_i))$ on E' which is linearly equivalent to 0, which is the case if and only if $Q \in K$. □

Example 6.7.2 *Take $Q_1, Q_2 \in E(k)$, $s_D = \prod_{T \in K} \frac{x - x(Q_1+T)}{x - x(Q_2+T)}$ (we use the convention that $x - x(0_E) := 1$). Its associated divisor is*

$$D = \sum_{T \in K} ((Q_1 + T) + (-Q_1 + T) - (Q_2 + T) - (-Q_2 + T)).$$

Then s_D is invariant by translation and descends to $\frac{x - f(Q_1)}{x - f(Q_2)}$ on E/K, x a Weierstrass coordinate. When $Q_2 = 0_E$, we recover a formula from [CH17; Ren18].

As illustrated by Example 6.7.2, we can use Lemma 6.7.1 to construct divisors associated to an invariant section. From such a divisor we can use Miller's algorithm to construct the associated section s. Since the isogeny is of odd degree, it preserves the 2-torsion, so by evaluating s on the ramification point of the Kummer model of E we can efficiently recover the Kummer model of E' given by s.

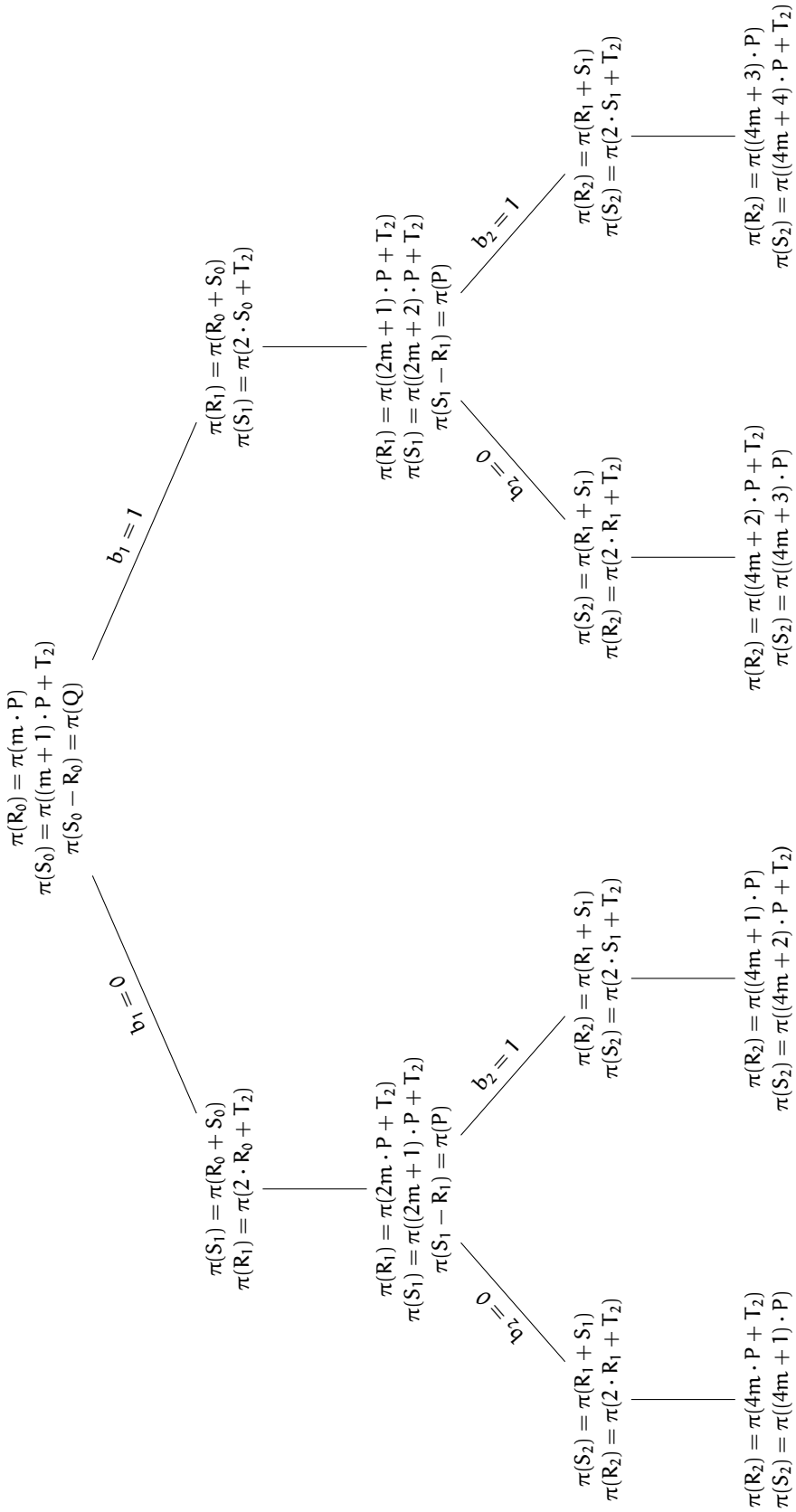

Figure 6.1: Two steps of scalar multiplication based on hybrid ladder

$$
\begin{tikzcd}
E_0 \arrow[r, "\varphi_0"] & E_1 \arrow[r, "\varphi_1"] & E_2 \arrow[r, "\varphi_2"] & E_3 \arrow[r, "\varphi_3"] & E_4 \arrow[r, "\varphi_4"] & E_5 \arrow[r, "\varphi_5"] & E_6 \arrow[r, "\varphi_6"] & E_7 \\
P_0 \arrow[r] \arrow[d] & P_1 \arrow[r] & P_2 \arrow[r] & P_3 \arrow[r] & P_4 \arrow[r] & P_5 \arrow[r] & P_6 & \\
2 \cdot P_0 \arrow[r] \arrow[d] & 2 \cdot P_1 \arrow[r] & 2 \cdot P_2 \arrow[r] & 2 \cdot P_3 \arrow[r] & 2 \cdot P_4 \arrow[r] & 2 \cdot P_5 & & \\
2^2 \cdot P_0 \arrow[r] \arrow[d] & 2^2 \cdot P_1 \arrow[r] & 2^2 \cdot P_2 \arrow[r] & 2^2 \cdot P_3 \arrow[r] & 2^2 \cdot P_4 & & & \\
2^3 \cdot P_0 \arrow[r] \arrow[d] & 2^3 \cdot P_1 \arrow[r] & 2^3 \cdot P_2 \arrow[r] & 2^3 \cdot P_3 & & & & \\
2^4 \cdot P_0 \arrow[r] \arrow[d] & 2^4 \cdot P_1 \arrow[r] & 2^4 \cdot P_2 & & & & & \\
2^5 \cdot P_0 \arrow[r] \arrow[d] & 2^5 \cdot P_1 & & & & & & \\
2^6 \cdot P_0 & & & & & & &
\end{tikzcd}
$$

(a) Images approach

$$
\begin{tikzcd}
E_0 \arrow[r, "\varphi_0"] & E_1 \arrow[r, "\varphi_1"] & E_2 \arrow[r, "\varphi_2"] & E_3 \arrow[r, "\varphi_3"] & E_4 \arrow[r, "\varphi_4"] & E_5 \arrow[r, "\varphi_5"] & E_6 \arrow[r, "\varphi_6"] & E_7 \\
P_0 \arrow[r] \arrow[d] & P_1 \arrow[r] \arrow[d] & P_2 \arrow[r] \arrow[d] & P_3 \arrow[r] \arrow[d] & P_4 \arrow[r] \arrow[d] & P_5 \arrow[r] \arrow[d] & P_6 & \\
2 \cdot P_0 \arrow[d] & 2 \cdot P_1 \arrow[d] & 2 \cdot P_2 \arrow[d] & 2 \cdot P_3 \arrow[d] & 2 \cdot P_4 \arrow[d] & 2 \cdot P_5 & & \\
2^2 \cdot P_0 \arrow[d] & 2^2 \cdot P_1 \arrow[d] & 2^2 \cdot P_2 \arrow[d] & 2^2 \cdot P_3 \arrow[d] & 2^2 \cdot P_4 & & & \\
2^3 \cdot P_0 \arrow[d] & 2^3 \cdot P_1 \arrow[d] & 2^3 \cdot P_2 \arrow[d] & 2^3 \cdot P_3 & & & & \\
2^4 \cdot P_0 \arrow[d] & 2^4 \cdot P_1 \arrow[d] & 2^4 \cdot P_2 & & & & & \\
2^5 \cdot P_0 \arrow[d] & 2^5 \cdot P_1 & & & & & & \\
2^6 \cdot P_0 & & & & & & &
\end{tikzcd}
$$

(b) Doublings approach

$$
\begin{tikzcd}
E_0 \arrow[r, "\varphi_0"] & E_1 \arrow[r, "\varphi_1"] & E_2 \arrow[r, "\varphi_2"] & E_3 \arrow[r, "\varphi_3"] & E_4 \arrow[r, "\varphi_4"] & E_5 \arrow[r, "\varphi_5"] & E_6 \arrow[r, "\varphi_6"] & E_7 \\
P_0 \arrow[r] \arrow[d] & P_1 \arrow[r] & P_2 \arrow[r] & P_3 \arrow[r] & P_4 \arrow[r] \arrow[d] & P_5 \arrow[d] & P_6 & \\
2 \cdot P_0 \arrow[d] & & & & 2 \cdot P_4 \arrow[d] & 2 \cdot P_5 & & \\
2^2 \cdot P_0 \arrow[d] & & & & 2^2 \cdot P_4 & & & \\
2^3 \cdot P_0 \arrow[r] & 2^3 \cdot P_1 \arrow[r] & 2^3 \cdot P_2 \arrow[r] & 2^3 \cdot P_3 & & & & \\
2^4 \cdot P_0 \arrow[d] & 2^4 \cdot P_1 \arrow[d] & 2^4 \cdot P_2 & & & & & \\
2^5 \cdot P_0 \arrow[d] & 2^5 \cdot P_1 & & & & & & \\
2^6 \cdot P_0 & & & & & & &
\end{tikzcd}
$$

(c) Optimized approach

Figure 6.2: 2^7-isogeny $\varphi = \varphi_6 \circ \cdots \circ \varphi_0$ with kernel P_0. Images are represented horizontally and doublings vertically.

7 HALF DIFFERENTIAL ADDITIONS ON KUMMER LINES

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This chapter corresponds to [RS25] presented at Eurocrypt 2025. It is a direct sequel to Chapter 6, where we looked for 2-isogenies formulas in order to factor the doubling. Here, we extend this framework to find differential addition formulas on models of Kummer lines, with a particular focus on formulas (called half differential additions) that factor through a 2-isogeny φ. The core idea is to re-use the computation of $\varphi(P)$ and $\varphi(Q)$ which happens during the doubling to also determine $P+Q$ on the Kummer line. A half doubling is the special case where $P = Q$, in which case we have $2 \cdot P = \widetilde{\varphi}(\varphi(P))$, so half doubling amounts to applying the dual isogeny. Combined with a time-memory trade-off to compute all needed isogeny images once, this leads to a new ladder — called half ladder — which is competitive with the Montgomery one. Instead of performing doublings and differential additions at the same time, we first pre-compute images of our base point P via 2-isogenies $\varphi_1, \dots, \varphi_\ell$, and we then go backwards with duals and half differential addition formulas to recover $n \cdot P$. A comparison is available in Table 7.1 below, where we use notations introduced in Sections 2.1 and 2.3 to count operations. We stress that the half ladder formulas cost that we give are only available on Montgomery curves with full rational 2-torsion, whereas the standard ladder formulas are available on all Montgomery curves. On the other hand, for these curves, our half ladder allows us to

gain a $1\mathbf{m}_0 - 1\mathbf{m}$ trade-off on a normalized base point P, and a $1\mathbf{m}_0 - 2\mathbf{M}$ trade-off on a non normalized base point.

	Montgomery ladder	Half ladder, our contribution
Non-normalized base point	$6\mathbf{M} + 4\mathbf{S} + 1\mathbf{m}_0$	$4\mathbf{M} + 4\mathbf{S} + 2\mathbf{m}_0$
Normalized base point	$4\mathbf{M} + 4\mathbf{S} + 1\mathbf{m} + 1\mathbf{m}_0$	

Table 7.1: Ladder costs per bit with no pre-computation

As in the previous chapter, we will use Mumford's theta group theory (see Chapter 5) to prove the existence of formulas and to determine them. If $\varphi : E \to E'$ is a 2-isogeny on elliptic curves, we relate sections above the divisor $2(0_{E'}) * 2(0_{E'})$ of $E' \times E'$ compatible with the diagonal isogeny $\Phi = (\varphi, \varphi)$ with sections above the divisor $2(0_E) * 2(0_E)$ of $E \times E$ compatible with the differential addition isogeny $F : (P, Q) \mapsto (P + Q, P - Q)$. Generators of the relations between the sections are what we call half differential addition formulas.

Related work

In [Oli+17, Alg. 4], the authors provide a variant of the Montgomery ladder, performing the operation from right-to-left (RtL), instead of the traditional left-to-right (LtR). This approach implies a pre-computation of points of the form $2^i \cdot P$. Since our half ladder also contains a form of pre-computation, it is more relevant to compare to this version. Tables 7.2 and 7.3 compare the pre-computation of Montgomery ladder right-to-left and our half ladder with our best formulas, which happens over a Montgomery curve with full rational 2-torsion. It appears that on each step we lose $1\mathbf{m}_0$, but our pre-computation is significantly faster in both cases, saving $2\mathbf{M}$. Moreover, our approach generalizes well in higher dimension as explained in Section 7.6, whereas the natural generalization of the RtL ladder is not interesting, to the best of our knowledge, in dimension $g > 1$.

Algorithm	Pre-computation	Step
Montgomery ladder LtR	—	$4\mathbf{M} + 4\mathbf{S} + 1\mathbf{m} + 1\mathbf{m}_0 + 8\mathbf{a}$
Montgomery ladder RtL	$2\mathbf{M} + 2\mathbf{S} + 1\mathbf{m}_0 + 4\mathbf{a}$	$4\mathbf{M} + 2\mathbf{S} + 6\mathbf{a}$
Half ladder, our contribution	$2\mathbf{S} + 1\mathbf{m}_0 + 2\mathbf{a}$	$4\mathbf{M} + 2\mathbf{S} + 1\mathbf{m}_0 + 4\mathbf{a}$

Table 7.2: Ladder costs per bit with a pre-computation but no normalization

Algorithm	Pre-computation	Normalization	Step
Montgomery Ladder LtR	—	—	$4\mathbf{M} + 4\mathbf{S} + 1\mathbf{m} + 1\mathbf{m}_0$
Montgomery Ladder RtL	$2\mathbf{M} + 2\mathbf{S} + 1\mathbf{m}_0$	$1\mathbf{I} + 1\mathbf{M} \overset{\text{asy}}{=} 4\mathbf{M}$	$3\mathbf{M} + 2\mathbf{S}$
Half ladder, our contribution	$2\mathbf{S} + 1\mathbf{m}_0$	$1\mathbf{I} + 1\mathbf{M} \overset{\text{asy}}{=} 4\mathbf{M}$	$3\mathbf{M} + 2\mathbf{S} + 1\mathbf{m}_0$

Table 7.3: Ladder costs per bit when normalizing the pre-computation, the addition costs are the same as in Table 7.2. $1\mathbf{I}$ reduces to $3\mathbf{M}$ thanks to Lemma 2.1.6.

Our contributions

In summary, our contributions are first a systematic way to derive half differential addition formulas using Mumford's theory of the theta group, and secondly the existence of a new kind of pre-computed Montgomery like ladder: the half ladder.

As described in more details in Section 7.3, the key principle behind the half ladder is as follows. Using half doubling and half differential addition formulas, one way to compute the Montgomery ladder is at each step to start with $U_{i-1} = m_{i-1} \cdot P, V_{i-1} = (m_{i-1} + 1) \cdot P$, compute $\varphi(U_{i-1}), \varphi(V_{i-1})$, and use one half doubling and one half differential addition to recover $2 \cdot U_i$ and $U_i + V_i$ (or $U_i + V_i$ and $2 \cdot V_i$ depending on the current bit). This costs two isogeny images, one half doubling, and one half differential addition per step.

Our idea is that rather than interleaving the isogeny images and half doublings and differential additions at each step, we can instead pre-compute several iterated isogeny images P_i (pre-computation which only depends on the base point and bit-length of the scalar m), and then 'unstack' these images at each step by doing one half doubling and one half differential addition. The key point in changing the order, is that one of the two isogeny images we need to compute is the image of the neutral point 0_E, which is 'free'. This helps us save one isogeny image by bit, at the cost of a slightly more expensive half differential addition, because the differences will be given by the isogeny images P_i rather than by the same base point P, hence are not normalized any more even if P was.

We provide an implementation[1] to find and check formulas.

Roadmap

In Section 7.1, we introduce the Kummer lines we are interested in. In Section 7.2, we discuss the main isogeny of interest of this chapter, $F : (P, Q) \mapsto (P + Q, P - Q)$, and we define half differential addition formulas. Assuming we have explicit half differential addition formulas, we then introduce our half ladder in Section 7.3. Finally, Section 7.4 details Mumford's theta group theory and how to find these half differential addition formulas in practice, altogether with an example. In Section 7.5, we use the context of `Curve25519` as another example where to find half differential addition formulas. In Section 7.6, we extend the half ladder to abelian varieties endowed with the level 2 theta model.

Sections 7.5 and 7.6 were part of an appendix, the core result being the half ladder and the way to derive half differential addition formulas.

In this chapter, we work over a perfect field k of characteristic different from 2.

1 https://gitlab.inria.fr/nsarkis/half-diff-add

7.1 KUMMER LINES MODELS

We recall in this section the different models of Kummer lines we are interested in. As introduced in Section 4.3.2, a Kummer line is a degree 2 covering $\pi : E \to \mathbb{P}^1$ from an elliptic curve E to the projective line given by 4 ramification points, corresponding to the 2-torsion $E[2]$. More details on how these are derived can be found in Example 5.3.8 and Sections 8.1 and 8.2.

Lemma 4.3.8 helps to compute the 4-torsion points coordinates on the Kummer line.

Example 7.1.1 (Montgomery model) *The Kummer line associated to a Montgomery curve $\beta y^2 = x(x^2 + \mathcal{A}x + 1)$ has the following ramification:*

$$0_E = (1:0)^*, \quad T_1 = (0:1), \quad T_2 = (a:b), \quad T_3 = (b:a),$$

where $\mathcal{A} = -\frac{a}{b} - \frac{b}{a}$. $\frac{a}{b}$ may not be rational, however we always have $\mathcal{A} \in k$. We denote it $M(a:b)$. The translation by T_1 is given by $t_{T_1} : (X:Z) \mapsto (Z:X)$. There is a rational 4-torsion point $T_1' = (1:1)$ and $T_1'' = (-1:1)$ above T_1. (Indeed, because $3 \cdot T_1' = T_1' + T_1 = -T_1'$ on the curve, we can find $\pi(T_1')$ via the equation $t_{T_1}(\pi(T_1')) = \pi(T_1')$, which becomes $(X_0 : Z_0) = (Z_0 : X_0)$ on the Montgomery line).

Example 7.1.2 (Theta model) *Let $\frac{a}{b} \in k$, the theta model $\theta(a:b)$ has the following ramification points:*

$$0_E = (a:b)^*, \quad T_1 = (-a:b), \quad T_2 = (b:a), \quad T_3 = (-b:a).$$

The translation by T_1 is given by $t_{T_1} : (X:Z) \mapsto (-X:Z)$, there are 4-torsion points above T_1 given by $T_1' = (1:0)$ and $T_1'' = (0:1)$, as well as above T_2 given by $T_2' = (1:1)$ and $T_2'' = (-1:1)$.

Example 7.1.3 (Theta squared model) *Let $\frac{a}{b} \in k$, the theta squared model $\theta_s(a:b)$ has the following ramification points:*

$$0_E = (a:b)^*, \quad T_1 = (b:a), \quad T_2 = (1:0), \quad T_3 = (0:1).$$

The translation by T_1 is $t_{T_1} : (X:Z) \mapsto (Z:X)$ and the 4-torsion above T_1 is given by $T_1' = (1:1)$ and $T_1'' = (-1:1)$. Given the shape of the ramification, it is isomorphic to $M(a:b)$ via the involution $(X:Z) \mapsto (aX - bZ : bX - aZ)$.

Example 7.1.4 (Theta twisted model) *Consider $\frac{a}{b} \in k$, the theta twisted model $\theta_t(a:b)$ has the following ramification points:*

$$0_E = (a:b)^*, \quad T_1 = (-a:b), \quad T_2 = (1:1), \quad T_3 = (-1:1).$$

The translation by T_1 is $t_{T_1} : (X:Z) \mapsto (-X:Z)$ and the 4-torsion above T_1 is given by $T_1' = (1:0)$ and $T_1'' = (0:1)$. It is isomorphic to $\theta_s(a':b')$ via the Hadamard transform $H : (X:Z) \mapsto (X+Z : X-Z)$ where $(a':b') = (a+b : a-b)$, and therefore to $M(a':b')$. The isomorphism to the Montgomery model $M(a':b')$ is given by $(X:Z) \mapsto (b'X + a'Z : a'Z - b'X)$. Its inverse is $(X:Z) \mapsto (a(X-Z) : b(X+Z))$. Hence, a Montgomery curve has a theta squared or equivalently a theta twisted model if and only if it has full rational 2-torsion.

Remark 7.1.5 *The names 'theta squared' and 'theta twisted' come from the fact that given a theta model $\theta(a:b)$, there is a 2-isogeny $f : \theta(a:b) \to \theta_s(a^2 : b^2)$ given by $f : (X:Z) \mapsto (X^2 : Z^2)$ and an isomorphism $g : \theta(a:b) \to \theta_t(a^2 : b^2)$ given by $g : (X:Z) \mapsto (aX : bZ)$.*

7.2 HALF DIFFERENTIAL ADDITIONS

Doubling formulas on Kummer lines are essential to perform scalar multiplication. In Chapter 6, we detailed how to find such formulas by computing a 2-isogeny $\varphi : E \to E'$ and by composing it with its dual $\widetilde{\varphi}$ such that $[2] = \widetilde{\varphi} \circ \varphi$. This is often more efficient, as computing directly the doubling — which is a degree 4 isogeny — can be slower than splitting it into two degree 2 isogenies.

The other important operation on a Kummer line is the differential addition. In this section we study the map

$$F : E \times E \to E \times E,\ (P, Q) \mapsto (P + Q, P - Q).$$

It is a $(2, 2)$-isogeny with kernel $K_F = \{(T, T) \mid T \in E[2]\}$, the diagonal of the 2-torsion. Having formulas for F on the product of Kummer lines yields differential addition ones since we can then recover $P + Q$ from P, Q and $P - Q$ on the Kummer line. Similarly to the case of doubling with 2-isogenies, we would like to factor it.

Let $\varphi : E \to E'$ be a 2-isogeny with kernel $\{0_E, T\}$. We will consider the $(2, 2)$-diagonal isogeny

$$\Phi : E \times E \to E' \times E',\ (P, Q) \mapsto (\varphi(P), \varphi(Q)),$$

its kernel is $K_\Phi = \langle T \rangle \times \langle T \rangle = \langle (0_E, T), (T, 0_E) \rangle$. Ideally, one would like to factor F through Φ, unfortunately this is not possible because there is no inclusion of the kernels, in fact $K_F \cap K_\Phi = \{(0_E, 0_E), (T, T)\}$.

Definition 7.2.1 *Let $\varphi : E \to E'$ be a 2-isogeny of elliptic curves, $\mathcal{K}$ and $\mathcal{K}'$ the Kummer lines corresponding respectively to E and E', $P, Q \in E$. Formulas that can recover $[P + Q] \in \mathcal{K}$ from the data of $[\varphi(P)], [\varphi(Q)] \in \mathcal{K}'$ and $[P - Q] \in \mathcal{K}$ will be called half differential addition formulas. We will denote such algorithm* $\texttt{HalfDiffAdd}_\varphi(\varphi(P), \varphi(Q), P - Q)$.

If we have $\varphi(P)$, applying the dual isogeny $\widetilde{\varphi}$ to it yields $2 \cdot P = \widetilde{\varphi} \circ \varphi(P)$. By analogy with the half differential additions, we will also denote this operation as $2 \cdot P = \texttt{HalfDouble}_\varphi(\varphi(P))$.

In Section 7.4 we will explain how to find explicit half differential formulas using the theta group theory. We will first discuss an application of such formulas to build a half ladder in the next section.

7.3 HALF LADDER

In this section, assuming we know how to compute half differential addition formulas — which will be discussed in Section 7.4 —, we explain how to build a new ladder based on those, and we compare it to the Montgomery one.

7.3.1 Cost of the Montgomery ladder

We first recall some costs of the Montgomery ladder which was introduced in Section 2.3. Each step of the ladder is exactly one differential addition and one doubling. Table 7.4 gives the cost of these on the different models studied in this chapter, as well as the total Montgomery ladder cost.

	Montgomery curve [Mon87]	Theta model [GL09, § 6.2]	Theta squared / twisted [GL09, § 6.2, Thm. 2]
Diff. add.	$2\mathbf{M} + 2\mathbf{S} + 1\mathbf{m}$	$2\mathbf{M} + 4\mathbf{S} + 1\mathbf{m} + 1\mathbf{m}_0$	$2\mathbf{M} + 2\mathbf{S} + 1\mathbf{m} + 1\mathbf{m}_0$
Doubling	$2\mathbf{M} + 2\mathbf{S} + 1\mathbf{m}_0$	$4\mathbf{S} + 2\mathbf{m}_0$	$4\mathbf{S} + 2\mathbf{m}_0$
Total cost	$4\mathbf{M} + 4\mathbf{S} + 1\mathbf{m} + 1\mathbf{m}_0$	$2\mathbf{M} + 8\mathbf{S} + 1\mathbf{m} + 3\mathbf{m}_0$	$2\mathbf{M} + 6\mathbf{S} + 1\mathbf{m} + 3\mathbf{m}_0$

Table 7.4: Montgomery ladder cost on several Kummer lines. Differential additions take $6\mathbf{a}$ and doublings take $4\mathbf{a}$.

Remark 7.3.1 *The Montgomery ladder on theta squared / twisted models was slightly improved in Chapter 6 via a hybrid ladder combining differential addition of Montgomery curves and theta doubling, saving* $1\mathbf{m}_0$ *for a total cost of* $2\mathbf{M} + 6\mathbf{S} + 1\mathbf{m} + 2\mathbf{m}_0 + 13\mathbf{a}$.

As discussed in the introduction, there is a variant of the Montgomery ladder including pre-computations described in [Oli+17, Alg. 4]. The authors give an algorithm going through the binary decomposition of n from right-to-left (RtL), whereas Algorithm 2.6 goes from left-to-right (LtR). The main difference is that for each bit, only one differential addition is needed, given the pre-computation of the points $P_i = 2^i \cdot P$. However, the difference of the points U and V involved in the differential addition in their algorithm is stored in an accumulator and can change throughout the algorithm, hence a differential addition costs $4\mathbf{M} + 2\mathbf{S} + 6\mathbf{a}$. By going further with the pre-computation by normalizing the points P_i, the differential addition can be reduced to $3\mathbf{M} + 2\mathbf{S} + 6\mathbf{a}$, this corresponds to the discussion in Section 5.2 of their article. To summarize:

- The Montgomery ladder right-to-left requires the pre-computation of points $2^i \cdot P_i$, at the cost of one doubling per bit, which is $2\mathbf{M} + 2\mathbf{S} + 1\mathbf{m}_0 + 4\mathbf{a}$.

- If for each $P_i = (X_i : Z_i)$, we further pre-compute a constant $\mu_i = \frac{X_i+Z_i}{X_i-Z_i}$, the cost per bit in the main loop is the one of a differential addition performed in $3\mathbf{M} + 2\mathbf{S} + 6\mathbf{a}$. This pre-computation is $1\mathbf{I} + 1\mathbf{M}$ per bit, which can be reduced to $4\mathbf{M}$ asymptotically thanks to Lemma 2.1.6.

- Otherwise, each differential addition is performed in $4\mathbf{M} + 2\mathbf{S} + 6\mathbf{a}$.

The two situations are described in Tables 7.2 and 7.3 at the beginning of the chapter. Since our approach also involves pre-computations, we will also compare to this version of the Montgomery ladder.

$$\begin{array}{ccccccc} \mathcal{K}_0 & \xrightarrow{\varphi_1} & \mathcal{K}_1 & \xrightarrow{\varphi_2} & \cdots & \xrightarrow{\varphi_\ell} & \mathcal{K}_\ell \\ P_0 & \longmapsto & P_1 & \longmapsto & \cdots & \longmapsto & P_\ell \end{array}$$

Figure 7.1: Half ladder context

7.3.2 Half ladder principle

We will work in the following context, we want to compute $n \cdot P_0$ where n is an ℓ-bit integer and P_0 is a point on the Kummer line $\mathcal{K}_0$. The i-th bit of n is written b_i. Assume we have Kummer lines $\mathcal{K}_1, \dots, \mathcal{K}_\ell$ and 2-isogenies $\varphi_1, \dots, \varphi_\ell$ where $\varphi_i : \mathcal{K}_{i-1} \to \mathcal{K}_i$ for $1 \leqslant i \leqslant \ell$. We also denote $P_i = \varphi_i(P_{i-1})$. The situation is represented in Fig. 7.1. In practice, we will simply use a 2-isogeny $\varphi : \mathcal{K}_0 \to \mathcal{K}_1$ and its dual $\widetilde{\varphi}$ alternatively: $\varphi_{2i+1} = \varphi$ and $\varphi_{2i} = \widetilde{\varphi}$.

Finally, we assume that for each isogeny φ_i, we have half differential addition formulas which given $\varphi_i(P)$, $\varphi_i(Q)$ and $P - Q$, computes $P + Q$ on the Kummer line $\mathcal{K}_{i-1}$. As introduced in Definition 7.2.1, we will denote such algorithm $\texttt{HalfDiffAdd}_{\varphi_i}(\varphi_i(P), \varphi_i(Q), P - Q)$.

The main idea is that instead of computing at each step the doubling of a point via a 2-isogeny and its dual as well as the differential addition of these two points, we will first pre-compute every image $P_1, \dots, P_\ell$ of our base point $P = P_0$, and then go backwards with half doublings and half differential addition formulas.

Assume we know $R_i = u_i \cdot P_i$ and $S_i = (u_i + 1) \cdot P_i$ on $\mathcal{K}_i$, with $1 \leqslant i \leqslant \ell$. In particular, we have $R_i = \varphi_i(u_i \cdot P_{i-1})$ and $S_i = \varphi_i((u_i + 1) \cdot P_{i-1})$, since P_i is $\varphi_i(P_{i-1})$. With the knowledge of P_{i-1}, using $\texttt{HalfDiffAdd}_{\varphi_i}(R_i, S_i, P_{i-1})$, we can compute $(2u_i + 1) \cdot P_{i-1}$. With the dual $\widetilde{\varphi}_i$, we can also compute either $2u_i \cdot P_{i-1} = \texttt{HalfDouble}_{\varphi_i}(R_i)$ or $2(u_i + 1) \cdot P_{i-1} = \texttt{HalfDouble}_{\varphi_i}(S_i)$. We set $u_{i-1} = 2u_i + b_{i-1}$ such that we can recover $R_{i-1} = u_{i-1} \cdot P_{i-1}$ and $S_{i-1} = (u_{i-1} + 1) \cdot P_{i-1}$ using one computation with $\texttt{HalfDiffAdd}_{\varphi_i}$ and one with $\texttt{HalfDouble}_{\varphi_i}$.

With the initial situation being $R_\ell = 0_\ell$ the neutral element on $\mathcal{K}_\ell$ and $S_\ell = P_\ell$, this process can be iterated and one can derive the formula

$$\forall 0 \leqslant i < \ell,\ u_i = b_{\ell-1}2^{\ell-1-i} + b_{\ell-2}2^{\ell-2-i} + \cdots + b_i 2^0.$$

A corollary is that $u_0 = n$ and consequently $R_0 = n \cdot P_0$, which is the point we were looking for. The generic algorithm is described in Algorithm 7.1.

In terms of cost, we first need to compute the images via $\varphi_1, \dots, \varphi_\ell$, then when going backwards we require a computation with each $\texttt{HalfDouble}_{\varphi_i}$ and one $\texttt{HalfDiffAdd}_{\varphi_i}$ for all $1 \leqslant i \leqslant \ell$. Similarly to the Montgomery right-to-left ladder, we could pre-compute when possible the images, and even normalize them if this is meaningful, depending on the context. We will look at an example in the following section.

Algorithm 7.1: Scalar multiplication with the half ladder

Input: $n = (b_{\ell-1}, b_{\ell-2}, \dots, b_0)$ an ℓ-bits integer, P a point on $\mathcal{K}_0$
Output: $n \cdot P$
Data: $\mathcal{K}_i$ is a Kummer line, $\varphi_i : \mathcal{K}_{i-1} \to \mathcal{K}_i$ a 2-isogeny for $1 \leqslant i \leqslant \ell$

```
def HalfLadder(n, P):
    P_0 ← P;
    for i ← 1 to ℓ do                // Potentially a pre-computation
        P_i ← φ_i(P_{i-1});
    end
    R ← O_ℓ;                                  // Neutral point on K_ℓ
    S ← P_ℓ;
    for i ← ℓ down to 1 do
        if b_{i-1} = 1 then
            R ← HalfDiffAdd_{φ_i}(R, S, P_{i-1});
            S ← HalfDouble_{φ_i}(S);
        else if b_{i-1} = 0 then
            S ← HalfDiffAdd_{φ_i}(R, S, P_{i-1});
            R ← HalfDouble_{φ_i}(R);
        end
    end
    return U;
```

7.3.3 Application to the theta model

In this section, we focus on the theta model $\theta(a : b)$ described in Example 7.1.2 and given by ramification points

$$(a : b)^*, \quad (-a : b), \quad (b : a), \quad (-b : a),$$

where $\frac{a}{b} \in k$. We will start with the case where there is a 8-torsion point $(r : s)$ above $(-a : b)$. As seen in Example 6.5.4, if $(A : B) := (r^2 + s^2 : r^2 - s^2)$, then $(A^2 : B^2) = (a^2 + b^2 : a^2 - b^2)$ and the following 2-isogeny ends on the theta model $\theta(A : B)$:

$$\varphi : (X : Z) \in \theta(a : b) \mapsto (B(X^2 + Z^2) : A(X^2 - Z^2)).$$

Its dual is simply given by $\widetilde{\varphi} : (X : Z) \in \theta(A : B) \mapsto (b(X^2 + Z^2) : a(X^2 - Z^2))$.

To build our half ladder, if n is an ℓ-bit integer, we set $\mathcal{K}_{2i} = \theta(a : b)$, $\mathcal{K}_{2i+1} = \theta(A : B)$, and $\varphi_{2i} = \widetilde{\varphi}$, $\varphi_{2i+1} = \varphi$ for $0 \leqslant 2i, 2i+1 \leqslant \ell$. The framework of Section 7.4 gives the following half differential addition formulas:

- $\texttt{HalfDiffAdd}_{\varphi}(\varphi(P), \varphi(Q), P - Q)$ $(P, Q \in \theta(a : b))$:

$$(X_{P+Q}X_{P-Q} : Z_{P+Q}Z_{P-Q}) = \begin{pmatrix} X_{\varphi(P)}X_{\varphi(Q)} + Z_{\varphi(P)}Z_{\varphi(Q)} \\ X_{\varphi(P)}X_{\varphi(Q)} - Z_{\varphi(P)}Z_{\varphi(Q)} \end{pmatrix}.$$

- $\texttt{HalfDiffAdd}_{\widetilde{\varphi}}(\widetilde{\varphi}(P), \widetilde{\varphi}(Q), P - Q)$ $(P, Q \in \theta(A : B))$:

$$(X_{P+Q}X_{P-Q} : Z_{P+Q}Z_{P-Q}) = \begin{pmatrix} X_{\widetilde{\varphi}(P)}X_{\widetilde{\varphi}(Q)} + Z_{\widetilde{\varphi}(P)}Z_{\widetilde{\varphi}(Q)} \\ X_{\widetilde{\varphi}(P)}X_{\widetilde{\varphi}(Q)} - Z_{\widetilde{\varphi}(P)}Z_{\widetilde{\varphi}(Q)} \end{pmatrix}.$$

Each operation has the following cost:

- A φ evaluation is $2\mathbf{S} + 1\mathbf{m}_0 + 2\mathbf{a}$, as well as a $\widetilde{\varphi}$ evaluation.
- A half differential addition with respect to φ is $4\mathbf{M}+2\mathbf{a}$, as well as a half differential addition with respect to $\widetilde{\varphi}$. We can save $1\mathbf{M}$ by normalizing the points $P_1, \dots, P_\ell$.

The costs are completely symmetric whether we work with φ or $\widetilde{\varphi}$, hence an image is $2\mathbf{S} + 1\mathbf{m}_0 + 2\mathbf{a}$ and a half differential addition is $4\mathbf{M} + 2\mathbf{a}$. This leads to the following costs per bit:

- If we do not perform any sort of pre-computation, the cost per bit is $4\mathbf{M} + 4\mathbf{S} + 2\mathbf{m}_0 + 6\mathbf{a}$, which is the best case scenario of Montgomery ladder left-to-right on a Montgomery curve where the base point is normalized and has a small x-coordinate, and is in general better than the Montgomery ladder left-to-right on a theta squared model.
- If we pre-compute the images $P_1, \dots, P_\ell$ but we don't normalize them, the pre-computation costs $2\mathbf{S} + 1\mathbf{m}_0 + 2\mathbf{a}$ per bit, and the main loop is $4\mathbf{M}+2\mathbf{S}+1\mathbf{m}_0+4\mathbf{a}$. The pre-computation saves $2\mathbf{M}$ over Montgomery ladder right-to-left whereas the main loop loses $1\mathbf{m}_0$.
- If we moreover normalize the points, the pre-computation cost raises to $4\mathbf{M} + 2\mathbf{S} + 1\mathbf{m}_0 + 2\mathbf{a}$ thanks to Lemma 2.1.6, and the main loop is then $3\mathbf{M}+2\mathbf{S}+1\mathbf{m}_0+4\mathbf{a}$. The difference with the Montgomery ladder right-to-left is the same as above, saving $2\mathbf{M}$ on the pre-computation but losing $1\mathbf{m}_0$ in the main loop.

We see that the number of additions in the half ladder is similar to the one in the Montgomery ladder, since these additions are negligible compared to multiplications and squarings, we will not write them in the sequel. The complete comparison in Table 7.2 includes the number of additions.

If one wants to work with a theta model with no additional assumptions, the isogenies can be chosen differently. We still set $(A^2 : B^2) = (a^2 + b^2 : a^2 - b^2)$. The 2-isogeny given by $\varphi : (X : Z) \in \theta(a : b) \mapsto (X^2 : Z^2)$ ends on the theta squared model $\theta_s(a^2 : b^2)$ with ramification

$$(a^2 : b^2)^*, \quad (b^2 : a^2), \quad (1 : 0), \quad (0 : 1).$$

The dual isogeny is then

$$\begin{aligned}\widetilde{\varphi} : (X : Z) &\in \theta_s(a^2 : b^2) \\ &\mapsto (b(B^2(X+Z)^2 + A^2(X-Z)^2) : a(B^2(X+Z)^2 - A^2(X-Z)^2)).\end{aligned}$$

The isogenies were computed using Example 6.5.3 and the isomorphism between Montgomery model and theta squared model from Example 7.1.3.

If n is an ℓ-bit integer, we set $\mathcal{K}_{2i} = \theta(a : b)$, $\mathcal{K}_{2i+1} = \theta_s(a^2 : b^2)$ for $0 \leqslant 2i, 2i+1 \leqslant \ell$, and $\varphi_{2i} = \widetilde{\varphi}$, $\varphi_{2i+1} = \varphi$. The differential addition formulas are the following according to Section 7.4.2:

- $\texttt{HalfDiffAdd}_{\varphi}(\varphi(P), \varphi(Q), P - Q)$:

$$(X_{P+Q}X_{P-Q} : Z_{P+Q}Z_{P-Q}) = \begin{pmatrix} B^2(X_{\varphi(P)} + Z_{\varphi(P)})(X_{\varphi(Q)} + Z_{\varphi(Q)}) + A^2(X_{\varphi(P)} - Z_{\varphi(P)})(X_{\varphi(Q)} - Z_{\varphi(Q)}) \\ B^2(X_{\varphi(P)} + Z_{\varphi(P)})(X_{\varphi(Q)} + Z_{\varphi(Q)}) - A^2(X_{\varphi(P)} - Z_{\varphi(P)})(X_{\varphi(Q)} - Z_{\varphi(Q)}) \end{pmatrix}.$$

- $\texttt{HalfDiffAdd}_{\widetilde{\varphi}}(\widetilde{\varphi}(P), \widetilde{\varphi}(Q), P - Q)$:

$$\begin{pmatrix} (X_{P+Q} + Z_{P+Q})(X_{P-Q} + Z_{P-Q}) \\ (X_{P+Q} - Z_{P+Q})(X_{P-Q} - Z_{P-Q}) \end{pmatrix} = \begin{pmatrix} A^2(X_{\widetilde{\varphi}(P)}X_{\widetilde{\varphi}(Q)} + Z_{\widetilde{\varphi}(P)}Z_{\widetilde{\varphi}(Q)}) \\ B^2(X_{\widetilde{\varphi}(P)}X_{\widetilde{\varphi}(Q)} - Z_{\widetilde{\varphi}(P)}Z_{\widetilde{\varphi}(Q)}) \end{pmatrix}.$$

As we can see:

- A φ evaluation is $2\mathbf{S}$.
- A $\widetilde{\varphi}$ evaluation is $2\mathbf{S} + 2\mathbf{m}_0$.
- A half differential addition with respect to φ is $4\mathbf{M} + 1\mathbf{m}_0$, as well as a half differential addition with respect to $\widetilde{\varphi}$.

Since half of the images are via φ and the other is via $\widetilde{\varphi}$, an image costs $2\mathbf{S} + 1\mathbf{m}_0$ on average. We see that using these isogenies is a bit less efficient. We will tackle this issue in the next section.

7.3.4 A variant on the theta twisted model

In this section, we work on the theta twisted model $\theta_t(a : b)$ as described in Example 7.1.4, with ramification points

$$(a : b)^*, \quad (-a : b), \quad (1 : 1), \quad (-1 : 1)$$

with $\frac{a}{b} \in k$, and set $(a' : b') = (a + b : a - b)$. If $P = (X : Z) \in \theta_t(a : b)$, we set $P^\times = (bX : aZ)$. The following 2-isogeny from $\theta_t(a : b)$ to $\theta_t(a' : b')$ can be derived from Theorem 6.3.3 and the isomorphisms between theta squared, twisted and Montgomery models from Examples 7.1.3 and 7.1.4. We get

$$\varphi : (X : Z) \mapsto (bX^2 + aZ^2 : bX^2 - aZ^2) \tag{7.1}$$

Its dual is $\widetilde{\varphi} : (X : Z) \mapsto (b'X^2 + a'Z^2 : b'X^2 - a'Z^2)$. The half differential addition formulas are then:

- $\texttt{HalfDiffAdd}_{\varphi}(\varphi(P), \varphi(Q), P - Q)$:

$$(X_{P+Q}X_{P-Q} : Z_{P+Q}Z_{P-Q}) = \begin{pmatrix} a(b'X_{\varphi(P)}X_{\varphi(Q)} + a'Z_{\varphi(P)}Z_{\varphi(Q)}) \\ b(b'X_{\varphi(P)}X_{\varphi(Q)} - a'Z_{\varphi(P)}Z_{\varphi(Q)}) \end{pmatrix}.$$

- $\texttt{HalfDiffAdd}_{\widetilde{\varphi}}(\widetilde{\varphi}(P), \widetilde{\varphi}(Q), P - Q)$:

$$(X_{P+Q}X_{P-Q} : Z_{P+Q}Z_{P-Q}) = \begin{pmatrix} a'(bX_{\varphi(P)}X_{\varphi(Q)} + aZ_{\varphi(P)}Z_{\varphi(Q)}) \\ b'(bX_{\varphi(P)}X_{\varphi(Q)} - aZ_{\varphi(P)}Z_{\varphi(Q)}) \end{pmatrix}.$$

$P+Q$ can be computed with $4\mathbf{M}+2\mathbf{m}_0$. However, if we use the point $\varphi(P)^\times$, we can recover $(P+Q)^\times$ in $4\mathbf{M}$ thanks to the following formula:

$$(X_{(P+Q)^\times}X_{P-Q} : Z_{(P+Q)^\times}Z_{P-Q}) = \begin{pmatrix} X_{\varphi(P)^\times}X_{\varphi(Q)} + Z_{\varphi(P)^\times}Z_{\varphi(Q)} \\ X_{\varphi(P)^\times}X_{\varphi(Q)} - Z_{\varphi(P)^\times}Z_{\varphi(Q)} \end{pmatrix}.$$

The roles of $\varphi(P)^\times$ and $\varphi(Q)$ are interchangeable, as well as those of φ and $\widetilde{\varphi}$ because of the symmetries of the formulas. Moreover, it is also possible to compute $\widetilde{\varphi}(P)$ with the knowledge of $P^\times$ because

$$\widetilde{\varphi}(P) = (b'X_P^2 + a'Z_P^2 : b'X_P^2 - a'Z_P^2) = (a'X_{P^\times}^2 + b'Z_{P^\times}^2 : a'X_{P^\times}^2 - b'Z_{P^\times}^2).$$

The cost is $2\mathbf{S}+1\mathbf{m}_0$ whether we use P or $P^\times$. Hence, we can adapt Algorithm 7.1 into Algorithm 7.2 below for theta twisted models by storing either $(R^\times, S)$ or $(R, S^\times)$ and by keeping track of this information.

Evaluation by φ and $\widetilde{\varphi}$ always costs $2\mathbf{S}+1\mathbf{m}_0$, and in this context half differential addition cost $4\mathbf{M}$, so the cost per bit of Algorithm 7.2 is the same as the case of a theta model with a 8-torsion point. This is the variant we compare to in Tables 7.1 to 7.3 in the introduction.

Hence, we can always achieve the best case scenario of the Montgomery ladder left-to-right if we have a Montgomery curve with full rational 2-torsion, even if the base point is not normalized, and we can significantly improve the pre-computation of the Montgomery ladder right-to-left at the cost of $1\mathbf{m}_0$ in the main loop. In the last section on half ladder, we discuss the case where the 2-torsion of the Montgomery curve is not completely rational, but there is a rational 8-torsion point.

7.3.5 Scalar multiplication on `Curve25519`

`Curve25519`[Ber06] is a Montgomery curve over $\mathbb{F}_p$ with $p = 2^{255}-19$ and equation $E : y^2 = x(x^2+\mathcal{A}x+1)$ where $\mathcal{A} = 486662$. It is a well-known curve used in several implementations of cryptographic protocols (OpenSSH, GnuPG, Signal, WireGuard, ...). Its 2-torsion is not rational, however it has a rational 8-torsion point above $(0:1)$. We will then work in the following context: let $M(A:B)$ be a Kummer line associated to a Montgomery curve with ramification

$$(1:0)^*, \quad (0:1), \quad (A:B), \quad (B:A),$$

as described in Example 7.1.1. We will not assume $\frac{A}{B} \in k$, however we suppose there is a rational 8-torsion point $\widetilde{T} = (r:s)$ above $T' = (1:1)$, itself above $T = (0:1)$. According to Theorem 6.3.10, we have a 2-isogeny from this curve to a Montgomery curve E' with full rational 2-torsion. We can then compose it with the isomorphism from Example 7.1.4. We set the constants $(\gamma:\delta) = (4rs : (r-s)^2)$, $(a:b) = (\gamma-\delta : \gamma+\delta)$ and $(a':b') = (a+b : a-b) = (-\gamma:\delta)$. We have the following 2-isogeny $\psi : M(A:B) \to \theta_t(a:b)$ given by

$$\psi : (X:Z) \mapsto (ab(X-Z)^2 - a\delta(X+Z)^2 : ab(X-Z)^2 + b\delta(X+Z)^2).$$

Algorithm 7.2: Scalar multiplication on theta twisted model with the twisted half ladder

Input: $n = (b_{\ell-1}, b_{\ell-2}, \dots, b_0)$ an ℓ-bits integer, P a point on $\theta_t(a : b)$
Output: $n \cdot P$
Data: $\mathcal{K}_{2i} = \theta_t(a : b)$, $\mathcal{K}_{2i+1} = \theta_t(a' : b')$, $\varphi_{2i} = \varphi$, $\varphi_{2i+1} = \widetilde{\varphi}$, with $0 \leqslant 2i, 2i+1 \leqslant \ell$

```
def TwistedHalfLadder(n, P):
    P_0 ← P;
    for i ← 1 to ℓ do                    // Potentially a pre-computation
        P_i ← φ_i(P_{i−1});
    end
    R^× ← 0_ℓ^× = (1 : 1);               // Neutral point on K_ℓ
    S ← P_ℓ;
    b_ℓ ← 1;
    for i ← ℓ down to 1 do
        if b_{i−1} = 1 then
            if b_i = 1 then              // Known points: R^×, S
                R^× ← HalfDiffAdd_{φ_i}(R^×, S, P_{i−1});
                S ← HalfDouble_{φ_i}(S);
            else if b_i = 0 then         // Known points: R, S^×
                R^× ← HalfDiffAdd_{φ_i}(R, S^×, P_{i−1});
                S ← HalfDouble_{φ_i}(S^×);
            end
        else if b_{i−1} = 0 then
            if b_i = 1 then              // Known points: R^×, S
                S^× ← HalfDiffAdd_{φ_i}(R^×, S, P_{i−1});
                R ← HalfDouble_{φ_i}(R^×);
            else if b_i = 0 then         // Known points: R, S^×
                S^× ← HalfDiffAdd_{φ_i}(R, S^×, P_{i−1});
                R ← HalfDouble_{φ_i}(R)
            end
        end
    end
    return R;                            // Derived from R^× if b_0 = 1
```

The dual is given by

$$\widetilde{\psi} : (X : Z) \mapsto (aZ^2 - bX^2 + 2\delta XZ : aZ^2 - bX^2 - 2\delta XZ).$$

We can afford to have these two isogenies being a bit slower than usual since they will only intervene once during the computations. The details on how we obtain the half differential addition formulas below are available in Section 7.5.

We then set $\mathcal{K}_0 = M(A : B)$, $\varphi_1 = \psi$, $\mathcal{K}_{2i} = \theta_t(a' : b')$, $\mathcal{K}_{2i+1} = \theta_t(a : b)$, $\varphi_{2i} = \varphi$ and $\varphi_{2i+1} = \widetilde{\varphi}$ when $1 \leqslant 2i, 2i+1 \leqslant \ell$, where φ is the 2-isogeny of Eq. (7.1).

Using the tools from Section 7.4, we can derive half differential addition formulas for ψ. If $P, Q \in \mathcal{K}_0$, $\texttt{HalfDiffAdd}_\psi(\psi(P), \psi(Q), S)$ is given by

$$\begin{pmatrix} (X_{P+Q} + Z_{P+Q})(X_{P-Q} + Z_{P-Q}) \\ (X_{P+Q} - Z_{P+Q})(X_{P-Q} - Z_{P-Q}) \end{pmatrix} = \begin{pmatrix} aZ_{\psi(P)}Z_{\psi(Q)} - bX_{\psi(P)}X_{\psi(Q)} \\ \delta(X_{\psi(P)}Z_{\psi(Q)} + Z_{\psi(P)}X_{\psi(Q)}) \end{pmatrix}.$$

Again, we can afford to spend a bit more time on this step because it is used only once. The steps on $\theta_t(a : b)$ and $\theta_t(a' : b')$ can be done using Algorithm 7.2, and the last step to go back to $M(A : B)$ is done via $\texttt{HalfDiffAdd}_\psi$ and $\widetilde{\psi}$ as in Algorithm 7.1, so the cost per step is the same as Algorithm 7.2.

Hence, the conclusion from previous section also holds on a Montgomery curve with a rational 8-torsion point above $(0 : 1)$, like `Curve25519`. There is however one major caveat: using the half ladder, the $\mathbf{m}_0$ corresponds to the curve constant on E' rather than on E. In the specific case of `Curve25519`, although the curve constant on E is small, this is unfortunately not the same[2] for its isogenous curve E'. Hence, this reduces the utility of the half ladder for `Curve25519`, at least when the pre-computations are not reused.

7.4 A FRAMEWORK TO DERIVE HALF DIFFERENTIAL ADDITION FORMULAS

We now give more details on the framework yielding half differential addition formulas. The whole theory is based on the theta group defined by Mumford in [Mum66].

7.4.1 Theta group on a product of elliptic curves

In Section 6.2.1, we studied the theta group on an elliptic curve E, especially the descents of the divisors $2(0_E)$ and $4(0_E)$. In this section, we will extend naturally these results to the abelian surface $E \times E$. Recall from Section 7.2 that our goal is to study the isogeny $F : (P, Q) \mapsto (P + Q, P - Q)$.

Remark 7.4.1 *For the sake of simplicity and because this is the context we are working on, the results in this section are only stated on $E \times E$, however most of those still hold on $X \times Y$ where X and Y are abelian varieties.*

2 The Montgomery constant on the isogenous curve is 24664115514453686528306332564023014466748285880603927312212995014455267148607 in $\mathbb{F}_p$.

Product divisor

Let $\pi_1 : E \times E \to E$ and $\pi_2 : E \times E \to E$ be the projection on the first and the second component respectively, $\pi_1 : (P, Q) \mapsto P$ and $\pi_2 : (P, Q) \mapsto Q$.

Definition 7.4.2 *Let D_1 and D_2 be divisors on E. $\pi_1^* D_1$ and $\pi_2^* D_2$ are divisors on $E \times E$, we define the product divisor on $E \times E$ as $D_1 * D_2 := \pi_1^* D_1 + \pi_2^* D_2$.*

This is the correct notion of product because it has good compatibility with the tools from Chapter 5. If $f, g \in k(E)$, we define $f \otimes g \in k(E \times E)$ as:

$$\forall (P, Q) \in E \times E, f \otimes g(P, Q) := f(P)g(Q). \tag{7.2}$$

Lemma 7.4.3 *Let $f, g \in k(E)$, then $\operatorname{div}(f \otimes g) = (\operatorname{div} f) * (\operatorname{div} g)$.*

Proof *We have $f \otimes g = (\pi_1^* f)(\pi_2^* g)$ where $\pi_1^* f$ and $\pi_2^* g$ are elements of $k(E \times E)$. Hence, $\operatorname{div}(f \otimes g) = \operatorname{div}(\pi_1^* f) + \operatorname{div}(\pi_2^* g) = \pi_1^*(\operatorname{div} f) + \pi_2^*(\operatorname{div} g) = (\operatorname{div} f) * (\operatorname{div} g)$.* □

Let D_1, D_2 be divisors on E. We can relate the global sections of D_1 and D_2 over E to the global sections of $D_1 * D_2$ over $E \times E$:

Lemma 7.4.4 *Let D_1 and D_2 be divisors on E. The following canonical map is an isomorphism of vector spaces:*

$$\begin{aligned} \Gamma(D_1) \otimes \Gamma(D_2) &\xrightarrow{\sim} \Gamma(D_1 * D_2) \\ f \otimes g &\mapsto f \otimes g \end{aligned}$$

where on the left side, $f \otimes g$ is an element of $k(E) \otimes k(E)$ and on the right side $f \otimes g$ is the element defined in Eq. (7.2).

Proof *This is a particular case of Künneth formula where $n = 0$, see [The25, Lemma oBED].* □

Since we want to study the theta group on $E \times E$ for product divisors, we have to study the associated polarization. We recall that we have a canonical identification $\operatorname{Pic}^0(X \times Y) \simeq \operatorname{Pic}^0(X) \times \operatorname{Pic}^0(Y)$ via $(D_1, D_2) \mapsto D_1 * D_2$, and that we can see $\operatorname{Pic}^0(X)$ inside $\operatorname{Pic}^0(X \times Y)$ via the pullback π_1^*. Modulo these identifications applied to $X = Y = E$, we have

$$\Lambda(D_1 * D_2) : E \times E \to \operatorname{Pic}^0(E) \times \operatorname{Pic}^0(E) : (P, Q) \mapsto (\Lambda_{D_1}(P), \Lambda_{D_2}(Q))$$

Unravelling the identifications, we need to check that if D_1 and D_2 are divisors on E, the polarization associated to $D_1 * D_2$ is

$$\begin{aligned} \Lambda(D_1 * D_2) : E \times E &\to \operatorname{Pic}^0(E \times E) \\ (P, Q) &\mapsto [t_{(P,Q)}^*(D_1 * D_2) - D_1 * D_2]. \end{aligned}$$

If $(P, Q) \in E \times E$, since $\pi_1 \circ t_{(P,Q)} = t_P \circ \pi_1$ and $\pi_2 \circ t_{(P,Q)} = t_Q \circ \pi_2$, we get:

$$t_{(P,Q)}^*(D_1 * D_2) = t_{(P,Q)}^* \pi_1^* D_1 + t_{(P,Q)}^* \pi_2^* D_2 = \pi_1^* t_P^* D_1 + \pi_2^* t_Q^* D_2 = (t_P^* D_1) * (t_Q^* D_2).$$

Because of this, $\Lambda(D_1 * D_2)(P, Q) = \Lambda(D_1)(P) * \Lambda(D_2)(Q)$. The kernel is thus given by:

Lemma 7.4.5 *Let* D_1 *and* D_2 *be divisors on* E*, then* $H(D_1 * D_2) = H(D_1) \times H(D_2)$.

We recover a statement similar to Lemma 7.4.4 on the theta group (see [Mum66, § 3, Lem. 1, p. 323])

Lemma 7.4.6 *Let* D_1 *and* D_2 *be divisors on* E*. The following canonical map is a surjective morphism of groups:*

$$\begin{aligned} G(D_1) \times G(D_2) &\longrightarrow G(D_1 * D_2) \\ (f, g) &\longmapsto f \otimes g. \end{aligned}$$

Its kernel is given by $\{(\lambda, \lambda^{-1}) \mid \lambda \in k^*\} \simeq k^*$*. Moreover, if* $f \in G(D_1)$ *lies above* $P \in H(D_1)$ *and* $g \in G(D_2)$ *lies above* $Q \in H(D_2)$*, then* $f \otimes g$ *lies above* $(P, Q) \in H(D_1 * D_2)$.

Proof *The surjectivity and the computation of the kernel correspond to [Mum66, § 3, Lem. 1, p. 323]. Consider* f, g, P *and* Q *as in the statement, then, by Lemma 7.4.3* $\operatorname{div} f \otimes g = \operatorname{div} f * \operatorname{div} g$*. As seen above,* $\operatorname{div} f * \operatorname{div} g = t^*_{(P,Q)}(D_1 * D_2) - D_1 * D_2$*, hence* $f \otimes g$ *lies above* (P, Q). □

Finally, a straight-forward computation shows that the action of $G(D_1 * D_2)$ on $\Gamma(D_1 * D_2)$ is compatible with these maps. Let $s_1 \in \Gamma(D_1)$, $s_2 \in \Gamma(D_2)$, $g_1 \in G(D_1)$ and $g_2 \in G(D_2)$, then:

$$\rho_{D_1 * D_2}(g_1 \otimes g_2)(s_1 \otimes s_2) = (\rho_{D_1}(g_1)s_1) \otimes (\rho_{D_2}(g_2)s_2).$$

In the sequel, to avoid having too cumbersome notations, we will denote the representation via a dot, and the above equation becomes

$$(g_1 \otimes g_2) \cdot (s_1 \otimes s_2) = (g_1 \cdot s_1) \otimes (g_2 \cdot s_2).$$

A commutative diagram for half differential additions

We set the following divisors on E and E': $D_2 = 2(0_E)$, $D'_2 = 2(0_{E'})$ and $D_4 = 4(0_E)$. Recall that F is the differential addition isogeny and Φ is the diagonal isogeny of φ, where φ is a 2-isogeny on E, as in Section 7.2.

Lemma 7.4.4 gives natural basis of $\Gamma(D_2 * D_2)$. Consider two points $R, S \in E$ with coordinates $(X_R : Z_R)$ and $(X_S : Z_S)$, these coordinates are section above D_2. Because of the lemma, a basis of $\Gamma(D_2 * D_2)$ is given by:

$$\Gamma(D_2 * D_2) = \left\langle \begin{matrix} X_R X_S & X_R Z_S \\ Z_R X_S & Z_R Z_S \end{matrix} \right\rangle.$$

By general theory, because of the uniqueness of totally symmetric line bundles in their algebraic equivalent classes, $D_4 * D_4$ descends to $D_2 * D_2$ via F (since both are totally symmetric). This descent corresponds to a lift of the kernel $\widetilde{K}$; then we obtain an isomorphism $\Gamma(D_4 * D_4)^{\widetilde{K}} \simeq \Gamma(D_2 * D_2)$ by Theorem 5.1.7 (3), from which we can express a basis of $\Gamma(D_2 * D_2)$ via a basis of $\Gamma(D_4 * D_4)$, which we can also compute. This give differential addition formulas, and with a more thorough study of those descents that are compatible with a suitable descent of $D_4 * D_4$ through Φ, we are able to find half differential addition formulas. We now give more details.

$$\begin{tikzcd} E \times E \arrow[r, "\Phi"] \arrow[d, "F"'] \arrow[dr, "G"] & E' \times E' \arrow[d, "F_0"] \\ E \times E \arrow[r, "\Phi_0"'] & X \end{tikzcd}$$

Figure 7.2: Factoring $G = F_0 \circ \Phi = \Phi_0 \circ F$ through F and Φ

As discussed in Section 7.2, if K_F and K_Φ are the kernels of F and Φ respectively, then $K_F \cap K_\Phi = \{(0_E, 0_E), (T, T)\}$, so we can't factor Φ through F or the converse. We then consider the isogeny $G : E \times E \to X$ with kernel $K_G = K_F + K_\Phi$, where X is an abelian surface, with a polarization of type $(1, 2)$ (hence which is not principal). Since $K_F, K_\Phi \subset K_G$, we can factor F and Φ through G. Let $\Phi_0 : E \times E \to X$ and $F_0 : E' \times E' \to X$ be such that $G = F_0 \circ \Phi = \Phi_0 \circ F$. The situation is summarized in Fig. 7.2.

The kernel of Φ_0 is $K_{\Phi_0} = F(K_G) = F(K_\Phi) = \{(0_E, 0_E), (T, T)\}$. Similarly, the kernel of F_0 is $K_{F_0} = \Phi(K_G) = \Phi(K_F) = \{(0_{E'}, 0_{E'}), (T', T')\}$, where $T' = \varphi(T_0)$ for $T_0 \in E[2] \setminus \{0_E, T\}$.

Descents of $D_4 * D_4$ *with respect to* Φ

We have seen in Section 6.2.1 that D_4 descends to D_2' via φ, the lift of the kernel $\widetilde{K}_\varphi$ is generated by a symmetric element $\widetilde{g_T}^{\otimes 2}$ of order 2. Let $\alpha \in k(E)$ with divisor $D_4 - \varphi^* D_2'$, then $\widetilde{g_T}^{\otimes 2} = \frac{t_T^* \alpha}{\alpha}$.

If we look at the image of $\widetilde{K}_\varphi \times \widetilde{K}_\varphi$ via the map of Lemma 7.4.6, we get a subgroup

$$\widetilde{K} = \{1 \otimes 1,\ \widetilde{g_T}^{\otimes 2} \otimes 1,\ 1 \otimes \widetilde{g_T}^{\otimes 2},\ \widetilde{g_T}^{\otimes 2} \otimes \widetilde{g_T}^{\otimes 2}\}.$$

If we look for instance at the second element, for $(P, Q) \in E \times E$:

$$(\widetilde{g_T}^{\otimes 2} \otimes 1)(P, Q) = \frac{t_T^* \alpha(P)}{\alpha(P)} = \frac{\alpha(P + T)\alpha(Q)}{\alpha(P)\alpha(Q)} = \frac{t_{(T, 0_E)}^* \alpha \otimes \alpha(P, Q)}{\alpha \otimes \alpha(P, Q)}.$$

With a similar computation,

$$\widetilde{K} = \left\{ \frac{t_{(0_E, 0_E)}^* \alpha \otimes \alpha}{\alpha \otimes \alpha}, \frac{t_{(T, 0_E)}^* \alpha \otimes \alpha}{\alpha \otimes \alpha}, \frac{t_{(0_E, T)}^* \alpha \otimes \alpha}{\alpha \otimes \alpha}, \frac{t_{(T, T)}^* \alpha \otimes \alpha}{\alpha \otimes \alpha} \right\},$$

and we check that it is a lift of K_Φ to $G(D_4 * D_4)$ according to Theorem 5.1.7 (1). We just have to compute $\operatorname{div} \alpha \otimes \alpha$, which can be done using Lemma 7.4.3:

$$\operatorname{div} \alpha \otimes \alpha = \pi_1^*(D_4 - \varphi^* D_2') + \pi_2^*(D_4 - \varphi^* D_2') = D_4 * D_4 - (\pi_1^* \varphi^* D_2' + \pi_2^* \varphi^* D_2').$$

For $i = 1, 2$, if π_i' is the projection from $E' \times E'$ on the i-th component of E', then $\pi_i' \circ \Phi = \varphi \circ \pi_i$ by construction, hence $\pi_i^* \varphi^* D_2' = \Phi^* \pi_i'^* D_2'$ which leads to $\operatorname{div} \alpha \otimes \alpha = D_4 * D_4 - \Phi^*(D_2' * D_2')$.

To summarize $D_4 * D_4$ descends to $D_2' * D_2'$ with respect to Φ and the lift of the kernel $\widetilde{K} =: \widetilde{K}_\Phi$ is naturally constructed as the product of the lift $\widetilde{K}_\varphi$ with itself.

This gives an isomorphism $\Gamma(D_4 * D_4)^{\widetilde{K}_\Phi} \simeq \Gamma(D_2' * D_2')$ by Theorem 5.1.7 (3).

Descents of $D_4 * D_4$ ***with respect to*** F

The isogeny F is not diagonal this time. Fortunately, if D is a symmetric divisor on E, [Mum66, § 3, Prop. 1, p. 320] states that $F^*(D * D) \sim (2D) * (2D)$. Considering $D = D_2$, we get $F^*(D_2 * D_2) \sim D_4 * D_4$. So $D_4 * D_4$ descends to $D_2 * D_2$ with respect to F, we denote by $\widetilde{K}_F$ the lift of K_F to $G(D_4 * D_4)$. Moreover, the elements of $\widetilde{K}_F$ are of the shape $g_P \otimes g_P$ for $P \in E[2]$ and $g_P \in G(D_4)$. This implies $\widetilde{g_T}^{\otimes 2} \otimes \widetilde{g_T}^{\otimes 2} \in \widetilde{K}_F$ is the element above (T, T).

This gives an isomorphism $\Gamma(D_4 * D_4)^{\widetilde{K}_F} \simeq \Gamma(D_2 * D_2)$, which permits to recover differential addition formulas by expressing sections of $D_2 * D_2$ — which are coordinates — in terms of sections of $D_4 * D_4$ invariant by F.

Descents of $D_2 * D_2$ ***and*** $D_2' * D_2'$ ***with respect to*** Φ_0 ***and*** F_0

Since T is a rational 2-torsion point on E, we have seen in Section 6.2.1 that there are elements $g_T \in G(D_2)$ such that $g_T^2 = \lambda_T$ where the class of λ_T modulo squares is the type of T (Definition 6.2.4). Via the morphism of Lemma 7.4.6, we consider $g_{(T,T)} := \frac{1}{\lambda_T} g_T \otimes g_T \in G(D_2 * D_2)$. This element does not depend on the choice of g_T, lies above (T, T), and is of order 2 because

$$\left(\frac{1}{\lambda_T} g_T \otimes g_T\right)^2 = \frac{1}{\lambda_T^2} g_T^2 \otimes g_T^2 = 1.$$

Hence, $\widetilde{K}_{\Phi_0} = \{1, g_{(T,T)}\}$ is a lift of K_{Φ_0} to $G(D_2 * D_2)$, and by Theorem 5.1.7 (1) there is a divisor D_X on the abelian variety X on which $D_2 * D_2$ descends with respect to Φ_0. This also means that $D_4 * D_4$ descends to D_X with respect to G because $\Phi_0^* D_X \sim D_2 * D_2$ and $F^*(D_2 * D_2) \sim D_4 * D_4$, which implies $G^* D_X \sim D_4 * D_4$. This produces a lift of the kernel $\widetilde{K}_X$.

Similarly, $\widetilde{K}_{F_0} = \{1, g_{(T',T')}\}$ where $g_{(T',T')} = \frac{1}{\lambda_{T'}} g_{T'} \otimes g_{T'}$, $g_{T'} \in G(D_2')$ above T', is a lift of K_{F_0} that descends $D_2' * D_2'$ to some divisor D_X' on the abelian variety X with respect to F_0. Similarly, $D_4 * D_4$ descends to D_X' with respect to G, via a lift of the kernel $\widetilde{K}_X'$.

We need to show that $\widetilde{K}_X = \widetilde{K}_X'$. Set $\widetilde{K}_0 = \widetilde{K}_F \cdot \widetilde{K}_\Phi$, the subgroup of $G(D_4 * D_4)$ generated by elements of $\widetilde{K}_F$ and $\widetilde{K}_\Phi$. It is a level subgroup because if $\lambda = g_{(P,Q)} \cdot g_{(P',Q')}^{-1} \in \widetilde{K}_0 \cap k^*$, with $g_{(P,Q)} \in \widetilde{K}_F$ and $g_{(P',Q')} \in \widetilde{K}_\Phi$, then $(P - P', Q - Q') = (0_E, 0_E)$, and $(P, Q) = (P', Q') \in K_F \cap K_\Phi = \{(0_E, 0_E), (T, T)\}$. In both cases, $g_{(P,Q)} = g_{(P',Q')}$, proving that $\widetilde{K}_0 \cap k^* = \{1\}$. It then corresponds to another descent of $D_4 * D_4$ with respect to G by Theorem 5.1.7 (1).

We will denote by $C_{\widetilde{K}}$ the centralizer of a level subgroup $\widetilde{K}$ in $G(D_2 * D_2)$ (this corresponds to $C_{\widetilde{K}}(D_2 * D_2)$ in Theorem 5.1.7 (2)). The elements of $\widetilde{K}_F$ commutes with the elements of $\widetilde{K}_\Phi$, hence $\widetilde{K}_0 \subseteq C_{\widetilde{K}_F}, C_{\widetilde{K}_\Phi}$ and via the isomorphism $C_{\widetilde{K}_F}/\widetilde{K}_F \simeq G(D_2 * D_2)$ of Theorem 5.1.7 (2), $\widetilde{K}_X$ must maps to $\widetilde{K}_{\Phi_0}$ by construction, but $\widetilde{K}_0$ also maps to $\widetilde{K}_{\Phi_0}$. This forces $\widetilde{K}_0 = \widetilde{K}_X$. Similarly, $\widetilde{K}_0 = \widetilde{K}_X'$. Hence, D_X and D_X' are linearly equivalent and corresponds to the same lift $\widetilde{K}_G := \widetilde{K}_0$.

Using Theorem 5.1.7 (3), $\Gamma(D_4 * D_4)^{\widetilde{K}_G} \simeq \Gamma(D_X)$ and using the previous results we have

$$\Gamma(D_2 * D_2)^{\widetilde{K}_{\Phi_0}} \simeq \Gamma(D_4 * D_4)^{\widetilde{K}_G} \simeq \Gamma(D_2' * D_2')^{\widetilde{K}_{F_0}}.$$

This is the main ingredient to derive half differential addition formulas, since from our commutative diagram, $\Gamma(D_2 * D_2)^{\widetilde{K}_{\Phi_0}}$ are precisely the subspace of the differential addition formulas, expressed in terms of coordinates $P+Q, P-Q$, that factorize through $\Phi(P,Q)$. The isomorphism with $\Gamma(D_4 * D_4)^{\widetilde{K}_G}$ allows us to write these formulas in terms of coordinates of P, Q, invariant under the action of $\widetilde{K}_G$. Since we know the action of $\widetilde{K}_G$ and the other lifted kernel on sections, basic linear algebra now gives us formulas to find a basis of invariants.

Even coordinates

Recall that over an abelian variety X, $\iota : x \in X \mapsto -x$ is an involution, giving an automorphism $\iota^* : k(X) \to k(X)$. If D is a symmetric divisor, the restriction of ι^* to $\Gamma(D)$ (resp. $G(D)$) is an automorphism of vector spaces (resp. of groups). A section $f \in k(X)$ is said to be even if $\iota^* f = f$ and is said to be odd if $\iota^* f = -f$, as defined in Remark 6.2.2.

Over E, if $\Gamma(D_2) = \langle X, Z\rangle$, a basis of $\Gamma(D_4)$ is $\langle X^2, Z^2, XZ, T\rangle$ where $T = YZ_0$ is a fourth odd section. The isomorphism of Lemma 7.4.4 then yields basis of $\Gamma(D_2 * D_2)$ and $\Gamma(D_4 * D_4)$:

$$\Gamma(D_2 * D_2) = \left\langle \begin{matrix} X_P X_Q & X_P Z_Q \\ Z_P X_Q & Z_P Z_Q \end{matrix} \right\rangle,$$

$$\Gamma(D_4 * D_4) = \left\langle \begin{matrix} X_P^2 X_Q^2 & X_P^2 Z_Q^2 & X_P^2 X_Q Z_Q & X_P^2 T_Q \\ Z_P^2 X_Q^2 & Z_P^2 Z_Q^2 & Z_P^2 X_Q Z_Q & Z_P^2 T_Q \\ X_P Z_P X_Q^2 & X_P Z_P Z_Q^2 & X_P Z_P X_Q Z_Q & X_P Z_P T_Q \\ T_P X_Q^2 & T_P Z_Q^2 & T_P X_Q Z_Q & T_P T_Q \end{matrix} \right\rangle.$$

The issue that arises when looking for differential addition formulas via $\Gamma(D_4 * D_4)^{\widetilde{K}_F} \simeq \Gamma(D_2 * D_2)$ is that, if we only know the Kummer line coordinates $(X : Z)$ of P, Q, we have no information about the section T. We have to restrict our study to even coordinates. We will denote, for any symmetric divisor D, $\Gamma(D)^+ = \Gamma(D)^{\iota^*}$ the subspace of even sections. We have $\Gamma(D_2)^+ = \Gamma(D_2)$ and $\Gamma(D_4)^+ = \langle X^2, Z^2, XZ\rangle$ and there is a canonical surjection $\Gamma(D_2)^+ \otimes \Gamma(D_2)^+ \to \Gamma(D_4)^+,\ f \otimes g \mapsto fg$.

The ι maps over E and $E \times E$ are compatible: if $\iota_1 : E \to E$ and $\iota_2 : E \times E \to E \times E$ are the maps on the corresponding varieties, then $\iota_2^* = \iota_1^* \otimes \iota_1^*$ on $k(E) \otimes k(E) \subseteq k(E \times E)$. Via the isomorphism of Lemma 7.4.4, we get an injection $\Gamma(D_4)^+ \otimes \Gamma(D_4)^+ \to \Gamma(D_4 * D_4)^+$.

However, one can compute the following basis for $\Gamma(D_4 * D_4)^+$:

$$\Gamma(D_4 * D_4)^+ = \left\langle \begin{matrix} X_P^2 X_Q^2 & X_P^2 Z_Q^2 & X_P^2 X_Q Z_Q & Z_P^2 X_Q^2 & Z_P^2 Z_Q^2 \\ Z_P^2 Z_Q Z_Q & X_P Z_P X_Q^2 & X_P Z_P Z_Q^2 & X_P Z_P X_Q Z_Q & T_P T_Q \end{matrix} \right\rangle.$$

The last section is even as it is the product of two odd sections, but the image of the injection is only of dimension 9 since $\Gamma(D_4)^+$ is of dimension 3. We denote the image of the injection $\Gamma(D_4 * D_4)^{++} \subsetneq \Gamma(D_4 * D_4)^+$:

$$\Gamma(D_4 * D_4)^{++} = \left\langle \begin{matrix} X_P^2 X_Q^2 & X_P^2 Z_Q^2 & X_P^2 X_Q Z_Q \\ Z_P^2 X_Q^2 & Z_P^2 Z_Q^2 & Z_P^2 Z_Q Z_Q \\ X_P Z_P X_Q^2 & X_P Z_P Z_Q^2 & X_P Z_P X_Q Z_Q \end{matrix} \right\rangle.$$

This is the set of sections we would like to work with. One can check immediately that $\Gamma(D_4 * D_4)^{++}$ is the set of sections above $D_4 * D_4$ invariants by $\iota^* \otimes 1$ and $1 \otimes \iota^*$. We will use the following lemma:

Lemma 7.4.7 *Let* D *be a symmetric divisor over* E, $g_T \in G(D)$ *a symmetric element above a 2-torsion point* $T \in E[2]$. *Then for any* $f \in \Gamma(D)$, $\iota^*(g_T \cdot f) = g_T \cdot (\iota^* f)$.

Proof *Let* $P \in E$, $f \in \Gamma(D)$, *we have* $\iota^*(g_T \cdot f)(P) = (g_T \cdot f)(-P) = g_T(-P)f(-P + T)$. *Because* T *is a 2-torsion point,* $-T = T$ *and since* g_T *is symmetric,* $g_T(-P) = g_T(P)$:

$$\iota^*(g_T \cdot f)(P) = g_T(P)f(-P - T) = g_T(P)\iota^* f(P + T) = g_T \cdot (\iota^* f)(P). \qquad \square$$

Since the kernels $\widetilde{K}_F$, $\widetilde{K}_\Phi$, $\widetilde{K}_{F_0}$, $\widetilde{K}_{\Phi_0}$ and $\widetilde{K}_G$ are all composed of symmetric elements above 2-torsion points, the action of $\iota^* \otimes 1$, $1 \otimes \iota^*$ and $\iota^* \otimes \iota^*$ commutes with the one of the kernels by the above lemma. Denote by $G_\iota = \{\iota^* \otimes 1, 1 \otimes \iota^*, \iota^* \otimes \iota^*\}$, such that $\Gamma(D_4 * D_4)^{++} = \Gamma(D_4 * D_4)^{G_\iota}$.

We start by studying the isomorphism $\Gamma(D_4 * D_4)^{\widetilde{K}_F} \simeq \Gamma(D_2 * D_2)$. What is the action of $\iota^* \otimes 1$ on $\Gamma(D_2 * D_2)$ through the isomorphism? Let $(R, S) \in E \times E$ and $(P, Q) \in E \times E$ such that $F(P, Q) = (R, S) = (P + Q, P - Q)$, $s \in \Gamma(D_2 * D_2)$. Then $(\iota^* \otimes 1) \cdot F^* s(P, Q) = F^* s(-P, Q)$, and $F(-P, Q) = (-P + Q, -P - Q) = (-S, -R)$, hence $(\iota^* \otimes 1) \cdot F^* s(P, Q) = s(-S, -R)$. Similarly, $(1 \otimes \iota^*) \cdot F^* s(P, Q) = s(S, R)$ and $(\iota^* \otimes \iota^*) \cdot F^* s(P, Q) = s(-R, -S)$. Therefore, firstly we have

$$\left(\Gamma(D_4 * D_4)^{\widetilde{K}_F}\right)^{G_\iota} = \left(\Gamma(D_4 * D_4)^{G_\iota}\right)^{\widetilde{K}_F} = \left(\Gamma(D_4 * D_4)^{++}\right)^{\widetilde{K}_F}$$

by Lemma 7.4.7, and secondly since all elements of $\Gamma(D_2 * D_2)$ are even, if $\tau : (R, S) \mapsto (S, R)$, we have

$$\left(\Gamma(D_4 * D_4)^{+}\right)^{\widetilde{K}_F} \simeq \Gamma(D_2 * D_2) \text{ and } \left(\Gamma(D_4 * D_4)^{++}\right)^{\widetilde{K}_F} \simeq \Gamma(D_2 * D_2)^{\tau^*}.$$

In particular, when looking for differential addition formulas with only X and Z coordinates, we cannot choose whichever section of $\Gamma(D_2 * D_2)$ we want, it must be invariant by permutation of R and S.

The situation is easier regarding Φ because it is a diagonal isogeny, hence one can check

$$\left(\Gamma(D_4 * D_4)^{++}\right)^{\widetilde{K}_\Phi} \simeq \Gamma(D_2' * D_2').$$

If we go down one more level however, the invariance by τ is automatic:

$$\Gamma(D_2 * D_2)^{\widetilde{K}_{\Phi_0}} \simeq \left(\Gamma(D_4 * D_4)^{++}\right)^{\widetilde{K}_G} \simeq \Gamma(D_2' * D_2')^{\widetilde{K}_{F_0}}.$$

In summary, the full set of differential addition formulas $\Gamma(D_4 * D_4)^{\widetilde{K}_F}$ is of dimension 4 and is automatically given by even sections. However, if we want to remove the section $T_P T_Q$ which cannot be computed from X_P, Z_P, X_Q, Z_Q, we need to work with the dimension 3 subspace $\left(\Gamma(D_4 * D_4)^{++}\right)^{\widetilde{K}_F}$. On the codomain of F, this corresponds to level 2 sections of $(R = P + Q, S = P - Q)$ that are also invariant by the permutation of R, S; for instance $X_R X_S, Z_R Z_S, X_R Z_S + Z_R X_S$, but not $X_R Z_S$. Finally, the half differential addition formulas correspond to sections in $\Gamma(D_4 * D_4)^{\widetilde{K}_G} \simeq \Gamma(D_2 * D_2)^{\widetilde{K}_{\Phi_0}}$, this is a space of dimension 2 which is automatically inside $\Gamma(D_4 * D_4)^{++}$.

Finding formulas

This is all we required to find our half differential addition formulas. Assume $E[2] = \{0_E, T_1, T_2, T_3\}$ with T_1 being rational, the method is then as follows:

1. Set $T = T_1$ such that $\ker \varphi = \{0_E, T\}$, we first compute the translation by T on the Kummer line associated to E, denoted t_T. If $T' = \varphi(T_2) = \varphi(T_3)$, we compute the translation by T' on the Kummer line associated to E', denoted $t_{T'}$. By considering the affine lifts of these translations, we derive the types of T and T', denoted λ_T and $\lambda_{T'}$.

2. Compute the invariant subspaces $\Gamma(D_2 * D_2)^{\widetilde{K}_{\Phi_0}}$ and $\Gamma(D_2' * D_2')^{\widetilde{K}_{F_0}}$.

3. Find the coefficients relating bases of $\Gamma(D_2 * D_2)^{\widetilde{K}_{\Phi_0}}$ and $\Gamma(D_2' * D_2')^{\widetilde{K}_{F_0}}$.

Remark 7.4.8 *If $\Gamma(D_2 * D_2)^{\widetilde{K}_{\Phi_0}} = \langle u_1, u_2 \rangle$ and $\Gamma(D_2' * D_2')^{\widetilde{K}_{F_0}} = \langle v_1, v_2 \rangle$, we then know there are relations*

$$\begin{cases} u_1(P+Q, P-Q) = \alpha_1 v_1(\varphi(P), \varphi(Q)) + \alpha_2 v_2(\varphi(P), \varphi(Q)), \\ u_2(P+Q, P-Q) = \beta_1 v_1(\varphi(P), \varphi(Q)) + \beta_2 v_2(\varphi(P), \varphi(Q)). \end{cases}$$

For our purpose, we consider these relations projectively, so we can multiply u_1 and u_2 by a same factor.

In the first step, we compute $t_T : \mathbb{P}^1 \to \mathbb{P}^1$ projectively, let $\widetilde{t}_T : k^2 \to k^2$ be an affine lift. Then $\widetilde{t}_T^2 = \lambda_T \operatorname{id}$. The map $\tau_T = \frac{1}{\lambda_T}(\widetilde{t}_T^* \otimes \widetilde{t}_T^*)$ is an involution of $k(E) \otimes k(E)$. Assume we have $\widetilde{u} \in \Gamma(D_2 * D_2)$ such that $\tau_T(\widetilde{u}) = \widetilde{u}$. Then, for $g_{(T,T)} \in \widetilde{K}_{\Phi_0}$ above (T, T), we have $g_{(T,T)} \cdot \widetilde{u} = g_{(T,T)}\lambda_T\tau_T(\widetilde{u}) = g_{(T,T)}\lambda_T\widetilde{u}$, and an element u invariant by $g_{(T,T)}$ is given by

$$u := (1 + \lambda_T g_{(T,T)})\widetilde{u}.$$

This is how we deal with the second step, we compute two invariants by τ_T, which projectively will lead to the same result as invariants by $g_{(T,T)}$ would have. The same holds for the point T'.

7.4.2 Example

In this example, we will consider the theta model $\theta(a : b)$ associated to E given by 2-torsion points

$$0_E = (a : b)^*, \quad T_1 = (-a : b), \quad T_2 = (b : a), \quad T_3 = (-b : a),$$

where $\frac{a}{b} \in k$. The 2-isogeny will be $\varphi : (X : Z) \mapsto (X^2 : Z^2)$. Its kernel is given by $T = T_1$. The Kummer line on the image has ramification points

$$0_{E'} = (a^2 : b^2)^*, \quad T_1' = (b^2 : a^2), \quad T_2' = (1 : 0), \quad T_3' = (0 : 1),$$

this is the theta squared model $\theta_s(a^2 : b^2)$, the interesting 2-torsion point on this model is $T' = \varphi(T_2) = \varphi(T_3) = T_1'$.

As given in Examples 7.1.2 and 7.1.3, the translations by T and T' on their respective model are $t_T : (X : Z) \mapsto (-X : Z)$ and $t_{T'} : (X : Z) \mapsto (Z : X)$. We

consider the affine lifts $\widetilde{t}_T : (X, Z) \mapsto (-X, Z)$ and $\widetilde{t}_{T'} : (X, Z) \mapsto (Z, X)$. The types are then $\lambda_T = \lambda_{T'} = 1$, we set $\tau_T = \widetilde{t}_T^* \otimes \widetilde{t}_T^*$ and $\tau_{T'} = \widetilde{t}_{T'}^* \otimes \widetilde{t}_{T'}^*$.

We will work with the following basis on $\Gamma(D_2 * D_2)$, where $(P, Q) \in E \times E$:

$$\Gamma(D_2 * D_2) = \left\langle \begin{matrix} X_{P+Q}X_{P-Q} & X_{P+Q}Z_{P-Q} \\ Z_{P+Q}X_{P-Q} & Z_{P+Q}Z_{P-Q} \end{matrix} \right\rangle.$$

τ_T acts as follows on $\Gamma(D_2 * D_2)$:

- $\tau_T(X_{P+Q}X_{P-Q}) = X_{P+Q+T}X_{P-Q+T} = (-X_{P+Q})(-X_{P-Q}) = X_{P+Q}X_{P-Q}$,
- $\tau_T(X_{P+Q}Z_{P-Q}) = X_{P+Q+T}Z_{P-Q+T} = -X_{P+Q}Z_{P-Q}$,
- $\tau_T(Z_{P+Q}X_{P-Q}) = Z_{P+Q+T}X_{P-Q+T} = -Z_{P+Q}X_{P-Q}$,
- $\tau_T(Z_{P+Q}Z_{P-Q}) = Z_{P+Q+T}Z_{P-Q+T} = Z_{P+Q}Z_{P-Q}$.

$X_{P+Q}X_{P-Q}$ and $Z_{P+Q}Z_{P-Q}$ are invariant sections, and since $\Gamma(D_2 * D_2)^{\widetilde{K}_{\Phi_0}}$ is of dimension 2, we get

$$\Gamma(D_2 * D_2)^{\widetilde{K}_{\Phi_0}} = \langle X_{P+Q}X_{P-Q}, Z_{P+Q}Z_{P-Q} \rangle.$$

Regarding $\Gamma(D_2' * D_2')$, we will use the following basis instead:

$$\left\langle \begin{matrix} (X_{\varphi(P)} + Z_{\varphi(P)})(X_{\varphi(Q)} + Z_{\varphi(Q)}), & (X_{\varphi(P)} + Z_{\varphi(P)})(X_{\varphi(Q)} - Z_{\varphi(Q)}) \\ (X_{\varphi(P)} - Z_{\varphi(P)})(X_{\varphi(Q)} + Z_{\varphi(Q)}), & (X_{\varphi(P)} - Z_{\varphi(P)})(X_{\varphi(Q)} - Z_{\varphi(Q)}) \end{matrix} \right\rangle.$$

The reason is that the computations with $\tau_{T'}$ are easier. We get two invariants of $\Gamma(D_2 * D_2)^{\widetilde{K}_{F_0}}$:

$$\langle (X_{\varphi(P)} + Z_{\varphi(P)})(X_{\varphi(Q)} + Z_{\varphi(Q)}), (X_{\varphi(P)} - Z_{\varphi(P)})(X_{\varphi(Q)} - Z_{\varphi(Q)}) \rangle.$$

We now set, for $(P, Q) \in E \times E$:

- $u_1(P, Q) = X_{P+Q}X_{P-Q}$,
- $u_2(P, Q) = Z_{P+Q}Z_{P-Q}$,
- $v_1(P, Q) = (X_{\varphi(P)} + Z_{\varphi(P)})(X_{\varphi(Q)} + Z_{\varphi(Q)})$,
- $v_2(P, Q) = (X_{\varphi(P)} - Z_{\varphi(P)})(X_{\varphi(Q)} - Z_{\varphi(Q)})$,

By the theory, there are constants $\alpha_1, \alpha_2, \beta_1, \beta_2 \in k$ such that for any $(P, Q) \in E \times E$:

$$(u_1(P, Q) : u_2(P, Q)) = (\alpha_1 v_1(P, Q) + \alpha_2 v_2(P, Q) : \beta_1 v_1(P, Q) + \beta_2 v_2(P, Q)).$$

We then evaluate at specific points to get relations on the coefficients. On the theta model $\theta(a : b)$, we denote by $\overline{T_1} = (1 : 0)$ and $\underline{T_1} = (0 : 1)$ the 4-torsion points above T_1, and the usual $(A^2 : B^2) = (a^2 + b^2 : a^2 - b^2)$.

1. If $(P, Q) = (0_E, 0_E)$, then $(P + Q, P - Q) = (0_E, 0_E)$ and $(\varphi(P), \varphi(Q)) = (0_{E'}, 0_{E'})$:
$$(a^2 : b^2) = (\alpha_1 A^4 + \alpha_2 B^4 : \beta_1 A^4 + \beta_2 B^4).$$

2. If $(P, Q) = (\overline{T_1}, 0_E)$, then $(P + Q, P - Q) = (\overline{T_1}, \overline{T_1})$ and $(\varphi(P), \varphi(Q)) = (T'_2, 0_{E'})$:
$$(1 : 0) = (\alpha_1 A^2 + \alpha_2 B^2 : \beta_1 A^2 + \beta_2 B^2).$$

3. If $(P, Q) = (\underline{T_1}, 0_E)$, then $(P + Q, P - Q) = (\underline{T_1}, \underline{T_1})$ and $(\varphi(P), \varphi(Q)) = (T'_3, 0_{E'})$:
$$(0 : 1) = (\alpha_1 A^2 - \alpha_2 B^2 : \beta_1 A^2 - \beta_2 B^2).$$

The second and third relations give $\beta_1 A^2 = -\beta_2 B^2$ and $\alpha_1 A^2 = \alpha_2 B^2$. When injected in the first one, we get

$$(a^2 : b^2) = (\alpha_2 B^2(A^2 + B^2) : -\beta_2 B^2(A^2 - B^2)) = (\alpha_2 a^2 : -\beta_2 b^2).$$

Hence, $\alpha_2 = -\beta_2$ and we can derive the general formula from these relations:

$$\begin{aligned}&(\alpha_1 v_1 + \alpha_2 v_2 : \beta_1 v_1 + \beta_2 v_2) = (\alpha_1 A^2 v_1 + \alpha_2 A^2 v_2 : \beta_1 A^2 v_1 + \beta_2 A^2 v_2)\\ &= (\alpha_2(B^2 v_1 + A^2 v_2) : \beta_2(-B^2 v_1 + A^2 v_2)) = (B^2 v_1 + A^2 v_2 : B^2 v_1 - A^2 v_2).\end{aligned}$$

Thus, the formulas $\texttt{HalfDiffAdd}_\varphi(\varphi(P), \varphi(Q), P - Q)$ associated to φ are:

$$\begin{aligned}&(X_{P+Q}X_{P-Q} : Z_{P+Q}Z_{P-Q}) =\\ &\begin{pmatrix} B^2(X_{\varphi(P)} + Z_{\varphi(P)})(X_{\varphi(Q)} + Z_{\varphi(Q)}) + A^2(X_{\varphi(P)} - Z_{\varphi(P)})(X_{\varphi(Q)} - Z_{\varphi(Q)})\\ B^2(X_{\varphi(P)} + Z_{\varphi(P)})(X_{\varphi(Q)} + Z_{\varphi(Q)}) - A^2(X_{\varphi(P)} - Z_{\varphi(P)})(X_{\varphi(Q)} - Z_{\varphi(Q)})\end{pmatrix}.\end{aligned}$$

Remark 7.4.9 (Montgomery differential addition) *On a Montgomery Kummer line given by*
$$(1 : 0)^*, \quad (0 : 1), \quad (a : b), \quad (b : a)$$
where $\frac{a}{b}$ *may not be rational, the translation by* $T = (0 : 1)$ *is* $t_T : (X : Z) \mapsto (Z : X)$. *As we have seen above,* $X_{P+Q}X_{P-Q}$ *and* $Z_{P+Q}Z_{P-Q}$ *are not invariant by this translation. This means that we cannot factor the traditional differential addition formulas into half differential addition formulas. We can still find new formulas like the ones given in Section 7.3 on* `Curve25519`*, but they are not as efficient as the usual ones.*

7.5 ANOTHER EXAMPLE ON CURVE25519

In this section, we detail another example on how to obtain half differential addition formulas over a Montgomery Kummer line $M(A : B)$ with ramification
$$(1 : 0)^*, \quad (0 : 1), \quad (A : B), \quad (B : A),$$
but where $\frac{A}{B}$ may not be rational. We further assume there is a 8-torsion point $(r : s)$ above $(1 : 1)$, itself above $(0 : 1)$. `Curve25519` for instance verifies such hypotheses. We set the following additional constants: $(\gamma : \delta) = (4rs : (r - s)^2)$ and $(a : b) = (\gamma - \delta : \gamma + \delta)$. The 2-isogeny with kernel $T = (0 : 1)$ we are interested in is

$$\psi : (X : Z) \mapsto (ab(X - Z)^2 - a\delta(X + Z)^2 : ab(X - Z)^2 + b\delta(X + Z)^2).$$

Its dual is given by

$$\widetilde{\psi} : (X : Z) \mapsto (aZ^2 - bX^2 + 2\delta XZ : aZ^2 - bX^2 - 2\delta XZ).$$

The translation by T on $M(A : B)$ is simply $t_T : (X : Z) \mapsto (Z : X)$ as explained in Example 7.1.1, with affine lift $\widetilde{t}_T : (X, Z) \mapsto (Z, X)$. The image is the theta twisted model $\theta_t(a : b)$ with ramification

$$(a : b)^*, \quad (-a : b), \quad (1 : 1), \quad (-1 : 1).$$

Recall these are derived from Theorem 6.3.10 and Proposition 6.3.11 and the composition with isomorphisms from Example 7.1.4 between theta twisted and Montgomery models.

We have:

- $\psi(1 : 0) = \psi(0 : 1) = (a : b)$,
- $\psi(A : B) = \psi(B : A) = (-1 : 1)$,
- $\psi(1 : 1) = (-a : b)$,
- $\psi(-1 : 1) = (1 : 1)$.

The point of interest on $\theta_t(a : b)$ is then $T' = (-1 : 1)$. We want to compute $t_{T'}$. It is a homography of $\mathbb{P}^1$, which must verify:

- $t_{T'}(a : b) = (-1 : 1)$;
- $t_{T'}(-1 : 1) = (a : b)$;
- $t_{T'}(-a : b) = (1 : 1)$;
- $t_{T'}(1 : 1) = (-a : b)$.

This leads to $t_{T'} : (X : Z) \mapsto (-aZ : bX)$, its affine lift is $\widetilde{t}_{T'} : (X, Z) \mapsto (-aZ, bX)$.

We have $\widetilde{t}_T^2 = \mathrm{id}$ so $\lambda_T = 1$ and $\widetilde{t}_{T'}^2 = -ab\,\mathrm{id}$, so $\lambda_{T'} = -ab$. We set $\tau_T = \widetilde{t}_T^* \otimes \widetilde{t}_T^*$ and $\tau_{T'} = \frac{1}{-ab}\widetilde{t}_{T'}^* \otimes \widetilde{t}_{T'}^*$. This is part of the reason we wanted to illustrate this example where the type has no reason to be a square, and we have to consider it in the computations. We work with the usual bases of $\Gamma(D_2 * D_2)$ and $\Gamma(D_2' * D_2')$:

$$\Gamma(D_2 * D_2) = \begin{pmatrix} X_{P+Q}X_{P-Q} & X_{P+Q}Z_{P-Q} \\ Z_{P+Q}X_{P-Q} & Z_{P+Q}Z_{P-Q} \end{pmatrix},$$

$$\Gamma(D_2' * D_2') = \begin{pmatrix} X_{\psi(P)}X_{\psi(Q)} & X_{\psi(P)}Z_{\psi(Q)} \\ Z_{\psi(P)}X_{\psi(Q)} & Z_{\psi(P)}Z_{\psi(Q)} \end{pmatrix}.$$

τ_T acts as follows on $\Gamma(D_2 * D_2)$:

- $\tau_T(X_{P+Q}X_{P-Q}) = Z_{P+Q}Z_{P-Q}$,
- $\tau_T(X_{P+Q}Z_{P-Q}) = Z_{P+Q}X_{P-Q}$.

This gives two invariants $X_{P+Q}X_{P-Q}+Z_{P+Q}Z_{P-Q}$ and $X_{P+Q}Z_{P-Q}+Z_{P+Q}X_{P-Q}$. With some linear algebra, we can derive the two invariants

$$(X_{P+Q}+Z_{P+Q})(X_{P-Q}+Z_{P-Q}),\ (X_{P+Q}-Z_{P+Q})(X_{P-Q}-Z_{P-Q}).$$

Therefore, we recover the following basis:

$$\Gamma(D_2 * D_2)^{\widetilde{K}_{\Phi_0}} = \langle (X_{P+Q}+Z_{P+Q})(X_{P-Q}+Z_{P-Q}), (X_{P+Q}-Z_{P+Q})(X_{P-Q}-Z_{P-Q})\rangle.$$

In the same manner, we look at invariants for $\tau_{T'}$:

- $\tau_{T'}(X_{\psi(P)}X_{\psi(Q)}) = \frac{a^2}{-ab}Z_{\psi(P)}Z_{\psi(Q)} = -\frac{a}{b}Z_{\psi(P)}Z_{\psi(Q)}$,
- $\tau_{T'}(X_{\psi(P)}Z_{\psi(Q)}) = \frac{-ab}{-ab}Z_{\psi(P)}X_{\psi(Q)} = Z_{\psi(P)}X_{\psi(Q)}$.

By rescaling the first invariant, we get the following basis:

$$\Gamma(D_2' * D_2')^{\widetilde{K}_{F_0}} = \langle aZ_{\psi(P)}Z_{\psi(Q)} - bX_{\psi(P)}X_{\psi(Q)}, X_{\psi(P)}Z_{\psi(Q)} + Z_{\psi(P)}X_{\psi(Q)}\rangle.$$

We then set:

- $u_1(P,Q) = (X_{P+Q}+Z_{P+Q})(X_{P-Q}+Z_{P-Q})$,
- $u_2(P,Q) = (X_{P+Q}-Z_{P+Q})(X_{P-Q}-Z_{P-Q})$,
- $v_1(P,Q) = aZ_{\psi(P)}Z_{\psi(Q)} - bX_{\psi(P)}X_{\psi(Q)}$,
- $v_2(P,Q) = X_{\psi(P)}Z_{\psi(Q)} + Z_{\psi(P)}X_{\psi(Q)}$.

We are looking for the constants $\alpha_1, \alpha_2, \beta_1, \beta_2 \in k$ such that:

$$(u_1(P,Q) : u_2(P,Q)) = (\alpha_1 v_1(P,Q) + \alpha_2 v_2(P,Q) : \beta_1 v_1(P,Q) + \beta_2 v_2(P,Q)).$$

Consider $\overline{T} = (1:1)$ and $\underline{T} = (-1:1)$ the 4-torsion points above $T = (0:1)$, then:

- $F(\overline{T}, 0_E) = (\overline{T}, \overline{T})$, $\psi(\overline{T}) = (-a:b)$, $\psi(0_E) = (a:b)$, giving the equation
$$(1:0) = (\alpha_1 ab(b-a) + 0 : \beta_1 ab(b-a) + 0) \implies \beta_1 = 0.$$
- $F(\underline{T}, 0_E) = (\underline{T}, \underline{T})$, $\psi(\underline{T}) = (1:1)$, $\psi(0_E) = (a:b)$, giving the equation
$$(0:1) = (0 + \alpha_2(a+b) : \beta_2(a+b)) \implies \alpha_2 = 0.$$
- $F(\underline{T}, \underline{T}) = (T, 0_E)$, $\psi(\underline{T}) = (1:1)$, giving the equation
$$(1:-1) = (\alpha_1(a-b) : 2\beta_2) \implies \alpha_1(b-a) = 2\beta_2 \implies \beta_2 = \delta\beta_1.$$

The half differential addition formulas $\texttt{HalfDiffAdd}_\psi(\psi(P), \psi(Q), P-Q)$ are then:

$$\begin{pmatrix}(X_{P+Q}+Z_{P+Q})(X_{P-Q}+Z_{P-Q})\\(X_{P+Q}-Z_{P+Q})(X_{P-Q}-Z_{P-Q})\end{pmatrix} = \begin{pmatrix}aZ_{\psi(P)}Z_{\psi(Q)} - bX_{\psi(P)}X_{\psi(Q)}\\ \delta(X_{\psi(P)}Z_{\psi(Q)} + Z_{\psi(P)}X_{\psi(Q)})\end{pmatrix}.$$

7.6 HALF DIFFERENTIAL ADDITIONS IN HIGHER DIMENSION

The arithmetic of theta function comes from the duplication formula, which can naturally be expressed as `HalfDiffAdd` operations. In this section, we give half differential addition formulas in any dimension, for level 2 theta coordinates. We then look at the impact of half ladder on the efficiency for abelian surfaces, compared to the standard ladder as introduced in [CC86; GL09].

Let (X, Θ_X) be a principally polarized abelian variety with a symmetric level 2 theta structure. We let $\theta_i, i \in (\mathbb{Z}/2\mathbb{Z})^g$ be the basis of theta functions of level 2. We also let $\varphi : X \to X'$ be the canonical isogeny induced from the theta structure, and a choice of compatible level 2 theta structure on X', with dual theta functions θ'_i.

Then we have the key formula, used in [Dar+24, § 2.2] to derive efficient 2^n-isogeny formulas.

$$(\theta_i(P+Q))_i \circ (\theta_i(P-Q))_i = H((\theta'_i(\varphi(P))) \circ (\theta'_i(\varphi(Q)))) \tag{7.3}$$

Here $(\theta_i(P))_i \circ (\theta_i(Q))_i$ denotes the component wise product $(\theta_i(P)\theta_i(Q))_i$, and H is the Hadamard transform.

From the knowledge of the theta constant $\theta_i(0_X)$ on X and the dual theta constants $\theta'_i(0_{X'})$ on X', it is possible to both compute the dual theta coordinates $\theta'_i(\varphi(P))$ from the theta coordinates of P (by setting $Q = 0_X$ in Eq. (7.3)), and to use Eq. (7.3) as a half differential addition formula.

7.6.1 The theta model

In this subsection, we assume that these theta constants are rational. Assuming that the theta null points of X and X' have been normalized and their inverse computed (this is a pre-computation which does not depend on the choice of base point P for the ladder), then an isogeny image costs $2^g\mathbf{S} + (2^g - 1)\mathbf{m}_0$, a half doubling costs $2^g\mathbf{S} + (2^g - 1)\mathbf{m}_0$, and a half differential addition costs $2^g\mathbf{M} + 2^g\mathbf{I}$. In dimension 1, since we are working with projective coordinates, these $2\mathbf{I}$ could be easily replaced by $2\mathbf{M}$. This is not the case any more in dimension greater than 1: the $2^g\mathbf{I}$ can only be replaced by $(2^{g+2} - 6)\mathbf{M}$. This means that in the half ladder, it is quite expensive to use non-normalized pre-computed points P_i, and so it will become interesting only when doing a full normalized pre-computation.

In the standard ladder, at each step we compute two isogeny images, one half doubling and one half differential addition. The differential addition is always done with the same base point P as difference, so via a pre-computation at the start to normalize P these $2^g\mathbf{I}$ become $(2^g - 1)\mathbf{M}$. The total cost is then $(2^{g+1} - 1)\mathbf{M} + 3.2^g\mathbf{S} + 3(2^g - 1)\mathbf{m}_0$ by bit. Sharing some of the operations between the doubling and differential addition, we can have a $(2^g - 1)(\mathbf{M} - \mathbf{S} - \mathbf{m}_0)$ trade-off, which gives a complexity of $(3 \cdot 2^g - 2)\mathbf{M} + (2^{g+1} + 1)\mathbf{S} + 2(2^g - 1)\mathbf{m}_0$ by bit.

In the half ladder, we first compute iterated isogeny images $P_i = \varphi_i(P_{i-1})$ (one by bit), and then at each step we do one half doubling and one half differential additions. Then we use one half doubling to recover $2 \cdot R_{i-1} = \texttt{HalfDouble}_{\varphi_i}(\varphi_i(R_{i-1}))$ or $2 \cdot S_{i-1} = \texttt{HalfDouble}_{\varphi_i}(\varphi_i(S_{i-1}))$ and one half differential addition $R_{i-1} + S_{i-1} = \texttt{HalfDiffAdd}_{\varphi_i}(\varphi_i(R_{i-1}), \varphi_i(S_{i-1}), P_{i-1})$, where $P_{i-1} = R_{i-1} - S_{i-1}$ has been pre-computed.

This time, the pre-computation to normalize each P_i, or rather to compute the inverse of its normalized coordinates is much more expensive: we need $(2^g - 1)$ divisions. Writing the divisions as $(2^g - 1)$ inversions followed by $(2^g - 1)$ multiplications, taking into account that these inversions cost $1\mathbf{I}$ and $(3(2^g-1)-3)\mathbf{M}$ by Lemma 2.1.6, we obtain a overall cost of $1\mathbf{I}+(4(2^g-1)-3)\mathbf{M}$ to compute these inverted coordinates. However, we note that the P_i only depend on P (and the scalar bit length), not on the actual scalar, so this pre-computation can be reused whenever we do several scalar multiplications with the same base point P. Taking into account the isogeny images, the global cost is one global inversion and $1\mathbf{I}+(4(2^g-1)-3)\mathbf{M}+2^g\mathbf{S}+(2^g-1)\mathbf{m}_0$ by bit for the pre-computations, which only depends on the base point P, and then $(2^{g+1}-1)\mathbf{M}+2^g\mathbf{S}+(2^g-1)\mathbf{m}_0$ by bits for the scalar exponentiations $m \mapsto m \cdot P$.

We remark that in the right-to-left Montgomery ladder, we have the same normalization problem, except that here the base points used in the differential addition formulas depend on the scalar n, so it is not possible to share the pre-computation once and for all. Indeed, the right-to-left ladder uses normal differential addition of the form $R_i + S_i = \texttt{DiffAdd}(R_i, S_i, R_i - S_i)$ where this time it is $R_i = 2^i \cdot P$ which has been pre-computed, rather than the difference $R_i - S_i$.

In Table 7.5, we put a comparison of the cost between the ladder and half ladder for abelian surfaces. We remark that the $3\mathbf{m}$ in the standard Montgomery ladder assume that the base point P is normalized and that the $1/\theta_i(P)$ have been computed; otherwise these become $6\mathbf{M}$. If the theta constants are small, we can replace the $7\mathbf{M} + 9\mathbf{S} + 3\mathbf{m} + 6\mathbf{m}_0$ cost by $4\mathbf{M} + 12\mathbf{S} + 3\mathbf{m} + 9\mathbf{m}_0$.

As we see from this table, unlike the case of dimension 1 where the pre-computation was so cheap the half ladder was still more efficient than the standard ladder even when including the pre-computation cost, in dimension 2 using the half ladder is only interesting when the same base point P will be reused several times. This will be the case for instance in signature schemes like μ-Kummer [Ren+16] or qDSA [RS17].

Algorithm	Pre-computation	Normalization	Step
Montgomery Ladder LtR	—	—	$7\mathbf{M} + 9\mathbf{S} + 3\mathbf{m} + 6\mathbf{m}_0$
Half ladder, our contribution	$4\mathbf{S} + 3\mathbf{m}_0$	$3\mathbf{I} + 3\mathbf{M} \overset{\text{asy}}{=} 12\mathbf{M}$	$7\mathbf{M} + 4\mathbf{S} + 3\mathbf{m}_0$

Table 7.5: Ladder costs per bit for the half ladder in dimension 2

7.6.2 The twisted theta model

We now explain how to adapt the half ladder to the twisted theta model, like we did in Algorithm 7.2. We recall that if $P = (\theta_i(P))_i$, its coordinates in the twisted theta model are given by $\theta_i(P)\theta_i(0)$. In particular, the twisted theta coordinates of the neutral point are the $\theta_i(0)^2$, so the twisted theta model only require these squares to be rational. It is customary to use the theta squared model instead, using the square of the standard theta coordinates. If $X \to X'$ is the 2-isogeny associated to the theta model, with the neutral point on X given by standard theta coordinates $\theta_i^X(0_X)$ and the one on X' given by the dual theta coordinates $\theta'^{X'}_i(0_{X'})$, then the twisted theta model on $(X, \theta_i^X(0_X))$ is the Hadamard transform of the squared theta model on $(X', \theta'^{X'}_i(0_{X'}))$.

We will also use the twisted$^\times$ theta model, which is given by $\theta_i(P)/\theta_i(0)$, this time the coordinates of the neutral point are given by $(1 : \cdots : 1)$.

The half ladder works as follows in the twisted theta model: we first pre-compute our series of 2-isogenies $P_i = \varphi_i(P_{i-1})$ in the twisted theta model. If the twisted coordinates are $X_j(P) = \theta_j(P)\theta_j(0)$, we can compute the isogeny via $(X_j(P_i))_j = H((X_j^2(P_{i-1})/X_j(0_{i-1}))_j)$. This costs $2^g\mathbf{S} + (2^g - 1)\mathbf{m}_0$, to which we need to add $(4(2^g - 1) - 3)\mathbf{M}$ for the precomputation of the inverse coordinates of the normalization of this point. The full precomputation cost is $(4(2^g - 1) - 3)\mathbf{M} + 2^g\mathbf{S} + (2^g - 1)\mathbf{m}_0$, like in the theta model.

Then we start with P_m in the twisted theta model, and 0 in the twisted$^\times$ theta model, and at each step we do a `HalfDiffAdd` and a `HalfDouble`, such that at the end the point that receive the `HalfDiffAdd` is in the twisted$^\times$ model, while the point that receives the `HalfDouble` is in the twisted model.

The formulas for the `HalfDiffAdd` are given by:

$$(X_j(R^\times)X_j(S))_j = H((X_j(\varphi(P)^\times)X_j(\varphi(Q)))_j)$$

and the `HalfDouble` formulas when the point is in twisted$^\times$ coordinates is given by $(X_j(\widetilde{\varphi}(\varphi(P))))_j = H((X_j^2(\varphi(P)^\times)X_j(0))_j)$.

The `HalfDiffAdd` costs $2^g\mathbf{M} + (2^g - 1)\mathbf{M}$, and the `HalfDouble` costs $2^g\mathbf{S} + (2^g - 1)\mathbf{m}_0$ for a total cost of $(2^{g+1} - 1)\mathbf{M} + 2^g\mathbf{S} + (2^g - 1)\mathbf{m}_0$ for the half ladder in twisted theta coordinates, which is the same cost as in theta coordinates, and so the cost of Table 7.5 still holds. We provide an implementation[3] in dimension 2.

3 https://gitlab.inria.fr/roberdam/kummer-line/-/tree/public/surface

8 CLASSIFICATION OF KUMMER LINES VIA GALOIS REPRESENTATION

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This chapter corresponds to [BRS25] available as a pre-print. It is a natural sequel to the previous chapters on Kummer lines. The line of investigations is the following:

- It is easy to say what the Kummer line of an elliptic curve in Montgomery form, or Edwards form or theta coordinates looks like. Can one define the Kummer lines in a unified and abstract manner to encompass any model, *without* going back to a model of the curve E itself? Can one make a complete list of Kummer models up to some equivalence that has to be specified?

- It is known that Montgomery curves have a rational 2-torsion point and a fast doubling formula. Similarly, it is known that Hessian curves have a 3-isogeny and have a fast tripling formula. It is part of the folklore that this is related to the fact that the doubling endomorphism factors as a composition of two isogenies and on a computer it is faster to evaluate

the composition of two polynomials when compared to evaluating a random polynomial of the same degree as their composition. Can one find efficient 2-isogeny formulas on Kummer lines in all models having a rational 2-torsion point?

- It is known that if P and Q are two points on an elliptic curve and P, Q and $P-Q$ are known on the Kummer line, then one can compute $P+Q$. Can we exploit a rational point of 2-torsion to speed up differential additions on Kummer lines, similar to how it was exploited to speed up doublings?

The last two items were the topic of Chapters 6 and 7, in this chapter we tackle the questions in the first one, i.e. we give a unified description of Kummer line models. Also, it allows one to find in a unified manner conversion formulas from one model to another. Finally, we revisit from a more geometric point of view classical models used in the literature: Montgomery models, theta, theta squared and theta twisted models.

Although this chapter gives a new and general point of view on old and new results on Kummer lines, it was the last article of the three we submitted. Indeed, having a good geometric understanding of the characterization of the isomorphism class of a Kummer line, as explained in Sections 4.3 and 8.7, was key to obtain the arithmetic results from Chapters 6 and 7 which bring further cryptographic applications of this chapter.

Some applications

In isogeny-based cryptography, a recent trend has been the use of higher dimensional isogenies, between product of elliptic curves. Currently, their implementation all use theta coordinates of level 2. By contrast, it is slightly more efficient to use the Montgomery models on elliptic curves. The change of coordinates from Section 8.2 have been used to convert between both models for the higher dimensional isogeny if required. Although conversion between Montgomery and theta coordinates where already given in [HR19], for the isogeny applications we really need to keep track of the associated level structure, in particular the extra level 4 information, which is easy to do with the tools we introduce in this chapter. As an example, the conversion formulas used in dimension 2 [Dar+24] or dimension 4 [Dar24] come from results obtained during this work.

For the round 2 submission of SQISign, the contributors needed a method to select a Montgomery model from a theta model (obtained via a 2d isogeny computation). In order not to leak any information from the extra level information contained in the choice of a Montgomery model, it was needed to select one uniformly among the 6 possible ones. This was done using the results of Example 8.4.11.

Recently, it was observed in [Rob24] that one could also use the 'cubical arithmetic' on abelian varieties, constructed via work of Mumford, Grothendieck and Breen, for algorithmic purpose. Indeed, the cubical arithmetic refines the standard arithmetic and give a unified framework for the computations of pairings and isogenies. It is easy to extend our results of Section 8.2

of isomorphisms of Kummer line to obtain isomorphism of the underlying cubical torsor structure. Indeed, by the unicity of the cubical torsor structure, it suffices to lift the isomorphism from a projective linear change of variable to an affine change of variable, such that the cubical neutral point of the domain is sent to the cubical neutral point of the codomain. For instance, the map $(X : Z) \mapsto (bX + aZ : aZ - bX)$ from Proposition 8.2.2 for the conversion from a theta model to a Montgomery model can be lifted to a map $(X, Z) \mapsto (X/a + Z/b, Z/b - X/a)$. This sends the cubical neutral point (a, b) to $(1, 0)$, hence is an isomorphism of cubical torsor structure. This isomorphism was originally used in [Rob24] to obtain the cubical formulas for the Montgomery model from the ones from the theta model. In the end, a more direct proof of the Montgomery cubical formulas was given in this chapter.

A brief presentation of the results

The chapter is relatively long and the reader can benefit from a brief abstract of the results. To do so we use a language which is more abstract than the rest of the chapter, where we introduce the concepts gradually.

Geometric description of Kummer lines

The Kummer line $E/\pm 1$ is isomorphic, as a scheme, to $\mathbb{P}^1$. However, it is not possible to recover E from the projective line, we need extra data. We first show that it is (almost) enough to give as extra marking the four ramification points on $\mathbb{P}^1$ (the image of the 2-torsion point on E), along with which point out of those four is the point 0_E. (It is customary to send 0_E to the point at infinity $(1 : 0)$ on the Kummer line, but some of the models we will study won't have this property). This model will be rational (i.e. correspond to the Kummer line of a rational elliptic curve) whenever the ramification is stable under the Galois action, and the image of 0_E is rational. We note that from this data, we can only recover E up to a quadratic twist.

Although our proofs of the above facts are completely elementary, their underlying structural reason is that the Kummer line ought to be taken as a stacky quotient $[E/\mu_2]$, where $\mu_2 = \{-1, 1\}$ is the group scheme of square roots of the unity, rather than as a schematic quotient E/μ_2 (the latter is the coarse space of the former). We explain this briefly in Section 8.7. This also explains why the Kummer line can 'see' E only up to a quadratic twist: the choice of E is determined by the stacky quotient $[E/\pm 1]$ along with a choice of map $[E/\mu_2] \to B\mu_2$, where $B\mu_2 = [k/\mu_2]$ is the classifying stack of étale μ_2-torsors. *We stress that we won't use this more abstract point of view in the main part of the chapter.*

Arithmetic of Kummer lines

The ramification data thus gives us a very simple description of the Kummer line, while still allowing us to find formulas for the arithmetic. Let us start with isomorphisms of Kummer lines: to find an isomorphism between the

Kummer coordinate $(X_1 : Z_1)$ of $E_1/\pm 1$ and $(X_2 : Z_2)$ of $E_2/\pm 1$, it suffices to find the homography that sends the ramification of $E_1/\pm 1$ to the ramification to $E_2/\pm 1$, and 0_{E_1} to 0_{E_2}; this boils down to linear algebra. Indeed, such a homography lifts to an isomorphism of elliptic curves $E_1 \to E_2$ (maybe up to replacing them by quadratic twists), isomorphism that sends 0_{E_1} to 0_{E_2} by assumption, hence which automatically preserve the group law. In particular, this gives a much simplified proof of the change of coordinates from [HR19], where the authors used a formal calculus software to check that their conversion formula preserved the group law. Instead, we only need to check that the image of the ramification matches.

Next, the ramification data is also enough to find doubling and differential addition formulas on the Kummer line. Doublings were worked out in Chapter 6, while differential additions in Chapter 7. As such, we won't expand on these arithmetic aspects in this chapter, but focus on the models themselves. In Section 8.1 we give a list of some standard Kummer models used in the literature, like the Montgomery, theta, squared theta (and maybe less standard like the twisted theta) models, and refer to the existing formulas in the literature for the arithmetic (which were also rebuilt 'from scratch' in previous chapters). Here our contribution is Section 8.2, where we give conversion map between these different models. We remark that most of these were already done in [HR19], descending the existing conversion formulas in the literature from elliptic curves to Kummer lines. What we do in Section 8.2 is to instead recover the conversion formulas from scratch, simply from the ramification of these different models.

In Chapter 6, using the general doubling framework on Kummer lines, it was observed that for some models doublings and translation by a point of 2-torsion could be faster than a doubling. This was used to adjust the usual Montgomery ladder by replacing doublings by doublings with translation. We mention that we can retrieve a particular case as follows: the Montgomery model has faster differential additions than the theta squared model, but the theta squared model has faster doublings. But, using the conversion formulas, we see in Proposition 8.2.7 that they have the exact same ramification, but with a different marked point for 0_E. This means that the identity $(X : Z) \mapsto (X : Z)$ behaves as translation by a point of 2-torsion when converting back and forth between these two models; and that using the Montgomery doubling formulas in the theta model behaves as a doubling + translation, and similarly using the theta squared differential additions in the Montgomery model behaves as a differential addition + translation. Since the extra translation is easy to keep track of (for instance it is killed by a subsequent doubling), it is easy to adjust the Montgomery ladder to use the translated formulas instead.

Interpretation with modular curves

In Section 8.3 we study when the models of Section 8.1 are rational, and how they are parametrized. This is where Galois representation comes in: we show that each of the family correspond to the Galois representation in the 2-Tate module T_2E taking a specific form. In fact for these families it is enough

to look at the Galois representation in $E[4]$. The Galois representation then gives matrices in $SL_2(\mathbb{Z}/4\mathbb{Z})$ and the families correspond to matrices living in some specific subgroup G. By Shimura theory (see e.g. [Zyw15]) our families can then be described by modular curves. More precisely, in Section 8.3 we show that Montgomery models correspond to $G = \Gamma^0(4)$, theta squared models to $G = \Gamma^0(4) \cap \Gamma(2)$, and theta models to $G = \Gamma(2,4) = \Gamma^0(4) \cap \Gamma_0(4)$. We also reinterpret the condition to have Galois representation in $\Gamma^0(4)$ in terms of the 2-Tate pairing. These results are certainly folklore.

For the theta model they are a consequence of the more general (analytic) result by Igusa that theta constants of level n in dimension g are modular forms for $\Gamma(n, 2n)$ (the algebraic interpretation of Igusa's result, as shown by Mumford in [Mum66], is having a rational symmetric theta structure of level n). However, we haven't seen explicitly stated in the literature the many equivalent conditions to having a rational Montgomery model given in Corollary 8.4.3. This section also allows us to understand the link between Montgomery and theta models better: in dimension 1 and level 2, having a rational theta structure is equivalent to having two different Montgomery models (if $E/\pm 1$ has a Montgomery model, $(X : Z) \mapsto (-X : Z)$ automatically gives another one, that leaves $(0 : 0)$ invariant. By two different Montgomery model we mean an isomorphism of models that sends $(0 : 0)$ to another 2-torsion point).

In Section 8.3, we consider the modular curves as schemes (hence as coarse spaces). For moduli problems the fine moduli spaces keep track of more information, but in presence of non-trivial automorphisms this requires to work with stacks rather than schemes. We give a brief survey of stacky modular curves in Section 8.8, and their parametrization of the Kummer models (and twists).

Interpretation with the theta group

In Section 8.4, we explain why the families studied in Section 8.1 are 'natural'. Recall that our Kummer lines are isomorphic as a scheme to $\mathbb{P}^1$. We certainly want to use a line as a model of $\mathbb{P}^1$, so use as coordinate a section of a divisor (t) of degree 1: either x (which has a pole at $\infty = (1 : 0)$) or a homography $(ax + b)/(cx + d)$ (which has a pole at $t = -d/c$). In practice, since we prefer to work with projective coordinates $(X : Z)$, this amounts to choosing a linear change of variable $(X : Z) \mapsto (aX + bZ : cX + dZ)$. From the elliptic curve point of view, if $\pi : E \to E/\pm 1 \simeq \mathbb{P}^1$ is the projection, (X, Z) can then be seen as sections of $\pi^*(t) \sim 2(0_E)$: they form coordinates of level 2. So for us a model of Kummer lines boils down to a choice of coordinate x, and looking at how the ramification behaves. This might seem rather boring, but there are actually some nice arithmetic considerations behind the choice of x.

We want to focus on models of Kummer lines which have at least one rational 2-torsion point T (because this allows us to split doublings into the composition of two 2-isogenies). Since our projective coordinates $(X : Z)$ are of level 2, the translation map $(X(P) : Z(P)) \mapsto (X(P + T) : Z(P + T))$ (which is well-defined on the Kummer line because T is of 2-torsion) is given by a linear change of variable $(X(P + T) : Z(P + T)) = M_T \cdot (X(P) : Z(P))$ for a

2×2 matrix M_T (well-defined up to a multiplication by $\lambda\,\mathrm{id}$ because $(X : Z)$ are projective coordinates). Since T is of 2-torsion, M_T^2 acts by the projective identity, so we have, $M_T^2 = \lambda_T\,\mathrm{id}$. A natural question is whether we can find M_T such that $M_T^2 = \mathrm{id}$, i.e. $\lambda_T = 1$? It is easy to see that changing M_T to λM_T changes λ_T to $\lambda^2\lambda_T$, so λ_T is well-defined up to squares: we call it the type of T (as defined in Definition 6.2.4). In particular, there is an *arithmetic obstruction* to finding such a M_T: namely whether the class of λ_T modulo squares is trivial (i.e. is λ_T a square)?

If this condition is satisfied, then up to a change of basis we have $M_T = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and the translation by T is $(X : Z) \mapsto (Z : X)$. Putting 0_E at infinity $(1 : 0)$, we then have that $T = (0 : 1)$, and the other 2-torsion points are given by $(\alpha : 1), (1 : \alpha)$ (since $\{T_1, T_2\}$ is stable under the action of M_T). We have rediscovered the Montgomery model!

An alternative is to take $M'_T = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, this matrix is conjugate to M_T and the conjugation gives the new variables $(X' : Z') = (X + Z : X - Z)$ (which we will call the Hadamard transform). Taking the Montgomery model, the ramification is then given in the $(X' : Z')$ coordinates by $0_E = (1 : 1)$, $T_1 = (-1 : 1)$, $T_2 = (\beta : 1)$, $T_3 = (-\beta : 1)$ with $\beta = (\alpha + 1)/(\alpha - 1)$.

Now assume that we have two rational 2-torsion points T_1 and T_2. We can try to further refine by asking that T_1, T_2 should both be of trivial type, so that we can find two matrices M_1, M_2 giving their translation map and such that $M_1^2 = M_2^2 = \mathrm{id}$. Then M_1 and M_2 have to anti-commute (this can be seen by working out the form they can take given the ramification, but also result from the deeper theory that the commutator bracket of M_1 and M_2 gives the Weil pairing $e_{W,2}(T_1, T_2) = -1$.) From general linear algebra, we can always find a basis (X, Z) such that $M_1 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $M_2 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$. We remark that such a basis is uniquely determined (up to a common scalar), in particular we do not have extra degrees of liberty to put 0_E at infinity any more. If we denote $(X(0_E) : Z(0_E)) = (a : b)$, then by constructions of M_1 and M_2, we must have $T_1 = (b : a), T_2 = (-a : b), T_1 + T_2 = (-b : a)$. We have just rediscovered the theta model! In other words: a Kummer line has a rational Montgomery model when its ramification is (up to a change of variable) symmetric for one of the two symmetries: $(X : Z) \mapsto (Z : X)$ or $(X : Z) \mapsto (-X : Z)$, while it admits a rational theta model if its ramification is (up to a change of variable) symmetric for both symmetries at the same time.

Hidden behind this story is Mumford's theta group $G(2(0_E))$: our matrices M_T are given by the action of elements of the theta groups on the sections $\Gamma(2(0_E))$, and being able to find a matrix M_T such that $M_T^2 = \mathrm{id}$ amounts to being able to find a symmetric element above T in the theta group. Unsurprisingly, we also show in Section 8.4 that the class of the type $\lambda_T \in k^*/k^{*,2}$ of T is given by the self Tate pairing $e_{T,2}(T, T)$. Yet another equivalent condition is to look at points of 4-torsion: a point T' on the Kummer line is of 4-torsion above T if and only if $(X(T') : Z(T'))$ are projectively invariant under the action of M_T, i.e. $(X(T'), Z(T'))$ is an eigenvector of M_T. There are two such eigenvectors, with eigenvalues $\pm\sqrt{\lambda_T} \in \overline{k}$ (because M_T is not projectively trivial, hence not diagonal), so these eigenvectors are rational if and only if λ_T is a square. We refer to Corollary 8.4.3 for other equivalent conditions. This

explains why in Section 8.3 we are interested in the Galois representation in $E[4]$.

Using these algebraic and arithmetic tools, we revisit the families of Section 8.1 from the theta group point of view. We also explain how we can reinterpret them as twisted variant of the theta model, and how to count their number of rational models.

Applications: isogeny volcanoes and ECM

In Sections 8.3 and 8.4, we look at whether a Kummer line admitted a Montgomery or theta model. In Section 8.5, we tackle whether it is *isogenous* to one such model. For that, we use the theory of isogeny volcanoes.

In Section 8.6 we study ECM with the level 2 theta model and some of its variants. Thanks to the modular curves from Section 8.3, we parametrize all the elliptic curves which are suitable for our coordinate systems. In particular, we can select curves with more torsion points and small parameters.

Roadmap

In Sections 8.1 and 8.2, we give more details on the Kummer lines built from Montgomery curves and certain theta functions of level 2, already introduced in Section 7.1. We also provide the differential addition and doubling formulas in every model, as well as the correlations between them and the conversion formulas. Section 8.3 covers the classification of elliptic curves via Shimura theory and the easier Galois representation criteria for the previously introduced Kummer lines. Finally, we relate the different models to isogeny volcanoes in Section 8.5 and describe how to find curves with theta squared model to use for instance in ECM, as well as some thoughts on stage 2 with Kummer lines in Section 8.6. Section 8.7 gives a more geometric interpretation of Kummer lines as stacks, while Section 8.8 gives a brief overview of stacky modular curves, generalizing the Shimura theory to fields different from $\mathbb{Q}$. Section 8.9 provides the list of all modular subgroups of $\mathrm{SL}_2(\mathbb{Z}/4\mathbb{Z})$.

Sections 8.7 to 8.9 were originally part of an appendix, giving more theory on Kummer lines but with more abstract geometrical notions.

8.1 SOME FAMILIES OF KUMMER LINES

We first recall the examples of Kummer lines we are interested in, including the classical Montgomery model as well as the theta model with squared and twisted variants. This section may sound a bit redundant with Section 7.1, however our approach to the theta models uses the analytic construction from Section 5.5 instead of the algebraic one earlier. Obviously, the ramification points have the same shape.

8.1.1 Montgomery curves

We have already studied the Kummer line of a Montgomery curve in Section 6.3, recall from Main example 6.2.1 that, if we consider a Montgomery curve

$$E : \beta y^2 = x(x^2 + \mathcal{A}x + 1),$$

then its ramification for the Kummer line derived from the x-coordinate is given by

$$0_E = (1 : 0)^*, \quad T_1 = (0 : 1), \quad T_2 = (A : B), \quad T_3 = (B : A), \tag{8.1}$$

where $\alpha = A/B \in \overline{k}$ is a root of $x^2 + \mathcal{A}x + 1$. The Kummer line will be denoted $M(A : B)$.

The arithmetic was recalled earlier in Algorithms 2.4 and 2.5.

Lastly, it will prove useful to determine when there are any 4-torsion point T' which is rational on the Kummer line. Of course, this implies that $T := 2 \cdot T'$ is rational. To do that, the approach is the one explained in Main example 6.2.10. We first determine the translation $t_T : P \mapsto P + T$ on the Kummer line, where T is a 2-torsion point, therefore t_T factors by $[-1]$ on both sides. Hence, it is well-defined and is a homography $t_T : \mathbb{P}^1 \xrightarrow{\sim} \mathbb{P}^1$. Then we look for a point $x(T') = (X(T') : Z(T'))$ that is invariant by t_T. We stress that although we only look for T' rational on the Kummer line (i.e. $x(T')$ rational); T' itself may only be defined over a quadratic extension on the elliptic curve. We note the following easy lemma:

Lemma 8.1.1 *A 4-torsion point $T' \in E$ above a rational 2-torsion point $T = 2 \cdot T'$ has $x(T') \in k$ if and only if $\sigma(T') = T'$ or $\sigma(T') = T' + T$ for all $\sigma \in \mathrm{Gal}(\overline{k}/k)$, if and only if the cyclic subgroup of order 4, $\langle T' \rangle$, is rational.*

In this case there is also a second rational cyclic subgroup of order 4 containing T, generated by $T' + T_0$ for a 2-torsion point $T_0 \neq T$.

Proof *Consider $\sigma \in \mathrm{Gal}(\overline{k}/k)$, then $2 \cdot \sigma(T') = T$ and $\sigma(T')$ must be a 4-torsion point above T. It is rational on the Kummer line if and only if $\sigma(-T')$ is T' or $-T' = T' + T$ for all σ, that is if and only if $\langle T' \rangle$ is rational.*

Note that if $\varphi : E \to E_2 = E/\langle T \rangle$ is the corresponding 2-isogeny, then the image of T_0 by φ is a rational 2-torsion point since it is the kernel generator of the dual isogeny. The image of T' is also a rational 2-torsion point by the first part, hence there is a third rational 2-torsion point on E_2, and composing the associated isogeny $E_2 \to E_3$ with φ gives a cyclic isogeny of degree 4 from E to E_3. □

On the Montgomery model, for $T = (0 : 1)$, $t_T : (X : Z) \mapsto (Z : X)$, and we find that there are always two points of 4-torsion above T: $T' = (1 : 1)$ and $T'' = (-1 : 1)$.

8.1.2 Theta models

It is easy to define from a Montgomery curve a Kummer line model with efficient arithmetic. Another way to build Kummer lines is via theta functions, which have already been used for instance in [GL09; KS20] and which

we already described in Example 5.3.8. Their arithmetic is also competitive with the Montgomery one, hence why we would like to study them. We will rely heavily on Section 5.5 in this section.

Recall that $\mathbb{H}$ is the Poincaré upper half-plane in $\mathbb{C}$. A period $\tau \in \mathbb{H}$ is set for the rest of this section.

Theta models and their duals

Recall that we defined the following theta functions of level 2 in Definition 5.5.4, and that the space of level 2 functions is denoted R_2^τ:

$$\theta_0^\tau(z) = \vartheta_2[0,0](z;\tau/2), \quad \theta_1^\tau(z) = \vartheta_2[0,1](z;\tau/2).$$

It is a basis of R_2^τ thanks to Proposition 5.5.5, so the map $\pi : z \in E_\tau \mapsto (\theta_0^\tau(z) : \theta_1^\tau(z))$ is a Kummer line by Theorem 5.5.6. For convenience, set $(a : b) := (\theta_0^\tau(0) : \theta_1^\tau(0))$, they are the theta constants. We have seen in Examples 5.3.8 and 7.1.2 that the ramification is the following:

$$0_E = (a : b)^*, \quad T_1 = (-a : b), \quad T_2 = (b : a), \quad T_3 = (-b : a). \tag{8.2}$$

Definition 8.1.2 *An elliptic curve E/k has a theta model with theta constants $(a : b) \in \mathbb{P}^1(k)$ if there exist a Kummer line $E \to \mathbb{P}^1$ with the following ramification points as in Eq. (8.2). The theta model will be written $\theta(a : b)$ generally, or $\theta^\tau(a : b)$ over E_τ.*

Another model that can be derived from this one is the dual one using the Hadamard transform.

Definition 8.1.3 *Let E/k be an elliptic curve with a theta model $\theta(a : b)$, the Kummer line is denoted π. The dual theta model is given by composing π with the Hadamard transform*

$$\begin{aligned} H : \mathbb{P}^1 &\to \mathbb{P}^1 \\ (X : Z) &\mapsto (X + Z : X - Z). \end{aligned} \tag{8.3}$$

This is the Hadamard transform from Example 5.3.9.

The theta constants are $(a' : b') = (a + b : a - b)$, and the ramification is:

$$0_E = (a' : b')^*, \quad T_1 = (b' : a'), \quad T_2 = (-a' : b'), \quad T_3 = (-b' : a').$$

The dual model is written $\theta'(a' : b')$ or $\theta'^\tau(a' : b')$ over E_τ.

The corresponding theta functions are:

$$\theta'^\tau_0 = \theta_0^\tau + \theta_1^\tau, \quad \theta'^\tau_1 = \theta_0^\tau - \theta_1^\tau.$$

It is a corollary from Proposition 5.5.5 that this is also a basis of R_2^τ.

Remark 8.1.4 *A computation with the analytic definition helps to express θ'^τ_0 and θ'^τ_1 as theta functions with characteristic, which were defined in Eq. (5.6):*

$$\theta'^\tau_0(z) = \vartheta_2[0,0](2z;2\tau), \quad \theta'^\tau_1(z) = \vartheta_2[1,0](2z;2\tau).$$

With these new notations, and using Proposition 5.5.7 with $(a_1, b_1) = (a_2, b_2)$, one can relate the theta functions to get a differential addition and doubling:

Proposition 8.1.5 *Let $\tau \in \mathbb{H}$, for every $u, v \in \mathbb{C}$:*

$$\begin{cases} 2\theta'^{\tau/2}_0(u+v)\theta'^{\tau/2}_0(u-v) = \theta^\tau_0(2u)\theta^\tau_0(2v) + \theta^\tau_1(2u)\theta^\tau_1(2v), \\ 2\theta'^{\tau/2}_1(u+v)\theta'^{\tau/2}_1(u-v) = \theta^\tau_0(2u)\theta^\tau_0(2v) - \theta^\tau_1(2u)\theta^\tau_1(2v), \\ \theta^\tau_0(u+v)\theta^\tau_0(u-v) = \theta'^{\tau/2}_0(u)\theta'^{\tau/2}_0(v) + \theta'^{\tau/2}_1(u)\theta'^{\tau/2}_1(v), \\ \theta^\tau_1(u+v)\theta^\tau_1(u-v) = \theta'^{\tau/2}_0(u)\theta'^{\tau/2}_0(v) - \theta'^{\tau/2}_1(u)\theta'^{\tau/2}_1(v). \end{cases} \tag{8.4}$$

It is convenient to give names to the theta constants on the 2-isogenous curve $E_{\tau/2}$. Let $(A : B) := (\theta^{\tau/2}_0(0) : \theta^{\tau/2}_1(0))$ and $(A' : B') := (A + B : A - B)$. Setting $u = v = 0$ in Proposition 8.1.5, one obtains[1] $(A'^2 : B'^2) = (a^2 + b^2 : a^2 - b^2)$.

Remark 8.1.6 *The formulas in Proposition 8.1.5, while obtained over $\mathbb{C}$, are purely algebraic in the end. This is what enables us to use them on more general fields, such as finite fields, thanks to Lefschetz' principle.*

Similarly to what Montgomery did, these formulas provide differential addition formulas as well as doubling ones, they are described in Algorithms 8.1 and 8.2. Differential addition costs $2\mathbf{M} + 4\mathbf{S} + 1\mathbf{m} + 1\mathbf{m}_0$, while doubling is $4\mathbf{S} + 2\mathbf{m}_0$.

Algorithm 8.1: Differential addition on theta model

Input: $[P] = (X_1 : Z_1)$, $[Q] = (X_2 : Z_2)$ and $[P - Q] = (X_0 : Z_0)$
Output: $[P + Q] = (X : Z)$
Data: On $\theta^\tau(a : b)$, $(A'^2 : B'^2) = (a^2 + b^2 : a^2 - b^2)$

```
def DiffAdd([P], [Q], [P − Q]):
    u ← (X_1^2 + Z_1^2)(X_2^2 + Z_2^2);
    v ← (A'^2/B'^2)(X_1^2 − Z_1^2)(X_2^2 − Z_2^2);
    X ← (u + v);
    Z ← (X_0/Z_0)(u − v);
    return (X : Z);
```

Algorithm 8.2: Doubling on theta model

Input: $[P] = (X_1 : Z_1)$
Output: $[2 \cdot P] = (X : Z)$
Data: On $\theta^\tau(a : b)$, $(A'^2 : B'^2) = (a^2 + b^2 : a^2 - b^2)$

```
def Doubling([P]):
    u ← (X_1^2 + Z_1^2)^2;
    v ← (A'^2/B'^2)(X_1^2 − Z_1^2)^2;
    X ← (u + v);
    Z ← (a/b)(u − v);
    return (X : Z);
```

We can also study 4-torsion points on the theta model $\theta^\tau(a : b)$. Recall that $T_1 = (-a : b)$ and $T_2 = (b : a)$ are two independent 2-torsion points,

1 This relation is also in [GL09, § 6.2], however there is a typo in the published version as squares are missing on A' and B'. The preprint is fixed.

then $t_{T_1} : (X : Z) \mapsto (-X : Z)$ is the translation by T_1 and $t_{T_2} : (X : Z) \mapsto (Z : X)$ is the translation by T_2. Then, on the Kummer line, if we have a 4-torsion point T_1' above T_1, it verifies $T_1' + T_1 = T_1'$. Hence, if $T_1' = (X : Z)$, we must have $(X : Z) = (-X : Z)$, that is $T_1' = (1 : 0)$ or $T_1' = (0 : 1)$. Similarly, if T_2' is above T_2, we find that $T_2' = (1 : 1)$ or $T_2' = (-1 : 1)$ (we already explained this step of determining the 4-torsion in Main example 6.2.10).

Proposition 8.1.7 *On the theta model $\theta^\tau(a : b)$, the 4-torsion is:*

$$T_1' = (1 : 0), \quad T_1'' = (0 : 1), \quad T_2' = (1 : 1), \quad T_2'' = (-1 : 1)$$

where $2 \cdot T_1' = 2 \cdot T_1'' = T_1$ and $2 \cdot T_2' = 2 \cdot T_2'' = T_2$. On the theta dual model $\theta'^\tau(a' : b')$, the 4-torsion is:

$$T_1' = (1 : 1), \quad T_1'' = (-1 : 1), \quad T_2' = (1 : 0), \quad T_2'' = (0 : 1).$$

On both model, the last points above T_3 may not be rational over a generic field k but are still given by $T_3' = (i : 1)$ and $T_3'' = (-i : 1)$ with $i^2 = -1$.

Theta squared model

It appears in Algorithms 8.1 and 8.2 that one only needs the square of the coordinates. This is what leads to the theta squared model. For a set $\tau \in \mathbb{H}$, we define the following theta functions:

$$\widehat{\theta}_0^{\tau/2}(z) = \vartheta_2[0,0](z;\tau/2)^2, \qquad \widehat{\theta}_1^{\tau/2}(z) = \vartheta_2[0,1](z;\tau/2)^2,$$
$$\breve{\theta}'^{\tau}_0(z) = \vartheta_2[0,0](z;\tau)^2, \qquad \breve{\theta}'^{\tau}_1(z) = \vartheta_2[1,0](z;\tau)^2.$$

Proposition 8.1.8 ([Mum83], § II.1, Prop. 1.3) *$(\breve{\theta}'^{\tau}_0, \breve{\theta}'^{\tau}_1)$ is a basis of R_2^τ and $(\widehat{\theta}_0^{\tau/2}, \widehat{\theta}_1^{\tau/2})$ is a basis of $R_2^{\tau/2}$.*

This gives new theta models, but more importantly they are related to the previous ones via the following relations:

$$\widehat{\theta}_i^{\tau/2}(z) = \theta_i^\tau(z)^2, \quad \breve{\theta}'^{\tau}_i(2z) = \theta'^{\tau/2}_i(z)^2.$$

The curve E_τ and $E_{\tau/2}$ are 2-isogenous, the isogeny and its dual are given by:

$$f : z \bmod \Lambda_\tau \mapsto z \bmod \Lambda_{\tau/2}, \quad \tilde{f} : z \bmod \Lambda_{\tau/2} \mapsto 2z \bmod \Lambda_\tau.$$

If we relate this to the formulas above, we have just expressed f and $\tilde{f}$ on certain theta models.

As for the ramification points, we use the additional 4-torsion given in Proposition 8.1.7.

Definition 8.1.9 *An elliptic curve E/k has a theta squared model with constants $(a^2 : b^2) \in \mathbb{P}^1(k)$ if there exist a Kummer line $E \to \mathbb{P}^1$ with the following ramification points:*

$$0_E = (a^2 : b^2)^*, \quad T_1 = (b^2 : a^2), \quad T_2 = (1 : 0), \quad T_3 = (0 : 1).$$

The theta squared model will be written $\widehat{\theta}(a^2 : b^2)$ generally, or $\widehat{\theta}^{\tau/2}(a^2 : b^2)$ over $E_{\tau/2}$, and $\breve{\theta}'^{\tau}(A'^2 : B'^2)$ over E_τ with the according constants.

Remark 8.1.10 *While as functions $\widehat{\theta}_i$ and $\check{\theta}'_i$ are not exactly the same, they both correspond to squared coordinates of a theta model and have ramification in the same shape, hence the name and the association. The ramification points are exactly the same as in the Montgomery model Eq. (8.1), but the marked point is different. This is what we used in Chapter 6 to find translated isogenies formulas, and we will develop this connection in the next section.*

Note that with this definition, an elliptic curve can admit a theta squared model $\widehat{\theta}(a^2 : b^2)$ without having a theta model $\theta(a : b)$: a/b may not be rational.

Proposition 8.1.11 *The rational 4-torsion on the theta squared model $\widehat{\theta}(a^2 : b^2)$ is given by $T'_1 = (1 : 1)$ and $T''_1 = (-1 : 1)$ where $2 \cdot T'_1 = 2 \cdot T''_1 = T_1$.*

By adapting Algorithms 8.1 and 8.2, we can recover a more efficient differential addition in $2\mathbf{M} + 2\mathbf{S} + 1\mathbf{m} + 1\mathbf{m}_0$ in this model with a doubling still in $4\mathbf{S} + 2\mathbf{m}_0$, they are in Algorithms 8.3 and 8.4.

Algorithm 8.3: Differential addition on theta squared model

Input: $[P] = (X_1 : Z_1)$, $[Q] = (X_2 : Z_2)$ and $[P - Q] = (X_0 : Z_0)$
Output: $[P + Q] = (X : Z)$
Data: On $\check{\theta}'^{\tau}(A'^2 : B'^2)$, $(a^2 : b^2) = (A'^2 + B'^2 : A'^2 - B'^2)$

```
def DiffAdd([P], [Q], [P − Q]):
    u ← (X1 + Z1)(X2 + Z2);
    v ← (a^2/b^2)(X1 − Z1)(X2 − Z2);
    X ← (u + v)^2;
    Z ← (X0/Z0)(u − v)^2;
    return (X : Z);
```

Algorithm 8.4: Doubling on theta squared model

Input: $[P] = (X_1 : Z_1)$
Output: $[2 \cdot P] = (X : Z)$
Data: On $\check{\theta}'^{\tau}(A'^2 : B'^2)$, $(a^2 : b^2) = (A'^2 + B'^2 : A'^2 - B'^2)$

```
def Doubling([P]):
    u ← (X1 + Z1)^2;
    v ← (a^2/b^2)(X1 − Z1)^2;
    X ← (u + v)^2;
    Z ← (A'^2/B'^2)(u − v)^2;
    return (X : Z);
```

Theta twisted model

While arithmetic on theta squared model is quite efficient, it is hard to go back to the original theta model if it exists because of the squares. We will then introduce a more natural model which is as efficient as the theta squared one and less restrictive to go back to the theta one.

If we have theta functions (θ_0, θ_1) which give a theta model, the associated theta constants are not zero, and we can then consider a new basis:

$$\tilde{\theta}_0(z) := \theta_0(0)\theta_0(z), \quad \tilde{\theta}_1(z) := \theta_1(0)\theta_1(z).$$

This is clearly a basis of R_2^τ from Proposition 5.5.5, and define a new theta model named theta twisted. The ramification is as follows if we start from the theta model $\theta(a : b)$:

Definition 8.1.12 *An elliptic curve E/k has a theta twisted model with constants $(a^2 : b^2) \in \mathbb{P}^1(k)$ if there exist a Kummer line $E \to \mathbb{P}^1$ with the following ramification points:*

$$0_E = (a^2 : b^2)^*, \quad T_1 = (-a^2 : b^2), \quad T_2 = (1 : 1), \quad T_3 = (-1 : 1).$$

The theta twisted model will be written $\tilde{\theta}(a^2 : b^2)$ generally, or $\tilde{\theta}^\tau(a^2 : b^2)$ over E_τ.

The 4-torsion above T_1 is always rational and is given by $T_1' = (1 : 0)$ and $T_1'' = (0 : 1)$. The 4-torsion above T_2 may not be rational, but is given by $T_2' = (a : b)$ and $T_2'' = (-a : b)$.

Remark 8.1.13 *Similarly to the theta squared model, an elliptic curve can admit a theta twisted model $\tilde{\theta}(a^2 : b^2)$ without having a theta model $\theta(a : b)$ if $(a : b)$ is not rational. However, it is much easier to recover the original model via $(X : Z) \in \tilde{\theta}(a^2 : b^2) \mapsto (bX : aZ) \in \theta(a : b)$ if $(a : b) \in \mathbb{P}^1(k)$.*

If (θ'_0, θ'_1) are the dual theta functions, we can twist this basis too:

$$\widetilde{\theta'}_0(z) := \theta'_0(0)\theta'_0(z), \quad \widetilde{\theta'}_1(z) := \theta'_1(0)\theta'_1(z).$$

It appears there are a lot of similarities between theta squared and theta twisted models. In fact, if we set $u = v$ in Proposition 8.1.5, we find:

$$(\widetilde{\theta'}_0^\tau : \widetilde{\theta'}_1^\tau) = (\widehat{\theta}_0^{\tau/2} + \widehat{\theta}_1^{\tau/2} : \widehat{\theta}_0^{\tau/2} - \widehat{\theta}_1^{\tau/2}), \quad (\tilde{\theta}_0^{\tau/2} : \tilde{\theta}_1^{\tau/2}) = (\check{\theta}'^\tau_0 + \check{\theta}'^\tau_1 : \check{\theta}'^\tau_0 - \check{\theta}'^\tau_1).$$

In the end, the theta twisted coordinates are just the Hadamard dual of the theta squared coordinates, hence why they behave similarly. But note in the above formula that they are coordinates for different period matrices τ and $\tau/2$, hence they are also quite useful to study 2-isogenies in theta coordinates.

The differential addition and doubling in theta twisted coordinates are given in Algorithms 8.5 and 8.6 respectively. They are derived from the ones in theta squared and by applying Eq. (8.3).

In terms of complexity, Algorithm 8.5 costs $2\mathbf{M} + 2\mathbf{S} + 1\mathbf{m} + 1\mathbf{m}_0$ and Algorithm 8.6 costs $4\mathbf{S} + 2\mathbf{m}_0$, which is the same as in the squared model.

A summary of the maps between various theta models is available in Fig. 8.1.

We remark that a similar figure first appeared in [RS17, § 4.3] (in dimension 2), and then in [HR19, Cor. 4, Thm. 5] in dimension 1. The 'intermediate Kummer' introduced in these work is our twisted theta model. The explanation for that name will be given in Section 8.4.2.

Algorithm 8.5: Differential addition in theta twisted coordinates

Input: $[P] = (X_1 : Z_1)$, $[Q] = (X_2 : Z_2)$ and $[P - Q] = (X_0 : Z_0)$
Output: $[P + Q] = (X : Z)$
Data: On $\widetilde{\theta}^{\tau}(a^2 : b^2)$, $(A'^2 : B'^2) = (a^2 + b^2 : a^2 - b^2)$

```
def DiffAdd([P], [Q], [P − Q]):
    u ← X_1 X_2;
    v ← (a^2/b^2) Z_1 Z_2;
    X̃ ← X_0 + Z_0;
    Z̃ ← X_0 − Z_0;
    w ← (u + v)^2;
    t ← (X̃/Z̃)(u − v)^2;
    X ← w + t;
    Z ← w − t;
    return (X : Z);
```

Algorithm 8.6: Doubling in theta twisted coordinates

Input: $[P] = (X_1 : Z_1)$ a point
Output: $[2 \cdot P] = (X : Z)$
Data: On $\widetilde{\theta}^{\tau}(a^2 : b^2)$, $(A'^2 : B'^2) = (a^2 + b^2 : a^2 - b^2)$

```
def Doubling([P]):
    u ← X_1^2;
    v ← (a^2/b^2) Z_1^2;
    w ← (u + v)^2;
    t ← (A'^2/B'^2)(u − v)^2;
    X ← w + t;
    Z ← w − t;
    return (X : Z);
```

8.2 CONVERSIONS BETWEEN THESE MODELS

As seen in Section 4.3.2, Kummer lines isomorphisms are given by homographies that preserve 2-torsion. This allows us to recover the existing formulas in the literature, notably those of [HR19].

Remark 8.2.1 *There is a less direct way to convert from Montgomery to theta squared and theta twisted, by chaining isomorphisms from Montgomery to Legendre and then from Legendre to theta squared, which can be found in [EH22, § 2.8] for the Montgomery to Legendre maps, and in [GL09, § 6; KS20, § 2.6] for the Legendre to theta squared maps.*

With the 2-torsion data, we can then define our isomorphisms:

$$\begin{tikzcd}
\theta(a:b) \arrow[r, shift left] \arrow[d, "f:[P]\mapsto[P]"'] & \tilde{\theta}(a^2:b^2) \arrow[l, dashed, shift left, "\text{if } a/b\in k"] \arrow[r, equal] & (\breve{\theta'})'(a^2:b^2) \arrow[r, shift left] & \breve{\theta'}(A'^2:B'^2) \arrow[l, shift left, "H"] \\
\widehat{\theta}(a^2:b^2) \arrow[r, shift left, "H"] & (\widehat{\theta})'(A'^2:B'^2) \arrow[l, shift left] \arrow[r, equal] & \widetilde{(\theta')}(A'^2:B'^2) \arrow[r, dashed, shift left, "\text{if } A'/B'\in k"] & \theta'(A':B') \arrow[l, shift left] \arrow[u, "\tilde{f}:[P]\mapsto[2\cdot P]"']
\end{tikzcd}$$

Figure 8.1: Relations among several models of Kummer lines. The first line consists of models of $\mathcal{K}_\tau$, whereas the second has models of $\mathcal{K}_{\tau/2}$.

Proposition 8.2.2 (Montgomery and theta) *Let $a, b \in k$ and set $(A'^2 : B'^2) = (a^2 + b^2 : a^2 - b^2)$. Let $M(A'^2 : B'^2)$ and $\theta(a : b)$ be the Kummer line models in Eq. (8.1) and Definition 8.1.2. There is a morphism φ given by:*

$$\begin{aligned}\varphi : \theta(a:b) &\to M(A'^2 : B'^2)\\ (X:Z) &\mapsto (bX + aZ : aZ - bX).\end{aligned}$$

Conversely, if $A'^2, B'^2 \in k$ and $(a^2 : b^2) = (A'^2 + B'^2 : A'^2 - B'^2)$ with $a, b \in \overline{k}$, the converse over $\overline{k}$ of φ is given by:

$$\begin{aligned}\varphi^{-1} : M(A'^2 : B'^2) &\to \theta(a:b)\\ (X:Z) &\mapsto (a(X-Z) : b(X+Z)).\end{aligned}$$

Proof *We are looking for a homography $\varphi : (X : Z) \mapsto (\alpha X + \beta Z : \gamma X + \delta Z)$ such that:*

- $\varphi(a : b) = (1 : 0)$, *that is* $\gamma a + \delta b = 0$,
- $\varphi(-a : b) = (0 : 1)$, *that is* $-\alpha a + \beta b = 0$,
- $\varphi(b : a) = (A'^2 : B'^2)$, *that is* $(\alpha b + \beta a : \gamma b + \delta a) = (A'^2 : B'^2)$.

The last equation can be reformulated as:

$$(A'^2 : B'^2) = (\alpha b^2 + \beta ab : \gamma b^2 + \delta ab) = (\alpha(b^2 + a^2) : \gamma(b^2 - a^2)).$$

Thus, $(A'^2 : B'^2) = (-\alpha A'^2 : \gamma B'^2)$, i.e. $\alpha = -\gamma$, $\beta = \delta$, and since $\gamma a = -\delta b$, we end up with the expression of φ. We can then check that with this expression $\varphi(-b : a) = (B'^2 : A'^2)$.

For the converse, we give another method which is to check that the 2-torsion is mapped correctly and the marked points correspond:

$$\begin{aligned}\varphi^{-1}(1:0) &= (a:b), & \varphi^{-1}(0:1) &= (-a:b),\\ \varphi^{-1}(A'^2:B'^2) &= (b:a), & \varphi^{-1}(B'^2:A'^2) &= (-b:a).\end{aligned}$$ □

We can compose these conversion maps with $M(A'^2 : B'^2) \to M(A'^2 : -B'^2)$, $(X : Z) \mapsto (-X : Z)$, to obtain $\theta(a : b) \to M(A'^2 : -B'^2)$, $(X : Z) \mapsto (X/a + Z/b : X/a - Z/b)$, the Hadamard transform of another twisted variant of theta coordinates: $\theta_0/\theta_0(0), \theta_1/\theta_1(0)$.

Remark 8.2.3 *In Proposition 8.2.2, we chose to map* $(A'^2 : B'^2)$ *and* $(-a : b)$ *together, but we could have done it differently. Set* $(a' : b') = (a + b : a - b)$. *Transposing 2-torsion is the same as working on the dual theta model, so we apply* φ *and* φ^{-1} *to* $\theta'(a' : b')$ *and then use a Hadamard transform to go back to* $\theta(a : b)$*:*

$$\begin{aligned}\varphi' : \theta(a : b) &\to M(\widetilde{A}^2 : \widetilde{B}^2)\\ (X : Z) &\mapsto (aX - bZ : bX - aZ),\end{aligned}$$

and:

$$\begin{aligned}\varphi'^{-1} : M(\widetilde{A}^2 : \widetilde{B}^2) &\to \theta(a : b)\\ (X : Z) &\mapsto (aX - bZ : bX - aZ).\end{aligned}$$

This time $(\widetilde{A}^2 : \widetilde{B}^2) = (a'^2 + b'^2 : a'^2 - b'^2) = (a^2 + b^2 : 2ab)$, $\varphi' = \varphi \circ H$, $\widetilde{A}^2, \widetilde{B}^2 \in k$ *and* $a, b \in \overline{k}$.

Remark 8.2.4 (Montgomery coordinates as theta functions) *The formulas in Proposition 8.2.2 also help to express Montgomery XZ-coordinates via theta functions. Indeed, let* $\tau \in \mathbb{H}$ *and consider the theta model* $\theta^\tau(a : b)$. *We write* $(X : Z) = (\theta_0^\tau(z) : \theta_1^\tau(z))$ *and* $(X_M : Z_M) = (bX + aZ : aZ - bX)$ *the corresponding Montgomery coordinates. Using Proposition 5.5.7, with* $\alpha = a_1 = a_2$ *and* $\beta = b_1 = b_2$, *one gets that* $2\vartheta_2[\alpha, \beta](z; \tau)^2$ *is equal to*

$$\vartheta_2[0, 0](0; \tau/2)\vartheta_2[0, \beta](z; \tau/2) + e^{-i\pi\alpha}\vartheta_2[0, 1](0; \tau/2)\vartheta_2[0, \beta + 1](z; \tau/2).$$

Applying this to $(\alpha, \beta) = (0, 1)$ *and* $(\alpha, \beta) = (1, 1)$, *one gets the two following formulas:*

$$\begin{cases} 2\vartheta_2[0, 1](z; \tau)^2 = \\ \qquad \vartheta_2[0, 0](0; \tau/2)\vartheta_2[0, 1](z; \tau/2) + \vartheta_2[0, 1](0; \tau/2)\vartheta_2[0, 0](z; \tau/2), \\ 2\vartheta_2[1, 1](z; \tau)^2 = \\ \qquad \vartheta_2[0, 0](0; \tau/2)\vartheta_2[0, 1](z; \tau/2) - \vartheta_2[0, 1](0; \tau/2)\vartheta_2[0, 0](z; \tau/2). \end{cases}$$

Hence, $(X_M : Z_M) = (aZ + bX : aZ - bX) = (\vartheta_2[0, 1](z; \tau)^2 : \vartheta_2[1, 1](z; \tau)^2)$ *expresses the Montgomery coordinates via theta functions of level 2.*

Proposition 8.2.5 (Montgomery and theta squared) *Consider* $a, b \in \overline{k}$ *with* $a^2, b^2 \in k$. *Let* $M(a^2 : b^2)$ *and* $\widehat{\theta}(a^2 : b^2)$ *be the Montgomery and theta squared models as in Eq. (8.1) and Definition 8.1.9. There is an isomorphism* $\widehat{\varphi}$ *given by:*

$$\begin{aligned}\widehat{\varphi} : \widehat{\theta}(a^2 : b^2) &\xrightarrow{\sim} M(a^2 : b^2)\\ (X : Z) &\mapsto (a^2X - b^2Z : b^2X - a^2Z).\end{aligned}$$

The converse is:

$$\begin{aligned}\widehat{\varphi}^{-1} : M(a^2 : b^2) &\xrightarrow{\sim} \widehat{\theta}(a^2 : b^2)\\ (X : Z) &\mapsto (a^2X - b^2Z : b^2X - a^2Z).\end{aligned}$$

Proof *Similarly to Proposition 8.2.2, we will determine* $\widehat{\varphi} : (X : Z) \mapsto (\alpha X + \beta Z : \gamma X + \delta Z)$ *using the 2-torsion. We want:*

- $\widehat{\varphi}(a^2 : b^2) = (1 : 0)$, *that is* $\gamma a^2 + \delta b^2 = 0$,

- $\widehat{\varphi}(1:0) = (a^2 : b^2)$, *that is* $(\alpha : \gamma) = (a^2 : b^2)$,
- $\widehat{\varphi}(0:1) = (b^2 : a^2)$, *that is* $(\beta : \delta) = (b^2 : a^2)$.

From the first two equations, we deduce $\alpha = -\delta$, *and because* $\beta a^2 = \delta b^2$, $\gamma a^2 = -\delta b^2$, *we have:*

$$\begin{aligned}(\alpha X + \beta Z : \gamma X + \delta Z) &= (\delta(-a^2 X + b^2 Z) : \delta(-b^2 X + a^2 Z)) \\ &= (a^2 X - b^2 Z : b^2 X - a^2 Z).\end{aligned}$$

It follows that $\widehat{\varphi}(b^2 : a^2) = (0:1)$.

For the converse, we will again just check that the point at infinity and the 2-torsion are correctly mapped:

$$\begin{aligned}\widehat{\varphi}^{-1}(1:0) &= (a^2 : b^2), & \widehat{\varphi}^{-1}(0:1) &= (b^2 : a^2), \\ \widehat{\varphi}^{-1}(a^2 : b^2) &= (1:0), & \widehat{\varphi}^{-1}(b^2 : a^2) &= (0:1).\end{aligned}$$

□

The last conversion between theta twisted and Montgomery models can be derived from Proposition 8.2.5 thanks to the Hadamard transform:

Proposition 8.2.6 (Montgomery and theta twisted) *Let* $a, b \in \overline{k}$ *such that* $a^2, b^2 \in k$, *and set* $(A'^2 : B'^2) = (a^2 + b^2 : a^2 - b^2)$. *Let* $M(A'^2 : B'^2)$ *and* $\widetilde{\theta}$ *be the Montgomery and twisted theta models as in Eq.* (8.1) *and Definition 8.1.12. There is an isomorphism* $\widetilde{\varphi}$ *given by:*

$$\begin{aligned}\widetilde{\varphi} : \widetilde{\theta}(a^2 : b^2) &\to M(A'^2 : B'^2) \\ (X : Z) &\mapsto (b^2 X + a^2 Z : a^2 Z - b^2 X).\end{aligned}$$

The converse is:

$$\begin{aligned}\widetilde{\varphi}^{-1} : M(A'^2 : B'^2) &\to \widetilde{\theta}(a^2 : b^2) \\ (X : Z) &\mapsto (a^2(X - Z) : b^2(X + Z)).\end{aligned}$$

Proof *Because of the isomorphism* $H : \widetilde{\theta}(a^2 : b^2) \to \widehat{\theta}(A'^2 : B'^2), (X : Z) \mapsto (X + Z : X - Z)$, *we just have to compose* $\widehat{\varphi} : \widehat{\theta}(A^2 : B'^2) \to M(A'^2 : B'^2)$ *with* H *to get* $\widetilde{\varphi} = \widehat{\varphi} \circ H$ *and* $\widetilde{\varphi}^{-1} = H^{-1} \circ \widehat{\varphi}^{-1}$. *We then have*

$$\begin{aligned}\widetilde{\varphi}(X : Z) &= (A'^2(X + Z) - B'^2(X - Z) : B'^2(X + Z) - A'^2(X - Z)) \\ &= (b^2 X + a^2 Z : a^2 Z - b^2 X),\end{aligned}$$

and

$$\begin{aligned}\widetilde{\varphi}^{-1}(X : Z) &= (A'^2 X - B'^2 Z + B'^2 X - A'^2 Z : A'^2 X - B'^2 - B'^2 X + A'^2 Z) \\ &= (a^2(X - Z) : b^2(X + Z)).\end{aligned}$$

□

To benefit from faster doubling formulas on the theta squared model while keeping the better differential addition on Montgomery model, one could do conversions, however this approach adds a lot of multiplications. However, there is one last conversion map that is interesting between theta squared and Montgomery models to tackle this issue:

Proposition 8.2.7 *Let* $a, b \in \overline{k}$ *such that* $a^2, b^2 \in k$. *Let* $[P_\theta] \in \widehat{\theta}(a^2 : b^2)$ *and* $[P_M] = \widehat{\varphi}([P_\theta])$ *from Proposition 8.2.5. Finally, let* $[T_{2,\theta}] = (1 : 0) \in \widehat{\theta}(a^2 : b^2)$ *and* $[T_{2,M}] = (a^2 : b^2) \in M(a^2 : b^2)$ *2-torsion points on their respective models.*

The map $\tau : \widehat{\theta}(a^2 : b^2) \to M(a^2 : b^2), [P_\theta] \mapsto [P_M + T_{2,M}]$ *is well-defined and given by* $(X : Z) \mapsto (X : Z)$. *It is a bijection and the converse is* $\tau^{-1} : [P_M] \mapsto [P_\theta + T_{2,\theta}]$.

Proof *The map is well-defined because* $T_{2,M}$ *is a 2-torsion point, so* $[P_M + T_{2,M}] = [P_M - T_{2,M}]$ *and the map* $P_\theta \mapsto P_M + T_{2,M}$ *on the elliptic curves factors through* $[-1]$, *giving* τ. *Recall the ramification on each model:*

$$0_{E,\theta} = (a^2 : b^2)^*, \quad T_{1,\theta} = (b^2 : a^2), \quad T_{2,\theta} = (1 : 0), \quad T_{3,\theta} = (0 : 1),$$

$$0_{E,M} = (1 : 0)^*, \quad T_{1,M} = (0 : 1), \quad T_{2,M} = (a^2 : b^2), \quad T_{3,M} = (b^2 : a^2).$$

It is then clear that $(X : Z) \mapsto (X : Z)$ *and* τ *match on the 2-torsion, hence on the whole model. The converse* τ^{-1} *also matches with* $(X : Z) \mapsto (X : Z)$. □

The idea to do the computations on several models up to a 2-torsion point has been developed in Section 6.2 with the hybrid ladder.

There are several possible conversions between the Montgomery and Legendre model, we give one choice below. Combining this with the other conversion maps helps to recover the conversion formulas between the theta model variants and the Legendre model.

Proposition 8.2.8 (Montgomery and Legendre) *Consider* $M(\alpha : 1)$ *a Montgomery model, the ramification is*

$$0_E = (1 : 0)^*, \quad T_1 = (0 : 1), \quad T_2 = (\alpha : 1), \quad T_3 = (1 : \alpha),$$

and so the map $(X : Z) \mapsto (\alpha X : Z)$ *gives an isomorphism (defined over the field of definition of* α*) with the Legendre Kummer line whose ramification is*

$$0_E = (1 : 0)^*, \quad T_1 = (0 : 1), \quad T_2 = (\lambda : 1), \quad T_3 = (1 : 1),$$

where $\lambda = \alpha^2$, *according to Example 4.3.7.*

In particular, combining with Proposition 8.2.2, given a theta model $\theta(a : b)$ *we see that it is isomorphic to the Legendre model with* $\lambda = A'^4/B'^4$.

8.3 INTERPRETATION IN TERMS OF GALOIS REPRESENTATION

Here we will give an interpretation of the models of Section 8.1 in terms of Galois representation. Let us introduce some notations first.

8.3.1 Weil and Tate pairings

k is as before a perfect field with $\operatorname{char} k \neq 2$, and we will consider an elliptic curve E defined over k. Let $n \geqslant 2$ be an integer, prime with $\operatorname{char} k$ if it

is positive. The rational n-torsion is denoted $E[n](k)$. As we have seen in Proposition 3.2.6, the structure of the n-torsion is well-known:

$$E[n] \simeq \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}.$$

If $P, Q \in E[n]$, one can define their Weil pairing $e_{W,n}(P, Q) \in \mu_n$ where μ_n are the n-th roots of unity, see for instance [Sil09, § III.8]. We will mainly use the following properties:

Proposition 8.3.1 $e_{W,n} : E[n] \times E[n] \to \mu_n$ *is:*

1. *Bilinear:* $\forall P, Q, R \in E[n]$,

$$\begin{cases} e_{W,n}(P + Q, R) = e_{W,n}(P, R)e_{W,n}(Q, R), \\ e_{W,n}(P, Q + R) = e_{W,n}(P, Q)e_{W,n}(P, R). \end{cases}$$

2. *Alternating:* $\forall P \in E[n],\ e_{W,n}(P, P) = 1$.

3. *Non-degenerate:* $(\forall P \in E[n], e_{W,n}(P, Q) = 1) \implies Q = 0_E$.

4. *Surjective: there are* $P, Q \in E[n]$ *such that* $e_{W,n}(P, Q)$ *is a primitive* n*-th root of unity.*

We already pointed out in Remark 5.2.3 that this is up to a sign the commutator pairing associated to some divisor.

There are many equivalent definitions of the Tate pairing. For our purposes, we will use the following definition (see for instance [Rob23, § 4.4, Eq. (10)]):

Definition 8.3.2 *Denote by* $\mathfrak{G}_k = \mathrm{Gal}(\overline{k}/k)$ *the absolute Galois group of* k *and assume that there is a rational* n*-torsion point on* E*, that is* $Q \in E[n](k)$*. Let* $P \in E(k)$ *and* $P_0 \in E$ *such that* $n \cdot P_0 = P$*. The Tate pairing is then defined as:*

$$\begin{aligned} e_{T,n}(P, Q) : \mathfrak{G}_k &\to \mu_n \\ \sigma &\mapsto e_{W,n}(\sigma(P_0) - P_0, Q) \end{aligned}$$

Notice that this definition does not depend on the choice of P_0.

Remark 8.3.3 *The Tate pairing can be evaluated up to an* n*-th power because* $H^1(\mathfrak{G}_k, \mu_n) \simeq k^*/(k^*)^n$*, more details can be found in [Rob23, § 4.4] to switch between the different points of view.*

On a finite field, the Tate pairing can even be reduced to studying only the case $\sigma = \pi_q$ *where* π_q *is the Frobenius map over* $\mathbb{F}_q$*. This is called the reduced Tate pairing (see [Rob23, § 4.5]), but we won't need it here.*

One use of the Tate pairing is to have a nice criterion about having a 4-isogeny (see also [Rob23, Ex. 5.9] which uses the geometric definition to obtain the following result as an immediate corollary):

Lemma 8.3.4 *Assume* E *has a non-trivial rational 2-torsion point* $P \in E[2](k)$ *and set* $P_0 \in E[4]$ *such that* $2 \cdot P_0 = P$*. The following conditions are equivalent:*

1. $e_{T,2}(P, P)$ *is trivial.*

2. *For all* $\sigma \in \mathfrak{G}_k$, $\sigma(P_0) = P_0$.

3. $K = \langle P_0 \rangle$ *is a rational cyclic subgroup of* $E[4]$ *of order* 4 *(i.e.* $\sigma(K) = K$ *for all* $\sigma \in \mathfrak{G}_k$*).*

4. E *has a* 4*-isogeny which is cyclic and rational.*

Moreover, E *has a rational cyclic* 4*-isogeny if and only if there is a rational* 4*-torsion point* $[P_0]$ *on the associated Kummer line* $\mathcal{K} = E/\{\pm 1\}$.

Proof *Let* $\sigma \in \mathfrak{G}_k$. *Since* $P \in E[2](k) \subseteq E[2]$, *we can consider* $Q \in E[2]$ *such that* $E[2] = \langle P, Q \rangle$. *The non-degeneracy of the Weil pairing imposes that* $e_{W,2}(P, Q)$ *is a primitive root of unity, i.e.* $e_{W,2}(P, Q) = -1$.

Because $\sigma(P_0) - P_0 \in E[2]$, *we write it as* $aP + bQ$ *with* $a, b \in \mathbb{Z}/2\mathbb{Z}$, *then* $e_{T,n}(P, P)(\sigma) = (-1)^b$. $e_{T,n}(P, P)$ *is trivial if and only if for all* $\sigma \in \mathfrak{G}_k$, $\sigma(P_0) - P_0 \in \langle P \rangle$, *and because* $P + P_0 = -P_0$:

$$e_{T,n}(P, P) = 1 \iff \forall \sigma \in \mathfrak{G}_k, \sigma(P_0) = \pm P_0.$$

The rest of the equivalence follows from Lemma 8.1.1. □

8.3.2 Galois representation and modular curves

For this section we refer to [Zyw15, § 2, § 3] which gives a modern point of view of Shimura's theory developed in [Shi94].

Galois representation

Let E be an elliptic curve defined over k a field with $\operatorname{char} k \neq 2$ and $n \geqslant 2$. As stated above, the structure of $E[n]$ is well known. We will set a basis of $E[n] = \langle P, Q \rangle$ for the rest of this section. $\mathfrak{G}_k$ acts on points of E coordinate-wise and is compatible with the addition law. Therefore, if $\sigma \in \mathfrak{G}_k$, $\sigma(E[n]) = E[n]$. So one only needs to know the image of P and Q by σ to get the full image of $E[n]$.

If $\sigma(P) = aP + cQ$ and $\sigma(Q) = bP + dQ$ with $a, b, c, d \in \mathbb{Z}/n\mathbb{Z}$, we can set:

$$M_\sigma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL_2(\mathbb{Z}/n\mathbb{Z}).$$

Definition 8.3.5 (Galois representation) *The following map defines a representation of* $\mathfrak{G}_k$ *on* $GL_2(\mathbb{Z}/n\mathbb{Z})$, *known as the Galois representation modulo* n:

$$\begin{array}{rccc} \rho_{E,n} : & \mathfrak{G}_k & \to & GL_2(\mathbb{Z}/n\mathbb{Z}) \\ & \sigma & \mapsto & M_\sigma. \end{array}$$

The morphism $\rho_{E,n}(\mathfrak{G}_k)$ plays a key role in the classification of elliptic curves E, but to explain why we first introduce modular curves.

Modular curves over $\mathbb{Q}$

Here we assume that $k = \mathbb{Q}$. Recall that $\mathbb{H}$ is the upper half-plane of $\mathbb{C}$, then $SL_2(\mathbb{Z})$ acts on $\mathbb{H}$ via $\gamma \cdot \tau = \frac{a\tau+b}{c\tau+d}$ if $\tau \in \mathbb{H}$ and $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in SL_2(\mathbb{Z})$.

Let $\Gamma(N)$ be the kernel of the modulo N map $SL_2(\mathbb{Z}) \twoheadrightarrow SL_2(\mathbb{Z}/N\mathbb{Z})$, that is the set of matrices of determinant 1 which reduce to the identity modulo N. The quotient $Y(N) = \Gamma(N)\backslash\mathbb{H}$ is a Riemann surface which can be compactified in a smooth compact Riemann surface denoted $X(N)$. If $\tau \in \mathbb{H}$, every meromorphic function f on $X(N)$ can be written in q-expansion ($q^{1/N} = e^{2i\pi\tau/N}$), that is there is a sequence of complex numbers $(c_n)_{n \geqslant n_0}$ with $n_0 \in \mathbb{Z}$ such that:

$$f(\tau) = \sum_{n \geqslant n_0} c_n q^{n/N}.$$

We denote by $\mathcal{F}_N$ the field of meromorphic functions over $X(N)$ that has q-expansion coefficients in $\mathbb{Q}(\zeta_N)$ with $\zeta_N = e^{2i\pi/N}$. Its elements are called the modular functions of level N. If $j = j(\tau)$ is the modular j-invariant, then $\mathcal{F}_1 = \mathbb{Q}(j)$. According to [Zyw15, Prop. 3.1], $\mathcal{F}_N$ is a Galois extension of $\mathbb{Q}(j)$ with Galois group isomorphic to $GL_2(\mathbb{Z}/N\mathbb{Z})/\{\pm I\}$.

For G a subgroup of $GL_2(\mathbb{Z}/N\mathbb{Z})$ containing $-I$ and such that $\det(G) = (\mathbb{Z}/N\mathbb{Z})^*$, we can then define $\mathcal{F}_N^G$ the subfield of $\mathcal{F}_N$ fixed by the action of G seen as a part of $\mathrm{Gal}(\mathcal{F}_N/\mathbb{Q}(j))$. By the assumption on $\det(G)$, $\mathbb{Q}$ is algebraically closed in $\mathcal{F}_N^G$ and $X_G = G\backslash X(N)$ is well-defined as the smooth projective curve over $\mathbb{Q}$ with function field $\mathcal{F}_N^G$. Because of the inclusion $\mathbb{Q}(j) \subseteq \mathcal{F}_N^G$, there is also a non-constant morphism of degree $[GL_2(\mathbb{Z}/N\mathbb{Z}) : G]$:

$$\pi_G : X_G \to \mathbb{P}^1(\mathbb{Q}).$$

The core of the classification of models relies on this theorem from [Zyw15, Prop. 3.3]:

Theorem 8.3.6 (Modular interpretation) *Let G be a subgroup of $GL_2(\mathbb{Z}/N\mathbb{Z})$ that contains $-I$ and satisfies $\det(G) = (\mathbb{Z}/N\mathbb{Z})^*$, and let E be an elliptic curve defined over $\mathbb{Q}$ such that its j-invariant is different from 0, 1728. Then $\rho_{E,N}(G_{\mathbb{Q}})$ is conjugate in $GL_2(\mathbb{Z}/N\mathbb{Z})$ to a subgroup of G if and only if $j_E \in \pi_G(X_G(\mathbb{Q}))$.*

Remark 8.3.7 *Theorem 8.3.6 can be generalized to more general fields, but requires developing the theory of modular curves over a generic field k. For completeness, this is done in Section 8.8.*

We will introduce some notations for specific curves, we have already seen $X(N)$. Let $T_+(N)$ be the set of upper triangular matrices in $GL_2(\mathbb{Z}/N\mathbb{Z})$ and $X_0(N) = X_{T_+(N)}$. Set also $T_-(N)$ be the set of lower triangular matrices in $GL_2(\mathbb{Z}/N\mathbb{Z})$ and $D(N) = T_+(N) \cap T_-(N)$ the diagonal matrices. Let $T_1(N) = \{\left(\begin{smallmatrix} 1 & * \\ 0 & * \end{smallmatrix}\right) \in GL_2(\mathbb{Z}/N\mathbb{Z})\}$ and $X_1(N) = X_{T_1(N)}$. Finally, if $M \geqslant 1$ is another integer, we set:

$$H(M, MN) = \{g \in T_+(MN) \mid g \equiv I \bmod M\} \text{ and } X(M, MN) = X_{H(M,MN)}.$$

Definition 8.3.8 *Γ is said to be a congruence subgroup if there exists $N \geqslant 1$ such that $\Gamma(N) \subset \Gamma$.*

Modular curves over $\mathbb{C}$ can be seen as the compactification of $Y(\Gamma) = \Gamma\backslash\mathbb{H}$, i.e. $X(\Gamma) = \Gamma\backslash\mathbb{H}^*$ where $\mathbb{H}^* = \mathbb{H} \cup \mathbb{Q} \cup \{\infty\}$ is the completed upper half plane, as done in [DS05]. Congruence subgroups then help to describe complex points of such curves. The following result can be computed with a computer algebra software like Magma and is tabulated in LMFDB[2].

Proposition 8.3.9 *When $N = 4$, the 19 possible images of congruence subgroups in $SL_2(\mathbb{Z}/4\mathbb{Z})$ have been represented in Fig. 8.5 in Section 8.9.*

Because of Theorem 8.3.6, some of these subgroups give families of elliptic curves with Kummer lines we would like to classify, as we will see in the next section.

8.3.3 Classification of models

Here we give description of the models of Section 8.1 in terms of modular curves. k is a field with $\operatorname{char} k \neq 2$.

It is an open question whether all the efficient families of curves in cryptography are associated to congruence subgroups, especially those in Fig. 8.5. In Section 8.6, we will come back to this classification where we will want both good arithmetic, such as Montgomery or theta coordinates, and also good torsion properties, so we may need to intersect some of these families.

An example of such classification is Legendre curves:

Proposition 8.3.10 *Let E be an elliptic curve over k with j-invariant different from $0, 1728$. Then E is a Legendre curve if and only if it has full rational 2-torsion or equivalently if $j \in \pi_G(X(2))$.*

Montgomery curves

To describe Montgomery curves, we will first need a criterion shown in [OKS00, Prop. 1]:

Lemma 8.3.11 *Let E be an elliptic curve over k and written in Weierstrass form $y^2 = h(x)$. Then E can be put in Montgomery form if and only if there exists $\alpha \in k$ such that $h(\alpha) = 0$ and $h'(\alpha)$ is a square different from zero.*

The first condition means there is a non-trivial rational 2-torsion point, which is clear with the equation of a Montgomery curve. We will explicit the second condition using the Tate pairing:

Proposition 8.3.12 *Let E be an elliptic curve over k with j-invariant different from $0, 1728$. Then E is a Montgomery curve if and only if it admits a rational cyclic subgroup K of order 4. Every Montgomery curve is described by a rational point $x \in X_0(4)(k)$ and vice-versa.*

Proof *We will write $E : y^2 = h(x)$ in Weierstrass form.*

First assume E is a Montgomery curve. By Lemma 8.3.11, $P = (\alpha, 0)$ is a rational 2-torsion point in Weierstrass coordinates with $\alpha \in k$ and such that $h'(\alpha) \in (k^*)^2$.

2 https://beta.lmfdb.org/

Now recall Remark 8.3.3, the Tate pairing $e_{T,2}(P,P) \in H^1(G,\mu_2)$ *is trivial if and only if it is a square. But if* P *is a rational 2-torsion point, thanks to some computation done in [Rob23, Ex. 5.11.1], we have* $e_{T,2}(P,P) \equiv h'(\alpha)$*, which is a square. So* $e_{T,2}(P,P)$ *is trivial and using Lemma 8.3.4 we have a rational subgroup* $K \simeq \mathbb{Z}/4\mathbb{Z}$*.*

Conversely, assume that E *has such a subgroup* K*, and denote its generator by* $P_0 \in E[4]$*. By Lemma 8.3.4,* $\sigma(P_0) = \pm P_0$ *for all* $\sigma \in \mathfrak{G}_k$ *because* K *is rational. Setting* $P = 2 \cdot P_0$*, we then have* $\sigma(P) = P$*, that is* P *is a non-trivial rational 2-torsion point on* E*. We write it* $(\alpha, 0)$ *in Weierstrass coordinates with* $\alpha \in k$*. Again, by Lemma 8.3.4,* $e_{T,2}(P,P)$ *is trivial, i.e. it is a square. This implies that* $h'(\alpha)$ *is also a square because of* $e_{T,2}(P,P) \equiv h'(\alpha)$*, and we conclude with Lemma 8.3.11.*

If we set the basis of $E[4] = \langle P_0, Q\rangle$ *where* $K = \langle P_0\rangle$*, then for all* $\sigma \in \mathfrak{G}_k$*,* $\sigma(P_0) = \pm P_0$*. Therefore,* $\rho_{E,4}(\sigma)$ *is upper triangular, i.e.* $\rho_{E,4}(\mathfrak{G}_k) \subset T_+(4)$*. Thanks to Theorem 8.3.6 applied to* $G = T_+(4)$*,* E *being in Montgomery form is equivalent to its* j*-invariant* j_E *being in* $\pi_G(X_0(4)(k))$*.* □

Remark 8.3.13 *In terms of congruence subgroups, we are on the curve associated to* $T_+(4) = \left\langle \Gamma_0(4), \left(\begin{smallmatrix} -1 & 0 \\ 0 & 1 \end{smallmatrix}\right)\right\rangle$ *which is group* `4.6.0.c.1` *in Fig. 8.5.*

Theta and theta squared models

As seen before, rational torsion is really helpful to determine on which modular curve we are. In addition to that, as shown in Lemma 8.3.4 the Tate pairing also gives useful information on the Galois representation. We will consider the theta model $\theta(a : b)$ and the theta squared model $\widehat{\theta}(a^2 : b^2)$. These two models have full 2-torsion, and some rational 4-torsion points, as already seen in Propositions 8.1.7 and 8.1.11.

We have the following proposition:

Proposition 8.3.14 *Let* E *be an elliptic curve over* k*, with* j*-invariant* j_E *different from* 0*,* 1728*.*

1. E *admits a theta model if and only if it has two rational subgroups* $K_1, K_2 \simeq \mathbb{Z}/4\mathbb{Z}$ *with trivial intersection. Every theta model can then be described by a rational point* $j_E \in \pi_{D(4)}(X_{D(4)}(k))$ *and vice-versa.*

2. E *admits a theta squared model if and only if* $E[2](k) = E[2]$ *and* E *has a rational subgroup* $K \simeq \mathbb{Z}/4\mathbb{Z}$*. Every theta squared model can then be described by a rational point* $j_E \in \pi_{H(2,4)}(X(2,4)(k))$ *and vice-versa.*

Proof *If* E *admits a theta model* $\theta(a : b)$*, we have two rational independent 4-torsion points on the Kummer line which are* $(1 : 0)$ *and* $(1 : 1)$ *according to Proposition 8.1.7. They can be lifted to points* $P_0, Q_0 \in E[4]$ *which generates two rational cyclic kernels* $K_1 = \langle P_0\rangle$ *and* $K_2 = \langle Q_0\rangle$ *thanks to Lemma 8.3.4 (the points* P_0*,* Q_0 *may not be rational however). The kernels now have trivial intersection because the torsion points are independent, they do not lie above the same 2-torsion point. We also recover that* E *has full rational 2-torsion generated by* $P = 2 \cdot P_0$ *and* $Q = 2 \cdot Q_0$*, which are rational because the kernels are too.*

If E *only has a theta squared model, we still have the full rational 2-torsion by lifting* $(b^2 : a^2)$ *and* $(1 : 0)$ *to* E*. The points are rational on the curve too because*

they are of order 2. We then set $K = \langle P_0 \rangle$ *where* P_0 *is a lift of the rational 4-torsion point* $(1:1)$ *on the Kummer line. Then* K *is rational because of Lemma 8.3.4.*

Conversely, assume that E *has a rational kernel* $K \simeq \mathbb{Z}/4\mathbb{Z}$ *and full rational 2-torsion. Then* E *can be put in Montgomery form using Proposition 8.3.12:*

$$E : By^2 = x(x^2 + \mathcal{A}x + 1).$$

Moreover, non-trivial 2-torsion points on a Montgomery curve are given by $y = 0$*, i.e.* $x = 0$ *or* $x^2 + \mathcal{A}x + 1 = 0$*. Because the 2-torsion is rational,* $x^2 + \mathcal{A}x + 1$ *is split, and we have* $\alpha \in k^*$ *such that:*

$$E : By^2 = x(x - \alpha)(x - \alpha^{-1}).$$

From that point, we know that the Kummer line associated to E *is* $M(\alpha : 1)$*, as defined in Section 8.1. This model is isomorphic to* $\widehat{\theta}(\alpha : 1)$ *via Proposition 8.2.5, hence* E *admits a theta squared model.*

If we have the stronger hypothesis that E *has two rational kernels* $K_1, K_2 \simeq \mathbb{Z}/4\mathbb{Z}$ *with trivial intersection, we write* $K_1 = \langle P_0 \rangle$ *and* $K_2 = \langle Q_0 \rangle$*,* $P = 2 \cdot P_0$ *and* $Q = 2 \cdot Q_0$*. Since for all* $\sigma \in \mathfrak{G}_k$*,* $\sigma(K_i) = K_i$*, when looking at the order we must have* $\sigma(P) = P$ *and* $\sigma(Q) = Q$*, so these points are rational, and because they are distinct they are generators of the 2-torsion. We then are in the previous case with* $\alpha \in k^*$ *such that:*

$$E : By^2 = x(x - \alpha)(x - \alpha^{-1}) = h(x).$$

We would like to prove that $\frac{\alpha+1}{\alpha-1}$ *is a square. Up to transposing* P *and* Q*, assume that* $Q = (\alpha, 0)$*. Because* K_2 *is cyclic rational of degree 4,* $e_{T,2}(Q, Q)$ *is trivial thanks to Lemma 8.3.4. With the computation done in [Rob23, Ex. 5.8.1], we know that* $e_{T,2}(Q, Q) \equiv h'(\alpha)$ *is a square.*

Because $h'(x) = 3x^2 + 2\mathcal{A}x + 1$ *where* $\mathcal{A} = -\alpha - \alpha^{-1}$*, when evaluated in* α*, we get:*

$$h'(\alpha) = \alpha^2 - 1 = \frac{\alpha + 1}{\alpha - 1}(\alpha - 1)^2.$$

Thus, $\beta = \sqrt{\frac{\alpha+1}{\alpha-1}} \in k$*, and* $M(\alpha : 1)$ *is isomorphic to* $\theta(\beta : 1)$ *thanks to Proposition 8.2.2, so* E *admits a theta model.*

The interpretation in terms of modular curves is straight-forward:

- *If* E *has a theta model, denote* $K_1 = \langle P_0 \rangle$ *and* $K_2 = \langle Q_0 \rangle$ *with* $P_0, Q_0 \in E[4]$ *the two rational kernels. We have* $E[4] = \langle P_0, Q_0 \rangle$*. Because both* K_1 *and* K_2 *are rational, for all* $\sigma \in \mathfrak{G}_k$*,* $\sigma(P_0) = \pm P_0$ *and* $\sigma(Q_0) = \pm Q_0$*, so* $\rho_{E,4}(\mathfrak{G}_k) \subset D(4)$*, which is equivalent to* $j_E \in \pi_{D(4)}(X_{D(4)}(k))$ *with Theorem 8.3.6.*

- *If* E *has a theta squared model, let* $K = \langle P_0 \rangle$ *with* $P_0 \in E[4]$ *be the rational kernel. Set* $P = 2 \cdot P_0 \in E[2](k)$*, write* $E[2](k) = \langle P, Q \rangle$ *and take* $Q_0 \in E[4]$ *such that* $2 \cdot Q_0 = Q$ *and* $E[4] = \langle P_0, Q_0 \rangle$*. Then for all* $\sigma \in \mathfrak{G}_k$*,* $\sigma(P_0) = \pm P_0$*,* $\sigma(P) = P$ *and* $\sigma(Q) = Q$*, so* $\rho_{E,2}(\sigma) = I$ *and* $\rho_{E,4}(\sigma) \in T_+(4)$*, i.e.* $\rho_{E,4}(\mathfrak{G}_k) \subset H(2,4)$*. This is again equivalent to* $j_E \in \pi_{H(2,4)}(X(2,4)(k))$ *with Theorem 8.3.6.* □

We remark that Proposition 8.3.14, for the theta model, is a particular case of a general result from Igusa that states that the theta constants of level n are modular forms for the subgroup $\Gamma(n, 2n) \subset \mathrm{Sp}_{2g}(\mathbb{Z})$. Indeed, in the particular case when $n = 2$ and $g = 1$, we have $\Gamma(2, 4)$ which is conjugated to $D(4) = T_+(4) \cap T_-(4)$.

Remark 8.3.15 *By Propositions 8.3.12 and 8.3.14, we see that the theta squared (or equivalently the theta twisted) model correspond to a Montgomery model with full rational 2-torsion (i.e. in Montgomery $\cap$ Legendre). This can be seen directly from the conversion formulas from Section 8.2.*

8.4 INTERPRETATION IN TERMS OF THE THETA GROUP

In this section, we reinterpret the results of Section 8.3 in terms of the theta group.

8.4.1 The theta group and the type of a point

We introduced the theta group in Chapter 5, and specified the study to an elliptic curve and a product of elliptic curves in Sections 6.2.1 and 7.4.

We only need to work with theta groups of level $n = 2$, consider $D = 2(0_E)$ and recall that there is a canonical faithful linear action of $G(D)$ on $\Gamma(D) = \langle X, Z\rangle$ denoted ρ_D, see Definition 5.2.17.

If $P \in E[2]$, we have the translation map $R \mapsto R + P$ which induces a linear map on $\Gamma(D) = \langle X, Z\rangle$, hence a homography γ_P on the Kummer line $E/\pm 1$. We can represent the homography γ_P as an element of PGL_2. A choice of a theta group element g_P is then equivalent to the choice of a matrix $M_P \in \mathrm{GL}_2$ above γ_P by the discussion between Eqs. (5.3) and (5.4). This is what we already used in Section 6.2.1.

Since P is a point of 2-torsion, we have $M_P^2 = \lambda_P$. If P is rational and we take $M_P \in \mathrm{GL}_2(k)$, then changing M_P by a scalar changes λ_P by a square; we see that the class of $\lambda_P \in k^*/k^{*,2}$ depends only on P, not on g_P, and is called the type of P as defined in Definition 6.2.4. Since the type of P can be recovered from the equation $g_P^2 = \lambda_P$ for a rational element g_P in the theta group above P, we see that it does not depend on the choice of basis of $\Gamma(D)$.

Theorem 8.4.1 *Assume that $P \neq 0_E \in E[2](k)$. The type $\lambda_P \in k^*/k^{*,2}$ of P is also the class of the non-reduced self Tate pairing $e_{T,2}(P, P) \in k^*/k^{*,2}$. If we take a representative λ_P of the type, then there is a model of the Kummer line $E/\pm 1$ where 0_E is sent to ∞, P to 0_E, and the two remaining 2-torsion points are given by $(\alpha : 1), (1/(\lambda_P\alpha) : 1)$ (where $\alpha \in \overline{k}$).*

Conversely, if we have a model of the Kummer line $E/\pm 1$ such that the ramification is stable by $(X : Z) \mapsto (Z : \lambda_P X)$ with 0_E sent to P, then P is of type λ_P. In particular, P is of type λ_P if and only if there is a rational Weierstrass equation of E of the form $\beta y^2 = x(x^2 + \mathcal{A}x + 1/\lambda_P)$ with $x(P) = 0$.

This is the kind of equation we were studying in Main example 6.2.1.

Proof *Recall that the type of P is given by $g_P^2 = \lambda_P$ where g_P is any theta group element above P. By the definition of the theta group law, we have $\lambda_P = e_{T,2}(P, P)$.*

Now looking at the matrix M_P *given by the action of* g_P *on* $\Gamma(2)$*, we also have* $M_P^2 = \lambda_P$*. This is the minimal polynomial of* M_P *because* M_P *cannot be a diagonal matrix since* $P \neq 0_E$*. So we can find a basis of sections of* D *such that* $M_P = \left(\begin{smallmatrix} 0 & \lambda_P \\ 1 & 0 \end{smallmatrix}\right)$.

If $0_E = (a : b)$ *with this basis, then* $P = \gamma_P \cdot 0_E = (b : \lambda_P a)$*. The change of basis* $(X, Z) \mapsto (X - \frac{b}{\lambda_P a} Z : Z - \frac{b}{a} X)$ *sends* 0_E *to* $(1 : 0)$ *and* P *to* $(0 : 1)$*, while* M_P *is invariant by the corresponding conjugation. It follows that if one other 2-torsion point as coordinates* $T_2 = (\alpha : 1)$*, then* $T_3 = \gamma_P \cdot T_2 = (1 : \lambda_P \alpha)$.

Conversely, a homography of order 2 that stabilizes the ramification and sends 0_E *to* P *has to be the one induced by the translation by* P*, and by our assumptions we can take* $M_P = \left(\begin{smallmatrix} 0 & \lambda_P \\ 1 & 0 \end{smallmatrix}\right)$. □

Remark 8.4.2 *Pick up* g_P *such that* $g_P^2 = \lambda_P$*. One way to find a basis* $(X, Z) \in \Gamma(D)$ *so that the ramification is as Theorem 8.4.1 is as follows. First let* $Z_0 \in \Gamma((0_E))$ *be a section, and let* $Z = Z_0^2$*. Then we let* $X = g_P \cdot Z$*. Since* $Z(0_E) = 0$*, we have* $0_E = (1 : 0)$*, and* $P = (0 : 1)$*. If* T_2 *is another 2-torsion point,* $T_3 = g_P \cdot T_2$*, where the action of* g_P *is simply the action of* $\gamma_P : (X : Z) \mapsto (Z : \lambda_P X)$.

Theorem 8.4.1 explains how the structure of the ramification on the Kummer line, the action of the theta group on $\Gamma(D)$, and the Tate self-pairing are all linked together. A particularly important case is when the type of P is a square, in which case we say that P is of *Montgomery type*.

Corollary 8.4.3 (Montgomery type) *Let* $P \in E[2](k)$ *be a rational point of 2-torsion, the following conditions are equivalent:*

1. *the type of* P *is a square;*
2. E *has a Montgomery model* $\beta y^2 = x(x^2 + \mathcal{A}x + 1)$ *with* $x(P) = 0$*;*
3. *there is a model of the Kummer line of* E *which has its ramification invariant by* $(X : Z) \mapsto (Z : X)$ *with* 0_E *sent to* P*;*
4. *there is a model of the Kummer line of* E *such that* $0_E = (1 : 0)$*,* $P = (0 : 1)$*, and the other points of 2-torsion are given by* $T_2 = (\alpha : 1), T_3 = (1 : \alpha)$*;*
5. *there is a model of the Kummer line of* E *which has its ramification invariant by* $(X : Z) \mapsto (-X : Z)$ *with* 0_E *sent to* P*;*
6. *there is a model of the Kummer line of* E *such that* $0_E = (1 : 1)$*,* $P = (-1 : 1)$*, and the other points of 2-torsion are given by* $T_2 = (\beta : 1), T_3 = (-\beta : 1)$*;*
7. *there exists a rational element* $g_P \in G(D)$ *such that* $g_P^2 = 1$ *(equivalently: the two symmetric elements* $\pm g_P \in G(D)$ *above* P *are rational);*
8. *the reduced Tate self pairing is trivial:* $e_{T,2}(P, P) = 1$*;*
9. *there exists a rational cyclic subgroup* K *of degree 4 containing* P*;*
10. *there exists a point of 4-torsion* P' *above* P *which is rational on the Kummer line (i.e. such that if* $E : \beta y^2 = x^3 + a_2 x^2 + a_4 x + a_6$*,* $x(P')$ *is rational). Equivalently,* $\sigma(P') = \pm P'$ *for all* $\sigma \in \mathrm{Gal}(\overline{k}/k)$*;*
11. $E/\langle P \rangle$ *has its full 2-torsion rational (i.e. admits a rational Legendre model);*

12. *the isogeny* $\varphi_P : E \to E/\langle P\rangle$ *can be extended to a rational cyclic isogeny of degree* 4 *(automatically in two different ways).*

Proof *The first eight equivalences are consequences of the definition of the type of* P *and of Theorem 8.4.1: if* P *has type a square, we can take* $\lambda_P = 1$ *and* $M_P = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. *Then the Hadamard change of basis* $(X, Z) \mapsto (X + Z, X - Z)$ *shows that we can also take* $M_P = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$.

The link between the Tate pairing and having a rational cyclic group is given by Lemma 8.3.4, but we can use the above result to give an alternative proof: if P *is a 2-torsion point, then an element* $g_P \in G(D)$ *such that* $g_P^2 = 1$ *is completely determined by a choice of a 4-torsion point* P' *above* P *(more precisely, the choice of* P' *determines a symmetric element* g_P*, but for* $n = 2$*,* g_P *is symmetric if and only if* $g_P^2 = 1$ *by Lemma 6.2.3). If* $P'' = P' + T$ *is a different choice of 4-torsion point, so that* $T \in E[2]$*, the element induced by* P'' *is the same one as the one by* P' *if and only if* $e_{W,2}(T, P) = 1$*. It follows that we can find a rational* g_P *if and only if we can find* P' *such that* $\sigma(P') = \pm P'$ *for all* $\sigma \in \mathrm{Gal}(\overline{k}/k)$.

Yet another alternative proof is to use descent theory: we can find a rational symmetric g_P *if and only if* $2(0_E)$ *descends to a symmetric divisor on* $E' = E/\langle P\rangle$*, i.e. there is a rational divisor* D' *linearly equivalent to* $D = 2(0_E)$ *of the form* $\varphi_P^*((T'))$ *for* $T' \in E'[2]$ *where* $\varphi_P : E \to E'$ *is the 2-isogeny. But then* $D' = (P') + (P' + P)$*, for some point of 4-torsion* P'*, and since* D' *is rational we recover that* $\sigma(P') = \pm P'$ *for all* $\sigma \in \mathrm{Gal}(\overline{k}/k)$.

We already saw that (9) is equivalent to (10) in Lemma 8.1.1.

For the last two statements, notice that on $E' = E/\langle P\rangle$ *there is always a rational 2-torsion point* T_1' *giving the dual isogeny* $E' \to E$*. The isogeny* φ_P *can be extended to a cyclic isogeny of degree 4 if and only if there is another rational point of 2-torsion* T_2' *on* E'*, but then* $T_3' = T_1' + T_2'$ *is rational too, so this is equivalent to* E' *having full rational 2-torsion.* □

If $k = \mathbb{F}_q$ is a finite field, then $\mathbb{F}_q^*/\mathbb{F}_q^{*,2}$ is of cardinal two. Pick up a $\lambda_- \in \mathbb{F}_q^*$ that is not a square, for instance if $q \equiv 3 \bmod 4$ we will pick $\lambda_- = -1$. Then if $P \in E[2](\mathbb{F}_q)$, it is either of type a square (i.e. of Montgomery type) in which case we can choose $\lambda_P = 1$, or not a square, in which case we can choose $\lambda_P = \lambda_-$, and in that case we say that P is of anti-Montgomery type. The pendant of Corollary 8.4.3 is:

Corollary 8.4.4 *If* $P \in E[2](k)$ *is a rational 2-torsion point, the following conditions are equivalent:*

1. *the type of* P *is not a square;*
2. E *has an 'anti-Montgomery' model* $\beta y^2 = x(x^2 + \mathcal{A}x + 1/\lambda_-)$ *with* $x(P) = 0$*;*
3. *there is a model of the Kummer line of* E *which has its ramification invariant by* $(X : Z) \mapsto (\lambda_- Z : X)$ *with* 0_E *sent to* P*;*
4. *there is a model of the Kummer line of* E *such that* $0_E = (1 : 0)$*,* $P = (0 : 1)$*, and the other points of 2-torsion are given by* $T_2 = (\alpha : 1), T_3 = (\lambda_- : \alpha)$*;*
5. *there does not exist a rational element* $g_P \in G(D)$ *such that* $g_P^2 = 1$ *(equivalently: the two symmetric elements* $\pm g_P \in G(D)$ *above* P *are not rational);*

6. *the reduced Tate self pairing is not trivial:* $e_{T,2}(P, P) = -1$;
7. *there does not exist a rational cyclic subgroup* K *of degree* 4 *containing* P;
8. $E/\langle P\rangle$ *has only one rational point of* 2*-torsion;*
9. *the isogeny* $\varphi_P : E \to E/\langle P\rangle$ *cannot be extended to a rational cyclic isogeny of degree* 4.

The results of Corollaries 8.4.3 and 8.4.4 recover the discussion on the Montgomery and Montgomery$^-$ models of [CD20].

8.4.2 Twisted theta models

We continue our reinterpretation of the results of Section 8.3 in terms of the theta group. This section is also related to finding rational theta functions as it was done in Section 5.4.

Let E/k be an elliptic curve and denote, as before, $D = 2(0_E)$. As defined in Definition 5.3.1, a theta structure on (E, D) is an isomorphism between the theta group $G(D)$ and the Heisenberg group $G(2)$ of level 2. Such an isomorphism always exists on the algebraic closure $\overline{k}$ and boils down to:

- Choosing a basis of $E[2] = \langle T_1, T_2\rangle$;
- Choosing two symmetric elements $g_1, g_2 \in G(D)$ above T_1 and T_2 respectively. (We recall that in our case g is symmetric if and only if $g^2 = 1$).

The elements being symmetric is a particular case of level 2*, in that case we say that the theta structure is symmetric.*

In which case, the canonical basis of theta functions θ_0, θ_1 can be defined as follows: we pick up θ_0 as a generator of $\Gamma(D)$ invariant under the action of g_2, and $\theta_1 := g_1 \cdot \theta_0$. This is what we explained earlier in Eq. (5.2) and Example 5.3.7.

Remark 8.4.5 *In the original article, we swapped conventions where* θ_0 *is invariant under* g_1 *and* $\theta_1 := g_2 \cdot \theta_0$*. This does not matter in the sequel, however, since we used Mumford convention in Section 5.3, we try to keep these here. Hopefully the two conventions are not mixed together in the rest of this section.*

The basis of theta functions is well-defined over $\overline{k}$, however the theta structure associated may not be rational.

Lemma 8.4.6 E *admits a rational symmetric theta structure of level* 2 *if and only if there exist two rational points* $T_1, T_2 \in E[2](k)$ *that are both of Montgomery type.*

In particular, by Corollary 8.4.3, we recover the first part of Proposition 8.3.14. In fact, we can refine it: if E has such a rational theta structure, let $T_1, T_2 \in E[2](k)$, $g_1, g_2 \in G(D)$ be rational elements given by the rational isomorphism of $G(D)$ with $G(2)$. Then $T_1 + T_2$ is certainly rational, g_1g_2 is above $T_1 + T_2$, and we have $(g_1g_2)^2 = e_D(T_1, T_2) = -1$, hence $T_1 + T_2$ is of type -1. It follows that $T_1 + T_2$ is also of Montgomery type if and only if -1 is a square in k (and then we will denote by i a square root).

Corollary 8.4.7 *Assume that E/k has a rational symmetric theta model of level 2. Then if -1 is a square in k, E/k has 24 different theta models; otherwise it has 8.*

We implicitly count theta models with multiplicities if E has non-trivial automorphisms.

Proof *If -1 is a square, all three points of 2-torsion are of Montgomery type. The choice of a theta model amount to choosing two points (T_1, T_2) out of the three (where the ordering count), for a total of 6 choices, and then for each T_i to fix a choice of symmetric element $\pm g_1, \pm g_2$, for a total of 4 choices. This gives 24 possible choices. If -1 is not a square, only two points are of Montgomery type, and now we have $2 \times 4 = 8$ choices.* □

Example 8.4.8 (Number of rational Montgomery models) *More generally, if we have two rational 2-torsion points T_1, T_2, of type λ_1, λ_2, then $T_1 + T_2$ is rational of type $-\lambda_1\lambda_2$.*

Also, if T_1 is of type λ_1 and we choose $g_1 \in G(D)$ such that $g_1^2 = \lambda_1$ to construct the (X, Z) coordinates as in Remark 8.4.2, which gives the equation $E : \beta y^2 = x^3 + \mathcal{A}x^2 + x/\lambda_1$, then picking up $-g_1$ instead changes X to $-X$, and gives the equation $E : -\beta y^2 = x^3 - \mathcal{A}x^2 + x/\lambda_1$

It follows that, if $k = \mathbb{F}_q$ is a finite field, and -1 is a square in $\mathbb{F}_q$:

- *If T_1, T_2 are of Montgomery type, $T_1 + T_2$ too, and there are 6 rational Montgomery models.*
- *If T_1 is of Montgomery type and T_2 of anti-Montgomery type, then $T_1 + T_2$ if of anti-Montgomery type, and there are 2 rational Montgomery models and 4 rational Montgomery$^-$ models.*
- *If T_1, T_2 are of anti-Montgomery type, then $T_1 + T_2$ if of Montgomery type, and there are 2 rational Montgomery models and 4 rational Montgomery$^-$ models.*

And if -1 is not a square in $\mathbb{F}_q$:

- *If T_1, T_2 are of Montgomery type, $T_1 + T_2$ is of anti-Montgomery type, and there are 4 rational Montgomery models and 2 rational Montgomery$^-$ models.*
- *If T_1 is of Montgomery type and T_2 of anti-Montgomery type, then $T_1 + T_2$ if of Montgomery type, and there are 4 rational Montgomery models and 2 rational Montgomery$^-$ models.*
- *If T_1, T_2 are of anti-Montgomery type, then $T_1 + T_2$ if of anti-Montgomery type, and there are 6 rational Montgomery$^-$ models.*

We want to expand on the rational theta functions discussed in Section 5.4. Now, if $G(D)$ is not isomorphic to $G(2)$ over k, we know that it is an (étale) twist of it (since it is isomorphic over the separable closure). Picking up such an automorphism γ, we see that for each $\sigma \in \mathrm{Gal}(\overline{k}/k)$ we obtain a (symmetric) automorphism $\xi(\sigma) = \gamma^\sigma\gamma^{-1}$ of $G(2)$, and $\sigma \mapsto \xi(\sigma)$ forms a cocycle.

Assume, for the sake of simplicity, that -1 is a square in k, so that the symmetric automorphisms of $G(2)$ are rational, and the coboundaries are trivial. We can identify these automorphisms with $\Gamma/\Gamma(2,4)$ where Γ is

$SL_2(\mathbb{Z}) = Sp_2(\mathbb{Z})$ and, in our special case, $\Gamma(2,4) = \Gamma^0(4) \cap \Gamma_0(4)$ (see the discussion after Proposition 8.3.14).

In particular, $\xi(\sigma)$ explains how σ acts on the canonical theta basis (θ_0, θ_1) given by γ: unless $\xi(\sigma) = \mathrm{id}$ for all σ, this basis is not rational in general. Pick a subgroup G such that $\Gamma(2,4) \subset G \subset \Gamma$. We say that $G(D)$ is of type G if $\xi(\sigma) \in G$ for all $\sigma \in \mathrm{Gal}(\overline{k}/k)$. In that case, we can try to build a basis θ'_0, θ'_1 as a linear combination of θ_0, θ_1 such that θ'_0, θ'_1 is invariant under the action of the automorphisms in G. It follows that we obtain a rational basis of sections on E, and we call (θ'_0, θ'_1) a twisted theta model. We stress that in a 'twisted theta model', we still describe a model of E (or its Kummer line), rather than a twist E' of E. Here the twist refer instead to the fact that for these models the theta group is a twist of the standard Heisenberg group.

Example 8.4.9 (Automorphisms of the theta group) *We list all possible* 24 *automorphisms of* $G(2)$*, for each give a representative in* $\Gamma/\Gamma(2,4)$*, and how it acts on the theta coordinates* θ_0, θ_1 *(hence on the theta constants). We give six blocks of four automorphisms. Each block correspond to a choice of a representative of* $\Gamma/\Gamma(2)$ *(which is of cardinal* $3! = 6$*), and where the first block corresponds to* $\gamma = \mathrm{id}$*. Then for these choices of* $\gamma \in \Gamma/\Gamma(2)$*, we list* 4 *possible lifts of* γ *to* $\Gamma/\Gamma(2,4)$*. More precisely, once we have picked up one possible lift* γ'*, we then give the other ones via the action of the first block of automorphisms.*

The change of coordinates follows from the explicit construction of θ_0, θ_1 *once* g_1, g_2 *has been chosen. Analytically, it also follows from the theta transformation formula [Igu72; Mum83] (which is more precise since it also keep track of the theta constants as* modular forms*).*

More precisely, if $\gamma = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$*, consider* 4*-torsion points* T'_1, T'_2 *above* T_1, T_2 *respectively. Set also* $T'_3 = T'_1 + T'_2$*. We have* $\gamma \cdot T'_1 = aT'_1 + bT'_2$ *and* $\gamma \cdot T'_2 = cT'_1 + dT'_2$ *by the Galois representation. This helps to determine how* γ *acts on the* 2*-torsion, and above which* 2*-torsion point* $\gamma \cdot g_1$ *and* $\gamma \cdot g_2$ *are. Finally, the sign is determined by the* 4*-torsion point, with the following correspondence:*

$$\begin{array}{ccc} \pm T'_1 \leftrightarrow g_1, & \pm T'_2 \leftrightarrow g_2, & \pm T'_3 \leftrightarrow g_3 := ig_2g_1, \\ \pm T'_1 + T_2 \leftrightarrow -g_1, & \pm T'_2 + T_1 \leftrightarrow -g_2, & \pm T'_3 + T_1 \leftrightarrow -g_3. \end{array}$$

The action on the theta functions is then determined from $\theta_0 = g_2 \cdot \theta_0$ *and* $\theta_1 = g_1 \cdot \theta_0$*. The full explanation is given above Example 8.4.11. We also make the connection with Table 5.1, written in parentheses.*

1. $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (g_1, g_2)$ *(A.1), hence*
$$(\theta_0, \theta_1) \mapsto (\theta_0, \theta_1).$$
2. $\left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (-g_1, g_2)$ *(A.2), hence*
$$(\theta_0, \theta_1) \mapsto (\theta_0, -\theta_1).$$
3. $\left(\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (g_1, -g_2)$ *(A.3), hence*
$$(\theta_0, \theta_1) \mapsto (\theta_1, \theta_0).$$

4. $\left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (-g_1, -g_2)$ *(A.4), hence*

$$(\theta_0, \theta_1) \mapsto (\theta_1, -\theta_0).$$

5. $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (g_2, g_1)$ *(B.1), hence*

$$(\theta_0, \theta_1) \mapsto (\theta_0 + \theta_1, \theta_0 - \theta_1).$$

6. $\left(\begin{smallmatrix} 2 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (-g_2, g_1)$ *(B.2), hence*

$$(\theta_0, \theta_1) \mapsto (\theta_0 + \theta_1, -\theta_0 + \theta_1).$$

7. $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 2 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (g_2, -g_1)$ *(B.3), hence*

$$(\theta_0, \theta_1) \mapsto (\theta_0 - \theta_1, \theta_0 + \theta_1).$$

8. $\left(\begin{smallmatrix} 2 & 1 \\ -1 & 2 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (-g_2, -g_1)$ *(B.4), hence*

$$(\theta_0, \theta_1) \mapsto (-\theta_0 + \theta_1, \theta_0 + \theta_1).$$

9. $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (g_3, g_2)$ *(C.1), hence*

$$(\theta_0, \theta_1) \mapsto (i\theta_0, \theta_1).$$

10. $\left(\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (-g_3, g_2)$ *(C.2), hence*

$$(\theta_0, \theta_1) \mapsto (\theta_0, i\theta_1).$$

11. $\left(\begin{smallmatrix} 1 & 1 \\ 2 & -1 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (g_3, -g_2)$ *(C.3), hence*

$$(\theta_0, \theta_1) \mapsto (\theta_1, i\theta_0).$$

12. $\left(\begin{smallmatrix} 1 & -1 \\ 2 & -1 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (-g_3, -g_2)$ *(C.4), hence*

$$(\theta_0, \theta_1) \mapsto (i\theta_1, \theta_0).$$

13. $\left(\begin{smallmatrix} -1 & 0 \\ -1 & -1 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (g_1, g_3)$ *(D.1), hence*

$$(\theta_0, \theta_1) \mapsto (i\theta_0 + \theta_1, \theta_0 + i\theta_1).$$

This matrix is obtained by applying (9) then (5) then (9).

14. $\left(\begin{smallmatrix} 1 & 2 \\ -1 & -1 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (-g_1, g_3)$ *(D.2), hence*

$$(\theta_0, \theta_1) \mapsto (\theta_0 - i\theta_1, i\theta_0 - \theta_1).$$

15. $\left(\begin{smallmatrix} -1 & 0 \\ 1 & -1 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (g_1, -g_3)$ *(D.3), hence*

$$(\theta_0, \theta_1) \mapsto (\theta_0 + i\theta_1, i\theta_0 + \theta_1).$$

16. $\left(\begin{smallmatrix} 1 & 2 \\ 1 & -1 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (-g_1, -g_3)$ *(D.4), hence*

$$(\theta_0, \theta_1) \mapsto (i\theta_0 - \theta_1, \theta_0 - i\theta_1).$$

17. $\left(\begin{smallmatrix} 0 & 1 \\ -1 & -1 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (g_2, g_3)$ *(E.1), hence*

$$(\theta_0, \theta_1) \mapsto (i\theta_0 + \theta_1, i\theta_0 - \theta_1).$$

This matrix is obtained by applying (9) then (5).

18. $\left(\begin{smallmatrix} 2 & -1 \\ -1 & -1 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (-g_2, g_3)$ *(E.2), hence*

$$(\theta_0, \theta_1) \mapsto (i\theta_0 + \theta_1, -i\theta_0 + \theta_1).$$

19. $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (g_2, -g_3)$ *(E.3), hence*

$$(\theta_0, \theta_1) \mapsto (i\theta_0 - \theta_1, i\theta_0 + \theta_1).$$

20. $\left(\begin{smallmatrix} 2 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (-g_2, -g_3)$ *(E.4), hence*

$$(\theta_0, \theta_1) \mapsto (-i\theta_0 + \theta_1, i\theta_0 + \theta_1).$$

21. $\left(\begin{smallmatrix} -1 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (-g_3, g_1)$ *(F.2), hence*

$$(\theta_0, \theta_1) \mapsto (i(\theta_0 + \theta_1), \theta_0 - \theta_1).$$

This matrix is obtained by applying (5) then (9).

22. $\left(\begin{smallmatrix} 1 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (g_3, g_1)$ *(F.1), hence*

$$(\theta_0, \theta_1) \mapsto (\theta_0 + \theta_1, i(\theta_0 - \theta_1)).$$

23. $\left(\begin{smallmatrix} -1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (-g_3, -g_1)$ *(F.4), hence*

$$(\theta_0, \theta_1) \mapsto (\theta_0 - \theta_1, i(\theta_0 + \theta_1)).$$

24. $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ *acts by* $(g_1, g_2) \mapsto (g_3, -g_1)$ *(F.3), hence*

$$(\theta_0, \theta_1) \mapsto (i(\theta_0 - \theta_1), \theta_0 + \theta_1).$$

We can use Example 8.4.9 to recover the families from Section 8.1.

Example 8.4.10

- *The coordinates* $\theta'_0 = \theta_0\theta_0(0), \theta'_1 = \theta_1\theta_1(0)$ *are invariant under the action of the automorphisms (1, 2). They are thus rational when* E *is of type* $\Gamma(2) \cap \Gamma^0(4)$*, which is of index* 12 *in* Γ*. We recover the coordinates of the twisted theta model from Section 8.1.*

- *The coordinates* $\theta'_0 = \theta_0/\theta_0(0), \theta'_1 = \theta_1/\theta_1(0)$ *are invariant under the action of the automorphisms (1, 2, 9, 10). They are thus rational when* E *is of type* $\Gamma^0(4)$*, which is of index 6 in* Γ*. In that case we know by Section 8.3 that* E *has a Montgomery model, and indeed we saw in Section 8.2 that the Montgomery coordinates are given by the Hadamard transform of* θ'_0, θ'_1*.*

- *The coordinates*

$$\theta'_0 = \theta_0\theta_0(0) + \theta_1\theta_1(0) \text{ and } \theta'_1 = (\theta_0\theta_0(0) - \theta_1\theta_1(0))\frac{\theta_0(0)^2 + \theta_1(0)^2}{\theta_0(0)^2 - \theta_1(0)^2}$$

  are invariant under the action of the automorphisms (1, 2, 3, 4). They are thus rational when E *is of type* $\Gamma(2)$*.*

  Indeed, looking at the ramification with the $(\theta'_0 : \theta'_1)$ *coordinates, we get that* 0_E *is sent to* $(1 : 1)$ *and the other ramified points are* $(1 : 0)$, $(0 : 1)$, $((a^2 - b^2)^2 : (a^2 + b^2)^2)$*, so it is isomorphic to the Legendre model with* $\lambda = \frac{(a^2-b^2)^2}{(a^2+b^2)^2}$ *via the change of variable given by the 2-torsion translation between* $(1 : 1)$ *and* $(1 : 0)$*.*

We now explain how to, given an elliptic curve equation over k, compute the Galois action on a theta model defined over an extension of k. First, by Lemma 8.4.6, the theta model is defined by the choice of two rational points $T_1, T_2 \in E[2](k')$, both of Montgomery type over k', and the choice of two symmetric elements g_1, g_2 above T_1, T_2. By Section 8.4.1, The choice of g_i is itself equivalent to the choice of a cyclic subgroup K_i of degree 4 containing T_i defined over k', or (via its generator) a choice of four torsion point $\pm U'_i$ above T_i (i.e. by the image T'_i of U'_i on the Kummer). If $E : y^2 = x^3 + a_2x^2 + a_4x + a_6$, and we take the standard basis of $\Gamma(2(0_E))$ given by $Z = Z_0^2$ (Z_0 is a section of $\Gamma(0_E)$) and $X = xZ$, then the action of the symmetric theta group element g_i associated to $\pm T'_i$ is given by [Dar+24, p. 9] as the matrix

$$\frac{1}{x_iw_i - z_iu_i}\begin{pmatrix} u_iz_i & z_iw_i \\ w_ix_i^2/z_i - 2u_ix_i & -u_iz_i \end{pmatrix},$$

where on the Kummer line $T'_i = (x_i : z_i)$, $T_i = (u_i : w_i)$. Conversely, given a choice of g_i, then T'_i is the unique point of 4-torsion on the Kummer above T_i such that $g_i \cdot (x_i, z_i) = (x_i, z_i)$ rather than $(-x_i, -z_i)$. Recall also that the choice of theta coordinates is given by:

- θ_0 a non-trivial linear combination of X, Z invariant by g_2;

- $\theta_1 = g_1 \cdot \theta_0$.

From this data, looking at the Galois action of $\sigma \in \mathrm{Gal}(k'/k)$ on T'_1, T'_2, it is easy to check which automorphism from Example 8.4.9 it induces on the theta group and so the action of σ on the theta coordinates.

We consider every case here:

- if $\sigma(T'_1) = T'_1$, i.e. $\sigma(K_1) = K_1$, then $\sigma(g_1) = g_1$;

- if $\sigma(T'_1) = T''_1 = T'_1 + T_2$, i.e. $\sigma(K_1) \neq K_1$, then $\sigma(g_1) = -g_1$;

- if $\sigma(T_1') = T_2'$, i.e. $\sigma(K_1) = K_2$, then $\sigma(g_1) = g_2$;
- if $\sigma(T_1') = T_2'' = T_2' + T_1$, i.e. $\sigma(T_1) = T_2$ but $\sigma(K_1) \neq K_2$, then $\sigma(g_1) = -g_2$;
- let $i = e_{W,4}(U_1', U_2')$ (where we use the sign convention given by the commutator pairing of the theta group $G(4(0_E))$ of level 4), and $K_3 = \langle U_1' + U_2' \rangle$, T_3' the image of $U_1' + U_2'$ on the Kummer line. Then if $\sigma(T_1') = T_3'$, i.e. $\sigma(K_1) = K_3$, then $\sigma(g_1) = ig_2g_1$;
- otherwise, $\sigma(g_1) = -ig_2g_1 = ig_1g_2$.

We conclude this section by combining the change of coordinates from Section 8.2 with the automorphisms from Example 8.4.9 to give the conversion between a theta model and all 6 possible Montgomery models.

Example 8.4.11 (Conversions — theta and Montgomery models) *We first recall that a choice of rational symmetric element g_T above a 2-torsion point T is the same as a choice of 4-torsion point T' above T such that $x(T')$ is rational. The Montgomery model associated to g_T is the one that sends T to $(0:0)$ and T' to $(1:1)$.*

A choice of theta model is the choice of two rational symmetric elements g_1, g_2. Once we have chosen the theta model, the theta constant $(a:b)$ determines the 2-torsion: $T_1 = (b:a), T_2 = (-a:b), T_3 = (-b:a)$, but also the 4-torsion points T_1', T_2' that induces g_1, g_2 respectively. We can recover T_i' as the 4-torsion point such that its coordinates $(X(T_i'), Z(T_i'))$ are invariant under the action of g_i. Recall that by definition $X = \theta_0$ is invariant by g_2 while $Z = \theta_1 = g_1 \cdot \theta_0$, so $g_1 \cdot (X, Z) = (Z, X)$ and $g_2 \cdot (X, Z) = (-X, Z)$. We thus have (see Proposition 8.1.7) $T_1' = (1:1)$ (while $T_1'' = (1:-1)$ induces $-g_1$), and $T_2' = (1:0)$ (while $T_2'' = (0:1)$ induces $-g_2$). Finally, we also have, if -1 is a square, the additional four torsion points on the Kummer line: $T_3' = (1:i)$ and $T_3'' = (i:1)$.

We have to use this order when choosing X invariant by g_2 and $Z = g_1 \cdot X$.

Then we let M_1 be the Montgomery model where T_1' is sent to $(1:1)$, and T_1'' to $(1:-1)$, hence T_1 maps to $(0:1)$. We have

$$\theta(a:b) \to M_1 : (X:Z) \mapsto (aX - bZ : aZ - bX),$$

and the converse is

$$M_1 \to \theta(a:b) : (X:Z) \to (aX + bZ : aZ + bX).$$

We have

$$\begin{array}{lll} T_2 \mapsto (a^2+b^2 : -2ab), & T_2' \mapsto (a:-b), & T_2'' \mapsto (b:-a), \\ T_3 \mapsto (-2ab : a^2+b^2), & T_3' \mapsto (a-bi : ai-b), & T_3'' \mapsto (ai-b : a-bi). \end{array}$$

It follows that $M_1 : \beta y^2 = x(x-\alpha_1)(x-1/\alpha_1) = x^3 + \mathcal{A}_1 x^2 + x$ with

$$\alpha_1 = -\frac{a^2+b^2}{2ab}, \ \mathcal{A}_1 = \frac{a^4+6a^2b^2+b^4}{2a^3b+2ab^3}.$$

Of course, if we want to send T_1' to $(1:-1)$ instead, we just need to compose with $M_1 \to M_1' : (X:Z) \mapsto (-X:Z)$. We then have

$$\alpha_1' = \frac{a^2+b^2}{2ab} = -\alpha_1, \ \mathcal{A}_1' = -\frac{a^4+6a^2b^2+b^4}{2a^3b+2ab^3} = -\mathcal{A}_1.$$

The conversion is

$$\theta(a:b) \to M_1' : (X:Z) \mapsto (aX - bZ : bX - aZ),$$

and the converse is

$$M_1' \to \theta(a:b) : (X:Z) \to (aX - bZ : bX - aZ).$$

These are the maps from Remark 8.2.3.

Now, if we want to send T_2' to $(1:1)$ and T_2'' to $(-1:1)$, we must have T_2 sent to $(0:1)$, and we obtain the Montgomery curve M_2 with the change of variable

$$\theta(a:b) \to M_2 : (X:Z) \mapsto (bX + aZ : bX - aZ),$$

and the converse is

$$M_2 \to \theta(a:b) : (X:Z) \to (a(X+Z) : b(X-Z)).$$

This sends

$$\begin{array}{lll} T_1 \mapsto (a^2+b^2 : b^2-a^2), & T_1' \mapsto (b+a : b-a), & T_1'' \mapsto (b-a : b+a), \\ T_3 \mapsto (b^2-a^2 : a^2+b^2), & T_3' \mapsto (b+ai : b-ai), & T_3'' \mapsto (bi+a : bi-a). \end{array}$$

Then $M_2 : \beta y^2 = x(x-\alpha_2)(x-1/\alpha_2) = x^3 + \mathcal{A}_2 x^2 + x$ with

$$\alpha_2 = \frac{a^2+b^2}{b^2-a^2},\ \mathcal{A}_2 = 2\frac{a^4+b^4}{a^4-b^4}.$$

Similarly, we can also look at the map $\theta(a:b) \to M_2'$ where $\alpha_2' = -\alpha_2$ and $\mathcal{A}_2' = -\mathcal{A}_2$, these are the maps from Proposition 8.2.2.

Finally, if we want to send T_3' to $(1:1)$ and T_3'' to $(-1:1)$, we have to map T_3 to $(0:1)$, yielding the Montgomery curve M_3, with the following change of variable:

$$\theta(a:b) \to M_3 : (X:Z) \mapsto (i(aX+bZ) : aZ - bX),$$

and the converse is

$$M_3 \to \theta(a:b) : (X:Z) \to (aX + ibZ : iaZ - bX).$$

This leads to

$$\alpha_3 = \frac{2iab}{a^2-b^2},\ \mathcal{A}_3 = \frac{a^4-6a^2b^2+b^4}{2iab^3-2ia^3b},$$

and M_3' can be derived in the same manner as earlier.

We remark that if we start from an elliptic curve $E : y^2 = x^3 + a_2x^2 + a_4x + a_6$ in Weierstrass equation, picking a Montgomery model for E first involves finding a root of $x^3 + a_2x^2 + a_4x + a_6$ to find a 2-torsion point T_1, then another square root to find $x(T_1')$ for T_1' a 4-torsion point above T_1. This is coherent with the fact that $\Gamma^0(4)$ is of index 6 in Γ. To go from the Montgomery model to the theta model, we need another square root to find another 2-torsion point T_2, and a square root again to find $x(T_2')$. This is coherent with the fact that $\Gamma(2,4)$ is of index 24 in Γ. On the other hand,

since the theta model is given, from the moduli point of view, by the level structure information of $T_1, T_2, x(T_1'), x(T_2')$*, it contains enough level structure information to reconstruct* 4 *Montgomery models. If* -1 *is a square, this above level structure is enough to determine* $x(T_3')$*, which gives us the* 2 *other remaining Montgomery models. This explains why the explicit conversion formulas above are rational.*

Keeping track of this level structure across the change of theta models from Example 8.4.9, we see that:

- *applying the automorphisms (5,6,21,23) to* $\theta(a : b)$ *before the change of coordinates* $\theta(a : b) \to M_1$ *gives the same Montgomery curve* M_1*;*
- *the automorphisms (7,8,22,24) give* M_1'*;*
- *the automorphisms (1,2,9,10) give* M_2*;*
- *the automorphisms (3,4,11,12) give* M_2'*;*
- *the automorphisms (13,14,17,18) give* M_3*;*
- *the automorphisms (16,19,20,23) give* M_3'*.*

Example 8.4.12 (Conversions — theta and Legendre models) *The last formula of Example 8.4.10 enables us to look at the different conversions between the theta model and the Legendre model. Given* $\theta(a : b)$*, assume that we pick the conversion that gives the Legendre parameter* $\lambda = (a^2 - b^2)^2/(a^2 + b^2)^2$*.*

If we act by an automorphism before computing λ*, then:*

- *(1, 2, 3, 4) give* λ*;*
- *(5, 6, 7, 8) give* $1 - \lambda$*;*
- *(9, 10, 11, 12) give* $1/\lambda$*;*
- *(13, 14, 15, 16) give* $\lambda/(\lambda - 1)$*;*
- *(17, 18, 19, 20) give* $1 - 1/\lambda$*;*
- *(21, 22, 23, 24) give* $1/(1 - \lambda)$*.*

8.5 THE POINT OF VIEW OF ISOGENY VOLCANOES

In Sections 8.3 and 8.4, we saw conditions for when a Kummer line of E has a Montgomery model, a Montgomery$^-$ model, a theta model, a theta squared model etc. When that is not the case, a natural condition is whether E is isogenous to a curve that admits such a model.

In this section, we use the theory of volcanoes to answer this question: namely depending on the shape of the volcano and the position of an elliptic curve in its 2-isogeny volcano we can determine if it has a Montgomery model, a Montgomery$^-$ model or a theta model (and even how many).

One application is the following: if we need to use some fast arithmetic formulas that are only available on some models, and our current Kummer

line does not admit such a rational model, we may try to find an isogeny in the volcano that leads to a curve having a rational model of the suitable type. This is the strategy used in Chapter 7, where fast 'half' differential addition formulas are available on Montgomery models with full 2-torsion: if we start with a curve E with one rational point (on the Kummer) of 8-torsion P, then $E/\langle 4 \cdot P \rangle$ is such a Montgomery curve.

8.5.1 Volcanoes

Being a Montgomery curve or having a theta model are properties that can be easily read on isogeny volcanoes, which we will recall in this section. We refer to [IJ13; Sut13] for the definitions and properties of these structures. Assume in this section that $k = \mathbb{F}_q$ is a finite field of odd characteristic p.

Definition 8.5.1 *Let ℓ be a prime, an ℓ-volcano is a connected undirected graph V with vertices partitioned in $V_0, \dots, V_h$ and such that:*

1. *The subgraph V_0 is regular of degree at most 2.*
2. *For every $i > 0$, each vertex in V_i has exactly one neighbour in V_{i-1}, and every edge that is not on V_0 is of this form.*
3. *For every $i < h$, each vertex in V_i is of degree $\ell + 1$.*

V_0 is called the crater or the surface, V_h the floor and h the height or the depth. The height will be denoted either h or $h(V)$.

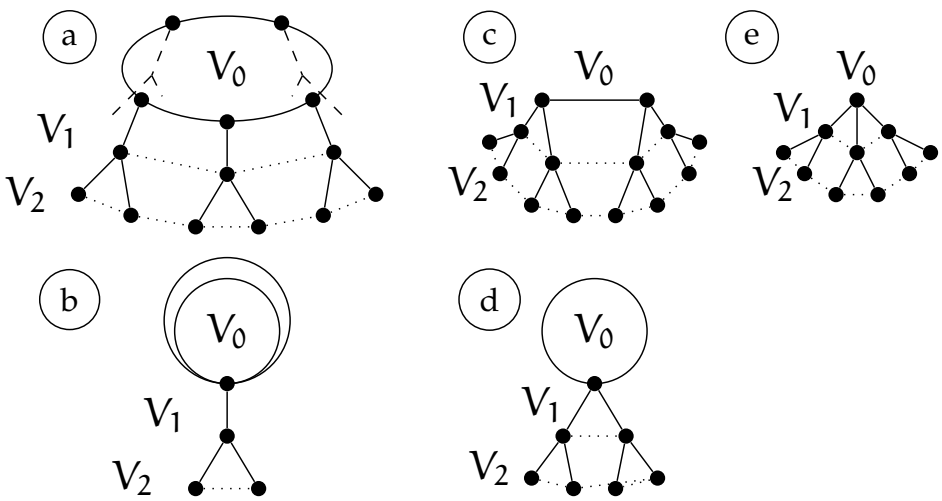


Figure 8.2: Examples of 2-volcanoes of height 2 with the possible crater shapes — dotted lines do not represent edges, only volcano levels.

Examples of 2-volcanoes are available in Fig. 8.2. To relate them to elliptic curves, we need to introduce a bit more material.

We will first look at elliptic curves up to isomorphism over $\overline{k}$, hence we will identify them by their j-invariant. If $\pi : E \to E$ is the Frobenius endomorphism given by $\pi(x, y) = (x^q, y^q)$ then its trace t verifies $t = q + 1 - \#E(\mathbb{F}_q)$. We

are only interested in ordinary elliptic curves in this section[3], i.e. curves E such that $t \neq 0$. For a given $t \neq 0$, we define the set of vertices as follows:

$$\mathrm{Ell}_t(\mathbb{F}_q) = \{j(E) \mid \#E(\mathbb{F}_q) = q + 1 - t\}.$$

Tate's theorem [Tat66] states that two elliptic curves E and E' over $\mathbb{F}_q$ are isogenous if and only if they have the same cardinal. Since we want to restrict to ℓ-isogenies where $\ell \neq p$ is prime, two elliptic curves in $\mathrm{Ell}_t(\mathbb{F}_q)$ are connected by an edge if they are ℓ-isogenous over $\mathbb{F}_q$. We denote the resulting graph by $G_{\ell,t}(\mathbb{F}_q)$.

Note that the data of the j-invariant and the cardinal help to fully recover the curve. Let us consider a curve E over $\mathbb{F}_q$ with trace $t \neq 0$. Its quadratic twist E' has the same j-invariant, however its trace will be $-t \neq t$ since we assumed the curve to be ordinary and the characteristic to be odd.

Assume E is a curve with j-invariant 0 or 1728, and that it has a rational 2-torsion point T_1. It then has non-trivial automorphisms φ and ψ such that $E[2](k) = \{0_E, T_1, T_2 = \varphi(T_1), T_3 = \psi(T_1)\}$, see [Sil09, Chap. III, Thm. 10.1]. If f is the 2-isogeny with kernel T_1, then $g = f \circ \varphi$ and $h = f \circ \psi$ are also 2-isogenies with kernel respectively T_2 and T_3, they all share the same codomain. However, if the codomain E' does not have j-invariant 0 or 1728, the dual isogenies all have the same kernel T', hence $\widetilde{f} = \widetilde{g} = \widetilde{h}$. This is a particular case where a 2-isogeny on E yields three rational 2-isogenies, but only one going backward.

When $j \notin \{0, 1728\}$, there are no non-trivial automorphism, and therefore to one rational 2-isogeny corresponds one backward rational 2-isogeny given by the dual. Therefore, we will not deal with connected components of $G_{\ell,t}(\mathbb{F}_q)$ containing j-invariants 0 and 1728, and we will consider the graphs undirected.

The last notion is the one of orders. If E is an ordinary elliptic curve, then $\mathrm{End}\, E$ is an order $\mathcal{O}$ in a quadratic imaginary field K (more precisely, $K = \mathcal{O} \otimes_{\mathbb{Z}} \mathbb{Q} = \mathbb{Q}(\pi)$). Let $\mathcal{O}_K$ be the ring of integers of K. Since π is an endomorphism, $\mathbb{Z}[\pi] \subseteq \mathcal{O} \subseteq \mathcal{O}_K$ holds. Let d_π, d and d_K be respectively the discriminant of $\mathbb{Z}[\pi]$, $\mathcal{O}$ and $\mathcal{O}_K$. Since the Frobenius satisfies $\pi^2 - t\pi + q = 0$, we know that $d_\pi = t^2 - 4q$. Let

$$\omega = \begin{cases} \sqrt{\frac{d_K}{4}} & \text{if } d_K \equiv 0 \bmod 4, \\ \frac{1+\sqrt{d_K}}{2} & \text{if } d_K \equiv 1 \bmod 4. \end{cases}$$

Then $\mathcal{O}_K = \mathbb{Z}[\omega]$, and we can write $\mathbb{Z}[\pi] = \mathbb{Z}[f_\pi \omega]$ and $\mathcal{O} = \mathbb{Z}[f\omega]$ with $f_\pi, f \geqslant 0$, they are respectively the conductor of $\mathbb{Z}[\pi]$ and $\mathcal{O}$. We also have $f_\pi = [\mathcal{O}_K : \mathbb{Z}[\pi]]$ and $f = [\mathcal{O}_K : \mathcal{O}]$, therefore $f \mid f_\pi$ and $d_\pi = f_\pi^2 d_K$, $d = f^2 d_K$.

The following result is due to Kohel who studied isogeny graphs in his thesis [Koh96, Prop. 21–23]:

Lemma 8.5.2 (Kohel, see e.g. Thm. 7 in [Sut13]) *Let V be a connected component of $G_{\ell,t}(\mathbb{F}_q)$ not containing j-invariants 0 and 1728. Then V is an ℓ-volcano with levels $V_0, \dots, V_h$ verifying the following properties:*

3 Although our results also hold for supersingular curves defined over the base field $\mathbb{F}_p$, because they also form a volcano structure, more details are available in Remark 8.5.13.

1. *Every curve in* V_i *has the same endomorphism ring* $\mathcal{O}_i$ *with discriminant* d_i *and conductor* f_i.

2. *The subgraph* V_0 *has degree* $1 + \left(\frac{d_0}{\ell}\right)$.

3. *If* $\left(\frac{d_0}{\ell}\right) \neq -1$, *then* ℓ *is split or ramified and* $\#V_0$ *is the order of* $[\mathfrak{p}_\ell]$ *in* $\mathrm{Cl}(\mathcal{O}_0)$ *where* $\mathfrak{p}_\ell \mid \ell$ *is prime, otherwise* $\#V_0 = 1$.

4. *The height* h *is the* ℓ*-valuation of the conductor of* $\mathbb{Z}[\pi]$.

5. $\ell \nmid [\mathcal{O}_K : \mathcal{O}_0]$, *and* $[\mathcal{O}_i : \mathcal{O}_{i+1}] = \ell$ *for* $0 \leqslant i < h$, *that is* $d_i = \ell^{2i} d_0$ *or* $f_i = \ell^i f_0$.

Using (2), since the discriminant d_i only differs from d_K by a square, the degree of V_0 depends only on whether ℓ is split, ramified or inert in $\mathcal{O}_K$. If ℓ is split, then V_0 has degree 2, if it is ramified V_0 has degree 1, and otherwise it is inert and V_0 is a point. From (4) and (5), we can also derive $h = v_\ell\left(\frac{d_\pi}{d_K}\right)/2 = v_\ell\left(\frac{d_\pi}{d_0}\right)/2$ since ℓ does not divide the conductor of $\mathcal{O}_0$.

Lemma 8.5.2 justifies the following terminology, let E and E' be two ℓ-isogenous elliptic curves via $f : E \to E'$, and $\mathcal{O}$ and $\mathcal{O}'$ there endomorphism rings. f is said to be horizontal if $\mathcal{O} = \mathcal{O}'$, descending if $[\mathcal{O} : \mathcal{O}'] = \ell$ and ascending if $[\mathcal{O}' : \mathcal{O}] = \ell$.

Formulas for the cardinal depending on its height and the size of the crater can be easily derived:

Proposition 8.5.3 *Let* V *be an* ℓ*-isogeny volcano with levels* $V_0, \dots, V_h$ *where* h *is the height. Then:*

$$\#V = \begin{cases} \#V_0\, \ell^h & \text{if } \deg V_0 = 2, \\ \#V_0 \left(1 + \ell \frac{\ell^h - 1}{\ell - 1}\right) & \text{if } \deg V_0 = 1, \\ 1 + (\ell + 1)\frac{\ell^h - 1}{\ell - 1} & \text{if } \deg V_0 = 0. \end{cases}$$

Proof *If* $h = 0$, *the formulas indeed give* $\#V = \#V_0$. *We will assume* $h \geqslant 1$. *Let* $u_i = \#V_i$. *We have* $\#V = u_0 + \cdots + u_h$. *If* $i = 1$, *then each vertex on the crater has degree* $\ell + 1$, *and taking into account the degree of the subgraph* V_0:

$$u_1 = \begin{cases} (\ell - 1)\#V_0 & \text{if } \deg V_0 = 2, \\ \ell\, \#V_0 & \text{if } \deg V_0 = 1, \\ (\ell + 1)\#V_0 & \text{if } \deg V_0 = 0. \end{cases}$$

Now if $1 \leqslant i < h$, *each vertex on level* V_i *has only one ascending isogeny, and because the degree is* $\ell + 1$, *there must be* ℓ *descending isogenies, hence* $u_{i+1} = \ell u_i$. *The result then comes from summing a geometric series.* □

From the theorem, one could wonder if every kind of volcano occurs. The answer is yes, even in the degenerate cases as we can see in Table 8.1. We consider a connected component V of $G_{\ell,t}(\mathbb{F}_p)$ where $\ell = 2$. These examples where found by looking for convenient quadratic fields and discriminant, using Lemma 8.5.2.

p	t	$j(E)$	h	$\deg(V_0)$	$\#V$	$\#V_0$	Fig. 8.2 example
450361	1094	300824	3	2	176	22	a
3049	102	591	4		16	1	b
155209	506	83458	3		30	2	c
521	6	354	4	1	31	1	d
464069	164	93759	0		2	2	—
182641	194	135674	3	0	22	1	e
159629	9	63979	0		1	1	—

Table 8.1: Some examples of 2-isogeny volcanoes

Information on the ℓ-torsion

Finally, we want to deduce information on a curve based on its position in a volcano, notably its ℓ-torsion. If E is an ordinary elliptic curve over $\mathbb{F}_q$ with endomorphism ring isomorphic to $\mathcal{O}$, it is known that $E(\mathbb{F}_q) \simeq \mathbb{Z}/M\mathbb{Z} \times \mathbb{Z}/N\mathbb{Z}$ with $N \mid M$, then $\#E(\mathbb{F}_q) = MN$ thanks to a Tate result stating that $E(\mathbb{F}_q) \simeq \mathcal{O}/(\pi - 1)$. We are interested in the ℓ-valuation of N. We can write $\pi = a + f_\pi \omega$ where:

$$a = \begin{cases} t/2 & \text{if } d_K \equiv 0 \bmod 4, \\ (t - f_\pi)/2 & \text{if } d_K \equiv 1 \bmod 4. \end{cases}$$

If f is the conductor of the endomorphism ring $\mathcal{O}$, then $N = \gcd(a-1, f_\pi/f)$. It is known [Mir+06, Thm. 1] (their proof holds for any ℓ) that $v_\ell(a-1) \geqslant \min(v_\ell(f_\pi), v_\ell(\#E(\mathbb{F}_q))/2)$. If we further assume that $v_\ell(N) < v_\ell(M)$, then $v_\ell(N) < v_\ell(\#E(\mathbb{F}_q))/2$, and because of the gcd, we must have $v_\ell(N) = v_\ell(f_\pi/f)$.

If we go down the volcano, the conductor is multiplied by ℓ, so the ℓ-valuation of N decreases by 1, whereas the one of M increases by 1. On the floor, because $h = v_\ell(f_\pi)$, ℓ does not divide N any more. Let $E[\ell^\infty](\mathbb{F}_q) \simeq \mathbb{Z}/\ell^m\mathbb{Z} \times \mathbb{Z}/\ell^n\mathbb{Z}$ denote the ℓ-Sylow of $E(\mathbb{F}_q)$, where $m = v_\ell(M)$ and $n = v_\ell(N)$.

8.5.2 Reading model of curves on a volcano

We will now focus on the case $\ell = 2$ in this section. Let $t \neq 0$, we consider a volcano V of a connected component of $G_{2,t}(\mathbb{F}_q)$ not containing j-invariants 0 and 1728. The levels will be denoted $V_0, \dots, V_h$. Preparing next proposition, we start with a useful lemma:

Lemma 8.5.4 *If* $\deg V_0 = 2$, *then* $h(V) \geqslant 1$.

Proof *Let* E *be a curve on* V_0. *Because* $\deg V_0 = 2$, *there are two distinct rational kernels of order* 2 *in* E. *But having a rational kernel of order* 2 *is equivalent to having a rational* 2*-torsion point, so* E *has complete* 2*-torsion generated by* T *and* R. *But then* $S = T + R$ *is also a rational* 2*-torsion point, giving a third rational* 2*-isogeny. Because* $\deg V_0 = 2$, *this isogeny must be a descending one, hence* $h(V) \geqslant 1$. □

Remark 8.5.5 *In the following proofs, we will be chaining isogenies in the volcano. One important thing to note is that, once we consider a descending isogeny, we must*

chain descending ones afterwards because there are no more horizontal isogenies at that level and the only ascending one available is going backwards.

Legendre curves are simple to classify since they are the one with full 2-torsion.

Proposition 8.5.6 (Legendre curves) *The elliptic curves in a 2-volcano are Legendre if and only if they are not on the floor i.e. they don't belong to V_h where h is the height of the volcano. In particular, a 2-volcano has no Legendre curve if and only if $h = 0$. In this case one also has $\deg V_0 \leqslant 1$.*

Proof *If $h \geqslant 1$, then for all $0 \leqslant i < h$, every curve in V_i has degree 3, hence 3 rational 2-torsion points, so the curve is in Legendre form. On the floor, the degree is 1, so they are not Legendre because they only have one rational 2-torsion point. If $h = 0$, because of Lemma 8.5.4, $\deg V_0 \leqslant 1$ and curves have at most one horizontal 2-isogeny, and no descending one, which does not give the full 2-torsion.* □

For Montgomery curves and onwards, we will need the following lemma:

Lemma 8.5.7 *Let f_1 and f_2 be two 2-isogenies that are not dual, then $f = f_1 \circ f_2$ is a cyclic 4-isogeny.*

Proof *f has degree 4, so its kernel is either isomorphic to $\mathbb{Z}/4\mathbb{Z}$ or $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. But in the second case, that would imply that $f = [2]$ and therefore f_1 and f_2 would be dual. Hence, f is cyclic.* □

Montgomery curves are described as curves with a rational cyclic 4-isogeny thanks to Proposition 8.3.12.

Proposition 8.5.8 (Montgomery curves) *Let V be a 2-isogeny volcano not containing curves with j-invariant 0 and 1728 and with levels $V_0, \ldots V_h$:*

1. *If $\deg V_0 = 2$, every curve on the volcano has a Montgomery form.*
2. *If $\deg V_0 = 1$, then either $h \geqslant 1$ and every curve on the volcano has a Montgomery form, otherwise $h = 0$ and no curve on the volcano has a Montgomery form.*
3. *If $\deg V_0 = 0$, then either $h \geqslant 2$ and every curve on the volcano has a Montgomery form. Otherwise, if the height is 1 then every curve on the floor V_1 has a Montgomery form and finally if $h = 0$ no curve has a Montgomery form.*

Proof *Lemma 8.5.7 helps to translate the property of having a rational 4-isogeny in terms of graph theory on the volcano: if we can follow a path of length 2 on the volcano, then the starting curve will have a 4-isogeny. Fig. 8.3 illustrates the different cases that can occur.*

Our goal is to build one rational 4-isogeny by chaining two 2-isogenies which are not dual thanks to Lemma 8.5.7. If we follow one direction on a volcano edge, the dual isogeny corresponds to the backward direction, which we would like to avoid.

- *If* $\deg V_0 = 2$ *and* E *is a curve on* V_0*, since the height is always greater than* 1 *with Lemma 8.5.4 and there is always a horizontal isogeny, it is possible to chain a horizontal and a descending one, see Fig. 8.3a.*

- *If* $\deg V_0 = 2$ *and* E *is a curve on* V_1*, it is possible to chain the ascending* 2*-isogeny with a horizontal one since we land on the crater after the first step.*

- *Assume* $\deg V_0 = 1$ *and* E *is a curve on* V_0*. Since there is only one horizontal isogeny, we must chain it with a descending one to get a* 4*-isogeny, so the height of the volcano must be greater than* 1 *in that case.*

- *If* $\deg V_0 = 0$ *and* E *is a curve on* V_0*, since there is no horizontal isogeny, we must consider two descending ones to build a* 4*-isogeny, this requires the height to be at least* 2*.*

- *If* $\deg V_0 = 0$ *or* 1 *and* E *is a curve on* V_1*, since curves on the crater have at least two descending isogenies, it is possible to chain first an ascending one starting from* E *and then a descending one, see Fig. 8.3d.*

- *Finally, assume that* $i \geqslant 2$ *and that* E *is a curve on* V_i*. It is then possible to chain two ascending* 2*-isogenies, giving a cyclic* 4*-isogeny.* □

Recall from Corollary 8.4.4 that a curve admits a Montgomery$^-$ model if and only if there exists a rational 2-isogeny that does not extend to a rational cyclic 4-isogeny. We leave to the reader how to adapt Proposition 8.5.8 to the Montgomery$^-$ case.

A direct consequence is the characterization of curves having a theta squared model:

Proposition 8.5.9 (Theta squared model) *Let* V *be a* 2*-isogeny volcano not containing curves with* j*-invariant* 0 *and* 1728 *and with levels* $V_0, \dots V_h$*:*

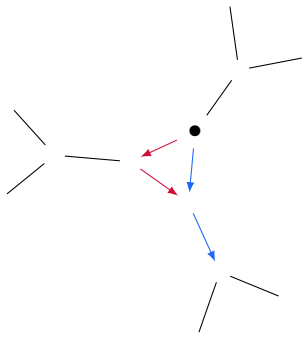
1. *If* $\deg V_0 = 2$*, every curve in* $V_0 \cup \dots \cup V_{h-1}$ *has a theta squared model.*

2. *If* $\deg V_0 = 1$*, then either* $h \geqslant 1$ *and every curve in* $V_0 \cup \dots \cup V_{h-1}$ *has a theta squared model, otherwise* $h = 0$ *and no curve on the volcano has a theta squared model.*

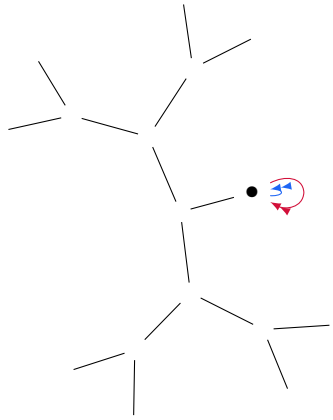
3. *If* $\deg V_0 = 0$*, then either* $h \geqslant 2$ *and every curve in* $V_0 \cup \dots \cup V_{h-1}$ *has a theta squared model. Otherwise,* $h \leqslant 1$ *and no curve on the volcano has a theta squared model.*

Proof *Thanks to Proposition 8.3.14, an elliptic curve* E *has a theta squared model if and only if it has a Montgomery form with full rational* 2*-torsion, hence it also has a Legendre form. The result is then deduced from Propositions 8.5.6 and 8.5.8.* □

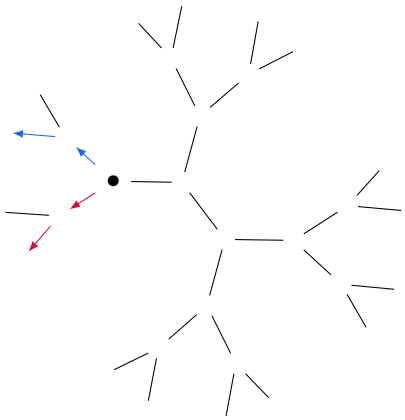
Finally, having a theta model is the most restrictive proposition since the curve requires two independent 4-isogenies thanks to Proposition 8.3.14. There is a particular case to address before looking at characterizing curves with a theta model.



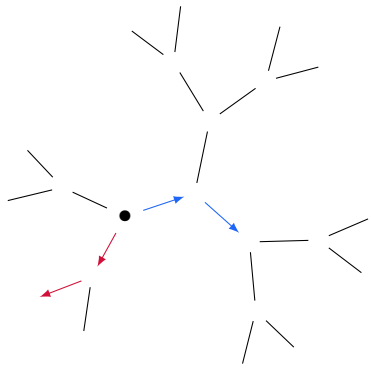
(a) $\deg V_0 = 2$, using horizontal isogenies



(b) $\deg V_0 = 0$, using endomorphisms



(c) $\deg V_0 = 1$, using only descending isogenies



(d) $\deg V_0 = 0$, going through the crater

Figure 8.3: Some examples of cyclic 4-isogenies on different volcanoes

Lemma 8.5.10 *Assume* $\#V_0 = 1$, *let* E *be the elliptic curve in* V_0. *If* $\deg V_0 = 2$, *then* E *admits exactly two non-trivial endomorphisms* f *and* $\tilde{f}$ *of degree* 2, *where* $\tilde{f}$ *is the dual of* f. *If* $\deg V_0 = 1$ *then* E *admits exactly one non-trivial endomorphism* f *of degree* 2 *which is self-dual, i.e.* $f = \tilde{f}$.

Proof *When* $\deg V_0 = 2$, *there are two* 2*-isogenies on the crater with domain and codomain* E. *Denote them* f *and* g, *and their kernel generators* P *and* Q. *They are distinct rational* 2*-torsion points, hence generators of* $E[2]$. *The last* 2*-isogeny with kernel* $P + Q$ *can't be an endomorphism because of the volcano structure, it must be descending. Because* $\tilde{f}$ *is also a non-trivial endomorphism, we must have* $\tilde{f} = f$ *or* $\tilde{f} = g$. *But in the first case, if* $\tilde{f} = f$, *looking at isogenies as ideals thanks to Deuring correspondence [Deu41], this would imply that* $2\mathcal{O}_K = \mathfrak{p}^2$ *where* $\mathfrak{p}$ *is the ideal corresponding to* f, *so* 2 *would be ramified in* $\mathcal{O}_K$. *But this is not the case according to Lemma 8.5.2,* 2 *is split when* $\deg V_0 = 2$. *Hence,* $g = \tilde{f}$.

When $\deg V_0 = 1$, *there is only one* 2*-isogeny on* E *that is a non-trivial endomorphism, we denote it by* f. *Then* $\tilde{f}$ *is also a non-trivial endomorphism and by uniqueness must verify* $\tilde{f} = f$. □

Proposition 8.5.11 (Theta model) *Let* V *be a* 2*-isogeny volcano not containing curves with* j*-invariant* 0 *and* 1728 *and with levels* $V_0, \dots V_h$:

- *If* $h \geqslant 2$, *then every curve in* $V_0 \cup V_1 \cup \dots \cup V_{h-2}$ *have a theta model.*
- *If* $\deg V_0 = 2$ *and* $h = 1$, *only curves on* V_0 *have a theta model.*
- *If* $\deg V_0 \leqslant 1$ *and* $h \leqslant 1$, *no curve on the volcano has a theta model.*

Proof *For* 4*-isogenies to be independent, they must not share an edge in the same direction, but they can share a vertex. Because of Lemma 8.5.7, they will be cyclic. We once again refer to Fig. 8.3 for several illustrations.*

- *If* E *is a curve not on the crater — say* E *is in* V_i *with* $i \geqslant 1$ *— since there is only one edge ascending, this implies that the second isogeny must start by descending, and by Remark 8.5.5, at least one of the two isogenies will descend two levels. In that case if* E *admits a theta model we must have* $h \geqslant i + 2$, *i.e.* $1 \leqslant i \leqslant h - 2$.
- *If* E *is in* V_0 *and* $\deg V_0 = 2$, *there are two horizontal* 2*-isogenies starting from* E.
  - *Assume* $\#V_0 \geqslant 2$, *in that case there are two horizontal* 2*-isogenies starting from* E. *Since the height is at least* 1 *thanks to Lemma 8.5.4, it is possible to build the first* 4*-isogeny by chaining a horizontal one and a descending one. The second* 4*-isogeny is obtained by chaining two horizontal* 2*-isogenies, which is always possible because* $\#V_0 \geqslant 2$, *see Fig. 8.3a.*
  - *Otherwise, if* $\#V_0 = 1$, *Lemma 8.5.10 shows that there are two* 2*-isogenies on the crater that are endomorphisms and dual one another. Hence, if* f *and* $\tilde{f}$ *are those two endomorphisms,* $f \circ f$ *and* $\tilde{f} \circ \tilde{f}$ *are two independent cyclic* 4*-isogenies, and* E *has a theta model, see Fig. 8.3b.*

- *If* E *is in* V_0 *and* $\deg V_0 \leqslant 1$*, there are two descending branches stemming from* E*. Moreover, there is at most one horizontal isogeny, therefore one of the 4-isogeny must start with a descending 2-isogeny. Hence, either* $h \geqslant 2$ *and* E *has a theta model, or* $h \leqslant 1$ *and it is not possible to construct both 4-isogenies,* E *does not have a theta model in that case.* □

All these properties are summarized in Table 8.2 below.

$\deg V_0$	Legendre curves h	Legendre curves Volcano Levels	Montgomery curves h	Montgomery curves Volcano Levels
2	$\geqslant 1$	$\{0, \dots, h-1\}$	$\geqslant 1$	$\{0, \dots, h\}$
1	$\geqslant 1$	$\{0, \dots, h-1\}$	$\geqslant 1$	$\{0, \dots, h\}$
	0	$\emptyset$	0	$\emptyset$
0	$\geqslant 1$	$\{0, \dots, h-1\}$	$\geqslant 2$	$\{0, \dots, h\}$
			1	$\{1\}$
	0	$\emptyset$	0	$\emptyset$

(a) Legendre and Montgomery curves

$\deg V_0$	Theta squared model h	Theta squared model Volcano Levels	Theta model h	Theta model Volcano Levels
2	$\geqslant 1$	$\{0, \dots, h-1\}$	$\geqslant 2$	$\{0, \dots, h-2\}$
			1	$\{0\}$
1	$\geqslant 1$	$\{0, \dots, h-1\}$	$\geqslant 2$	$\{0, \dots, h-2\}$
	0	$\emptyset$	$\leqslant 1$	$\emptyset$
0	$\geqslant 2$	$\{0, \dots, h-1\}$	$\geqslant 2$	$\{0, \dots, h-2\}$
	$\leqslant 1$	$\emptyset$	$\leqslant 1$	$\emptyset$

(b) Theta and theta squared models

Table 8.2: Curves on a volcano $V = V_0 \cup \dots \cup V_h$ with different models

Remark 8.5.12 (Number of Montgomery and Montgomery$^-$ models)
We can refine the discussion about whether a curve, depending on its position in the volcano, admits a Montgomery or Montgomery$^-$ or theta model, to how many such models it may admit using Example 8.4.8.

Consider the case when the volcano is not a single point, which happens when the cardinal of $E(\mathbb{F}_q)$ *is odd, and so there is not even a rational 2-torsion point.*

On the bottom of the volcano, the 2-Sylow is cyclic, so there is only one rational 2-torsion point T_1*. Depending on whether the isogeny generated by* T_1 *extends to a rational 4-isogeny (which can be read as above from the shape of the volcano), then* E *admits 2 Montgomery models or 2 Montgomery$^-$ models. And* E *does not have a rational theta model.*

When not on the bottom, then E *has full rational 2-torsion* T_1, T_2, T_3*. It admits 6 Montgomery + Montgomery$^-$ models, and the exact number of Montgomery vs*

Montgomery⁻ models depends on the type of T_1, T_2, T_3*, which again can be read from the volcano.*

Remark 8.5.13 (The supersingular case) *We conclude by a discussion about the supersingular case. If* $E/\mathbb{F}_p$ *is a supersingular curve its* 2*-isogeny graph behaves similarly to the ordinary curve and in particular it has a volcano structure.*

If $p \equiv 1 \bmod 4$*, the volcano has height* 1 *(and* 2 *split in the maximal order), and if* $p \equiv 3 \bmod 4$*, the volcano has height* 2*. If* $p \equiv 3 \bmod 8$*,* 2 *is inert in the maximal order, so the volcano consist of one node on the top and three nodes on the bottom (all* 2*-isogenies from the top are descending), whereas if* $p \equiv 7 \bmod 8$*,* 2 *splits in the maximal order, so we have a cycle at the top, and each curve on the top has two horizontal* 2*-isogenies, and one descending* 2*-isogeny.*

The arguments from Remark 8.5.12 counting the number of Montgomery and Montgomery⁻ models recover the results from [CD20, Figs. 1–2]. We also see that E *admits a rational theta model if and only if* $p \equiv 1 \bmod 4$*, or* $p \equiv 7 \bmod 8$ *and* E *is at the top of the volcano (and the* 2*-cycle is not degenerate).*

If $E/\mathbb{F}_{p^2}$ *is a maximal or minimal supersingular curve, then the* 2*-torsion is rational, and the self Tate pairings are trivial. It follows that* E *always has a rational theta model, and in fact all* 24 *possible theta models on* E *are rational.*

8.6 APPLICATION TO THE ELLIPTIC CURVE METHOD

As it was shown in Chapters 6 and 7, combining arithmetic of theta squared and Montgomery models can speed up the Montgomery ladder. However, while the ladder is useful when constant time is required — for instance in cryptographic protocols — PRAC is preferred in other contexts, like Elliptic Curve Method. If the arithmetic can be improved using theta squared model, we are interested in finding curves with small coefficients and 'good' torsion properties for ECM, which was introduced in Section 1.3.

8.6.1 Curves with fast arithmetic and large success probability

We are interested here in finding elliptic curves E which are ECM-friendly i.e. with a fast arithmetic and a large success probability (see e.g. [Mon92, Sec 6.3; Ber+13, Fig 9.1; Bar+13]). An analysis based on the Galois representation [BS22] estimated how the success probability of several families compares to each other.

We represent in Fig. 8.4 the subgroups of $GL_2(\mathbb{Z}/4\mathbb{Z})$ which contain $X_{sp}(4)$, in particular those which characterize elliptic curves having a Montgomery, theta and respectively theta squared model (the data is extracted from the LMFDB site). For the full diagram of subgroups of $SL_2(\mathbb{Z}/4\mathbb{Z})$, see Fig. 8.5.

Kubert (see e.g. [Mon92, Thm. 6.2.4]) parametrized the elliptic curves whose rational torsion contains $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$ or equivalently those corresponding to $X(2, 8)$, as depicted in Table 8.3.

We search for small integer values of a and b to exploit the fact that the multiplication by very small constants as 2 and 3 can be done by additions and is negligible compared to a full multiplication.

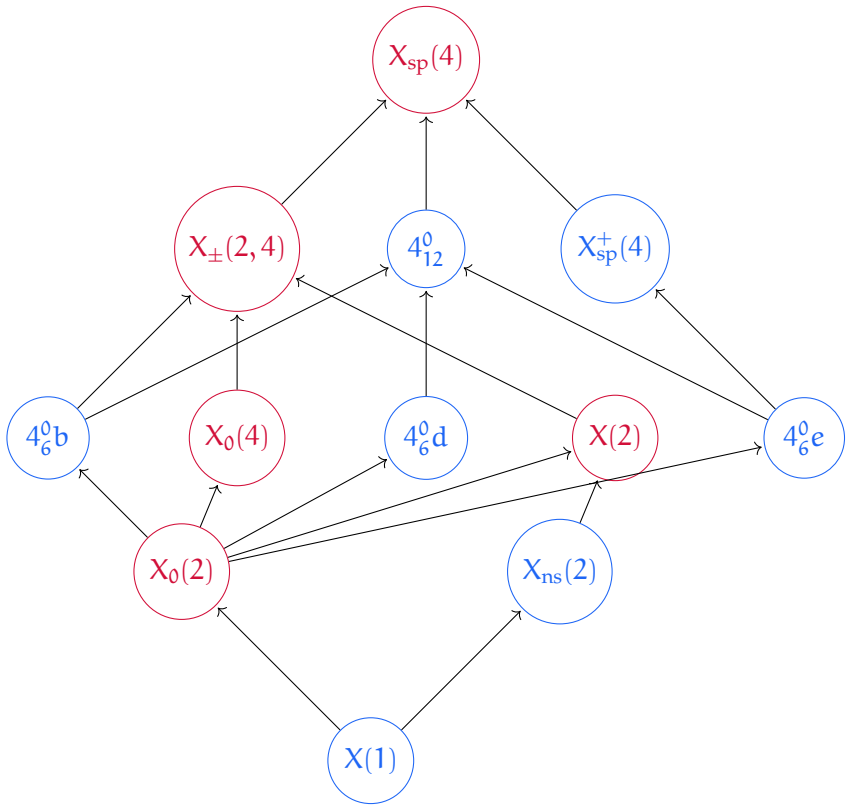


Figure 8.4: The diagram of the subgroups of $GL_2(\mathbb{Z}/4\mathbb{Z})$ containing $D(4) = \Gamma(2,4) = X_{sp}(4)$. The subgroups which characterize the elliptic curves having Kummer models studied in this chapter are as follows: $X(1)$ is the set of all the curves, $X_0(2)$ corresponds to the curves with a rational point of order 2, $X(2)$ corresponds to the Legendre curves (Proposition 8.3.10) $X_0(4)$ corresponds to the Montgomery model M (Proposition 8.3.12), $D(4) = X_{sp}(4)$ corresponds to the theta model θ (Proposition 8.3.14), $X(2,4)$ corresponds to the theta squared model $\widehat{\theta}$ (Proposition 8.3.14 (2)).

In a numerical experiment, we first generated a set of curves with a^2+b^2 a square, a starting point for ECM $P = (X : Y : Z)$, and with the constraint that a^2, b^2, X and Z are less than a computer word. We found around 3 million of Pythagorean triplets, but by testing only 120000 of them, we got 91 curves with small constants. We then dropped the constraint $a^2 + b^2$ is a square. This was giving a lot of tuples (a, b), so we tested only the first 15600 and got 2861 more curves. Finally, we tried to set $b^2 = 1$, and out of 60000 tuples (a, b), we found 90 more curves.

Given the number of curves found in the second case, we'd like them to be as efficient as the first family. To do so, we will compare the following quantity, where G is a subgroup of $\mathbb{Z}/2^n\mathbb{Z} \times \mathbb{Z}/2^n\mathbb{Z}$: $\mathbb{P}(E(\mathbb{F}_p)[2^n] \simeq G)$. If we have equal probabilities for 2 and 4-torsion, because we are working on level 4 modular curves, this will be sufficient to say that the efficiency will be similar for these two families. Thanks to Chebotarev density theorem and [Bar+13, Thm. 2.7.1], we know that this quantity is written (with $a \in \mathbb{N}$):

$$\mathbb{P}(E(\mathbb{F}_p)[2^n] \simeq G) = \frac{a}{\#GL_2(\mathbb{Z}/4\mathbb{Z})} = \frac{a}{96}.$$

Equation	Kummer model or torsion
$y^2 = x(x^2 + \mathcal{A}x + \gamma)$	Torsion $\mathbb{Z}/2\mathbb{Z}$
$y^2 = x(x^2 + \mathcal{A}x + 1)$	Montgomery model
$y^2 = x(x - \frac{a}{b})(x - \frac{b}{a})$, $a \neq b$	Theta squared model
$y^2 = x(x - \frac{a}{b})(x - \frac{b}{a})$, $a^2 - b^2 = \square$	Theta model
$y^2 = x(x - \frac{a^2}{b^2})(x - \frac{b^2}{a^2})$, $a^2 + b^2 = \square$	Torsion $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$

Table 8.3: Different elliptic curves and their corresponding Kummer line model or torsion

To achieve that, we compute an experimental probability over many primes for each family of curves, and then look for the closest fraction in the form $\frac{a}{96}$. If we choose a large enough number of primes — which is known thanks to an effective version under GRH of Chebotarev theorem by Lagarias and Odlyzko [LO77, Thm. 1.1] and with explicit constants by Winckler [Win13, Thm. 1.2] — we then know that this is the exact value.

Results are given in Table 8.4. We can conclude that the second family, i.e.

Subgroup G of $\mathbb{Z}/2^n\mathbb{Z} \times \mathbb{Z}/2^n\mathbb{Z}$	$\mathbb{P}$ with $a^2 + b^2 = \square$	$\mathbb{P}$ with generic $a^2 + b^2$
	2-torsion	
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	1	1
	4-torsion	
$\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	0.75	0.75
$\mathbb{Z}/4\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$	0.25	0.25

Table 8.4: Comparison of the 2 and 4-torsions of the two families of curves

torsion $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/8\mathbb{Z}$, has the same success probability as the first one, i.e. curves with a theta model. One can therefore use the theta model if it has a better arithmetic cost.

Remark 8.6.1 *We studied ECM with theta coordinates in the first place because of its genus 2 generalization HECM [Cos11, Ch. 4]. Indeed, we believe there is a small mistake in [Cos11, Tab. 3.2], the first column about genus 1 should be* $2\mathbf{M} + 6\mathbf{S} + 1\mathbf{m}_P + 3\mathbf{m}$ *on the first line, which is the sum of the cost of Algorithms 8.3 and 8.4. With this complexity, two instances of ECM should be a little bit faster than one pass of HECM, contrary to what is claimed in [Cos11].*

8.7 KUMMER LINES AS STACKY CURVES

In this section, we reinterpret the definitions of Section 4.3 in terms of stack quotients. We will assume that we are not in characteristic 2.

If E is an elliptic curve, the group scheme $\mu_2 = \{1, -1\}$ acts via $-1 \cdot P = -P$. The scheme quotient E/μ_2 (both as a geometric and categorical quotient) is the projective line $\mathbb{P}^1$.

This means that we cannot recover E directly from E/μ_2. In particular, although the action of μ_2 is ramified on the points of 2-torsion, E/μ_2 is still smooth. By the Nagata-Zariski purity theorem, this can happen only because the ramification locus is of codimension 1. And indeed, in dimension $g \geqslant 2$, the scheme quotient A/μ_2 of an abelian variety is no longer smooth: the non-smooth locus is precisely the image of the 2-torsion.

A way to remember the ramification points is to take the stack quotient $[E/\mu_2]$. Its coarse moduli space is E/μ_2. Hence, $[E/\mu_2]$ is a smooth stacky curve, and is therefore completely determined by the data of the coarse space $E/\mu_2 \simeq \mathbb{P}^1$, the ramification points on $\mathbb{P}^1$, and for each ramified points P_i its associated (non-trivial) inertia [VZ22, Lemma 5.3.10] (the tameness condition is automatic in our case since we are not in characteristic 2). In our case the inertia can only be μ_2, so it follows that $[E/\mu_2]$ is completely determined by the image of the 2-torsion points on $\mathbb{P}^1$. We recover the more elementary definition from Definition 4.3.4.

Now the stack quotient $[E/\mu_2]$ 'remembers' E. More precisely, the map $E \to k$ yields a canonical map $\varphi : [E/\mu_2] \to B\mu_2 = [k/\mu_2]$. Recall that $B\mu_2$ is the classifying stack of (étale) μ_2-torsors: a μ_2-torsor $T \to S$ is the same as a point $S \to B\mu_2$, where T can be recovered as the pullback of the trivial torsor $i_0 : \operatorname{Spec} k \to B\mu_2$ by $S \to B\mu_2$. Likewise, an S-point $S \to [E/\mu_2]$ corresponds to a μ_2-torsor $T \to S$ with a μ_2-equivariant map $T \to E$. Like above T is given by the pullback of $E \to [E/\mu_2]$ by the point $S \to [E/\mu_2]$. We remark also that T is also the μ_2-torsor associated to the S-point $S \to [E/\mu_2] \to B\mu_2$. If k is perfect, by Hilbert 90 and Kummer theory, we have $H^1_{et}(k, \mu_n) \simeq k^*/k^{*,n}$, and if $\mu_n \subset k$, to give a μ_n-torsor X is the same as giving the smallest field extension k'/k that trivializes it (i.e. the subfield of $\overline{k}$ invariant by the inertia of the map $\operatorname{Gal}(\overline{k}/k) \to \mu_n$ associated to the torsor), whose Galois group is isomorphic to a subgroup μ_d of μ_n, and also to give the induced isomorphism $\operatorname{Gal}(k'/k) \to \mu_d$. If $n = 2$, there are only two possibilities, either $d = 1$ and $k' = k$ (X is the trivial torsor), or k'/k and k' uniquely determines X, since $\operatorname{Aut} \mu_2 = \{\mathrm{id}\}$.

A scheme X/S can be recovered from its stack quotient $[X/G]$ via the pullback $X \simeq [X/G] \times_{BG} S$. Thus, we have that $E = [E/\mu_2] \times_{B\mu_2} k$ is the pullback of $[E/\mu_2] \to B\mu_2$ by the canonical map $i_0 : \operatorname{Spec} k \to B\mu_2$ associated to the trivial torsor.

It is important to note that the reconstruction of E depends on the forgetting map $\varphi : [E/\mu_2] \to B\mu_2$. For instance, consider k'/k the étale μ_2-torsor associated to a quadratic extension of k, then we have a corresponding map $i' : \operatorname{Spec} k \to B\mu_2$. This time, the pullback of i' by the canonical map φ gives a quadratic twist E' of E, precisely the quadratic twist associated to the μ_2-torsor k'/k.

Now, the stack quotient $[E'/\mu_2]$ is isomorphic to $[E/\mu_2]$. However, the isomorphism is not above their respective canonical maps φ, φ' to $B\mu_2$. Instead, we have the following diagram, where the leftmost and rightmost squares are pullbacks:

$$\begin{tikzcd}
E \times \mu_2 \arrow[r] \arrow[d] & E \arrow[d, "\pi"] & E' \arrow[d, "\pi'"'] & E' \times \mu_2 \arrow[l] \arrow[d] \\
E \arrow[r, "\pi"] \arrow[d] & {[E/\mu_2]} \arrow[r, dash, "\sim"] \arrow[d, "\varphi"] & {[E'/\mu_2]} \arrow[d, "\varphi'"'] & E' \arrow[l, "\pi'"'] \arrow[d] \\
\operatorname{Spec} k \arrow[r, "i_0"] & B\mu_2 & B\mu_2 \arrow[l, "\sim", "\alpha"'] & \operatorname{Spec} k \arrow[l, "i_0"']
\end{tikzcd}$$

Recall that if we have a G-torsor X/k corresponding to a map $x : \operatorname{Spec} k \to BG$, then x factors through $\operatorname{Spec} k \to BG \to BG$ where the first map is the canonical one $x_0 : \operatorname{Spec} k \to BG$ corresponding to the trivial torsor $\operatorname{Spec} k \times G \to \operatorname{Spec} k$, and the second one sends a G-torsor $T \to S$ to the G-torsor $T \times_G X_S \to S$. Here, $\alpha : B\mu_2 \to B\mu_2$ is the isomorphism constructed as above for the μ_2-torsor k'/k, so that $i' = \alpha \circ i_0$. Since pulling back the torsor $\operatorname{Spec} k' \to \operatorname{Spec} k$ to itself trivializes it, if we pull back the diagram above by $\operatorname{Spec} k' \to \operatorname{Spec} k$, α becomes trivial and we obtain an isomorphism $E' \to E$ over k'.

In elementary terms, given the description of $[E/\mu_2]$ in terms of the ramification locus $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ in $\mathbb{P}^1$ the choice of forgetting map $[E/\mu_2] \to B\mu_2$ (hence of elliptic curve E) amounts to the choice of $B \in k^*/k^{*,2}$: given $E : By^2 = (x - \alpha_1)(x - \alpha_2)(x - \alpha_3)(x - \alpha_4) = h(x)$, the map $[E/\mu_2] \to B\mu_2$ is simply the map that sends $[P] : \operatorname{Spec} k \to [E/\mu_2]$ to the μ_2-torsor defined by $By^2 = h(x(P))$ (i.e. by $\pi^{-1}(x(P))$ where $\pi : E \to E/\pm 1$ is the canonical projection), where $x(P)$ is the image of $[P]$ composed with the forgetting map $[E/\mu_2] \to E/\mu_2 \simeq \mathbb{P}^1$ (this assumes that $x(P)$ is not in the ramification locus, in which case $[E/\mu_2]$ is étale locally isomorphic to E/μ_2 around $[P]$).

Once we have fixed E, hence the map $[E/\mu_2] \to B\mu_2$, then given $[P] : \operatorname{Spec} k \to [E/\mu_2]$ (not above the ramification), the torsor $\pi^{-1}([P]) = \pi^{-1}(x(P))$ associated to the map $i : \operatorname{Spec} k \to [E/\mu_2] \to B\mu_2$ determines if $x(P)$ lifts to two rational points $\pm P$ in $E(k)$. This is the case precisely when the torsor is trivial, i.e. isomorphic to $\operatorname{Spec} k \times \mu_2$. In the other cases it corresponds to a quadratic extension k'/k, and $\pm P$ live in $E(k')$. Alternatively, by the diagram above, $x(P)$ lifts to a rational point in the quadratic twist E' defined by k' (because the map $i' : \operatorname{Spec} k \to [E/\mu_2] \to B\mu_2 \to B\mu_2$, where the last map is the isomorphism induced by the torsor k'/k, corresponds to the trivial torsor). In particular, this quadratic twist E' is the unique one to which $x(P)$ lifts (over k).

In summary, the fibres of $E \to [E/\mu_2] \to E/\mu_2$ are as follows. If P is not above the ramification, we have the following pullback diagram:

$$\begin{tikzcd}
\pi^{-1}([P]) = \pi^{-1}(x(P)) = \{\pm P\} \arrow[r] \arrow[d] & E \arrow[d, "\pi"'] \\
\operatorname{Spec} k \arrow[r, "{[P]}"] \arrow[d] & {[E/\mu_2]} \arrow[d] \\
\operatorname{Spec} k \arrow[r, "x(P)"] & E/\mu_2
\end{tikzcd}$$

And if $P = T$ is a 2-torsion point, we have instead the pullback diagram:

$$\begin{tikzcd}
\operatorname{Spec} k \arrow[r, "T"] \arrow[d] & E \arrow[d, "\pi"'] \\
B\mu_2 \arrow[r] \arrow[d] & [E/\mu_2] \arrow[d] \\
\operatorname{Spec} k \arrow[r, "x(T)"] & E/\mu_2
\end{tikzcd}$$

8.8 MODULAR CURVES

In Section 8.3.2, we used the construction of modular curves over $\mathbb{Q}$ from [Zyw15], using the Galois theory of the étale fundamental group.

In this section we briefly describe modular curves as stacks, and refer to [DR73] for more details. First we start with the modular curves $X(n)$ as the fine moduli space (Deligne-Mumford stack) parametrizing elliptic curves with a level-n structure. There are two definition in the literature: the first one is that of a basis (P_1, P_2) of $E[n]$ such that the Weil pairing $e_{W,n}(P_1, P_2) = \zeta$ for $\zeta \in \mu_n$ a fixed primitive n-th root of unity. The corresponding curve is then defined (and smooth) over $\mathbb{Z}[1/n](\mu_n)$.

The second definition, more general, is to instead ask for an isomorphism $E[n] \to \mathbb{Z}/n\mathbb{Z} \times \mu_n$. The resulting curve $X(n)$ is this time defined (and smooth) over $\mathbb{Z}[1/n]$, but it is not geometrically connected: each choice of primitive n-th root gives a connected geometric component.

Now, if $\Gamma \subset \Gamma(1)$ is some level subgroup of $\Gamma(1) = SL_2(\mathbb{Z})$, let n such that $\Gamma(n) \subset \Gamma$, and define $X(\Gamma)$ as the stacky quotient $[X(n)/(\Gamma/\Gamma(n))]$. It is not hard to see that $X(\Gamma)$ is independent of the choice of n.

In particular, this gives a construction of $X(\Gamma^0(4))$, $X(\Gamma^0(4) \cap \Gamma(2))$ and $X(\Gamma(2,4))$ valid over $\mathbb{Z}[1/2]$, as used in Section 8.3.

Since our moduli curves are now fine moduli space, we have a universal elliptic curve (stack) with Γ-level structure over $X(\Gamma)$. Its automorphism group (preserving the level structure) is either μ_2 if $-1 \notin \Gamma$, or trivial if $-1 \in \Gamma$. In the latter case, the generic inertia of $X(\Gamma)$ is then trivial, hence $X(\Gamma)$ is a scheme, étale locally at all points that do not admit extra automorphisms, so concretely for all elliptic curves (with level structure) with j-invariant different from $0, 1728$. If we require furthermore that $\det(\Gamma/\Gamma(n)) = (\mathbb{Z}/N\mathbb{Z})^*$, then $X(\Gamma)$ is geometrically connected. We recover the conditions from Theorem 8.3.6.

It remains to tackle the question of fields of moduli vs field of definition. In general, if X is a fine Deligne-Mumford moduli stack parametrizing a class of objects, then by definition to give such an object C/k is the same as giving a point $\operatorname{Spec} k \to X$. We say that k is a field of definition of C. The fine space X also admits a coarse space X_0 (an algebraic space in general, but for our modular curves their coarse spaces are schemes), and the point $\operatorname{Spec} k \to X$ induces a point x_0 on X_0; we call the field of moduli the residual field k_0 of

x_0. For instance for an elliptic curve E/k, its field of moduli is given by its j-invariant $j(E)$.

In general, the field of moduli is not a field of definition. Assume that we have two models C_1/k_1, C_2/k_2 that become isomorphic over a common algebraic closure of k_1, k_2 (so give points on X above the same point x_0 on X_0). Let us call $C/\overline{k}$ a representative of the isomorphism class of C_1, C_2 over $\overline{k}$, then by descent theory (assuming all our fields are perfect and our inertia tame to avoid complications with inseparability) C_1/k_1 is given by a descent data from $C/\overline{k}$. But the two descent data defining C_1, C_2 may not be compatible with each other, hence may not 'glue' to define a curve over $k_1 \cap k_2$.

From the stack point of view the way to understand the situation is as follows: we have a residual gerbe G_{x_0} above x_0; informally this gerbe is the category of all models C_i/k_i that are isomorphic to C over $\overline{k}$. If $C_1, C_1'/k_1$ correspond to two elements of the gerbe above the same field, then by definition they are twist of each others. So étale locally around our field of definition k_1, our category is equivalent to the category of $\mathrm{Aut}(C_1)$-torsors ($\mathrm{Aut}(C_1)$ is precisely the inertia of the corresponding point $\operatorname{Spec} k_1 \to X$): indeed the category of twists of C_1 is equivalent to the category of $\mathrm{Aut}(C_1)$-torsors. But our gerbe G_{x_0} itself may not be equivalent to a category of torsor (i.e. neutral), unless k_0 is also a field of definition, not only a field of moduli.

Now, a very nice feature of modular curves, is that all their residual gerbes are neutral:

Proposition 8.8.1 *Let $X(\Gamma)$ be the modular stack of elliptic curves with a Γ-level structure as above, and $x : \operatorname{Spec} k \to X(\Gamma)$ a point. Then the residual gerbe at x is trivial: the field of moduli is a field of definition.*

Proof *This is [DR73, Prop. 3.2, p. 274]. For the convenience of the reader we summarize the proof. First the result is well known for $\Gamma = \{\mathrm{id}\}$: we have explicit formulas that give an equation of an elliptic curve from its j-invariant (over the same field).*

Now if we add a level structure, then x corresponds to a curve with level structure, (E, G), and we distinguish several cases. If $\mathrm{Aut}(E, G) = \mathrm{Aut}(E)$, then the obstruction for the residual gerbe at (E, G) being neutral, which lives in $H^2(k, \mathrm{Aut}(E, G))$ is the same (via the isomorphism above) to the obstruction for the residual gerbe at E to be neutral, which lives in $H^2(k, \mathrm{Aut}(E))$. But the latter is trivial by the first point, so the former too.

Otherwise, we at least have an injection $\mathrm{Aut}(E, G) \subset \mathrm{Aut}(E)$, and a map $H^2(k, \mathrm{Aut}(E, G)) \to H^2(k, \mathrm{Aut}(E))$. The image of the obstruction related to (E, G) in $H^2(k, \mathrm{Aut}(E))$ is zero by the first point. But a difficulty is that the map on the H^2 need not be injective.

If $\mathrm{Aut}(E) = \mu_n$ is isomorphic to a group of roots of unity, then it is easy to see using the long exact sequence of cohomology and Hilbert's 90 that $H^2(k, \mathrm{Aut}(E, G)) \to H^2(k, \mathrm{Aut}(E))$ is injective in that case.

The other cases are all induced by $j(E) = 0, 1728$ and the characteristic is $p = 2, 3$, in which case E is defined over $\mathbb{F}_p$, hence (E, G) is defined over a finite field $\mathbb{F}_q$. But finite fields have cohomological dimension 1, hence do not admit non-trivial gerbes. $\square$

This finishes the generalization of Theorem 8.3.6. We stress that this feature is specific to the dimension 1 case, in [BL24] it is shown that moduli space of abelian varieties can have arbitrary bad obstruction: any gerbe (over one point) lives in the moduli stack $\mathcal{A}_g$ of principally polarized abelian variety of dimension g for g large enough.

We conclude this section wit a discussion about twists. If $E \to X(1)$ is the universal elliptic curve, then $\mathrm{Aut}(E) = \mu_2$: the generic inertia of a point $\mathrm{Spec}\, k \to X(1)$ is μ_2. One can then form (using the construction from [AOV08, App. A]) the quotient $E/\mu_2 \to X(1)/\mu_2$ to obtain the universal Kummer line; we remark that $X(1)$ and $X(1)/\mu_2$ have the same coarse moduli space (which is isomorphic to $\mathbb{P}^1$ via the j-invariant). A similar construction holds for the universal elliptic curve with level structure over $X(\Gamma)$ when $-1 \notin \Gamma$.

Consequently, for a generic Kummer line $E/\pm 1$, $\mathrm{Aut}(E/\pm 1) = \mathrm{Aut}(E)/\pm 1$ is trivial (this also holds in higher dimension) and a (generic) Kummer line does not have twists: if we have two models of Kummer lines defined over k and isomorphic over $\overline{k}$, then this isomorphism is automatically rational.

The exceptions are as always the Kummer lines associated to the elliptic curves of j-invariant $0, 1728$.

Example 8.8.2 (Twists of Kummer lines) *We first look at the possible twists of the Kummer line of $E : y^2 = x^3 - x$ of j-invariant 1728; we will assume that k is not of characteristic 2 for simplicity. Its ramification is given by, along the point at infinity, $x = -1, 0, 1$. The map $x \mapsto -x$ is a non-trivial automorphism of $E/\pm 1$, so the Kummer line admits quadratic twists.*

Explicitly, the curves $E_a : y^2 = x^3 + ax$ are all isomorphic to E over $\overline{k}$. The corresponding Kummer line has ramification given by $x = 0, a^{1/2}, -a^{1/2}$ (and the point at infinity).

If $\xi = (a_2/a_1)^{1/2}$, then the map $x \mapsto \xi x$ is an isomorphism between the Kummer line of E_{a_1} and the Kummer line of E_{a_2}. But this isomorphism is rational if and only if ξ is rational. So each element in $k^*/k^{*,2} \simeq H^1(k, \mu_2)$ gives a different quadratic twist.

$E/\pm 1$ has a Legendre form with $\lambda = -1$, the other Legendre invariants are $\lambda = 2, 1/2$ (we only have 3 invariants because of the extra automorphism). The other quadratic twists of this Kummer line do not have rational 2-torsion, hence no Legendre form (nor theta form). It also has a theta model (possibly over an extension) with theta constants $(\zeta_8 : 1)$, where ζ_8 is a primitive 8-th root of unity. The other theta constants are: $(\zeta_8^3 : 1), (\zeta_8^5 : 1), (\zeta_8^7 : 1), (\zeta_8 + 1 : \zeta_8 - 1), (\zeta_8^3 + 1 : \zeta_8^3 - 1), (\zeta_8^5 + 1 : \zeta_8^5 - 1), (\zeta_8^7 + 1 : \zeta_8^7 - 1), (\zeta_8(1 + \zeta_8) : \zeta_8(1 - \zeta_8)), (\zeta_8^3(1 + \zeta_8^3) : \zeta_8^3(1 - \zeta_8^3)), (\zeta_8^5(1 + \zeta_8^5) : \zeta_8^5(1 - \zeta_8^5)), (\zeta_8^7(1 + \zeta_8^7) : \zeta_8^7(1 - \zeta_8^7))$.

We can recover possible Montgomery models (possibly over an extension) associated to these three Legendre models of $E/\pm 1$ by letting $\alpha = \sqrt{\lambda}$ and $\mathcal{A} = -\alpha - 1/\alpha$. For $\lambda = -1$, we find $\alpha = i$ and $\mathcal{A} = 0$. For $\lambda = 2$, we find $\alpha = \sqrt{2}$ and $\mathcal{A} = -3\sqrt{2}/2$, or $\alpha = -\sqrt{2}$ and $\mathcal{A} = 3\sqrt{2}/2$. But $E/\pm 1$ can be isomorphic to the first Montgomery model over the base field only if -1 is a square in k, and to the second and third only when 2 is a square in k.

Now we look at the twists of the Kummer line of $E : y^2 = x^3 - 1$ of j-invariant 0; we will assume that k is not of characteristic 3 for simplicity. The ramification is given by the point at infinity and $x = 1, j, j^2$ where j is a 3-rd root of unity.

If $\mu_3 \subset k$, then $x \mapsto jx$ is an automorphism of the Kummer line, so it admits cubic twists. The curves $E_b : y^2 = x^3 + b$ are all isomorphic to E over $\overline{k}$. The corresponding Kummer line has ramification given by $b^{1/3}, jb^{1/3}, j^2b^{1/3}$ (and the point at infinity). If $\xi = (b_2/b_1)^{1/3}$, then the map $x \mapsto \xi x$ is an isomorphism between the Kummer line of E_{b_1} and the Kummer line of E_{b_2}. But this isomorphism is rational if and only if ξ is rational. So each element in $k^*/k^{*,3} \simeq H^1(k, \mu_3)$ gives a different cubic twist.

If $\mu_3 \not\subset k$, then over k the Kummer line does not have automorphisms and do not admit twists over k; but it will admit twists over $k(j)$.

$E/\pm 1$ has a Legendre form with $\lambda = \zeta_6$, where $\zeta_6^6 = 1$ is a primitive 6-th root of unity. The other Legendre invariant is ζ_6^5 (the automorphism is of order 3, so there are only two Legendre invariants).

8.9 GRAPH OF THE SUBGROUPS OF $SL_2(\mathbb{Z}/4\mathbb{Z})$

Let $f : SL_2(\mathbb{Z}) \twoheadrightarrow SL_2(\mathbb{Z}/4\mathbb{Z})$ be the reduction map modulo 4. For $M, N \geqslant 1$ two integers, we will deal with the following groups:

- $\Gamma(N) = \{g \in SL_2(\mathbb{Z}) \mid g \equiv I \bmod N\}$
- $\Gamma_1(N) = \{g \in SL_2(\mathbb{Z}) \mid g \equiv \left(\begin{smallmatrix} 1 & * \\ 0 & 1 \end{smallmatrix}\right) \bmod N\}$
- $\Gamma_0(N) = \{g \in SL_2(\mathbb{Z}) \mid g \equiv \left(\begin{smallmatrix} * & * \\ 0 & * \end{smallmatrix}\right) \bmod N\}$
- $\Gamma(M, MN) = \{g \in SL_2(\mathbb{Z}) \mid g \equiv \left(\begin{smallmatrix} * & * \\ 0 & * \end{smallmatrix}\right) \bmod MN, g \equiv I \bmod M\} = \Gamma(M) \cap \Gamma_0(MN)$

Column on the left list the order of the groups. The notations on the Fig. 8.5 are as follows for a group ${}_cG_{\texttt{idx}}$:

- The relevant G will be of the form $f(H)$ where H is a subgroup of $\Gamma(4)$.
- For the groups that are not of this form, they are written G_a where a is the index in Magma when listing subgroups of $SL_2(\mathbb{Z}/4\mathbb{Z})$.
- c is the number of conjugacy classes of G in $SL_2(\mathbb{Z}/4\mathbb{Z})$.
- If present, `idx` is the label of the corresponding modular curve on the LMFDB (beta feature as of writing).

Remark 8.9.1 *Note that, aside from level 2 groups which can be seen as subgroups of $GL_2(\mathbb{Z}/2\mathbb{Z})$ (and therefore trivially verifies $-I \in G$ and $\det G = (\mathbb{Z}/2\mathbb{Z})^*$), none of these respect the conditions of Theorem 8.3.6 (indeed, $\det G = \{1\} \neq (\mathbb{Z}/4\mathbb{Z})^*$, and some of them doesn't have $-I \in G$, like $\Gamma_1(4)$). It is then implicit that we consider the smallest subgroup H in $GL_2(\mathbb{Z}/4\mathbb{Z})$ that contains G, $-I$ and such that $\det H = (\mathbb{Z}/4\mathbb{Z})^*$.*

This amounts to looking at the following representation $\tilde{\rho}_{E,4} : \mathrm{Gal}(\mathbb{Q}/\mathbb{Q}(\zeta_4)) \to SL_2(\mathbb{Z}/4\mathbb{Z})$ rather than $\rho_{E,4} : \mathrm{Gal}(\mathbb{Q}/\mathbb{Q}) \to GL_2(\mathbb{Z}/4\mathbb{Z})$ as defined in Definition 8.3.5.

$SL_2(\mathbb{Z}/4\mathbb{Z})$ is labelled 48.30 on LMFDB and $GL_2(\mathbb{Z}/4\mathbb{Z})$ is 96.195 (beta feature as of writing).

Red nodes correspond to the models studied in this chapter, which are from top to bottom: Montgomery, Theta twisted, Theta.

We also have:

- $G_{18} = \langle \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right), \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right), \left(\begin{smallmatrix} 3 & 0 \\ 2 & 3 \end{smallmatrix}\right), \left(\begin{smallmatrix} 3 & 2 \\ 0 & 3 \end{smallmatrix}\right) \rangle$.
- $G_{16} = \langle \left(\begin{smallmatrix} 1 & 3 \\ 2 & 3 \end{smallmatrix}\right), \left(\begin{smallmatrix} 3 & 0 \\ 0 & 3 \end{smallmatrix}\right), \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right) \rangle$.
- $G_{15} = \langle \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right), \left(\begin{smallmatrix} 3 & 0 \\ 2 & 3 \end{smallmatrix}\right), \left(\begin{smallmatrix} 3 & 2 \\ 0 & 3 \end{smallmatrix}\right) \rangle$.
- $G_{13} = \langle \left(\begin{smallmatrix} 1 & 3 \\ 2 & 3 \end{smallmatrix}\right), \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right), \left(\begin{smallmatrix} 3 & 2 \\ 0 & 3 \end{smallmatrix}\right) \rangle$.
- $G_{11} = \langle \left(\begin{smallmatrix} 3 & 0 \\ 0 & 3 \end{smallmatrix}\right), \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right) \rangle$.
- $G_9 = \langle \left(\begin{smallmatrix} 1 & 3 \\ 2 & 3 \end{smallmatrix}\right), \left(\begin{smallmatrix} 3 & 0 \\ 0 & 3 \end{smallmatrix}\right) \rangle$.
- $G_8 = \langle \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right), \left(\begin{smallmatrix} 3 & 0 \\ 2 & 3 \end{smallmatrix}\right) \rangle$.
- $G_6 = \langle \left(\begin{smallmatrix} 3 & 0 \\ 2 & 3 \end{smallmatrix}\right), \left(\begin{smallmatrix} 3 & 2 \\ 0 & 3 \end{smallmatrix}\right) \rangle$.
- $G_5 = \langle \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right) \rangle$.
- $G_4 = \langle \left(\begin{smallmatrix} 3 & 0 \\ 2 & 3 \end{smallmatrix}\right) \rangle$.

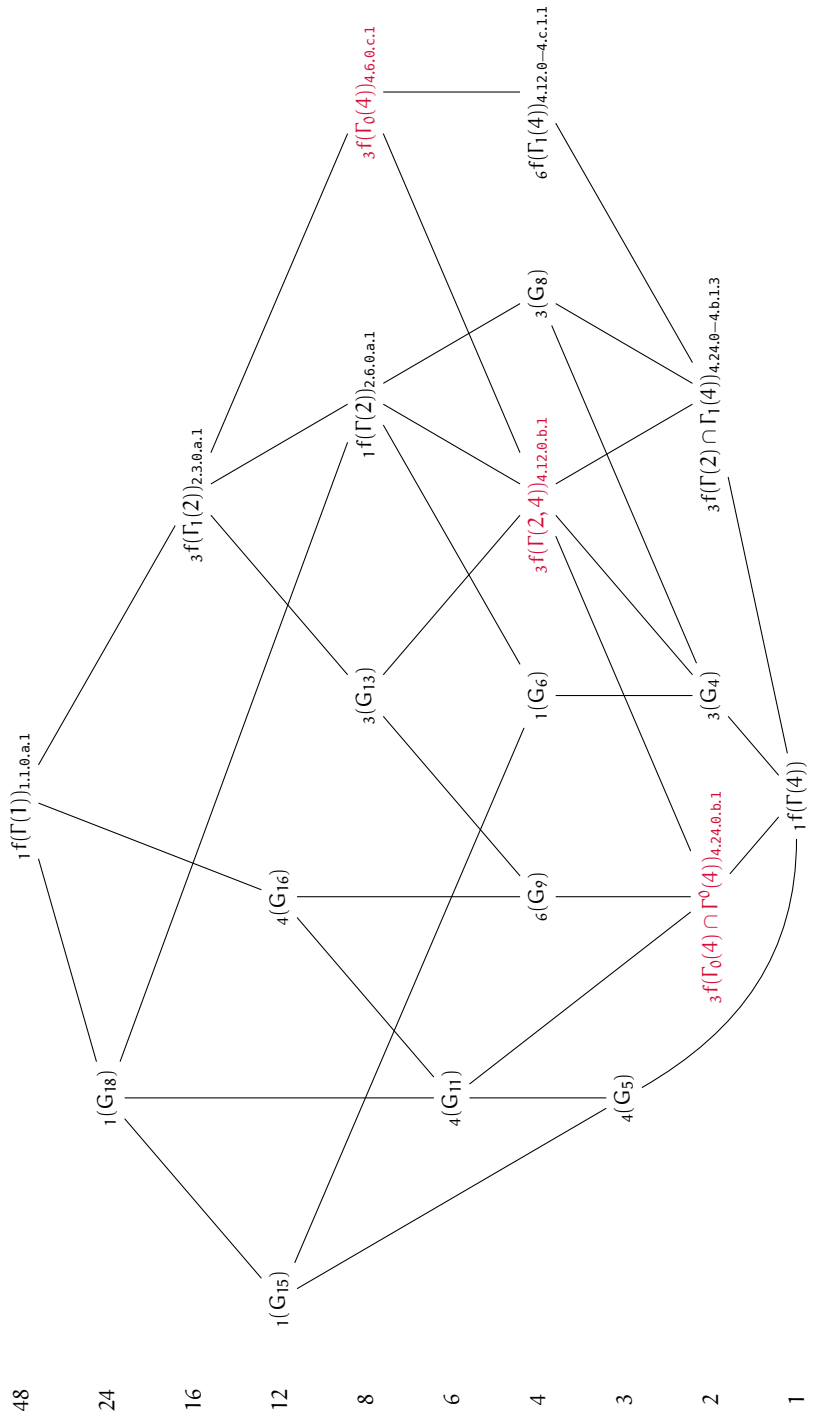


Figure 8.5: Subgroups of $SL_2(\mathbb{Z}/4\mathbb{Z})$

PERSPECTIVES

This thesis, through Chapters 6 to 8, aimed at studying various models of Kummer lines and their arithmetic, or equivalently models of elliptic curves of level 2, that is given by a choice of basis of $\Gamma(2(0_E))$. Since coordinates of level 2 are even, they descend to the Kummer line.

A unifying theme in this thesis was the use of the theta group $G(2(0_E))$ and its action on the sections $\Gamma(2(0_E))$, whether to derive arithmetic formulas, our to classify our models.

From the arithmetic aspect, both ladder we provided require a Montgomery curve with full rational 2-torsion. Unfortunately, we do not have one that has small constants and good cryptographic properties. `Curve25519` is 2-isogenous to a Montgomery curve with full rational 2-torsion, but the constants are quite huge as explained in Section 7.3.5. A first perspective would then be to look for such a curve.

There are then two natural generalizations to this work. The first one is to stay in dimension 1 but to increase the level, in particular to obtain models of the elliptic curve itself. It is shown in [LR16] that if $\pi : E \to E/\pm 1 \simeq \mathbb{P}^1$ is the Kummer line, and $P_0 \in E(k)$ is not a 2-torsion point, then $P \mapsto (\pi(P), \pi(P + P_0)) \in \mathbb{P}^1 \times \mathbb{P}^1$ is an embedding. If P_0 is a point of n-torsion, taking coordinates on $E/\pm 1$ of level 2, this gives coordinates of level $\mathrm{lcm}(2, n)$. The most well known example of this embedding is given by Edwards curves [BL07a], or more precisely the completed Edwards curve [Ber+13, § 2.7]. Indeed, up to a small change of variable, the completed Edwards model is precisely given by the embedding $E \to \mathbb{P}^1 \times \mathbb{P}^1$, $P \mapsto ((X(P) : Z(P)), X(P+T) : Z(P+T))$ where (X, Z) are the Kummer line coordinates of a Montgomery model, and $T = (1 : 1)$ is the canonical 4-torsion point (this seems to have been first noticed by Kohel in [Koh11]). It is therefore natural to look at similar level 4 embeddings via the other Kummer models studied in Section 8.1.

The other natural generalization will be to stay in level 2 but to move from dimension 1 to 2. First, as pointed out in Section 7.6, we have formulas which could be used as half differential addition ones, and therefore we could adapt our half ladder to dimension 2, with good hopes of improving the arithmetic. This could have some interest in cryptography as some protocols use abelian surfaces, see for instance qDSA [RS17].

Furthermore, the Kummer surface $X/\pm 1$ (where X is an abelian surface) can be described as a surface in $\mathbb{P}^3$. Similarly to Kummer lines, there is a combinatorial description of the Kummer surfaces [Gau07], namely via the $(16, 6, 2)$-design induced by their 2-torsion points and (translation) of their Θ-divisor. Unfortunately, this description is less tractable than the simple 4 ramification points in $\mathbb{P}^1$ we had in dimension 1. Still, the theta model is characterized by symmetry conditions on the (ramification locus of) Kummer surface: this time it has to satisfy four different symmetries at the same time, while we only required $(X : Z) \mapsto (-X : Z)$ and $(X : Z) \mapsto (Z : X)$ in dimension

1. It could then be useful to look at models of Kummer surfaces where the theta group is a twist of the Heisenberg group, as in Section 8.4.2.

Finally, Humbert surfaces are objects which classify principally polarized abelian surfaces, they have been extensively studied by Kani, and we discussed at some point to look into how to compute some invariants of these surfaces.

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